

Answers to Odd-Numbered Exercises and Tests

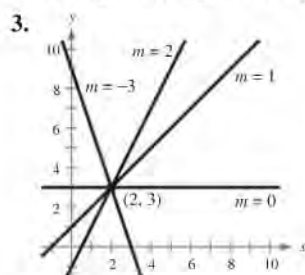
Chapter 1

Section 1.1 (page 11)

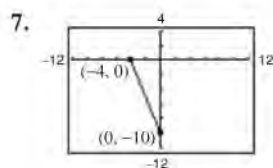
Vocabulary Check (page 11)

1. (a) iii (b) i (c) v (d) ii (e) iv
2. slope 3. parallel 4. perpendicular
5. linear extrapolation

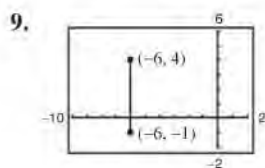
1. (a) L_2 (b) L_3 (c) L_1



5. $\frac{3}{2}$



$m = -\frac{5}{2}$



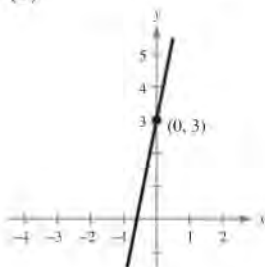
m is undefined.

11. $(0, 1), (3, 1), (-1, 1)$ 13. $(1, 4), (1, 6), (1, 9)$

15. $(-1, -7), (-2, -5), (-5, 1)$

17. $(3, -4), (5, -3), (9, -1)$

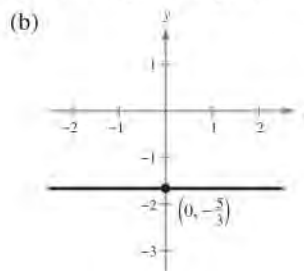
19. (a) $m = 5$;
y-intercept: $(0, 3)$
(b)



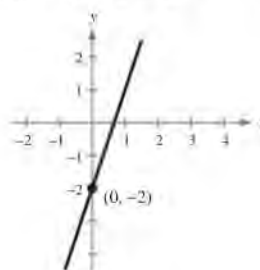
21. (a) Slope is undefined;
there is no y-intercept.
(b)



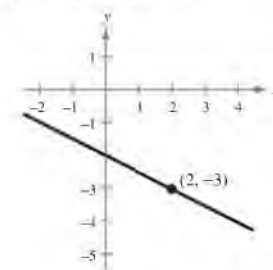
23. (a) $m = 0$; y-intercept: $(0, -\frac{5}{3})$



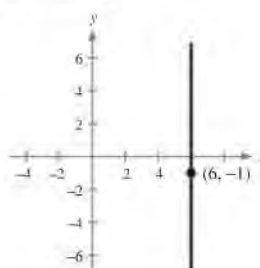
25. $3x - y - 2 = 0$



27. $y + \frac{1}{2}x + 2 = 0$



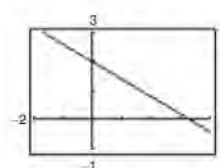
29. $x - 6 = 0$



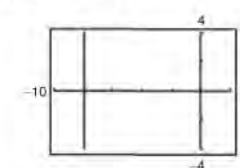
31. $y - \frac{3}{2} = 0$



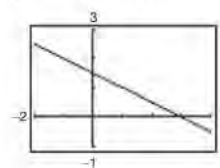
33. $y = -\frac{3}{5}x + 2$



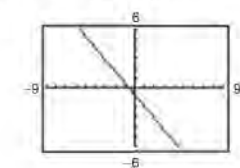
35. $x + 8 = 0$



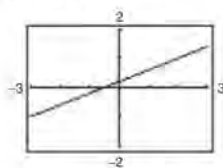
37. $y = -\frac{1}{2}x + \frac{3}{2}$



39. $y = -\frac{6}{5}x - \frac{18}{25}$



41. $y = \frac{2}{5}x + \frac{1}{5}$



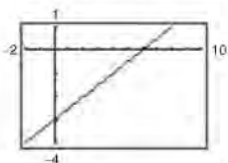
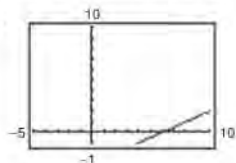
43. $y = 2x - 5$

45. \$37,300

47. $m = \frac{1}{2}$; y-intercept: $(0, -2)$; a line that rises from left to right

49. Slope is undefined; no y-intercept; a vertical line at $x = -6$

51.



The second setting gives a more complete graph, with a view of both intercepts.

53. Perpendicular

55. Parallel

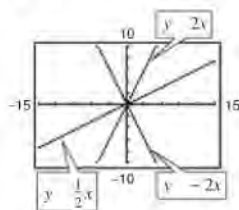
57. (a) $y = 2x - 3$ (b) $y = -\frac{1}{2}x + 2$

59. (a) $y = -\frac{3}{4}x + \frac{3}{8}$ (b) $y = \frac{4}{3}x + \frac{127}{72}$

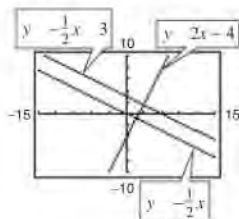
61. (a) $x = 3$ (b) $y = -2$

63. $y = 2x + 1$ 65. $y = -\frac{1}{2}x + 1$

67. The lines $y = \frac{1}{2}x$ and $y = -2x$ are perpendicular.



69. The lines $y = -\frac{1}{2}x$ and $y = -\frac{1}{2}x + 3$ are parallel. Both are perpendicular to $y = 2x - 4$.



71. (a) The greatest increase (+0.86) was from 1998 to 1999 and the greatest decrease (-0.78) was from 1999 to 2000.

(b) $y = -0.067t + 1.24$, $t = 5$ corresponds to 1995.

(c) There is a decrease of about \$0.067 per year.

(d) -0.1, Answers will vary.

73. 12 feet 75. $V = 125t + 1790$

77. $V = -2000t + 32,400$

79. b; slope = -10; the amount owed decreases by \$10 per week.

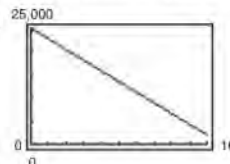
80. c; slope = 1.5; 1.50; the hourly wage increases by \$1.50 per unit produced.

81. a; slope = 0.35; expenses increase by \$0.35 per mile.

82. d; slope = -100; the value depreciates \$100 per year.

83. (a) $V = 25,000 - 2300t$

(b)



t	0	1	2	3	4
V	25,000	22,700	20,400	18,100	15,800

t	5	6	7	8	9	10
V	13,500	11,200	8900	6600	4300	2000

85. (a) $C = 16.75t + 36,500$ (b) $R = 27t$

(c) $P = 10.25t - 36,500$ (d) $t \approx 3561$ hours

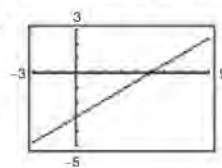
87. (a) Increase of about 341 students per year

(b) 72,962; 77,395; 78,418

(c) $y = 341x + 75,008$, where $x = 1$ corresponds to 1991; $m = 341$; the slope determines the average increase in enrollment.

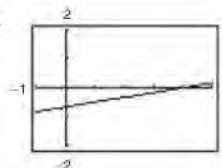
89. False. The slopes $(\frac{2}{7})$ and $(-\frac{11}{7})$ are not equal.

91.



a and b represent the x - and y -intercepts.

93.



a and b represent the x - and y -intercepts.

95. $3x + 2y - 6 = 0$ 97. $12x + 3y + 2 = 0$

99. a 101. c

103. No. Answers will vary. Sample answer: The line, $y = 2$, does not have an x -intercept.

105. Yes. Once a parallel line is established to the given line, there are an infinite number of distances away from that line, creating an infinite number of parallel lines.

107. Yes, $x + 20$ 109. No 111. No

113. $(x - 9)(x + 3)$ 115. $(2x - 5)(x + 8)$

117. Answers will vary.

Section 1.2 (page 24)

Vocabulary Check (page 24)

1. domain, range, function
2. independent, dependent 3. piecewise-defined
4. implied domain 5. difference quotient

1. Yes. Each element of the domain is assigned to exactly one element of the range.
3. No. The National League, an element in the domain, is assigned to three items in the range, the Cubs, the Pirates, and the Dodgers; the American League, an element in the domain, is also assigned to three items in the range, the Orioles, the Yankees, and the Twins.
5. Yes. Each input value is matched with one output value.
7. No. The inputs 7 and 10 are both matched with two different outputs.
9. (a) Function
(b) Not a function because the element 1 in A corresponds to two elements, -2 and 1 , in B .
(c) Function
(d) Not a function because the element 2 in A corresponds to no element in B .
11. Each is a function of the year. To each year there corresponds one and only one circulation.
13. Not a function 15. Function 17. Function
19. Not a function 21. Function 23. Not a function
25. (a) $\frac{1}{5}$ (b) 1 (c) $\frac{1}{4t + 1}$ (d) $\frac{1}{x + c + 1}$
27. (a) 7 (b) -11 (c) $3t + 7$
29. (a) 0 (b) -0.75 (c) $x^2 + 2x$
31. (a) 1 (b) 2.5 (c) $3 - 2|x|$
33. (a) $-\frac{1}{9}$ (b) Undefined (c) $\frac{1}{y^2 + 6y}$
35. (a) 1 (b) -1 (c) $\frac{|t|}{t}$
37. (a) -1 (b) 2 (c) 6

39. (a) 6 (b) 3 (c) 10

41. (a) 0 (b) 4 (c) 17

43.

t	-5	-4	-3	-2	-1
$h(t)$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1

45.

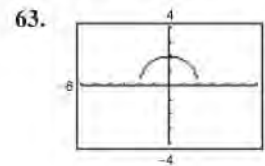
x	-2	-1	0	1	2
$f(x)$	5	$\frac{9}{2}$	4	1	0

47. 5 49. $\frac{4}{3}$ 51. 2, -1 53. All real numbers x

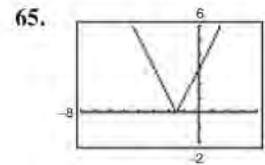
55. All real numbers t except $t = 0$ 57. All real numbers x

59. All real numbers x except $x = 0, -2$

61. All real numbers y such that $y > 10$



Domain: $[-2, 2]$; range: $[0, 2]$



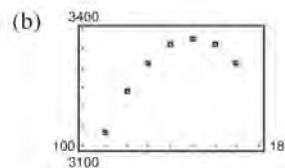
Domain: $(-\infty, \infty)$; range: $[0, \infty)$

67. $\{(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)\}$

69. $\{(-2, 4), (-1, 3), (0, 2), (1, 3), (2, 4)\}$

71. $A = \frac{C^2}{4\pi}$

73. (a) \$3375



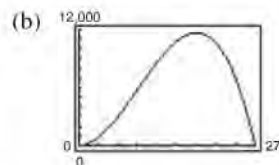
Yes, it is a function.

(c) $P(x) = \begin{cases} 30x, & x \leq 100 \\ 45x - 0.15x^2, & x > 100 \end{cases}$

75. $A = \frac{x^2}{2(x - 2)}, x > 2$

77. (a) $V = 108x^2 - 4x^3$

Domain: all real numbers x such that $0 < x < 27$



$x = 18$ inches, $y = 36$ inches

79. $7 \leq x \leq 12$, $1 \leq x \leq 6$; Answers will vary.

81. 4.63; \$4630 in monthly revenue in November

83.

t	0	1	2	3	4	5	6
$n(t)$	577	647	704	749	782	803	811

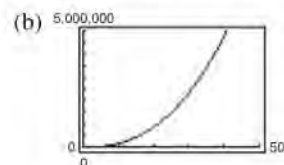
t	7	8	9	10	11	12	13
$n(t)$	846	871	896	921	946	971	996

85. (a)

y	5	10	20
$F(y)$	26,474	149,760	847,170

y	30	40
$F(y)$	2,334,527	4,792,320

Each time the depth is doubled, the force increases by more than 5.7 times.



Xmin = 0
Xmax = 50
Xscl = 10
Ymin = 0
Ymax = 5,000,000
Yscl = 500,000

(c) Depth ≈ 21.37 feet

Use the *trace* and *zoom* features on a graphing utility.

87. $2, c \neq 0$ 89. $3 + h, h \neq 0$ 91. $-\frac{1}{t}, t \neq 1$

93. False. The range is $[-1, \infty)$.

95. $f(x) = \begin{cases} x + 4, & x \leq 0 \\ 4 - x^2, & x > 0 \end{cases}$

97. $f(x) = \begin{cases} 2 - x, & x \leq -2 \\ 4, & -2 < x < 3 \\ x + 1, & x \geq 3 \end{cases}$

99. The domain is the set of input values of a function. The range is the set of output values.

101. $\frac{12x + 20}{x + 2}$ 103. $\frac{(x + 6)(x + 10)}{5(x + 3)}, x \neq 0, \frac{1}{2}$

Section 1.3 (page 38)

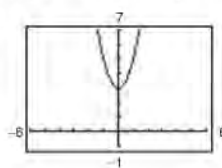
Vocabulary Check (page 38)

- ordered pairs
- Vertical Line Test
- decreasing
- minimum
- greatest integer
- even

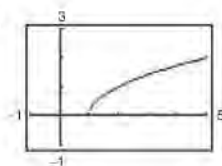
1. Domain: $(-\infty, \infty)$; range: $(-\infty, 1]$; $f(0) = 1$

3. Domain: $[-4, 4]$; range: $[0, 4]$; $f(0) = 4$

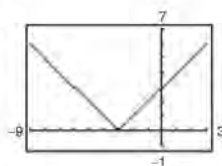
5. Domain: $(-\infty, \infty)$
Range: $[3, \infty)$



7. Domain: $[1, \infty)$
Range: $[0, \infty)$



9. Domain: $(-\infty, \infty)$
Range: $[0, \infty)$



11. (a) $(-\infty, \infty)$ (b) $-2, 3$ (c) x -intercepts
(d) -6 (e) y -intercept (f) $-6; (1, -6)$
(g) $-4; (-1, -4)$ (h) 6

13. (a) $(-\infty, \infty)$ (b) $-1, 3$ (c) x -intercepts
(d) -1 (e) y -intercept (f) $-2; (1, -2)$
(g) $0; (-1, 0)$ (h) 2

15. Function. Graph the given function over the window shown in the figure.

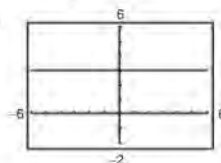
17. Not a function. Solve for y and graph the two resulting functions.

19. Increasing on $(-\infty, \infty)$

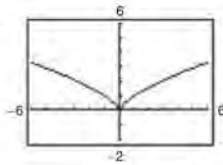
21. Increasing on $(-\infty, 0), (2, \infty)$

Decreasing on $(0, 2)$

23. (a) (b) Constant: $(-\infty, \infty)$

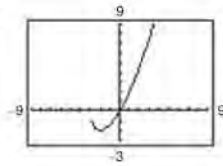


25. (a)



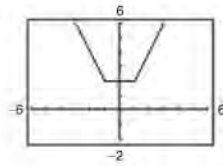
(b) Decreasing on $(-\infty, 0)$
Increasing on $(0, \infty)$

27. (a)



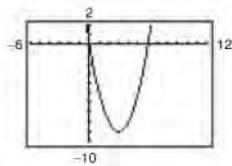
(b) Increasing on $(-2, \infty)$
Decreasing on $(-3, -2)$

29. (a)



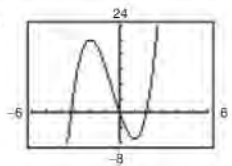
(b) Decreasing on $(-\infty, -1)$
Constant on $(-1, 1)$
Increasing on $(1, \infty)$

31.



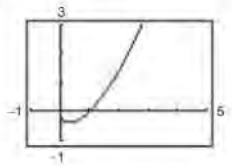
Relative minimum: $(3, -9)$

33.



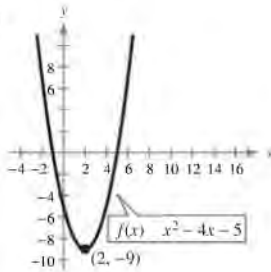
Relative minimum: $(1, -7)$
Relative maximum: $(-2, 20)$

35.



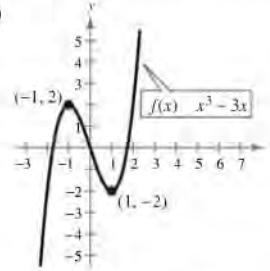
Minimum: $(0.33, -0.38)$

37. (a)



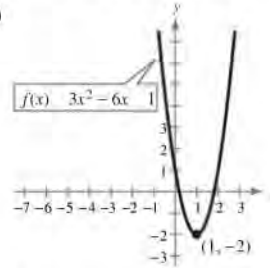
(b) Relative minimum
at $(2, -9)$
(c) Answers are the
same.

39. (a)



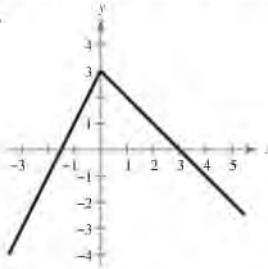
(b) Relative minimum at $(1, -2)$
Relative maximum at $(-1, 2)$
(c) Answers are the same.

41. (a)

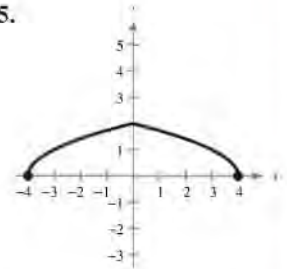


(b) Relative minimum at $(1, -2)$
(c) Answers are the same.

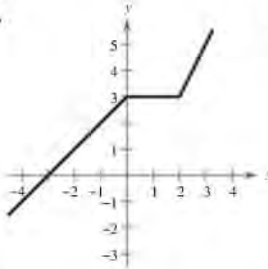
43.



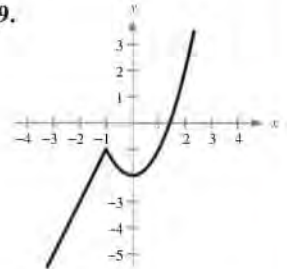
45.



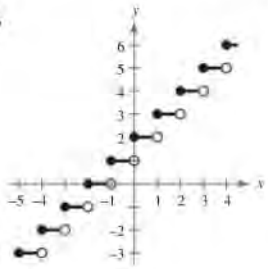
47.



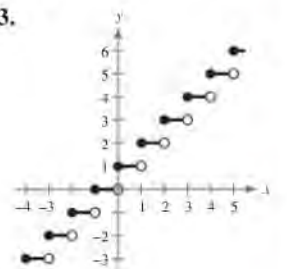
49.



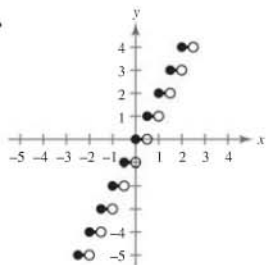
51.



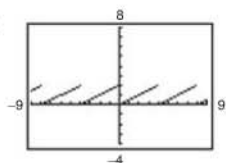
53.



55.



57.


Domain: $(-\infty, \infty)$

Range: $[0, 2)$

Sawtooth pattern

59. Neither even nor odd

61. Odd function

63. Odd function

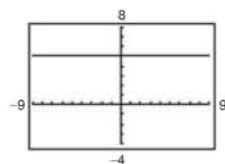
65. Even function

67. (a) $(\frac{3}{2}, 4)$ (b) $(\frac{3}{2}, -4)$

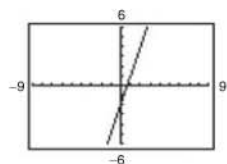
69. (a) $(-4, 9)$ (b) $(-4, -9)$

71. (a) $(-x, -y)$ (b) $(-x, y)$

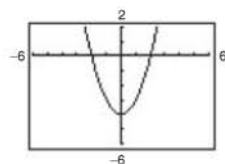
73. Even function



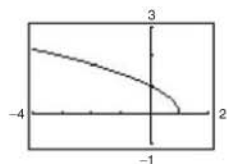
75. Neither even nor odd



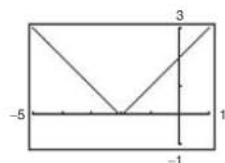
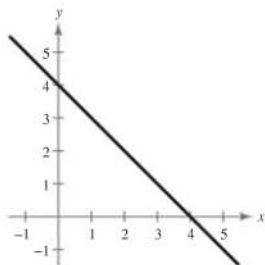
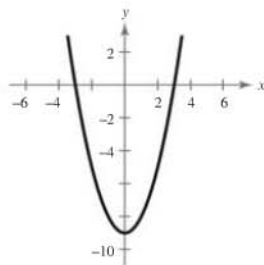
77. Even function



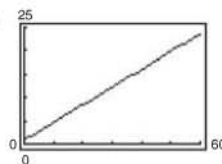
79. Neither even nor odd



81. Neither even nor odd


83. $(-\infty, 4]$

85. $(-\infty, -3], [3, \infty)$

87. (a) C_2 is the appropriate model. The cost of the first minute is \$1.05 and the cost increases \$0.38 when the next minute begins, and so on.

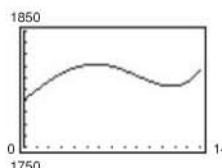
(b)



\$7.89

89. $h = -x^2 + 4x - 3$, $1 \leq x \leq 3$

91. (a)



(b) Increasing from 1990 to 1995 and from 2001 to 2004; decreasing from 1995 to 2001

(c) About 1,821,00.

93. False. Counterexample: $f(x) = \sqrt{1+x^2}$

95. c

96. d

97. b

98. e

99. a

100. f

101. Proof

103. (a) Even. g is a reflection in the x -axis.

(b) Even. g is a reflection in the y -axis.

(c) Even. g is a vertical shift downward.

(d) Neither even nor odd. g is shifted to the right and reflected in the x -axis.

105. No. x is not a function of y because horizontal lines can be drawn to intersect the graph twice. Therefore, each y -value corresponds to two distinct x -values when $-5 < y < 5$.

107. Terms: $-2x^2$, $8x$; coefficients: -2 , 8

109. Terms: $\frac{x}{3}$, $-5x^2$, x^3 ; coefficients: $\frac{1}{3}$, -5 , 1

111. (a) $d = 4\sqrt{5}$ (b) Midpoint: $(2, 5)$

113. (a) $d = \sqrt{41}$ (b) Midpoint: $(\frac{1}{2}, \frac{3}{2})$

115. (a) 29 (b) -6 (c) $5x - 16$

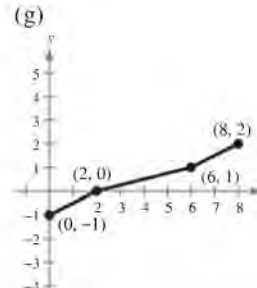
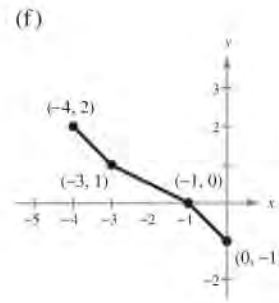
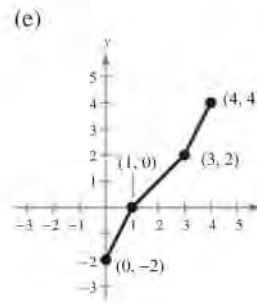
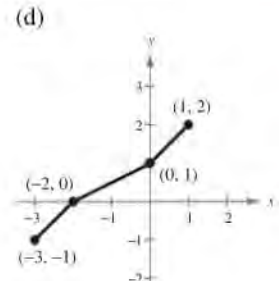
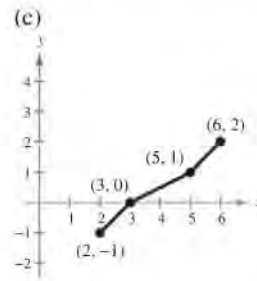
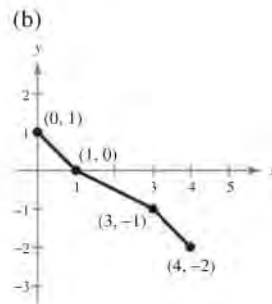
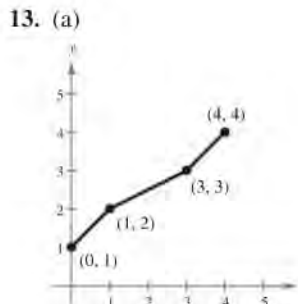
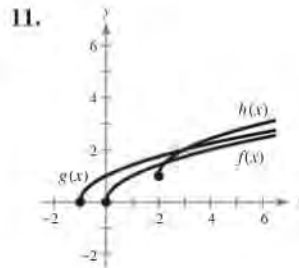
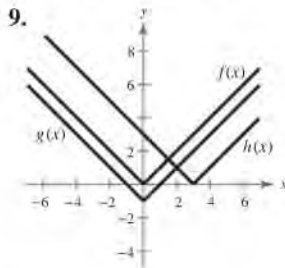
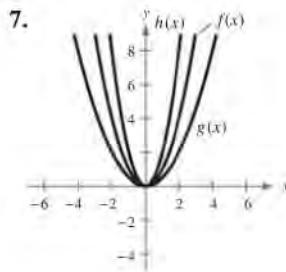
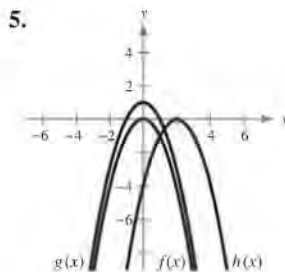
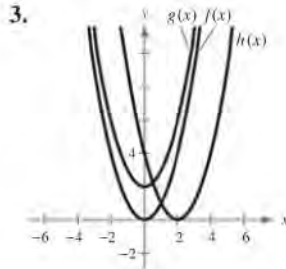
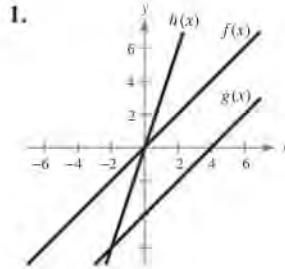
117. (a) 0 (b) 36 (c) $6\sqrt{3}$

119. $h + 4$, $h \neq 0$

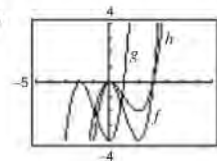
Section 1.4 (page 48)

Vocabulary Check (page 48)

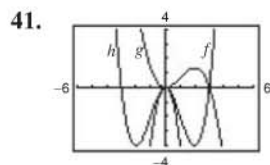
1. quadratic function
2. absolute value function
3. rigid transformations
4. $-f(x), f(-x)$
5. $c > 1, 0 < c < 1$
6. (a) ii (b) iv (c) iii (d) i



15. Vertical shift of $y = x$; $y = x + 3$
17. Vertical shift of $y = x^2$; $y = x^2 - 1$
19. Reflection in the x -axis and a vertical shift one unit upward of $y = \sqrt{x}$; $y = 1 - \sqrt{x}$
21. Reflection in the x -axis and vertical shift one unit downward
23. Horizontal shift two units to the right
25. Vertical stretch
27. Horizontal shift five units to the left
29. Reflection in the x -axis
31. Vertical stretch
33. Reflection in the x -axis and vertical shift four units upward
35. Horizontal shift two units to the left and vertical shrink
37. Horizontal stretch and vertical shift two units upward
- 39.

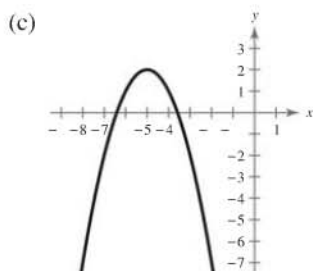


g is a horizontal shift and h is a vertical shrink.



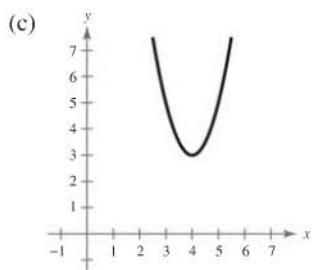
g is a vertical shrink and a reflection in the x -axis and h is a reflection in the y -axis.

43. (a) $f(x) = x^2$
 (b) Horizontal shift five units to the left, reflection in the x -axis, and vertical shift two units upward



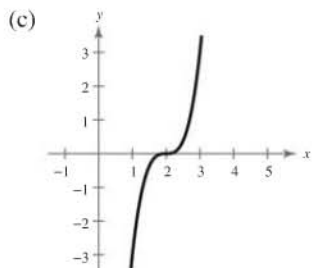
(d) $g(x) = 2 - f(x + 5)$

45. (a) $f(x) = x^2$
 (b) Horizontal shift four units to the right, vertical stretch, and vertical shift three units upward



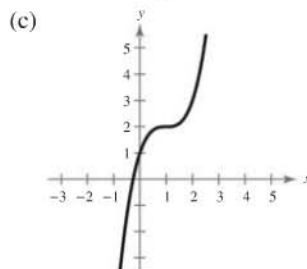
(d) $g(x) = 3 + 2f(x - 4)$

47. (a) $f(x) = x^3$
 (b) Horizontal shift two units to the right and vertical stretch



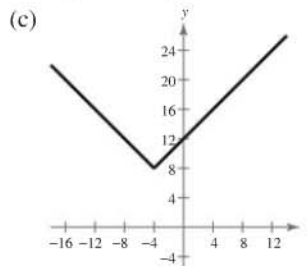
(d) $g(x) = 3f(x - 2)$

49. (a) $f(x) = x^3$
 (b) Horizontal shift one unit to the right and vertical shift two units upward



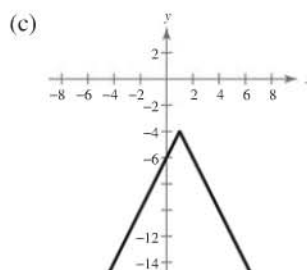
(d) $g(x) = f(x - 1) + 2$

51. (a) $f(x) = |x|$
 (b) Horizontal shift four units to the left and vertical shift eight units upward



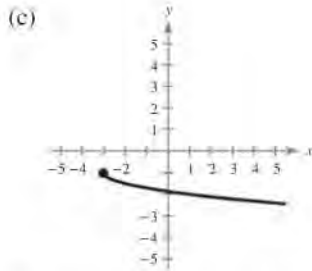
(d) $g(x) = f(x + 4) + 8$

53. (a) $f(x) = |x|$
 (b) Horizontal shift one unit to the right, reflection in the x -axis, vertical stretch, and vertical shift four units downward



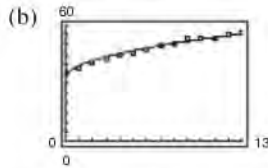
(d) $g(x) = -2f(x - 1) - 4$

55. (a) $f(x) = \sqrt{x}$
 (b) Horizontal shift three units to the left, reflection in the x -axis, vertical shrink, and vertical shift one unit downward



(d) $g(x) = -\frac{1}{2}f(x+3) - 1$

57. (a) Vertical stretch and vertical shift



(c) $G(t) = 33.0 + 6.2\sqrt{t+13}$, where $t = 0$ corresponds to 2003. Answers will vary.

59. False. When $f(x) = x^2$, $f(-x) = (-x)^2 = x^2$. Because $f(x) = f(-x)$ in this case, $y = f(-x)$ is not a reflection of $y = f(x)$ across the x -axis in all cases.

61. (a) $x = -2$ and $x = 3$

(b) $x = -3$ and $x = 2$

(c) $x = -3$ and $x = 2$

(d) Cannot be determined

(e) $x = 0$ and $x = 5$

63. (a) Increasing on the interval $(-\infty, -2)$ and decreasing on the interval $(-2, \infty)$

(b) Increasing on the interval $(2, \infty)$ and decreasing on the interval $(-\infty, 2)$

(c) Increasing on the interval $(-\infty, 2)$ and decreasing on the interval $(2, \infty)$

(d) Increasing on the interval $(-\infty, 2)$ and decreasing on the interval $(2, \infty)$

(e) Increasing on the interval $(-\infty, 1)$ and decreasing on the interval $(1, \infty)$

65. c 67. c 69. Neither

71. All real numbers x except $x = 9$

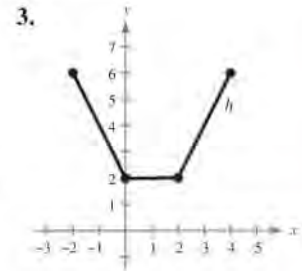
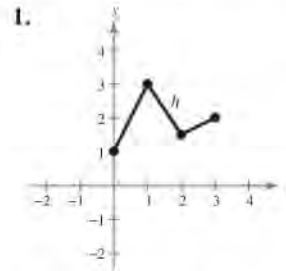
73. All real numbers x such that $-10 \leq x \leq 10$

Section 1.5 (page 58)

Vocabulary Check (page 58)

1. addition, subtraction, multiplication, division

2. composition 3. $g(x)$ 4. inner, outer



5. (a) $2x$ (b) 6 (c) $x^2 - 9$

(d) $\frac{x+3}{x-3}$, All real numbers x , except $x = 3$.

7. (a) $x^2 - x + 1$ (b) $x^2 + x - 1$ (c) $x^2 - x^3$

(d) $\frac{x^2}{1-x}$, All real numbers x , except $x = 1$.

9. (a) $x^2 + 5 + \sqrt{1-x}$ (b) $x^2 + 5 - \sqrt{1-x}$

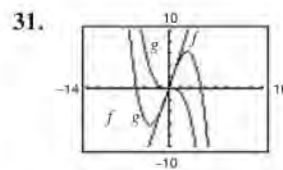
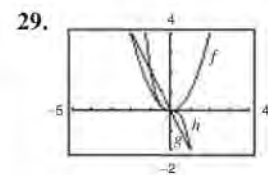
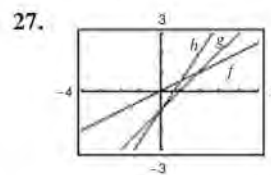
(c) $(x^2 + 5)\sqrt{1-x}$ (d) $\frac{x^2 + 5}{\sqrt{1-x}}$, $x < 1$

11. (a) $\frac{x+1}{x^2}$ (b) $\frac{x-1}{x^2}$ (c) $\frac{1}{x^3}$ (d) x , $x \neq 0$

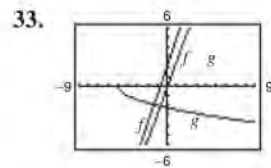
13. 9 15. 1 17. 30 19. $-\frac{24}{7}$

21. $4t^2 - 2t + 1$ 23. $-125t^3 - 50t^2 + 5t + 2$

25. $\frac{t^2 - 1}{-t - 2}$



$f(x)$, $0 \leq x \leq 2$;
 $g(x)$, $x > 6$



$f(x)$, $0 \leq x \leq 2$;
 $f(x)$, $x > 6$

35. (a) $(x-1)^2$ (b) $x^2 - 1$ (c) 1

37. (a) $20 - 3x$ (b) $-3x$ (c) 20

39. (a) All real numbers x such that $x \geq -4$

(b) All real numbers x (c) All real numbers x

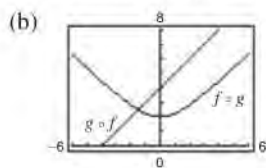
41. (a) All real numbers x
 (b) All real numbers x such that $x \geq 0$
 (c) All real numbers x such that $x \geq 0$

43. (a) All real numbers x except $x = 0$
 (b) All real numbers x
 (c) All real numbers x except $x = -3$

45. (a) All real numbers x (b) All real numbers x
 (c) All real numbers x

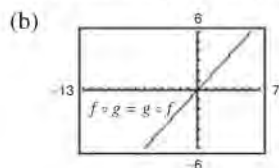
47. (a) All real numbers x
 (b) All real numbers x except $x = \pm 2$
 (c) All real numbers x except $x = \pm 2$

49. (a) $(f \circ g)(x) = \sqrt{x^2 + 4}$
 $(g \circ f)(x) = x + 4, x \geq -4$
 Domain of $f \circ g$: all real numbers x



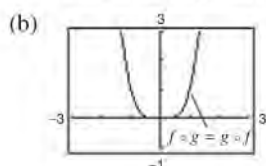
$$f \circ g \neq g \circ f$$

51. (a) $(f \circ g)(x) = x; (g \circ f)(x) = x$
 Domain of $f \circ g$: all real numbers x



$$f \circ g = g \circ f$$

53. (a) $(f \circ g)(x) = x^4; (g \circ f)(x) = x^4$
 Domain of $f \circ g$: all real numbers x



$$f \circ g = g \circ f$$

55. (a) $(f \circ g)(x) = 24 - 5x; (g \circ f)(x) = -5x$
 (b) $24 - 5x \neq -5x$

(c)

x	0	1	2	3
$g(x)$	4	3	2	1
$(f \circ g)(x)$	24	19	14	9

x	0	1	2	3
$f(x)$	4	9	14	19
$(g \circ f)(x)$	0	-5	-10	-15

57. (a) $(f \circ g)(x) = \sqrt{x^2 + 1}; (g \circ f)(x) = x + 1, x \geq -6$
 (b) $x + 1 \neq \sqrt{x^2 + 1}$

(c)

x	0	1	2	3
$g(x)$	-5	-4	-1	4
$(f \circ g)(x)$	1	$\sqrt{2}$	$\sqrt{5}$	$\sqrt{10}$

x	0	1	2	3
$f(x)$	$\sqrt{6}$	$\sqrt{7}$	$\sqrt{8}$	3
$(g \circ f)(x)$	1	2	3	4

59. (a) $(f \circ g)(x) = |2x + 2|; (g \circ f)(x) = 2|x + 3| - 1$
 (b) $(f \circ g)(x) = \begin{cases} 2x + 2, & x \geq -1 \\ -2x - 2, & x < -1 \end{cases}$
 $(g \circ f)(x) = \begin{cases} 2x + 5, & x \geq -3 \\ -2x - 7, & x < -3 \end{cases}$
 $(f \circ g)(x) \neq (g \circ f)(x)$

(c)

x	-2	-1	0	1	2
$g(x)$	-16	-2	0	2	16
$(f \circ g)(x)$	-16	2	0	2	16

x	-2	-1	0	1	2
$f(x)$	2	1	0	1	2
$(g \circ f)(x)$	-16	2	0	2	16

61. (a) 3 (b) 0 63. (a) 0 (b) 4

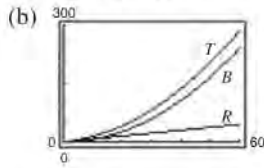
65. $f(x) = x^2, g(x) = 2x + 1$

67. $f(x) = \sqrt[3]{x}, g(x) = x^2 - 4$

69. $f(x) = \frac{1}{x}, g(x) = x + 2$

71. $f(x) = x^2 + 2x$, $g(x) = x + 4$

73. (a) $T = \frac{3}{4}x + \frac{1}{15}x^2$



(c) B. For example, $B(60) = 240$, whereas $R(60)$ is only 45.

75.

Year	1995	1996	1997	1998
y_1	140	151.4	162.8	174.2
y_2	325.8	342.8	364.4	390.6
y_3	458.8	475.3	497.9	526.5

Year	1999	2000	2001	2002
y_1	185.6	197	208.4	219.8
y_2	421.5	457	497.1	541.8
y_3	561.2	602	648.8	701.7

Year	2003	2004	2005
y_1	231.2	242.6	254
y_2	591.2	645.2	703.8
y_3	760.7	825.7	896.8

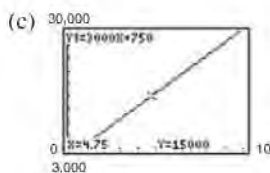
Answers will vary.

77. $(A \circ r)(t) = 0.36\pi t^2$

$(A \circ r)(t)$ represents the area of the circle at time t .

79. (a) $C(x(t)) = 3000t + 750$; $C(x(t))$ represents the production cost after t production hours.

(b) 200 units



$t = 4.75$, or 4 hours 45 minutes

81. (a) $N(T(t))$ or $(N \circ T)(t) = 40t^2 + 590$; $N(T(t))$ or $(N \circ T)(t)$ represents the number of bacteria after t hours outside the refrigerator.

(b) $(N \circ T)(6) = 2030$; There are 2030 bacteria in a refrigerated food product after 6 hours outside the refrigerator.

(c) About 2.3 hours

83. $g(f(x))$ represents 3% of an amount over \$500,000.

85. False. $(f \circ g)(x) = 6x + 1 \neq 6x + 6 = (g \circ f)(x)$

87. (a) $O(M(Y)) = 2(6 + \frac{1}{2}Y) = 12 + Y$

(b) Middle child is 8 years old, youngest child is 4 years old.

89 and 91. Proof

93. $(0, -5), (1, -5), (2, -7)$

95. $(0, 2\sqrt{6}), (1, \sqrt{23}), (2, 2\sqrt{5})$

97. $10x - y + 38 = 0$ 99. $30x + 11y - 34 = 0$

Section 1.6 (page 69)

Vocabulary Check (page 69)

- inverse, f^{-1}
- range, domain
- $y = x$
- one-to-one
- Horizontal

1. $f^{-1}(x) = \frac{x}{6}$ 3. $f^{-1}(x) = x - 7$

5. $f^{-1}(x) = \frac{1}{2}(x - 1)$ 7. $f^{-1}(x) = x^3$

9. (a) $f(g(x)) = f\left(-\frac{2x+6}{7}\right)$

$$= -\frac{7}{2}\left(-\frac{2x+6}{7}\right) - 3 = x$$

$$g(f(x)) = g\left(-\frac{7}{2}x - 3\right)$$

$$= -\frac{2(-\frac{7}{2}x - 3) + 6}{7} = x$$

(b)

x	0	2	-2	6
$f(x)$	-3	-10	4	-24

x	-3	-10	4	-24
$g(x)$	0	2	-2	6

11. (a) $f(g(x)) = f(\sqrt[3]{x-5}) = (\sqrt[3]{x-5})^3 + 5 = x$
 $g(f(x)) = g(x^3 + 5) = \sqrt[3]{x^3 + 5} - 5 = x$

(b)

x	0	1	-1	-2	4
$f(x)$	5	6	4	-3	69

x	5	6	4	-3	69
$g(x)$	0	1	-1	-2	4

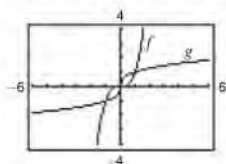
13. (a) $f(g(x)) = f(8 + x^2)$
 $= -\sqrt{(8 + x^2) - 8}$
 $= -\sqrt{x^2} = -(-x) = x, x \leq 0$
 $g(f(x)) = g(-\sqrt{x-8})$
 $= 8 + (-\sqrt{x-8})^2$
 $= 8 + (x-8) = x, x \geq 8$

(b)

x	8	9	12	15
$f(x)$	0	-1	-2	$-\sqrt{7}$

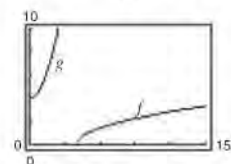
x	0	-1	-2	$-\sqrt{7}$
$g(x)$	8	9	12	15

15. $f(g(x)) = f(\sqrt[3]{x}) = (\sqrt[3]{x})^3 = x$
 $g(f(x)) = g(x^3) = \sqrt[3]{x^3} = x$



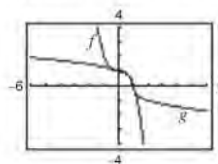
Reflections in the line $y = x$

17. $f(g(x)) = f(x^2 + 4), x \geq 0$
 $= \sqrt{(x^2 + 4) - 4} = x$
 $g(f(x)) = g(\sqrt{x-4})$
 $= (\sqrt{x-4})^2 + 4 = x$



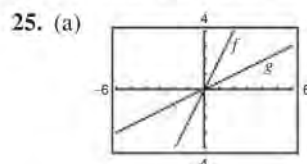
Reflections in the line $y = x$

19. $f(g(x)) = f(\sqrt[3]{1-x}) = 1 - (\sqrt[3]{1-x})^3 = x$
 $g(f(x)) = g(1 - x^3) = \sqrt[3]{1 - (1 - x^3)} = x$



Reflections in the line $y = x$

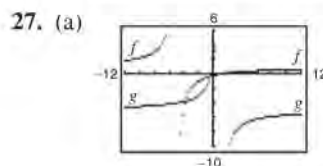
21. c 22. b 23. a 24. d



(b)

x	-4	-2	0	2	4
$f(x)$	-8	-4	0	4	8

x	-8	-4	0	4	8
$g(x)$	-4	-2	0	2	4



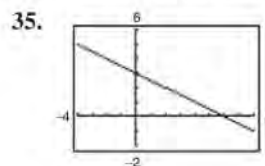
(b)

x	-3	-2	-1	0	2	3	4
$f(x)$	-2	-1	$-\frac{1}{2}$	$-\frac{1}{3}$	$\frac{1}{7}$	$\frac{1}{4}$	$\frac{1}{3}$

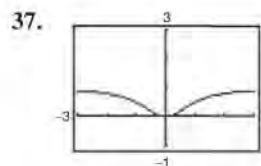
x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{3}$	$\frac{1}{7}$	$\frac{1}{4}$	$\frac{1}{3}$
$g(x)$	-3	-2	-1	0	2	3	4

29. Not a function 31. Function; one-to-one

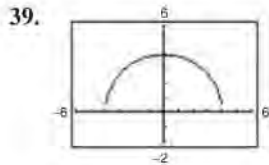
33. Function; one-to-one



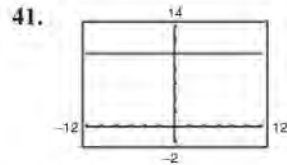
One-to-one



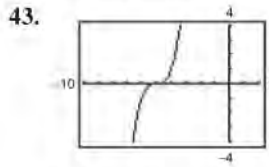
Not one-to-one



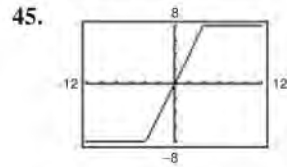
Not one-to-one



Not one-to-one



One-to-one



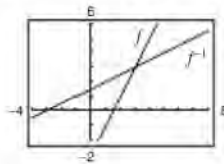
Not one-to-one

47. Not one-to-one 49. $f^{-1}(x) = \frac{5x - 4}{3}$

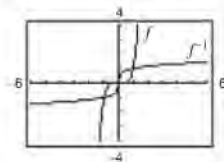
51. Not one-to-one 53. $f^{-1}(x) = \sqrt{x} - 3, x \geq 0$

55. $f^{-1}(x) = \frac{x^2 - 3}{2}, x \geq 0$ 57. $f^{-1}(x) = 2 - x, x \geq 0$

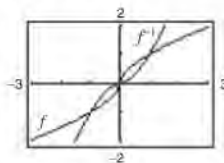
59. $f^{-1}(x) = \frac{x + 3}{2}$

Reflections in the line $y = x$

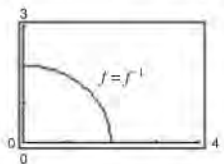
61. $f^{-1}(x) = \sqrt[5]{x}$

Reflections in the line $y = x$

63. $f^{-1}(x) = x^{5/3}$

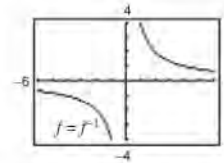
Reflections in the line $y = x$

65. $f^{-1}(x) = \sqrt{4 - x^2}, 0 \leq x \leq 2$



The graphs are the same.

67. $f^{-1}(x) = \frac{4}{x}$



The graphs are the same.

69. $f^{-1}(x) = \sqrt{x} + 2$

Domain of f : all real numbers x such that $x \geq 2$ Range of f : all real numbers y such that $y \geq 0$ Domain of f^{-1} : all real numbers x such that $x \geq 0$ Range of f^{-1} : all real numbers y such that $y \geq 2$

71. $f^{-1}(x) = x - 2$

Domain of f : all real numbers x such that $x \geq -2$ Range of f : all real numbers y such that $y \geq 0$ Domain of f^{-1} : all real numbers x such that $x \geq 0$ Range of f^{-1} : all real numbers y such that $y \geq -2$

73. $f^{-1}(x) = \sqrt{x} - 3$

Domain of f : all real numbers x such that $x \geq -3$ Range of f : all real numbers y such that $y \geq 0$ Domain of f^{-1} : all real numbers x such that $x \geq 0$ Range of f^{-1} : all real numbers y such that $y \geq -3$

75. $f^{-1}(x) = \frac{\sqrt{-2(x-5)}}{2}$

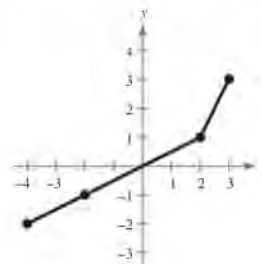
Domain of f : all real numbers x such that $x \geq 0$ Range of f : all real numbers y such that $y \leq 5$ Domain of f^{-1} : all real numbers x such that $x \leq 5$ Range of f^{-1} : all real numbers y such that $y \geq 0$

77. $f^{-1}(x) = x + 3$

Domain of f : all real numbers x such that $x \geq 4$ Range of f : all real numbers y such that $y \geq 1$ Domain of f^{-1} : all real numbers x such that $x \geq 1$ Range of f^{-1} : all real numbers y such that $y \geq 4$

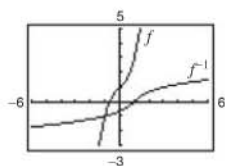
79.

x	-4	-2	2	3
$f^{-1}(x)$	-2	-1	1	3



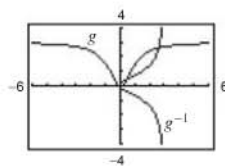
81. $\frac{1}{2}$ 83. -2 85. 0 87. 2

89. (a) and (b)



(c) Inverse function because it satisfies the Vertical Line Test

91. (a) and (b)



(c) Not an inverse function because it does not satisfy the Vertical Line Test

93. 32 95. 600 97. $2\sqrt[3]{x+3}$

99. $\frac{x+1}{2}$ 101. $\frac{x+1}{2}$

103. (a) f is one-to-one because no two elements in the domain (men's U.S. shoe sizes) correspond to the same element in the range (men's European shoe sizes).

(b) 45 (c) 10 (d) 41 (e) 13

105. (a) Yes

(b) $f^{-1}(t)$ represents the year new car sales totaled \$ t billion.

(c) 10 or 2000

(d) No. The inverse would not be a function because f would not be one-to-one.

107. False. For example, $y = x^2$ is even, but does not have an inverse.

109. Proof

111. f and g are not inverses of each other because one is not the graph of the other when reflected through the line $y = x$.

113. f and g are inverses of each other because one is the graph of the other when reflected through the line $y = x$.

115. This situation could be represented by a one-to-one function. The inverse function would represent the number of miles completed in terms of time in hours.

117. This function could not be represented by a one-to-one function because it oscillates.

119. $9x$, $x \neq 0$ 121. $-(x+6)$, $x \neq 6$

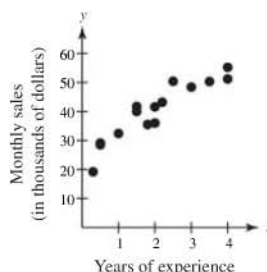
123. Function 125. Not a function 127. Function

Section 1.7 (page 78)

Vocabulary Check (page 78)

1. positive 2. negative 3. fitting a line to data
4. -1, 1

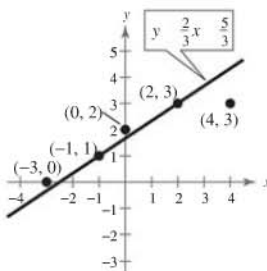
1. (a)



(b) Answers will vary.

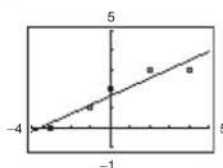
3. Negative correlation 5. No correlation

7. (a)



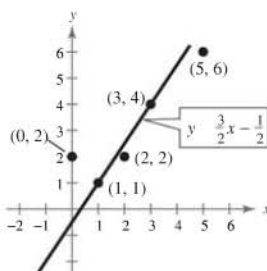
(b) $y = 0.46x + 1.62$; 0.95095

(c)



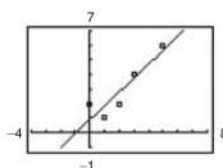
(d) The models appear valid.

9. (a)

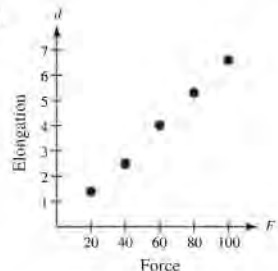


(b) $y = 0.95x + 0.92$; 0.90978

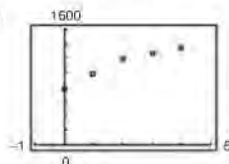
(c)

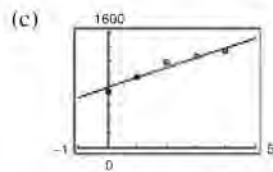


(d) The models appear valid.

11. (a)
- 
- (b)
- $d = 0.07F - 0.3$

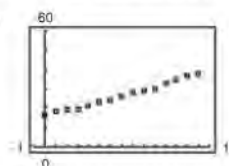
- (c)
- $d = 0.066F$
-
- (d) 3.63 centimeters

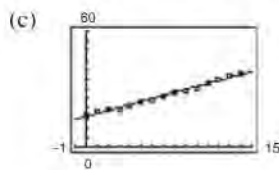
13. (a)
- 
- (b)
- $S = 136.1t + 836$



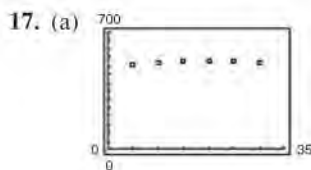
The model fits the data.

- (d) 2005: \$1,516,500; 2010: \$2,197,000;
-
- Answers will vary.
-
- (e) 136.1; The slope represents the average annual
-
- increase in salaries (in thousands of dollars).

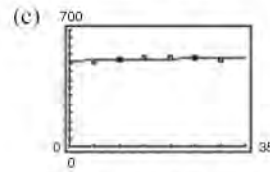
15. (a)
- 
- (b)
- $C = 1.552t + 15.70$
- ; 0.99544



- (d) Yes; answers will vary.
-
- (e) 2005: \$38.98; 2010: \$46.74
-
- (f) Answers will vary.



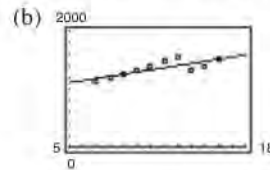
- (b)
- $P = 0.6t + 512$



The model is not the best fit for the data.

- (d) 542,000 people; answers will vary.

19. (a)
- $T = 36.7t + 926$
- ; 0.79495



- (c) The slope represents an increase of about 37 Target
-
- stores annually.

- (d) 2013

- (e)

Year	1997	1998	1999	2000	2001
Actual T -values (in thousands)	1130	1182	1243	1307	1381
T -values from model (in thousands)	1183	1220	1256	1293	1330

Year	2002	2003	2004	2005	2006
Actual T -values (in thousands)	1475	1553	1308	1400	1505
T -values from model (in thousands)	1366	1403	1440	1477	1513

Answers will vary.

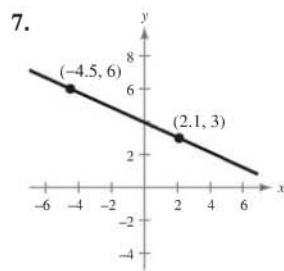
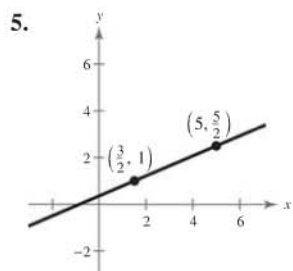
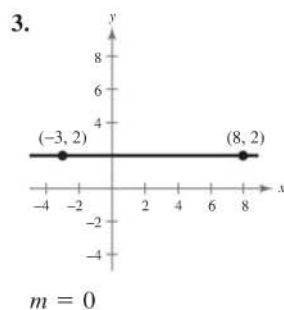
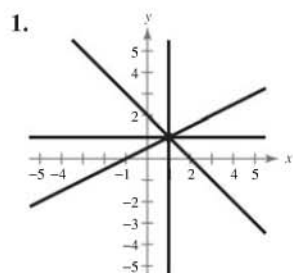
21. True. To have positive correlation, the
- y
- values tend to
-
- increase as
- x
- increases.

23. Answers will vary. 25. (a) 10 (b)
- $2w^2 + 5w + 7$

27. (a) 5 (b) 1 29.
- $-\frac{3}{5}$
- 31.
- $-\frac{1}{4}, \frac{3}{2}$

- 33.
- $\frac{7 \pm \sqrt{17}}{4}$

Review Exercises (page 83)



9. $x - 4y - 6 = 0$; $(6, 0)$, $(10, 1)$, $(-2, -2)$

11. $3x - 2y - 10 = 0$; $(4, 1)$, $(2, -2)$, $(-2, -8)$

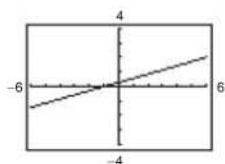
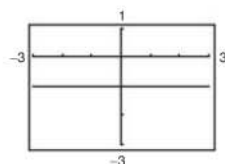
13. $5x + 5y + 24 = 0$; $(-5, \frac{1}{5})$, $(-4, -\frac{4}{5})$, $(-6, \frac{6}{5})$

15. $y - 6 = 0$; $(0, 6)$, $(1, 6)$, $(-1, 6)$

17. $x - 10 = 0$; $(10, 1)$, $(10, 3)$, $(10, -2)$

19. $y = -1$

21. $y = \frac{2}{7}x + \frac{2}{7}$



23. $V = 850t + 5700$

25. $V = 42.70t + 283.9$

27. \$210,000

29. (a) $y = \frac{5}{4}x - \frac{23}{4}$

31. (a) $x = -6$

(b) $y = -\frac{4}{5}x + \frac{2}{5}$

(b) $y = 2$

33. (a) Not a function because element 20 in A corresponds to two elements, 4 and 6, in B .

(b) Function

(c) Function

(d) Not a function because 30 in A corresponds to no element in B .

35. Not a function

37. Function

39. (a) 2 (b) 10 (c) $b^6 + 1$ (d) $x^2 - 2x + 2$

41. (a) -3 (b) -1 (c) 2 (d) 6

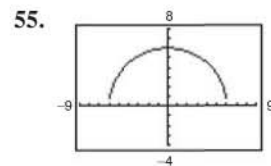
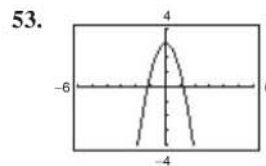
43. All real numbers x except $x = -2$

45. All real numbers x such that $-5 \leq x \leq 5$.

47. All real numbers s except $s = 3$

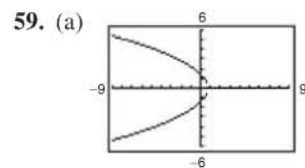
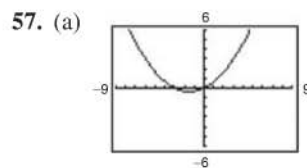
49. (a) $C = 5.35x + 16,000$ (b) $P = 2.85x - 16,000$

51. $2h + 4x + 3$, $h \neq 0$



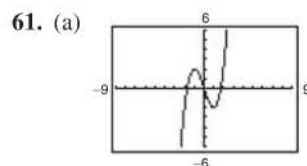
Domain: all real numbers x
Range: $(-\infty, 3]$

Domain: $[-6, 6]$
Range: $[0, 6]$

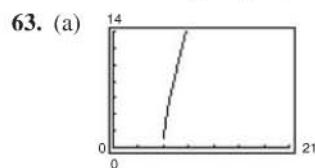


(b) Function

(b) Not a function



(b) Increasing on $(-\infty, -1)$, $(1, \infty)$
Decreasing on $(-1, 1)$

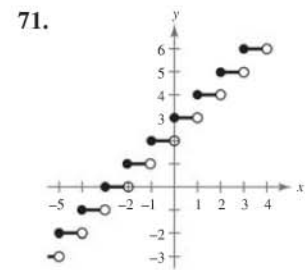
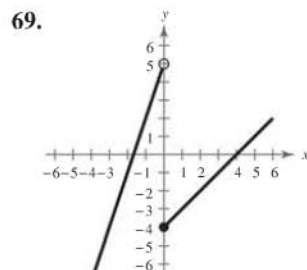


(b) Increasing on $(6, \infty)$

65. Relative maximum: $(0, 16)$

Relative minima: $(-2, 0)$, $(2, 0)$

67. Relative maximum: $(3, 27)$



73. Even 75. Even 77. Neither

79. Constant function $f(x) = c$

Vertical shift two units downward

$$g(x) = -2$$

81. Quadratic function $f(x) = x^2$

Vertical shift one unit upward, horizontal shift two units right

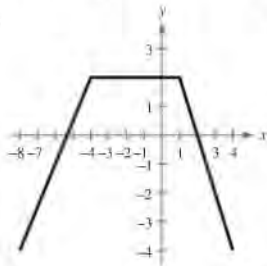
$$g(x) = (x - 2)^2 + 1$$

83. Absolute value function $f(x) = |x|$

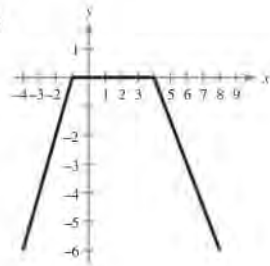
Vertical shift three units upward

$$g(x) = |x| + 3$$

85.



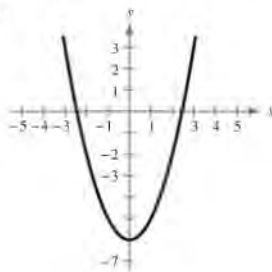
87.



89. (a) Quadratic function $f(x) = x^2$

(b) Vertical shift six units downward

(c)



(d) $h(x) = f(x) - 6$

91. (a) Cubic function $f(x) = x^3$

(b) Horizontal shift two units right, vertical shift five units upward

(c)

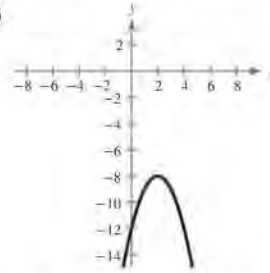


(d) $h(x) = f(x - 2) + 5$

93. (a) Quadratic function $f(x) = x^2$

(b) Horizontal shift two units to the right, reflection in the x -axis, vertical shift eight units downward

(c)



(d) $h(x) = -f(x - 2) - 8$

95. (a) Square root function $f(x) = \sqrt{x}$

(b) Reflection in the x -axis, vertical shift five units upward

(c)

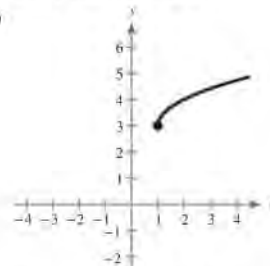


(d) $h(x) = -f(x) + 5$

97. (a) Square root function $f(x) = \sqrt{x}$

(b) Horizontal shift one unit to the right, vertical shift, three units upward

(c)



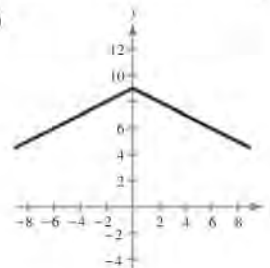
(d) $h(x) = f(x - 1) + 3$

99. (a) Absolute value function $f(x) = |x|$

(b) Reflection in the x -axis, vertical shrink by a factor of $\frac{1}{2}$, vertical shift nine units upward

(c)

(d) $h(x) = -\frac{1}{2}f(x) + 9$

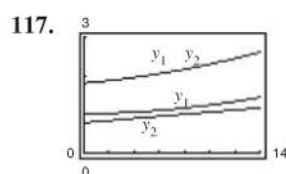


101. -7 103. -42 105. 5 107. 23

109. -97 111. $f(x) = x^2$, $g(x) = x + 3$

113. $f(x) = \sqrt{x}$, $g(x) = 4x + 2$

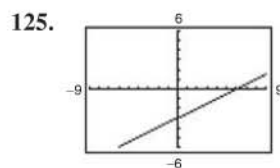
115. $f(x) = \frac{4}{x}$, $g(x) = x + 2$



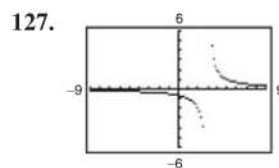
119. $f^{-1}(x) = \frac{x}{6}$

121. $f^{-1}(x) = 2x - 6$

123. Answers will vary.



One-to-one

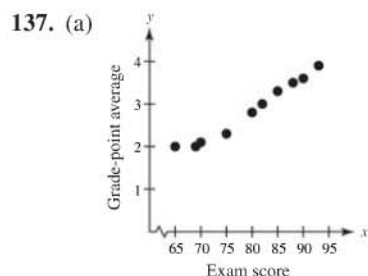


One-to-one

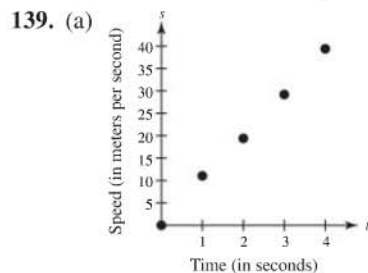
129. $f^{-1}(x) = 2x + 10$ 131. $f^{-1}(x) = \sqrt[3]{\frac{x+3}{4}}$

133. $f^{-1}(x) = x^2 - 10$, $x \geq 0$

135. Negative correlation



(b) Yes. Answers will vary.



(b) Answers will vary.

Sample answer: $s = 10t - 0.4$

(c) $s = 9.7t + 0.4$; 0.99933; This model fits the data better.

(d) 24.65 meters per second

141. $y = 95.174x - 458.42$

143. The model doesn't fit the data very well.

145. False. The point $(-1, 28)$ does not lie on the graph of the function $g(x) = -(x - 6)^2 - 3$.

147. False. For example, $f(x) = 4 - x = f^{-1}(x)$.

Chapter Test (page 88)

1. (a) $5x + 2y - 8 = 0$ (b) $2x - 5y + 20 = 0$

2. $y = -x + 1$

3. No. To some x there correspond more than one value of y .

4. (a) -9 (b) 1 (c) $|t - 4| - 15$

5. $(-\infty, 3]$ 6. $C = 5.60x + 24,000$

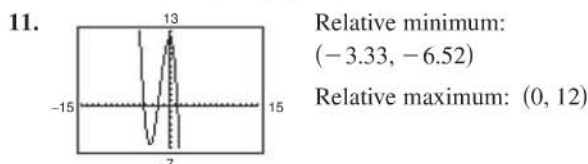
$P = 93.9x - 24,000$

7. Odd 8. Even

9. Increasing: $(-2, 0)$, $(2, \infty)$

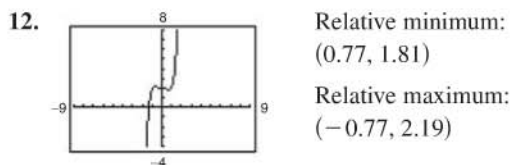
Decreasing: $(-\infty, -2)$, $(0, 2)$

10. Increasing: $(-2, 2)$

Constant: $(-\infty, -2)$, $(2, \infty)$


Relative minimum:

 $(-3.33, -6.52)$

Relative maximum: $(0, 12)$


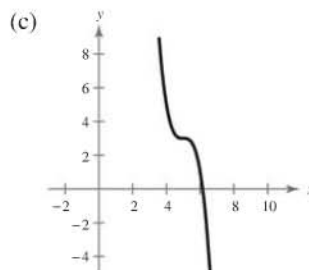
Relative minimum:

 $(0.77, 1.81)$

Relative maximum:

 $(-0.77, 2.19)$

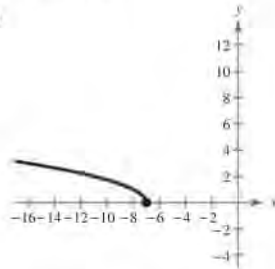
13. (a) Cubic function $f(x) = x^3$

(b) Horizontal shift five units to the right, reflection in the x -axis, vertical stretch, and vertical shift three units upward


14. (a) Square root function $f(x) = \sqrt{x}$

(b) Reflection in the y -axis and horizontal shift seven units to the left

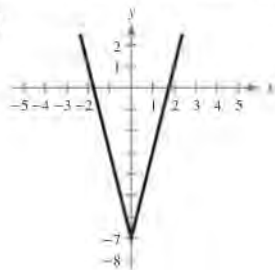
(c)



15. (a) Absolute value function $f(x) = |x|$

(b) Reflection in the y -axis (no effect), vertical stretch, and vertical shift seven units downward

(c)



16. (a) $x^2 - \sqrt{2-x}$, $(-\infty, 2]$ (b) $\frac{x^2}{\sqrt{2-x}}$, $(-\infty, 2]$

(c) $2-x$, $(-\infty, 2]$ (d) $\sqrt{2-x^2}$, $[-\sqrt{2}, \sqrt{2}]$

17. $f^{-1}(x) = \sqrt[3]{x-8}$ 18. No inverse

19. $f^{-1}(x) = (\frac{8}{3}x)^{2/3}$, $x \geq 0$

20. $S = 18.3t - 76.2$; 2005

Chapter 2

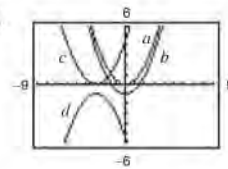
Section 2.1 (page 99)

Vocabulary Check (page 99)

1. nonnegative integer, real
2. quadratic, parabola
3. axis
4. positive, minimum
5. negative, maximum

1. c 2. d 3. b 4. a

5.



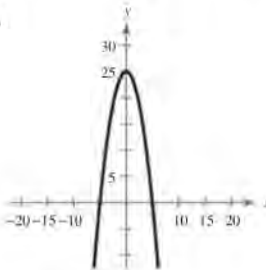
(a) Vertical shrink

(b) Vertical shrink and vertical shift one unit downward

(c) Vertical shrink and a horizontal shift three units to the left

(d) Vertical shrink, reflection in the x -axis, a horizontal shift three units to the left, and a vertical shift one unit downward

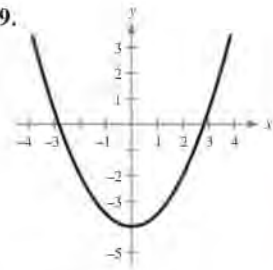
7.



Vertex: (0, 25)

x -intercepts: $(\pm 5, 0)$

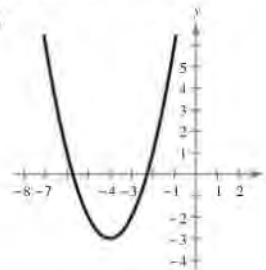
9.



Vertex: (0, -4)

x -intercepts: $(\pm 2\sqrt{2}, 0)$

11.

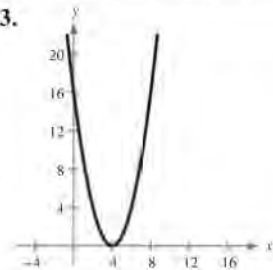


Vertex: (-4, -3)

x -intercepts:

$(\pm\sqrt{3}-4, 0)$

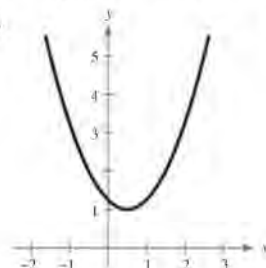
13.



Vertex: (4, 0)

x -intercept: (4, 0)

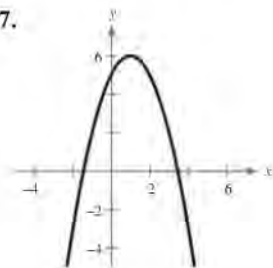
15.



Vertex: $(\frac{1}{2}, 1)$

x -intercept: None

17.

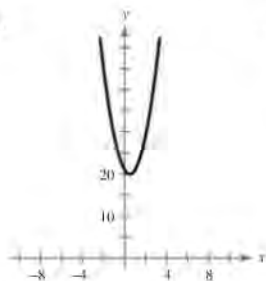


Vertex: (1, 6)

x -intercepts:

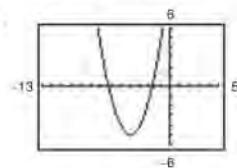
$(1 \pm \sqrt{6}, 0)$

19.



Vertex: $(\frac{1}{2}, 20)$
 x -intercept: None

23.



Vertex: $(-4, -5)$
 x -intercepts:
 $(-4 \pm \sqrt{5}, 0)$

27. $y = -(x + 1)^2 + 4$

29. $f(x) = (x + 2)^2 + 5$

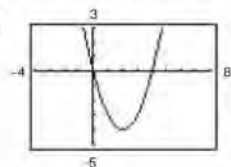
31. $y = 4(x - 1)^2 - 2$

33. $y = -\frac{104}{125}(x - \frac{1}{2})^2 + 1$

35. $(5, 0), (-1, 0)$; They are the same.

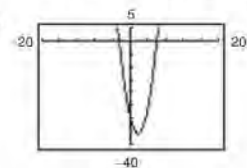
37. $(-4, 0)$; They are the same.

39.



$(0, 0), (4, 0)$; They are the same.

41.



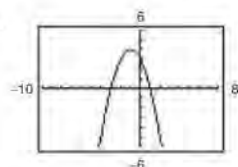
$(-\frac{5}{2}, 0), (6, 0)$;
 They are the same.

45. $f(x) = x^2 - 2x - 3$
 $g(x) = -x^2 + 2x + 3$

49. 55, 55

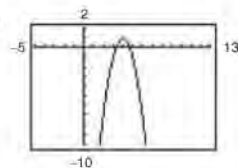
51. 12, 6

21.



Vertex: $(-1, 4)$
 x -intercepts:
 $(1, 0), (-3, 0)$

25.



Vertex: $(4, 1)$
 x -intercepts:
 $(4 \pm \frac{1}{2}\sqrt{2}, 0)$

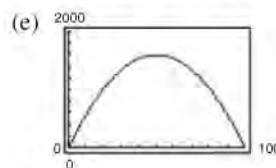
53. (a)



(b) $r = \frac{1}{2}y$; $d = y\pi$

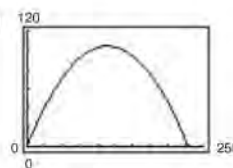
(c) $y = \frac{200 - 2x}{\pi}$

(d) $A = x\left(\frac{200 - 2x}{\pi}\right)$



$x = 50$ meters, $y = \frac{100}{\pi}$ meters

55. (a)



(b) $\frac{3}{2}$ feet

(c) About 104 feet

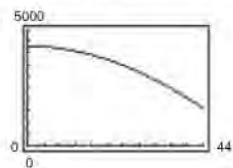
(d) About 228.6 feet

57. 20 fixtures

59. (a) \$14,000,000; \$14,375,000; \$13,500,000

(b) \$24 (c) \$14,400,000 (d) Answers will vary.

61. (a)



(b) 1960; 4306 cigarettes per person per year; Answers will vary.

(c) 8909 cigarettes per smoker per year; 24 cigarettes per smoker per day

63. True. The vertex is $(0, -1)$ and the parabola opens down.

65. c, d 67. $b = \pm 20$ 69. $b = \pm 8$

71. Model (a). The profits are positive and rising.

73. $(1.2, 6.8)$ 75. $(2, 5), (-3, 0)$ 77. $-5 - 3i$

79. $19 - 25i$ 81. Answers will vary.

Section 2.2 (page 112)

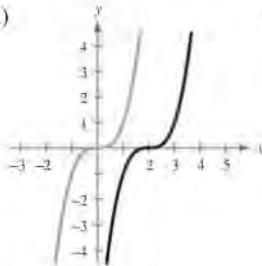
Vocabulary Check (page 112)

1. continuous
2. Leading Coefficient Test
3. $n, n - 1$, relative extrema
4. solution, $(x - a)$, x -intercept
5. touches, crosses
6. Intermediate Value

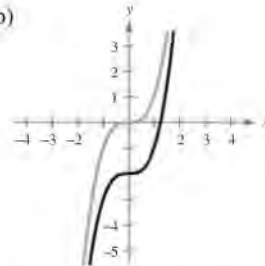
1. f 2. h 3. c 4. a 5. e 6. d

7. g 8. b

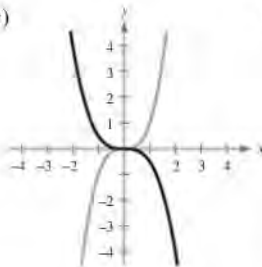
9. (a)



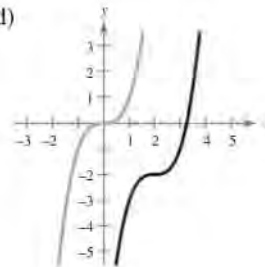
(b)



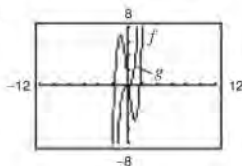
(c)



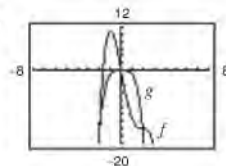
(d)



11.



13.



15. Rises to the left,
rises to the right

17. Falls to the left,
falls to the right

19. Falls to the left,
rises to the right

21. Falls to the left,
falls to the right

23. ± 5 (multiplicity 1)

25. 3 (multiplicity 2)

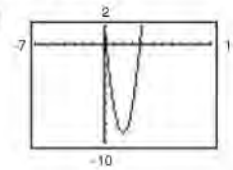
27. 1, -2 (multiplicity 1)

29. 2 (multiplicity 2), 0 (multiplicity 1)

31. $\frac{-5 \pm \sqrt{37}}{2}$ (multiplicity 1)

33. (a) $(2 \pm \sqrt{3}, 0)$

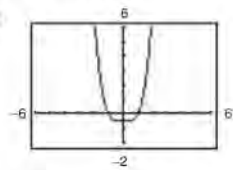
(b)



(c) $(0.27, 0), (3.73, 0)$; answers are approximately the same.

35. (a) $(-1, 0), (1, 0)$

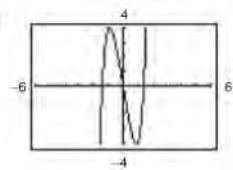
(b)



(c) $(-1, 0), (1, 0)$; answers are approximately the same.

37. (a) $(0, 0), (\pm\sqrt{2}, 0)$

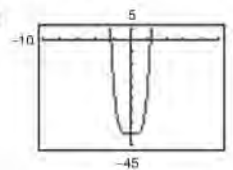
(b)



(c) $(-1.41, 0), (0, 0), (1.41, 0)$; answers are approximately the same.

39. (a) $(\pm\sqrt{5}, 0)$

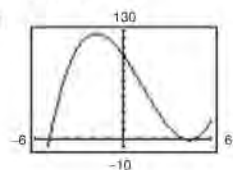
(b)



(c) $(-2.236, 0), (2.236, 0)$

41. (a) $(4, 0), (\pm 5, 0)$

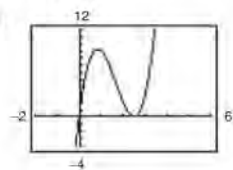
(b)



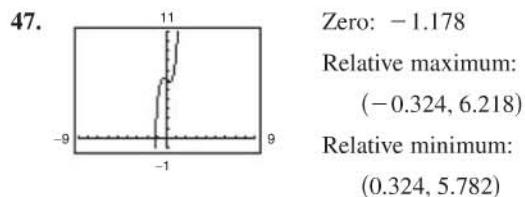
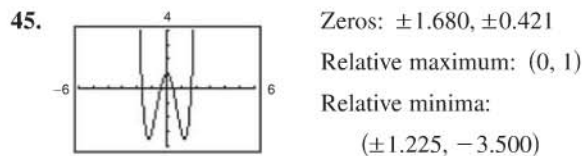
(c) $(-5, 0), (4, 0), (5, 0)$

43. (a) $(0, 0), (\frac{5}{2}, 0)$

(b)



(c) $(0, 0), (\frac{5}{2}, 0)$



49. $f(x) = x^2 - 4x$ 51. $f(x) = x^3 + 5x^2 + 6x$

53. $f(x) = x^4 - 4x^3 - 9x^2 + 36x$

55. $f(x) = x^2 - 2x - 2$

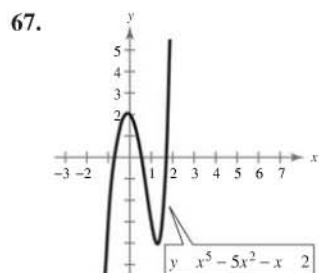
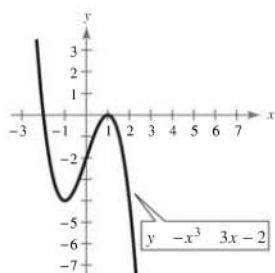
57. $f(x) = x^3 - 10x^2 + 27x - 22$

59. $f(x) = x^3 + 5x^2 + 8x + 4$

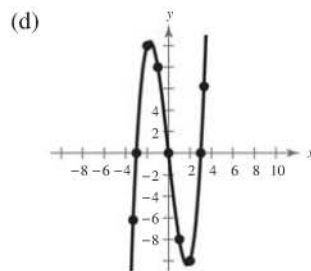
61. $f(x) = x^4 + 2x^3 - 23x^2 - 24x + 144$

63. $f(x) = -x^3 - 4x^2 - 5x - 2$

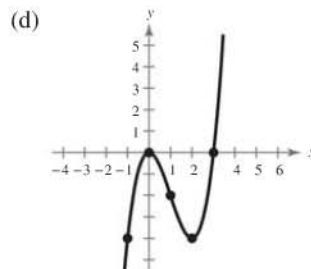
65. Not possible; odd-degree polynomials must have an odd number of real solutions.



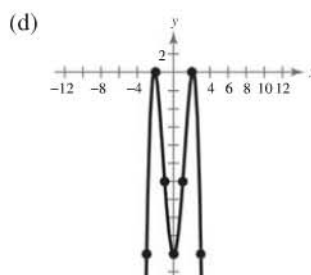
69. (a) Falls to the left, rises to the right
(b) $(0, 0), (3, 0), (-3, 0)$
(c) Answers will vary.



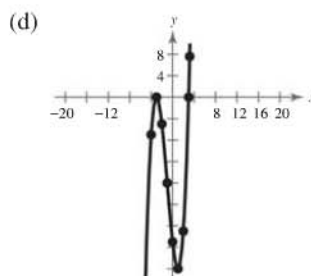
71. (a) Falls to the left, rises to the right
(b) $(0, 0), (3, 0)$ (c) Answers will vary.



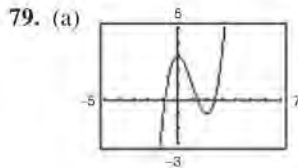
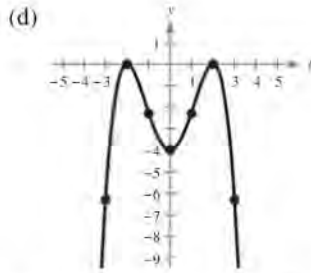
73. (a) Falls to the left, falls to the right
(b) $(\pm 2, 0), (\pm \sqrt{5}, 0)$ (c) Answers will vary.



75. (a) Falls to the left, rises to the right
(b) $(\pm 3, 0)$ (c) Answers will vary.

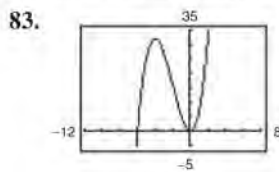


77. (a) Falls to the left, falls to the right
(b) $(-2, 0), (2, 0)$
(c) Answers will vary.

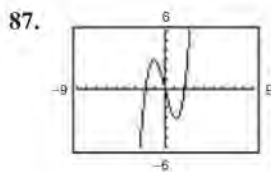


$(-1, 0), (1, 2), (2, 3)$

(b) $-0.879, 1.347, 2.532$



Two x -intercepts



Origin symmetry
Three x -intercepts

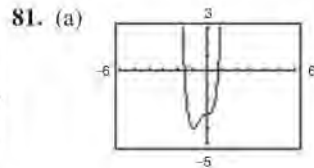
91. (a) Answers will vary.

(b) Domain: $0 < x < 18$

(c)

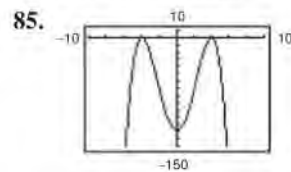
Height, x	Volume, V
1	1156
2	2048
3	2700
4	3136
5	3380
6	3456
7	3388

$5 < x < 7$

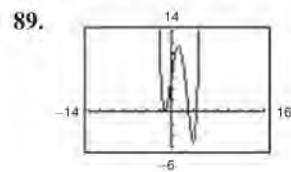


$(-2, -1), (0, 1)$

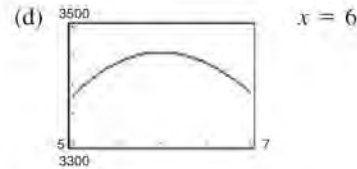
(b) $-1.585, 0.779$



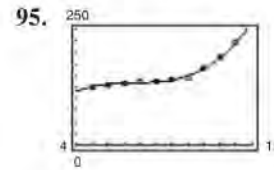
y -axis symmetry
Two x -intercepts



Three x -intercepts



93. $(200, 160)$



The model fits the data.

97. Northeast: \$730,200; South: \$285,000; Answers will vary.

99. True; $y = x^6$

101. False; the graph touches the x -axis at $x = 1$.

103. True 105. b 107. a 109. 33

111. $-\frac{4}{3} \approx -1.3$ 113. 72

115. $x \leq -\frac{1}{2}, x \geq 1$ 117. $x \leq -24, x \geq 8$



Section 2.3 (page 127)

Vocabulary Check (page 127)

- $f(x)$ is the dividend, $d(x)$ is the divisor, $q(x)$ is the quotient, and $r(x)$ is the remainder.
- improper, proper 3. synthetic division
- Rational Zero 5. Descartes's Rule, Signs
- Remainder Theorem
- upper bound, lower bound

1. $2x + 4, x \neq -3$ 3. $x^3 + 3x^2 - 1, x \neq -2$

5. $x^2 - 3x + 1, x \neq -\frac{5}{4}$ 7. $7x^2 - 14x + 28 - \frac{53}{x+2}$

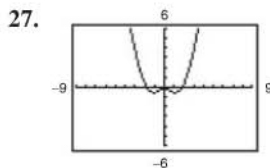
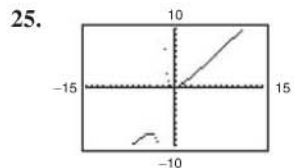
9. $3x + 5 - \frac{2x-3}{2x^2+1}$ 11. $x - \frac{x+9}{x^2+1}$

13. $2x - \frac{17x-5}{x^2-2x+1}$ 15. $3x^2 - 2x + 5, x \neq 5$

17. $6x^2 + 25x + 74 + \frac{248}{x-3}$ 19. $9x^2 - 16, x \neq 2$

21. $x^2 - 8x + 64, x \neq -8$

23. $4x^2 + 14x - 30, x \neq -\frac{1}{2}$



29. $f(x) = (x - 4)(x^2 + 3x - 2) + 3, f(4) = 3$

31. $f(x) = (x - \sqrt{2})[x^2 + (3 + \sqrt{2})x + 3\sqrt{2}] - 8, f(\sqrt{2}) = -8$

33. $f(x) = (x - 1 + \sqrt{3})[4x^2 - (2 + 4\sqrt{3})x - (2 + 2\sqrt{3})], f(1 - \sqrt{3}) = 0$

35. (a) -2 (b) 1 (c) $-\frac{1}{4}$ (d) 5

37. (a) -35 (b) -22 (c) -10 (d) -211

39. $(x - 2)(x + 3)(x - 1)$ 41. $(2x - 1)(x - 5)(x - 2)$
Zeros: 2, -3, 1 Zeros: $\frac{1}{2}, 5, 2$

43. (a) Answers will vary. (b) $(2x - 1), (x - 1)$
(c) $(x + 2)(x - 1)(2x - 1)$ (d) -2, 1, $\frac{1}{2}$

45. (a) Answers will vary. (b) $(x - 1), (x - 2)$
(c) $(x - 5)(x + 4)(x - 1)(x - 2)$ (d) -4, 1, 2, 5

47. (a) Answers will vary. (b) $(x + 7), (3x - 2)$
(c) $(2x + 1)(3x - 2)(x + 7)$ (d) -7, $-\frac{1}{2}, \frac{2}{3}$

49. $\pm 1, -3$

51. $\pm 1, \pm 3, \pm 5, \pm 9, \pm 15, \pm 45, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{5}{2}, \pm \frac{9}{2}, \pm \frac{15}{2}, \pm \frac{45}{2}; -1, \frac{3}{2}, 3, 5$

53. -1, 2 55. $-6, \frac{1}{2}, 1$ 57. $-3, -\frac{3}{2}, \frac{1}{2}, 4$

59. $-\frac{5}{2}, -2, \pm 1, \frac{3}{2}$

61. (a) -2, 0.268, 3.732 (b) -2

(c) $h(t) = (t + 2)(t - 2 + \sqrt{3})(t - 2 - \sqrt{3})$

63. (a) 0, 3, 4, -1.414, 1.414 (b) 0, 3, 4

(c) $h(x) = x(x - 3)(x - 4)(x + \sqrt{2})(x - \sqrt{2})$

65. 4, 2, or 0 positive real zeros, no negative real zeros

67. 2 or 0 positive real zeros, 1 negative real zero

69. (a) 1 positive real zero, 2 or 0 negative real zeros

(b) $\pm 1, \pm 2, \pm 4$

(c) (d) -2, -1, 2

71. (a) 3 or 1 positive real zeros, 1 negative real zero

(b) $\pm 1, \pm 2, \pm 4, \pm 8, \pm \frac{1}{2}$

(c) (d) $-\frac{1}{2}, 1, 2, 4$

73. (a) 2 or 0 positive real zeros, 1 negative real zero

(b) $\pm 1, \pm 3, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{1}{4}, \pm \frac{3}{4}, \pm \frac{1}{8}, \pm \frac{3}{8}, \pm \frac{1}{16}, \pm \frac{3}{16}, \pm \frac{1}{32}, \pm \frac{3}{32}$

(c) (d) $-\frac{1}{8}, \frac{3}{4}, 1$

75. Answers will vary; 1.937, 3.705

77. Answers will vary; ± 2 79. $\pm 2, \pm \frac{3}{2}$ 81. $\pm 1, \frac{1}{4}$

83. d 84. a 85. b 86. c 87. $-\frac{1}{2}, 2 \pm \sqrt{3}, 1$

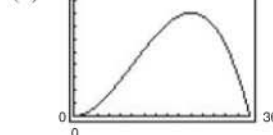
89. $-1, \frac{3}{2}, 4 \pm \sqrt{17}$

91. (a) (b) The model fits the data.

(c) 307.8; Answers will vary.

93. (a) Answers will vary.

(b) 20 × 20 × 40



(c) $15, \frac{15 \pm 15\sqrt{5}}{2};$

$\frac{15 - 15\sqrt{5}}{2}$ represents a negative volume.

95. False. If $(7x + 4)$ is a factor of f , then $-\frac{4}{7}$ is a zero of f .

97. $-2(x - 1)^2(x + 2)$

99. $-(x - 2)(x + 2)(x + 1)(x - 1)$ 101. 7

103. (a) $x + 1, x \neq 1$

(b) $x^2 + x + 1, x \neq 1$

(c) $x^3 + x^2 + x + 1, x \neq 1$

$\frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \dots + x^2 + x + 1, x \neq 1$

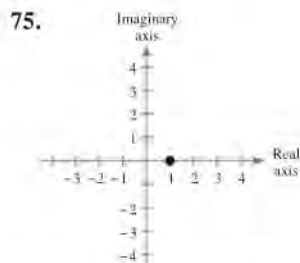
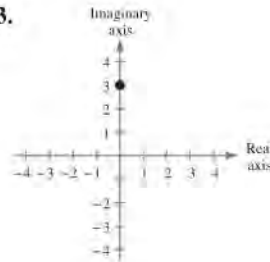
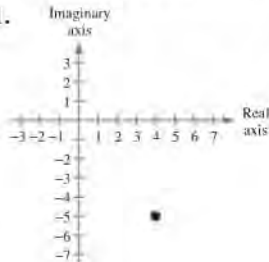
105. $\pm \frac{5}{3}$ 107. $\frac{-3 \pm \sqrt{3}}{2}$ 109. $f(x) = x^2 + 12x$
 111. $f(x) = x^4 - 6x^3 + 3x^2 + 10x$

Section 2.4 (page 137)

Vocabulary Check (page 137)

1. (a) ii (b) iii (c) i 2. $\sqrt{-1}$, -1
 3. complex, $a + bi$ 4. real, imaginary
 5. Mandelbrot Set

1. $a = -9$, $b = 4$ 3. $a = 6$, $b = 5$ 5. $5 + 4i$
 7. -6 9. $-1 - 5i$ 11. -75 13. $0.3i$
 15. $-3 + 3i$ 17. $7 - 3\sqrt{2}i$ 19. $-14 + 20i$
 21. $\frac{19}{6} + \frac{37}{6}i$ 23. $-4.2 + 7.5i$ 25. $-2\sqrt{3}$
 27. -10 29. $5 + i$ 31. $-20 + 32i$ 33. 24
 35. $80i$ 37. $4 - 3i$; 25 39. $-6 + \sqrt{5}i$; 41
 41. $-\sqrt{20}i$; 20 43. $3 + \sqrt{-2}$; 11
 45. $-6i$ 47. $\frac{8}{41} + \frac{10}{41}i$ 49. $\frac{3}{5} + \frac{4}{5}i$
 51. $-\frac{40}{1681} - \frac{9}{1681}i$ 53. $-\frac{1}{2} - \frac{5}{2}i$ 55. $\frac{62}{949} + \frac{297}{949}i$
 57. $-1 + 6i$ 59. $-375\sqrt{3}i$ 61. i
 63. (a) 8 (b) 8 (c) 8; Answers will vary.
 65. $4 + 3i$ 67. $5i$ 69. 2
 71. 73.



77. $0.5i$, $-0.25 + 0.5i$, $-0.1875 + 0.25i$, $-0.0273 + 0.4063i$,
 $-0.1643 + 0.4778i$, $-0.2013 + 0.3430i$; Yes, bounded
 79. $3.12 - 0.97i$
 81. False. Any real number is equal to its conjugate.

83. False. Example: $(1 + i) + (1 - i) = 2$, which is not an imaginary number.
 85. True 87. $16x^2 - 25$ 89. $3x^2 + \frac{23}{2}x - 2$

Section 2.5 (page 144)

Vocabulary Check (page 144)

1. Fundamental Theorem, Algebra
 2. Linear Factorization Theorem
 3. irreducible, reals 4. complex conjugate

1. $-3, 0, 0$ 3. $-9, \pm 4i$
 5. Zeros: $4, -i, i$. One real zero; they are the same.
 7. Zeros: $\sqrt{2}i, \sqrt{2}i, -\sqrt{2}i, -\sqrt{2}i$. No real zeros; they are the same.
 9. $2 \pm \sqrt{3}$
 $(x - 2 - \sqrt{3})(x - 2 + \sqrt{3})$
 11. $6 \pm \sqrt{10}$
 $(x - 6 - \sqrt{10})(x - 6 + \sqrt{10})$
 13. $\pm 5i$
 $(x + 5i)(x - 5i)$
 15. $\pm \frac{3}{2}, \pm \frac{3}{2}i$
 $(2x - 3)(2x + 3)(2x - 3i)(2x + 3i)$
 17. $\frac{1 \pm \sqrt{223}i}{2}$
 $\left(z - \frac{1 - \sqrt{223}i}{2}\right)\left(z - \frac{1 + \sqrt{223}i}{2}\right)$
 19. $\pm i, \pm 3i$
 $(x + i)(x - i)(x + 3i)(x - 3i)$
 21. $\frac{5}{3}, \pm 4i$
 $(3x - 5)(x - 4i)(x + 4i)$
 23. $-5, 4 \pm 3i$
 $(t + 5)(t - 4 + 3i)(t - 4 - 3i)$
 25. $1 \pm \sqrt{5}i, -\frac{1}{5}$
 $(5x + 1)(x - 1 + \sqrt{5}i)(x - 1 - \sqrt{5}i)$
 27. $2, 2, \pm 2i$
 $(x - 2)^2(x + 2i)(x - 2i)$
 29. (a) $7 \pm \sqrt{3}$ (b) $(x - 7 - \sqrt{3})(x - 7 + \sqrt{3})$
 (c) $(7 \pm \sqrt{3}, 0)$
 31. (a) $\frac{3}{2}, \pm 2i$ (b) $(2x - 3)(x - 2i)(x + 2i)$ (c) $(\frac{3}{2}, 0)$

33. (a) $-6, 3 \pm 4i$ (b) $(x+6)(x-3-4i)(x-3+4i)$
 (c) $(-6, 0)$

35. (a) $\pm 4i, \pm 3i$ (b) $(x+4i)(x-4i)(x+3i)(x-3i)$
 (c) None

37. $f(x) = x^3 - 2x^2 + x - 2$

39. $f(x) = x^4 - 12x^3 + 53x^2 - 100x + 68$

41. $f(x) = x^4 + 3x^3 - 7x^2 + 15x$

43. (a) $-(x-1)(x+2)(x-2i)(x+2i)$
 (b) $f(x) = -(x^4 + x^3 + 2x^2 + 4x - 8)$

45. (a) $-2(x+1)(x-2+\sqrt{5}i)(x-2-\sqrt{5}i)$
 (b) $f(x) = -2x^3 + 6x^2 - 10x - 18$

47. (a) $(x^2+1)(x^2-7)$
 (b) $(x^2+1)(x+\sqrt{7})(x-\sqrt{7})$
 (c) $(x+i)(x-i)(x+\sqrt{7})(x-\sqrt{7})$

49. (a) $(x^2-6)(x^2-2x+3)$
 (b) $(x+\sqrt{6})(x-\sqrt{6})(x^2-2x+3)$
 (c) $(x+\sqrt{6})(x-\sqrt{6})(x-1-\sqrt{2}i)(x-1+\sqrt{2}i)$

51. $-\frac{3}{2}, \pm 5i$ 53. $-3, 5 \pm 2i$ 55. $-\frac{2}{3}, 1 \pm \sqrt{3}i$

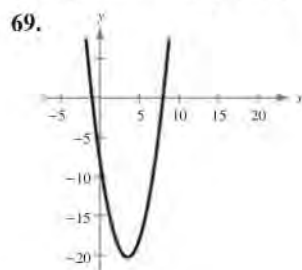
57. $\frac{3}{4}, \frac{1}{2}(1 \pm \sqrt{5}i)$ 59. (a) 1,000, 2,000 (b) $-3 \pm \sqrt{2}i$

61. (a) 0.750 (b) $\frac{1}{2} \pm \frac{\sqrt{5}}{2}i$

63. No. Setting $h = 64$ and solving the resulting equation yields imaginary roots.

65. False. A third-degree polynomial must have at least one real zero.

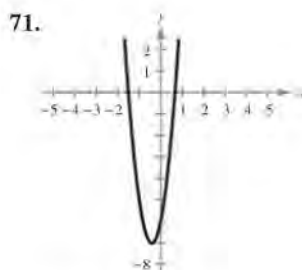
67. (a) $k = 4$ (b) $k < 0$



Vertex: $(\frac{7}{2}, -\frac{81}{4})$

Intercepts:

$(-1, 0), (8, 0), (0, -8)$



Vertex: $(-\frac{15}{12}, -\frac{169}{24})$

Intercepts:

$(-\frac{3}{2}, 0), (\frac{2}{3}, 0), (0, -6)$

Section 2.6 (page 152)

Vocabulary Check (page 152)

1. rational functions
2. vertical asymptote
3. horizontal asymptote

1. (a) Domain: all real numbers x except $x = 1$

(b)

x	$f(x)$	x	$f(x)$
0.5	-2	1.5	2
0.9	-10	1.1	10
0.99	-100	1.01	100
0.999	-1000	1.001	1000

x	$f(x)$	x	$f(x)$
5	0.25	-5	-0.16
10	0.1	-10	-0.09
100	0.01	-100	-0.0099
1000	0.001	-1000	-0.000999

(c) f approaches $-\infty$ from the left and ∞ from the right of $x = 1$.

3. (a) Domain: all real numbers x except $x = 1$

(b)

x	$f(x)$	x	$f(x)$
0.5	3	1.5	9
0.9	27	1.1	33
0.99	297	1.01	303
0.999	2997	1.001	3003

x	$f(x)$	x	$f(x)$
5	3.75	-5	-2.5
10	3.33	-10	-2.727
100	3.03	-100	-2.97
1000	3.003	-1000	-2.997

(c) f approaches ∞ from both the left and the right of $x = 1$.

5. (a) Domain: all real numbers
- x
- except
- $x = \pm 1$

x	$f(x)$	x	$f(x)$
0.5	-1	1.5	5.4
0.9	-12.79	1.1	17.29
0.99	-147.8	1.01	152.3
0.999	-1498	1.001	1502.3

x	$f(x)$	x	$f(x)$
5	3.125	-5	3.125
10	$3.\overline{03}$	-10	$3.\overline{03}$
100	$3.\overline{0003}$	-100	$3.\overline{0003}$
1000	3	-1000	3.000003

- (c)
- f
- approaches
- ∞
- from the left and
- $-\infty$
- from the right of
- $x = -1$
- .
- f
- approaches
- $-\infty$
- from the left and
- ∞
- from the right of
- $x = 1$
- .

7. a. 8. d. 9. c. 10. e. 11. b. 12. f

13. (a) Vertical asymptote:
- $x = 0$

 Horizontal asymptote: $y = 0$

- (b) Holes: none

15. (a) Vertical asymptote:
- $x = 2$

 Horizontal asymptote: $y = -1$

- (b) Hole at
- $x = 0$

17. (a) Vertical asymptote:
- $x = 0$

 Horizontal asymptote: $y = 1$

- (b) Hole at
- $x = -5$

19. (a) Domain: all real numbers
- x

- (b) Continuous

- (c) Vertical asymptote: none

 Horizontal asymptote: $y = 3$

21. (a) Domain: all real numbers
- x
- except
- $x = 0$

- (b) Not continuous

- (c) Vertical asymptote:
- $x = 0$

 Horizontal asymptotes: $y = \pm 1$

23. (a) Domain of
- f
- : all real numbers
- x
- except
- $x = 4$
- ;
-
- domain of
- g
- : all real numbers
- x

- (b)
- $x + 4, x \neq 4$
- ; Vertical asymptote: none (c)
- $x = 4$

x	1	2	3	4	5	6	7
$f(x)$	5	6	7	Undef.	9	10	11
$g(x)$	5	6	7	8	9	10	11

- (e) Values differ only where
- f
- is undefined.

25. (a) Domain of
- f
- : all real numbers
- x
- except
- $x = -1, 3$
- ;
-
- domain of
- g
- : all real numbers
- x
- except
- $x = 3$

- (b)
- $\frac{x-1}{x-3}, x \neq -1$
- ; Vertical asymptote:
- $x = 3$

- (c)
- $x = -1$

x	-2	-1	0	1	2	3	4
$f(x)$	$\frac{3}{5}$	Undef.	$\frac{1}{3}$	0	-1	Undef.	3
$g(x)$	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{1}{3}$	0	-1	Undef.	3

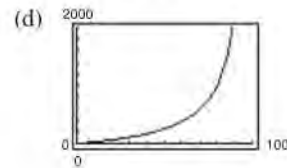
- (e) Values differ only where
- f
- is undefined and
- g
- is defined.

27. 4; less than; greater than 29. 2; greater than; less than

- 31.
- ± 2
33. 7 35. -1, 3 37. 2

39. (a) \$28.33 million (b) \$170 million

- (c) \$765 million



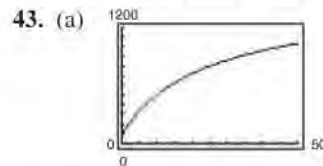
Answers will vary.

- (e) No. The function is undefined at the 100% level.

41. (a)
- $y = \frac{1}{0.445 - 0.007x}$

Age, x	16	32	44	50	60
Near point, y	3.0	4.5	7.3	10.5	40

- (c) No; the function is negative for
- $x = 70$
- .



- (b) 333 deer, 500 deer, 800 deer

- (c) 1500. Because the degrees of the numerator and the denominator are equal, the limiting size is the ratio of the leading coefficients,
- $60/0.04 = 1500$
- .

45. False. The degree of the denominator gives the maximum possible number of vertical asymptotes, and the degree is finite.

47. b. 49.
- $f(x) = \frac{x-1}{x^3-8}$
- 51.
- $f(x) = \frac{2x^2-18}{x^2+x-2}$

- 53.
- $x - y - 1 = 0$
- 55.
- $3x - y + 1 = 0$

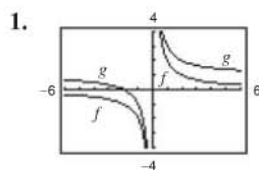
57. $x + 9 + \frac{42}{x-4}$

59. $2x^2 - 9 + \frac{34}{x^2 + 5}$

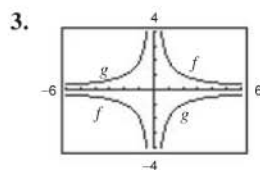
Section 2.7 (page 161)

Vocabulary Check (page 161)

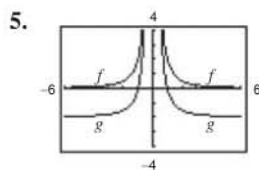
1. slant, asymptote 2. vertical



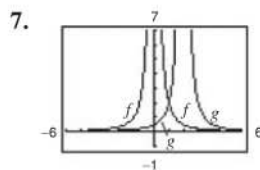
Vertical shift



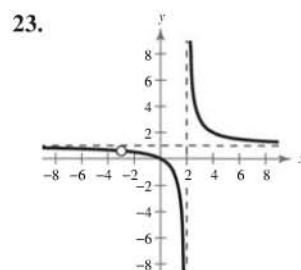
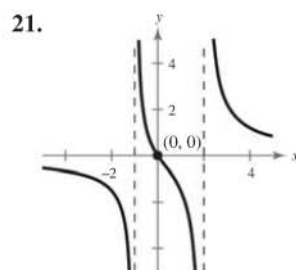
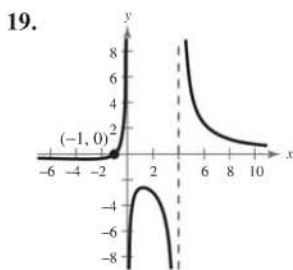
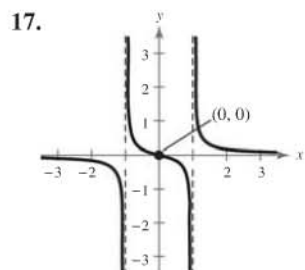
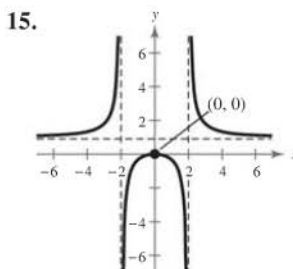
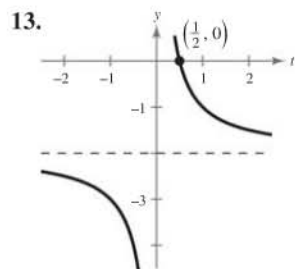
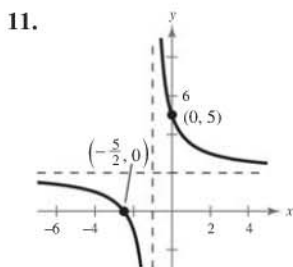
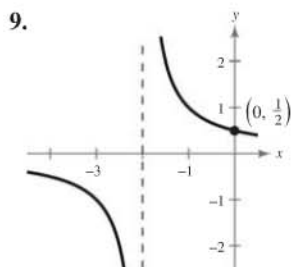
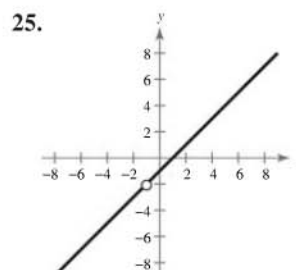
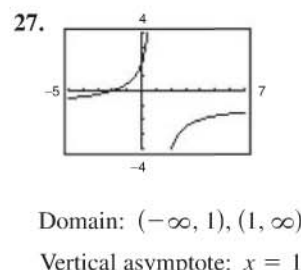
Reflection in the x-axis



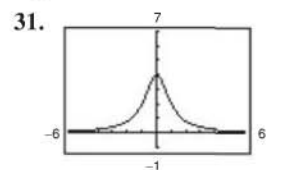
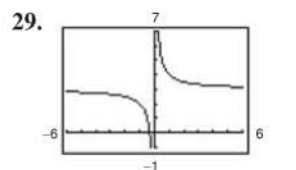
Vertical shift



Horizontal shift


There is a hole at $x = -3$.

There is a hole at $x = -1$.

Domain: $(-\infty, 1), (1, \infty)$

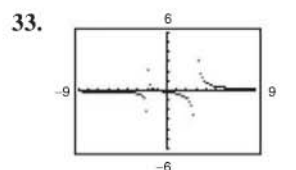
Vertical asymptote: $x = 1$

Horizontal asymptote: $y = -1$

Domain: $(-\infty, 0), (0, \infty)$

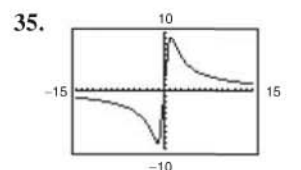
Vertical asymptote: $t = 0$

Horizontal asymptote: $y = 3$

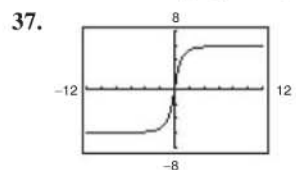
Domain: $(-\infty, \infty)$

Horizontal asymptote: $y = 0$

Domain: $(-\infty, -2), (-2, 3), (3, \infty)$

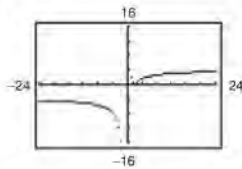
Vertical asymptotes: $x = -2, x = 3$

Horizontal asymptote: $y = 0$

Domain: $(-\infty, 0), (0, \infty)$

Vertical asymptote: $x = 0$

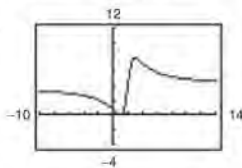
Horizontal asymptote: $y = 0$

There are two horizontal asymptotes, $y = \pm 6$.

39.



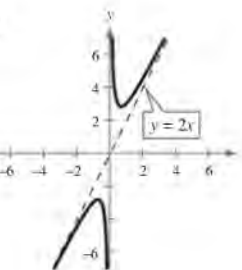
There are two horizontal asymptotes, $y = \pm 4$, and one vertical asymptote, $x = -1$.

41.

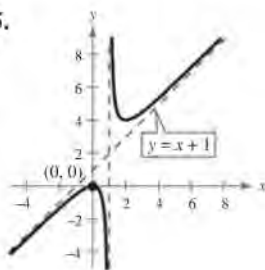


The graph crosses the horizontal asymptote, $y = 4$.

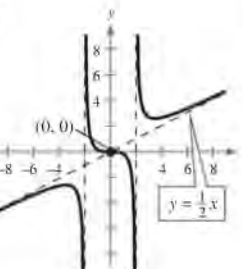
43.



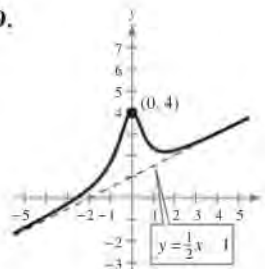
45.



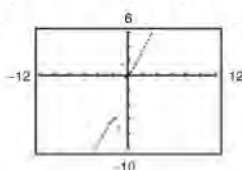
47.



49.

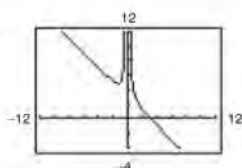

 51. $(-1, 0)$ 53. $(1, 0), (-1, 0)$

55.



Domain: $(-\infty, -1), (-1, \infty)$
Vertical asymptote: $x = -1$
Slant asymptote: $y = 2x - 1$

57.



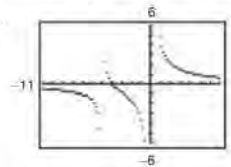
Domain: $(-\infty, 0), (0, \infty)$
Vertical asymptote: $x = 0$
Slant asymptote: $y = -x + 3$

59. Vertical asymptotes: $x = \pm 2$; horizontal asymptote: $y = 1$; slant asymptote: none; holes: none

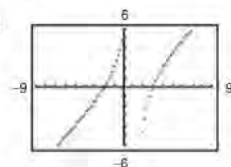
61. Vertical asymptote: $x = -\frac{3}{2}$; horizontal asymptote: $y = 1$; slant asymptote: none; hole at $x = 2$

63. Vertical asymptote: $x = -2$; horizontal asymptote: none; slant asymptote: $y = 2x - 7$; hole at $x = -1$

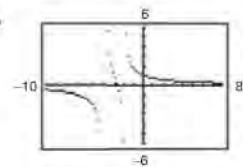
65.


 $(-4, 0)$

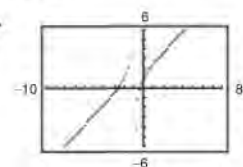
69.



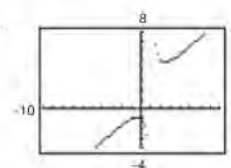
67.


 $(-\frac{8}{3}, 0)$

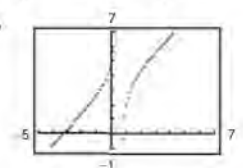
71.


 $(3, 0), (-2, 0)$

73.



75.

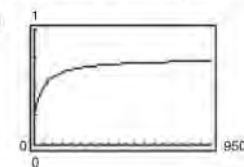


None

 $(\frac{-5 \pm \sqrt{65}}{4}, 0)$

77. (a) Answers will vary. (b) $[0, 950]$

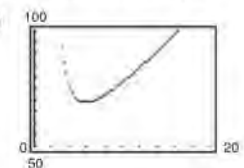
(c)



The concentration increases more slowly; the concentration approaches 75%.

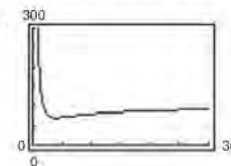
79. (a) Answers will vary. (b) $(2, \infty)$

(c)

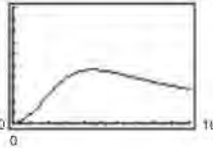


5.9 inches $\times$ 11.8 inches

81.

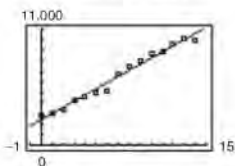

 $x \approx 40$

83. (a) $C = 0$. The chemical will eventually dissipate.

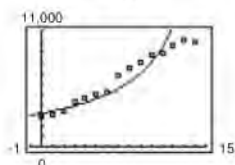
- (b)  $t \approx 4.5$ hours

- (c) Before ≈ 2.6 hours and after ≈ 8.3 hours

85. (a) $A = 583.8t + 2414$



(b)
$$A = \frac{1}{-0.0000185t + 0.000315}$$
$$= \frac{-54054.05}{t - 17.03}$$



Year, x	1990	1991	1992	1993
Original data, y	2777	3013	3397	4193
Model from (a), y	2414	2998	3582	4165
Model from (b), y	3174	3372	3596	3853

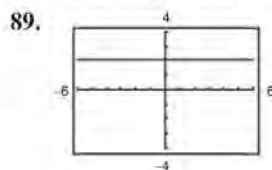
Year, x	1994	1995	1996	1997
Original data, y	4557	4962	5234	6734
Model from (a), y	4749	5333	5917	6501
Model from (b), y	4148	4493	4901	5389

Year, x	1998	1999	2000	2001
Original data, y	7387	8010	8698	8825
Model from (a), y	7084	7668	8252	8836
Model from (b), y	5986	6732	7689	8964

Year, x	2002	2003	2004
Original data, y	9533	10,164	10,016
Model from (a), y	9420	10,003	10,587
Model from (b), y	10,746	13,413	17,840

Answers will vary.

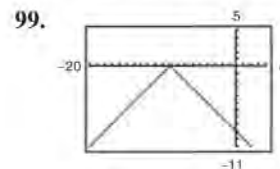
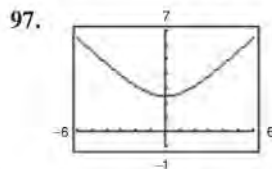
87. False. A graph with a vertical asymptote is not continuous.



The denominator is a factor of the numerator.

91. $f(x) = \frac{x^2 + 3x - 10}{x + 2}$

93. $\frac{512}{x^3}$ 95. 3



Domain: $(-\infty, \infty)$

Domain: $(-\infty, \infty)$

Range: $[\sqrt{6}, \infty)$

Range: $(-\infty, 0]$

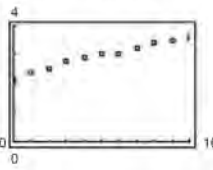
101. Answers will vary.

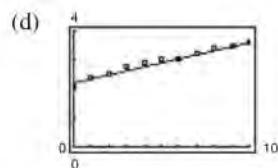
Section 2.8 (page 169)

Vocabulary Check (page 169)

1. linear 2. quadratic

1. Quadratic 3. Linear 5. Neither

7. (a)  (b) Linear
(c) $y = 0.14x + 2.2$

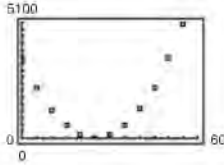


(e)

x	0	1	2	3	4	5
Actual, y	2.1	2.4	2.5	2.8	2.9	3.0
Model, y	2.2	2.3	2.5	2.6	2.8	2.9

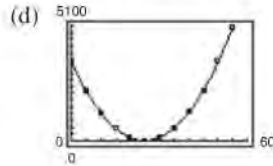
x	6	7	8	9	10
Actual, y	3.0	3.2	3.4	3.5	3.6
Model, y	3.0	3.2	3.3	3.5	3.6

9. (a)



(b) Quadratic

(c) $y = 5.55x^2 - 277.5x + 3478$

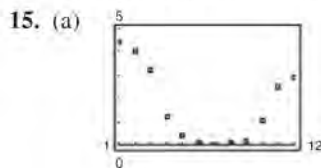


x	0	5	10	15	20
Actual, y	3480	2235	1250	565	150
Model, y	3478	2229	1258	564	148

x	25	30	35	40
Actual, y	12	145	575	1275
Model, y	9	148	564	1258

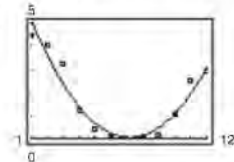
x	45	50	55
Actual, y	2225	3500	5010
Model, y	2229	3478	5004

11. (a) $y = 2.48x + 1.1$; $y = 0.071x^2 + 1.69x + 2.7$
 (b) 0.98995; 0.99519 (c) Quadratic
13. (a) $y = -0.89x + 5.3$; $y = 0.001x^2 - 0.90x + 5.3$
 (b) 0.99982; 0.99987 (c) Quadratic

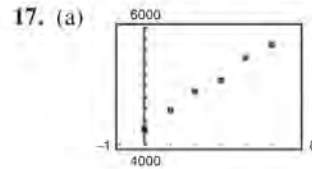


(b) $P = 0.1322t^2 - 1.901t + 6.87$

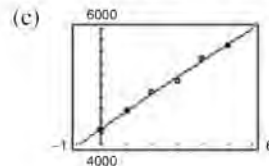
(c)



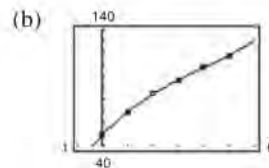
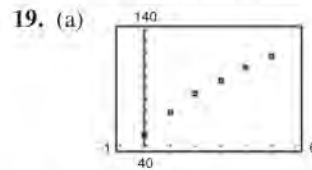
(d) July



(b) $S = -2.630t^2 + 301.74t + 4270.2$

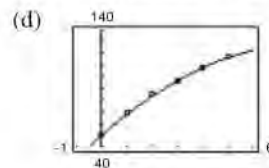


(d) 2024 (e) Answers will vary.



Answers will vary.

(c) $A = -1.1357t^2 + 18.999t + 50.30$; 0.99859



Answers will vary.

(e) Cubic model; the coefficient of determination is closer to 1.

Year	2006	2007	2008
$A^{\dagger}$	127.76	140.15	154.29
Cubic model	129.91	145.13	164.96
Quadratic model	123.40	127.64	129.60

Explanations will vary.

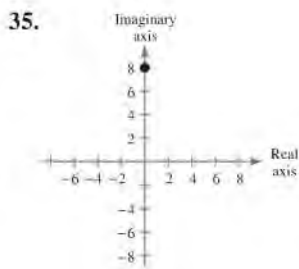
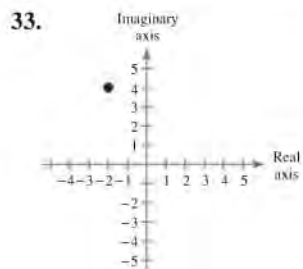
21. True 23. The model is consistently above the data.

25. (a) $(f \circ g)(x) = 10x^2 + 3$

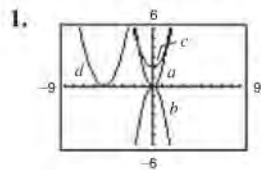
(b) $(g \circ f)(x) = 50x^2 + 160x + 127$

27. (a) $(f \circ g)(x) = x$ (b) $(g \circ f)(x) = x$

29. $f^{-1}(x) = 5x + 4$ 31. $f^{-1}(x) = \frac{\sqrt{2x+6}}{2}$



Review Exercises (page 173)

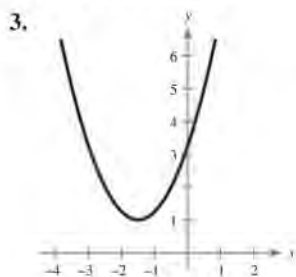


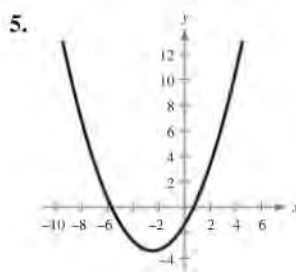
(a) Vertical stretch

(b) Vertical stretch, reflection in the x -axis

(c) Vertical shift

(d) Horizontal shift

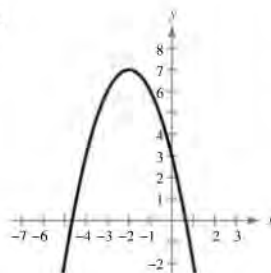

Vertex: $(-\frac{3}{2}, 1)$

Intercept: $(0, \frac{13}{4})$

Vertex: $(-\frac{5}{2}, -\frac{41}{12})$

Intercepts:

 $(0, -\frac{4}{3}), (-\frac{5 \pm \sqrt{41}}{2}, 0)$

7.


Vertex: $(-2, 7)$

Intercepts:

 $(0, 3), (-2 \pm \sqrt{7}, 0)$

9. $f(x) = (x - 1)^2 - 4$

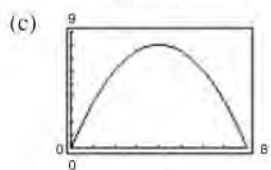
11. $f(x) = 2(x + 2)^2 - 2$

13. (a) $A = x\left(\frac{8-x}{2}\right), 0 < x < 8$

(b)

x	1	2	3	4	5	6
Area	$\frac{7}{2}$	6	$\frac{15}{2}$	8	$\frac{15}{2}$	6

$x = 4, y = 2$



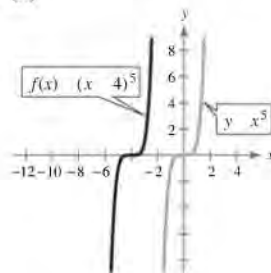
$x = 4, y = 2$

(d) $A = -\frac{1}{2}(x - 4)^2 + 8$

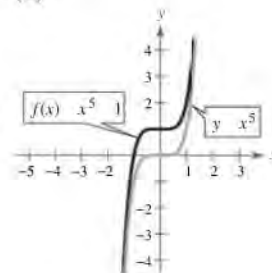
(e) Answers will vary.

15. $x = 125$ feet, $y = 187.5$ feet

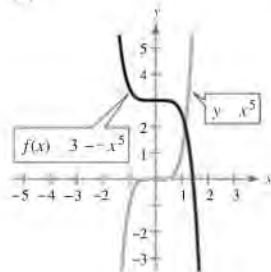
17. (a)



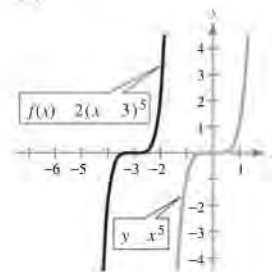
(b)



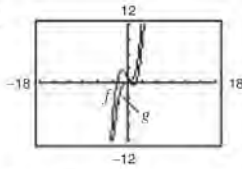
(c)



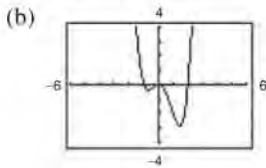
(d)

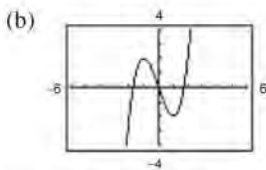


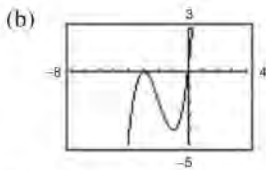
19.


 21. Falls to the left,
falls to the right

 23. Rises to the left,
rises to the right

 25. (a) $x = -1, 0, 0, 2$

 (c) $x = -1, 0, 0, 2$; They are the same.

 27. (a) $t = 0, \pm\sqrt{3}$

 (c) $t = 0, \pm 1.73$; They are the same.

 29. (a) $x = -3, -3, 0$

 (c) $x = -3, -3, 0$; They are the same.

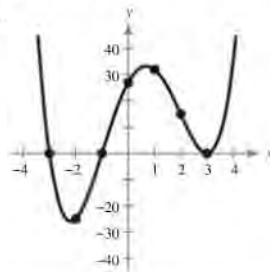
 31. $f(x) = x^4 - 5x^3 - 3x^2 + 17x - 10$

 33. $f(x) = x^3 - 7x^2 + 13x - 3$

35. (a) Rises to the left, rises to the right

 (b) $x = -3, -1, 3, 3$

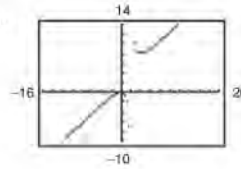
(c) and (d)


 37. (a) $(-3, -2), (-1, 0), (0, 1)$

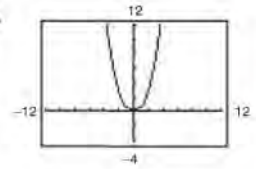
 (b) $x = -2.25, -0.56, 0.80$

 39. (a) $(-3, -2), (2, 3)$ (b) $x = -2.57, 2.57$

41.



43.


 45. $8x + 5 + \frac{2}{3x - 2}$

 47. $x^2 - 2, x \neq \pm 1$

 49. $5x + 2, x \neq \frac{3 \pm \sqrt{5}}{2}$

 51. $3x^2 + 5x + 8 + \frac{10}{2x^2 - 1}$

 53. $0.25x^3 - 4.5x^2 + 9x - 18 + \frac{36}{x + 2}$

 55. $6x^3 - 27x, x \neq \frac{2}{3}$

 57. $3x^2 + 2x + 20 + \frac{58}{x - 4}$

 59. (a) -421 (b) -156

61. (a) Answers will vary.

 (b) $(x + 1)(x + 7)$

 (c) $f(x) = (x - 4)(x + 1)(x + 7)$

 (d) $x = 4, -1, -7$

63. (a) Answers will vary.

 (b) $(x + 1)(x - 4)$

 (c) $f(x) = (x + 2)(x - 3)(x + 1)(x - 4)$

 (d) $x = -2, 3, -1, 4$

 65. $\pm 1, \pm 3, \pm \frac{3}{2}, \pm \frac{3}{4}, \pm \frac{1}{2}, \pm \frac{1}{4}$

 67. $x = \frac{5}{6}$

 69. $x = -1, \frac{3}{2}, 3, \frac{2}{3}$

71. 2 or 0 positive real zeros

1 negative real zero

73. Answers will vary.

 75. $6 + 5i$

 77. $2 + 7i$

 79. $3 + 7i$

 81. $40 + 65i$

 83. $-26 + 7i$

 85. $3 + 9i$

 87. $-4 - 46i$

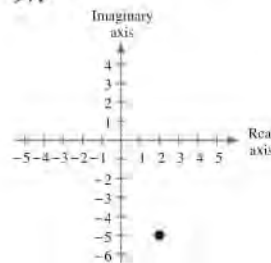
 89. -80

 91. $1 - 6i$

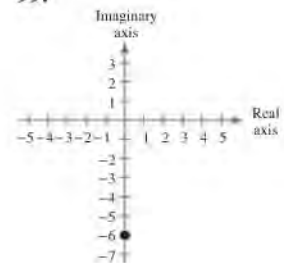
 93. $\frac{17}{26} + \frac{7}{26}i$

 95. $-3 - 2i$

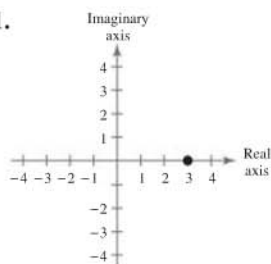
97.



99.



101.

103. $x = 0, 2, 2$ 105. $x = 2, -\frac{3}{2}, 1 \pm i$;

$$(x - 2)(2x + 3)(x - 1 + i)(x - 1 - i)$$

107. $x = 4, \frac{3 \pm \sqrt{15}i}{2}$;

$$(x - 4)\left(x - \frac{3 + \sqrt{15}i}{2}\right)\left(x - \frac{3 - \sqrt{15}i}{2}\right)$$

109. $x = 0, -1, \pm\sqrt{5}i$; $x^2(x + 1)(x + \sqrt{5}i)(x - \sqrt{5}i)$ 111. (a) $x = 2, 1 \pm i$

$$(b) (x - 2)(x - 1 - i)(x - 1 + i) \quad (c) (2, 0)$$

113. (a) $x = -6, -1, \frac{2}{3}$

$$(b) -(x + 1)(x + 6)(3x - 2)$$

$$(c) (-6, 0), (-1, 0), \left(\frac{2}{3}, 0\right)$$

115. (a) $\pm 3i, \pm 5i$

$$(b) (x - 3i)(x + 3i)(x - 5i)(x + 5i) \quad (c) \text{None}$$

$$117. f(x) = x^4 - 2x^3 + 17x^2 - 50x - 200$$

$$119. f(x) = x^4 + 9x^3 + 48x^2 + 78x - 136$$

$$121. (a) (x^2 + 9)(x^2 - 2x - 1)$$

$$(b) (x^2 + 9)(x - 1 + \sqrt{2})(x - 1 - \sqrt{2})$$

$$(c) (x + 3i)(x - 3i)(x - 1 + \sqrt{2})(x - 1 - \sqrt{2})$$

123. $x = -3, \pm 2i$ 125. (a) Domain: all real numbers x except $x = -3$

(b) Not continuous

(c) Vertical asymptote: $x = -3$ Horizontal asymptote: $y = -1$ 127. (a) Domain: all real numbers x except $x = 6, -3$

(b) Not continuous

(c) Vertical asymptotes: $x = 6, x = -3$ Horizontal asymptote: $y = 0$ 129. (a) Domain: all real numbers x except $x = 7$

(b) Not continuous

(c) Vertical asymptote: $x = 7$ Horizontal asymptote: $y = -1$ 131. (a) Domain: all real numbers x except $x = \pm \frac{\sqrt{6}}{2}$

(b) Not continuous

(c) Vertical asymptotes: $x = \pm \frac{\sqrt{6}}{2}$ Horizontal asymptote: $y = 2$ 133. (a) Domain: all real numbers x except $x = 5, -3$

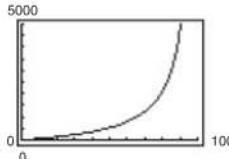
(b) Not continuous

(c) Vertical asymptote: $x = -3$ Horizontal asymptote: $y = 0$ 135. (a) Domain: all real numbers x

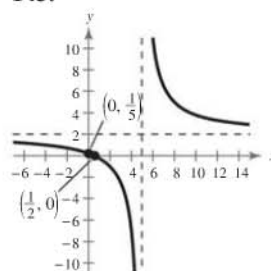
(b) Continuous

(c) Horizontal asymptotes: $y = \pm 1$

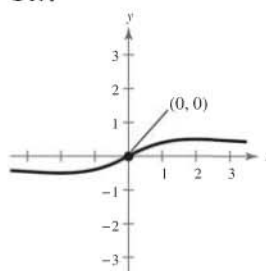
137. (a) \$176 million; \$528 million; \$1584 million

(b)  Answers will vary.(c) No. As $p \rightarrow 100$, the cost approaches ∞ .139. Vertical asymptote: $x = -1$ Horizontal asymptote: $y = 1$ Hole at $x = 1$ 141. Vertical asymptote: $x = -\frac{3}{2}$ Horizontal asymptote: $y = 1$ Hole at $x = 3$ 143. Vertical asymptote: $x = -1$ Slant asymptote: $y = 3x - 10$ Hole at $x = -2$

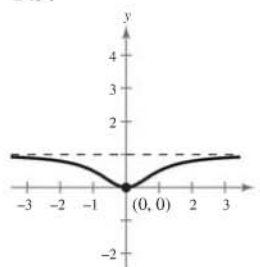
145.



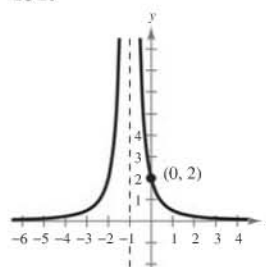
147.



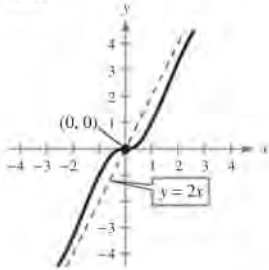
149.



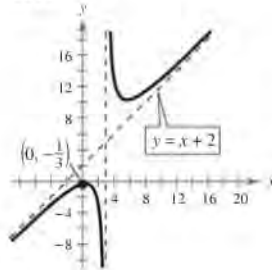
151.



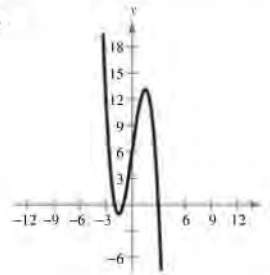
153.



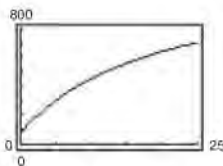
155.



4.



157. (a)

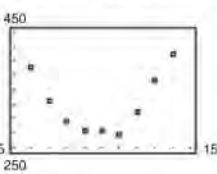


(b) 304,000; 453,333; 702,222

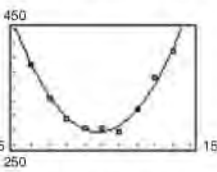
 (c) 1,200,000, because N has a horizontal asymptote at $y = 1200$.

159. Quadratic 161. Linear

163. (a)


 (b) $y = 8.03x^2 - 157.1x + 1041$

(c)



Answers will vary.

(d) 2005

(e) Answers will vary.

165. False. For the graph of a rational function to have a slant asymptote, the degree of its numerator must be exactly one more than the degree of its denominator.

 167. False. Example: $(1 + 2i) + (1 - 2i) = 2$.

169. Answers will vary.

 171. The first step is completed incorrectly: $\sqrt{-4} = 2i \neq 4i$

Chapter Test (page 179)

 1. Vertex: $(-2, -1)$

 Intercepts: $(0, 3)$, $(-3, 0)$, $(-1, 0)$

 2. $y = (x - 3)^2 - 6$

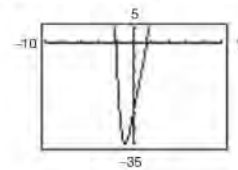
 3. 0, multiplicity 1; $-\frac{1}{2}$, multiplicity 2

5. $3x + \frac{x-1}{x^2+1}$

6. $2x^3 + 4x^2 + 3x + 6 + \frac{9}{x-2}$

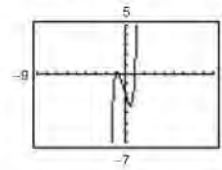
7. 13

8. $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12, \pm 24, \pm \frac{1}{2}, \pm \frac{3}{2}$



$$t = -2, \frac{3}{2}$$

9. $\pm 1, \pm 2, \pm \frac{1}{3}, \pm \frac{2}{3}$



$$x = \pm 1, -\frac{2}{3}$$

10. $x = -1, 4 \pm \sqrt{3}i$

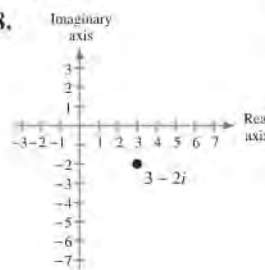
$$(x+1)(x-4+\sqrt{3}i)(x-4-\sqrt{3}i)$$

11. $-9 - 18i$ 12. $6 + (2\sqrt{5} + \sqrt{14})i$ 13. $13 + 4i$

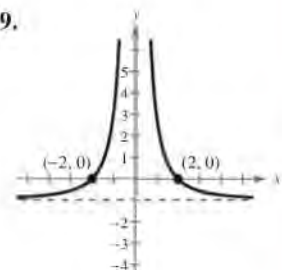
14. $-17 + 14i$ 15. $\frac{43}{37} + \frac{38}{37}i$ 16. $1 + 2i$

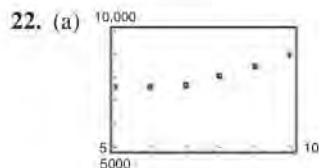
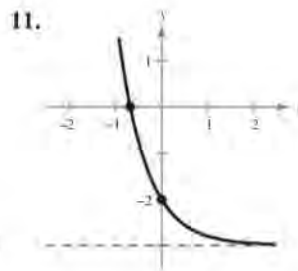
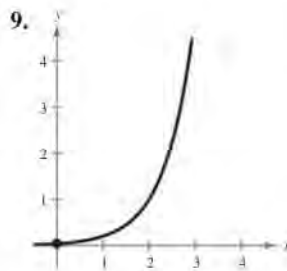
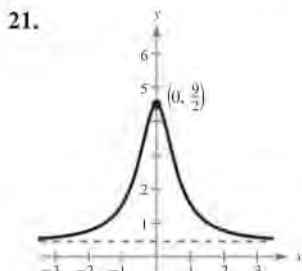
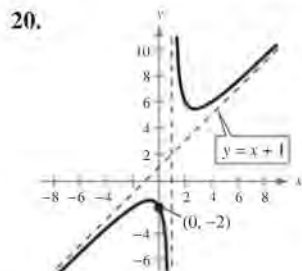
17. $\frac{4}{13} + \frac{7}{13}i$

18.

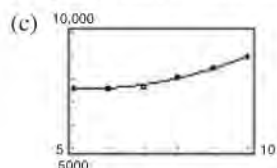


19.





(b) $y = 5.582x^2 - 85.53x + 602.0$



Answers will vary.

(d) \$574.96 billion; \$1124.17 billion

(e) Answers will vary.

Chapter 3

Section 3.1 (page 193)

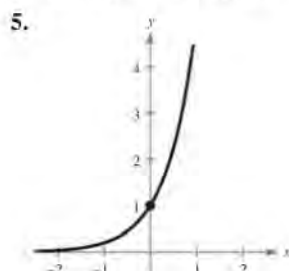
Vocabulary Check (page 193)

1. algebraic 2. transcendental

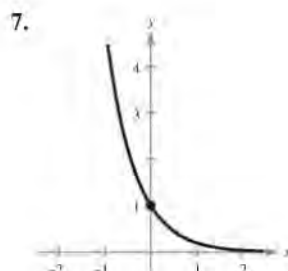
3. natural exponential, natural

4. $A = P\left(1 + \frac{r}{n}\right)^{nt}$ 5. $A = Pe^{rt}$

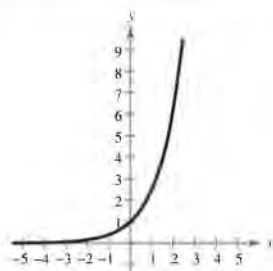
1. 4112.033 3. 0.006



$y = 0, (0, 1)$, increasing



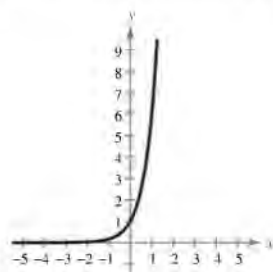
$y = 0, (0, 1)$, decreasing



Asymptote: $y = 0$

29.

x	-2	-1	0	1	2
$f(x)$	0.03	0.17	1	6	36



Asymptote: $y = 0$

$y = 0, (0, \frac{1}{25})$, increasing

$y = -3, (0, -2)$,
 $(-0.683, 0)$, decreasing

13. d 14. a 15. c 16. b

17. Right shift of five units

19. Left shift of four units and reflection in the x -axis

21. Right shift of two units and downward shift of three units

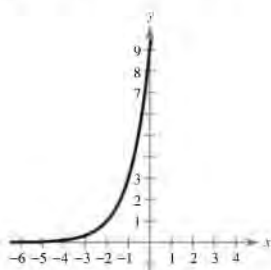
23. 9897.129 25. 54.164

27.

x	-2	-1	0	1	2
$f(x)$	0.16	0.4	1	2.5	6.25

31.

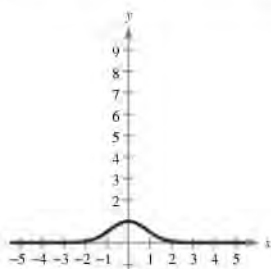
x	-3	-2	-1	0	1
$f(x)$	0.33	1	3	9	27



Asymptote: $y = 0$

33.

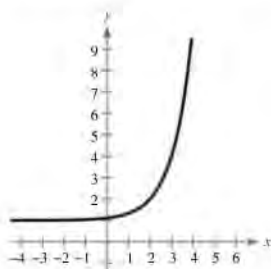
x	-2	-1	0	1	2
y	0.06	0.5	1	0.5	0.06



Asymptote: $y = 0$

35.

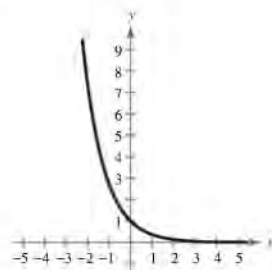
x	-1	0	1	2	3	4
y	1.04	1.11	1.33	2	4	10



Asymptote: $y = 1$

37.

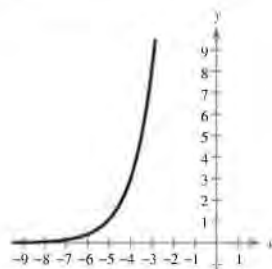
x	-2	-1	0	1	2
$f(x)$	7.39	2.72	1	0.37	0.14



Asymptote: $y = 0$

39.

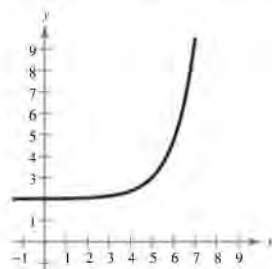
x	-6	-5	-4	-3	-2
$f(x)$	0.41	1.10	3	8.15	22.17



Asymptote: $y = 0$

41.

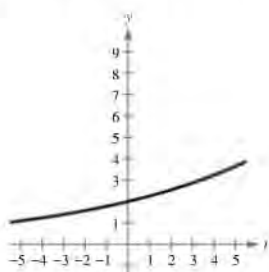
x	3	4	5	6	7
$f(x)$	2.14	2.37	3	4.72	9.39



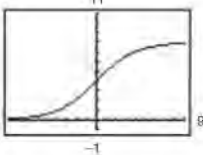
Asymptote: $y = 2$

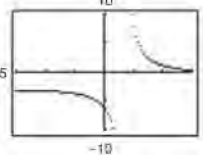
43.

t	-2	-1	0	1	2
$s(t)$	1.57	1.77	2	2.26	2.54

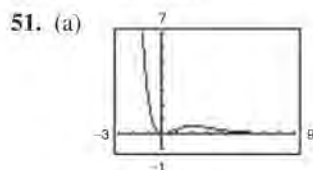


Asymptote: $y = 0$

45. (a)  (b) $y = 0, y = 8$

47. (a)  (b) $y = -3, y = 0, x \approx 3.47$

49. (86.350, 1500)



- (b) Decreasing on $(-\infty, 0), (2, \infty)$

Increasing on $(0, 2)$

- (c) Relative minimum: $(0, 0)$

Relative maximum: $(2, 0.54)$

53.

n	1	2	4	12
A	\$3200.21	\$3205.09	\$3207.57	\$3209.23

n	365	Continuous
A	\$3210.04	\$3210.06

55.

n	1	2	4	12
A	\$5477.81	\$5520.10	\$5541.79	\$5556.46

n	365	Continuous
A	\$5563.61	\$5563.85

57.

t	1	10	20
A	\$12,489.73	\$17,901.90	\$26,706.49

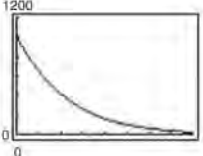
t	30	40	50
A	\$39,841.40	\$59,436.39	\$88,668.67

59.

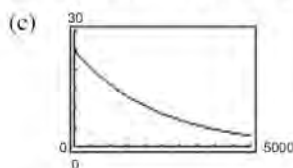
t	1	10	20
A	\$12,427.44	\$17,028.81	\$24,165.03

t	30	40	50
A	\$34,291.81	\$48,662.40	\$69,055.23

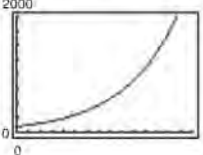
61. \$1530.57 63. \$17,281.77

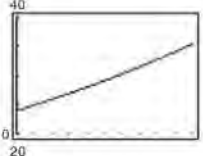
65. (a)  (b) \$421.12
(c) \$350.13

67. (a) 25 grams (b) 16.21 grams



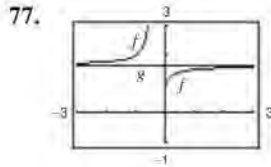
- (d) Never. The graph has a horizontal asymptote at $Q = 0$.

69. (a)  (b) 100; 300; 900

71. (a)  (b) and (c) \$35.45

73. True. The definition of an exponential function is $f(x) = a^x$, $a > 0$, $a \neq 1$.

75. d

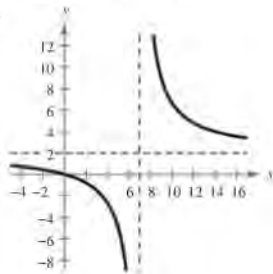


$f(x)$ approaches $g(x) \approx 1.6487$.

79. > 81. > 83. $f^{-1}(x) = \frac{x+7}{5}$

85. $f^{-1}(x) = x^3 - 8$

87.



89. Answers will vary.

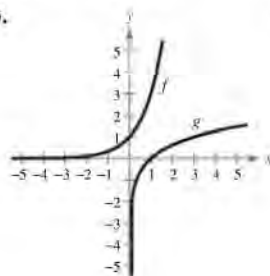
Section 3.2 (page 203)

Vocabulary Check (page 203)

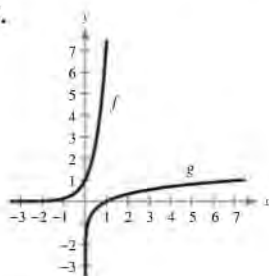
1. logarithmic function 2. 10
3. natural logarithmic 4. $a^{\log_a x} = x$ 5. $x = y$

1. $4^3 = 64$ 3. $7^{-2} = \frac{1}{49}$ 5. $32^{2/5} = 4$ 7. $e^0 = 1$
9. $e^1 = e$ 11. $e^{1/2} = \sqrt{e}$ 13. $\log_5 125 = 3$
15. $\log_{81} 3 = \frac{1}{4}$ 17. $\log_6 \frac{1}{36} = -2$
19. $\ln 20.0855 \dots = 3$ 21. $\ln 3.6692 \dots = 1.3$
23. $\ln 1.3596 \dots = \frac{1}{3}$ 25. 4 27. -3
29. 2.538 31. 7.022 33. 9 35. 2 37. $\frac{1}{10}$
39. $3x$ 41. -3

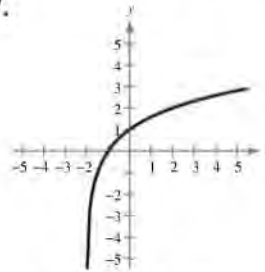
43.



45.



47.

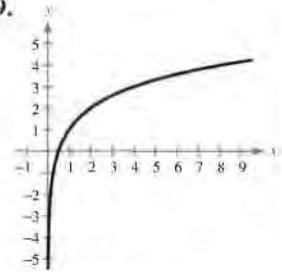


Domain: $(-2, \infty)$

Vertical asymptote: $x = -2$

x-intercept: $(-1, 0)$

49.

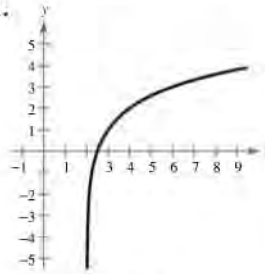


Domain: $(0, \infty)$

Vertical asymptote: $x = 0$

x-intercept: $(\frac{1}{2}, 0)$

51.



Domain: $(2, \infty)$

Vertical asymptote: $x = 2$

x-intercept: $(\frac{5}{2}, 0)$

53. b 54. c 55. d 56. a

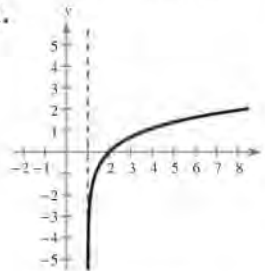
57. Reflection in the x -axis

59. Reflection in the x -axis, vertical shift four units upward

61. Horizontal shift three units to the left and vertical shift two units downward

63. 1.869 65. 0.693 67. 2 69. 1.8

71.

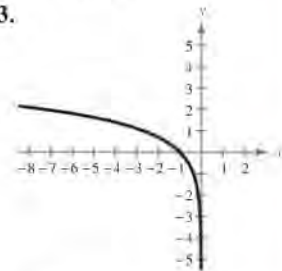


Domain: $(1, \infty)$

Vertical asymptote: $x = 1$

x-intercept: $(2, 0)$

73.



Domain: $(-\infty, 0)$

Vertical asymptote: $x = 0$

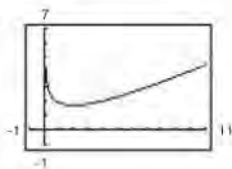
x-intercept: $(-1, 0)$

75. Horizontal shift three units to the left

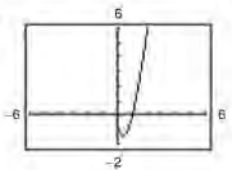
77. Vertical shift five units downward

79. Horizontal shift one unit to the right and vertical shift two units upward

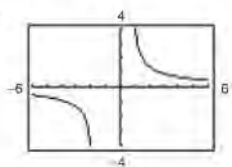
81. (a)

(b) Domain: $(0, \infty)$ (c) Decreasing on $(0, 2)$; increasing on $(2, \infty)$ (d) Relative minimum: $(2, 1.693)$

83. (a)

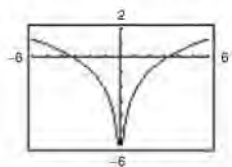
(b) Domain: $(0, \infty)$ (c) Decreasing on $(0, 0.368)$; increasing on $(0.368, \infty)$ (d) Relative minimum: $(0.37, -1.47)$, $(0.368, -1.472)$

85. (a)

(b) Domain: $(-\infty, -2)$, $(1, \infty)$ (c) Decreasing on $(-\infty, -2)$; $(1, \infty)$

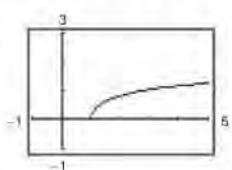
(d) No relative maxima or minima

87. (a)

(b) Domain: $(-\infty, 0)$, $(0, \infty)$ (c) Decreasing on $(-\infty, 0)$; increasing on $(0, \infty)$

(d) No relative maxima or minima

89. (a)

(b) Domain: $[1, \infty)$ (c) Increasing on $(1, \infty)$ (d) Relative minimum: $(1, 0)$

91. (a) 80 (b) 68.12 (c) 62.30

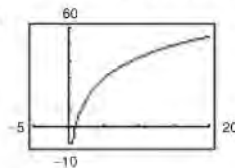
93. (a)

K	1	2	4	6
t	0	12.60	25.21	32.57

K	8	10	12
t	37.81	41.87	45.18

It takes 12.60 years for the principal to double.

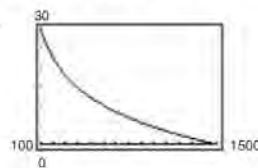
(b)



95. (a) 120 decibels (b) 100 decibels

(c) No. The function is not linear.

97.



17.66 cubic feet per minute

99. False. Reflect $g(x)$ about the line $y = x$.101. 2 103. $\frac{1}{4}$ 105. b107. $\log_a x$ is the inverse of a^x only if $0 < a < 1$ and $a > 1$, so $\log_a x$ is defined only for $0 < a < 1$ and $a > 1$.

109. (a) False (b) True (c) True (d) False

111. (a)

x	1	5	10	10^2
$f(x)$	0	0.32	0.23	0.046

x	10^4	10^6
$f(x)$	0.00092	0.0000138

(b) 0

113. $(x+3)(x-1)$ 115. $(4x+3)(3x-1)$ 117. $(4x+5)(4x-5)$ 119. $x(2x-9)(x+5)$

121. 15 123. 4300 125. 2.75 127. 27.67

Section 3.3 (page 211)

Vocabulary Check (page 211)

1. change-of-base 2. $\frac{\ln x}{\ln a}$ 3. $\log_a u^n$ 4. $\ln u + \ln v$

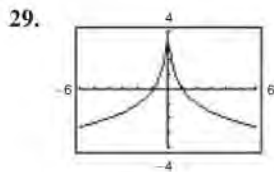
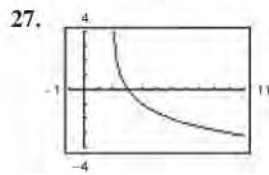
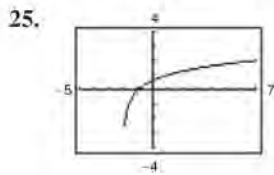
1. (a) $\frac{\log_{10} x}{\log_{10} 5}$ (b) $\frac{\ln x}{\ln 5}$ 3. (a) $\frac{\log_{10} x}{\log_{10} 5}$ (b) $\frac{\ln x}{\ln 5}$

5. (a) $\frac{\log_{10} \frac{3}{10}}{\log_{10} a}$ (b) $\frac{\ln \frac{3}{10}}{\ln a}$

7. (a) $\frac{\log_{10} x}{\log_{10} 2.6}$ (b) $\frac{\ln x}{\ln 2.6}$ 9. 1.771 11. -2

13. -0.102 15. 2.691 17. $\ln 5 + \ln 4$

19. $\ln 5 - 3 \ln 4$ 21. 1.6542 23. 0.2823



31. $\frac{3}{2}$ 33. $6 + \ln 5$ 35. Answers will vary.

37. $\log_{10} 5 + \log_{10} x$ 39. $\log_{10} 5 - \log_{10} x$

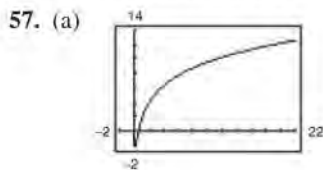
41. $4 \log_8 x$ 43. $\frac{1}{2} \ln z$ 45. $\ln x + \ln y + \ln z$

47. $2 \log_3 a + \log_3 b + 3 \log_3 c$

49. $2 \ln a + \frac{1}{2} \ln(a-1)$ 51. $\frac{1}{3} \ln x - \frac{1}{3} \ln y$

53. $\ln(x+1) + \ln(x-1) - 3 \ln x, x > 1$

55. $4 \ln x + \frac{1}{2} \ln y - 5 \ln z$


 (b)

x	1	2	3	4	5	6
y ₁	1.61	3.87	5.24	6.24	7.03	7.68
y ₂	1.61	3.87	5.24	6.24	7.03	7.68

x	7	8	9	10	11
y ₁	8.24	8.72	9.16	9.55	9.90
y ₂	8.24	8.72	9.16	9.55	9.90

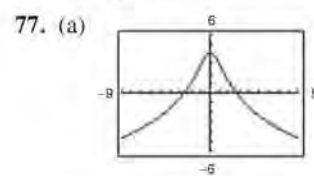
 (c) $y_1 = y_2$ for positive values of x .

59. $\ln 4x$ 61. $\log_4 \frac{z}{y}$ 63. $\log_2(x+3)^2$

65. $\ln \sqrt{x^2 + 4}$ 67. $\ln \frac{x}{(x+1)^3}$ 69. $\ln \frac{x-2}{x+2}$

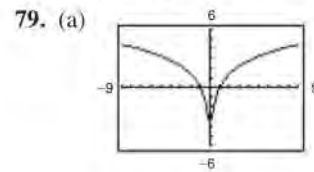
71. $\ln \frac{x}{(x^2-4)^2}$ 73. $\ln \sqrt[3]{\frac{x(x+3)^2}{x^2-1}}$

75. $\ln \frac{\sqrt[3]{y(y+4)^2}}{y-1}$


 (b)

x	-5	-4	-3	-2	-1
y ₁	-2.36	-1.51	-0.45	0.94	2.77
y ₂	-2.36	-1.51	-0.45	0.94	2.77

x	0	1	2	3	4	5
y ₁	4.16	2.77	0.94	-0.45	-1.51	-2.36
y ₂	4.16	2.77	0.94	-0.45	-1.51	-2.36

 (c) $y_1 = y_2$

 (b)

x	-5	-4	-3	-2	-1
y ₁	3.22	2.77	2.20	1.39	0
y ₂	Error	Error	Error	Error	Error

x	0	1	2	3	4	5
y ₁	Error	0	1.39	2.20	2.77	3.22
y ₂	Error	0	1.39	2.20	2.77	3.22

(c) No. The domains differ.

81. 2 83. 6.8

85. Not possible; -4 is not in the domain of $\log_2 x$.

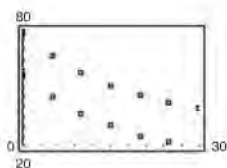
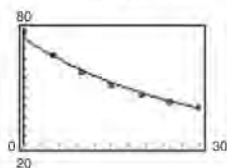
87. 2 89. -4 91. 8 93. $-\frac{1}{2}$

95. (a) $\beta = 120 + 10 \log_{10} I$

(b)

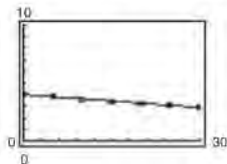
I	10^{-4}	10^{-6}	10^{-8}	10^{-10}	10^{-12}	10^{-14}
β	80	60	40	20	0	-20

97. (a)

(b) $T = 54.438(0.964)^t + 21$ 

The model fits the data.

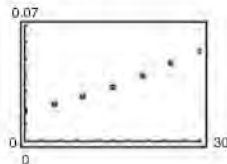
(c)



$$\ln(T - 21) = -0.037t + 3.997$$

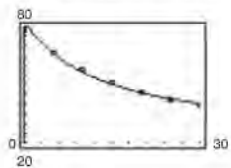
$$T = e^{(-0.037t + 3.997)} + 21$$

(d)

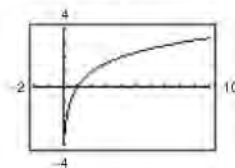


$$\frac{1}{T - 21} = 0.0012t + 0.0162$$

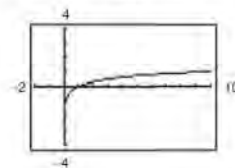
$$T = \frac{1}{0.0012t + 0.0162} + 21$$

99. True 101. False. $f\left(\frac{x}{a}\right) = f(x) - f(a)$, not $\frac{f(x)}{f(a)}$.103. False. $f(\sqrt{x}) = \frac{1}{2}f(x)$ 105. True. When $f(x) = 0$, $x = 1 < e$. 107. Proof

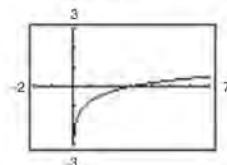
109. $f(x) = \frac{\log_{10} x}{\log_{10} 2}$



111. $f(x) = \frac{\log_{10} \sqrt{x}}{\log_{10} 3}$



113. $f(x) = \frac{\log_{10}(x/3)}{\log_{10} 5}$



115. $\ln 1 = 0$

$\ln 2 \approx 0.6931$

$\ln 3 \approx 1.0986$

$\ln 4 \approx 1.3862$

$\ln 5 \approx 1.6094$

$\ln 6 \approx 1.7917$

$\ln 8 \approx 2.0793$

$\ln 9 \approx 2.1972$

$\ln 10 \approx 2.3025$

$\ln 12 \approx 2.4848$

$\ln 15 \approx 2.7080$

$\ln 16 \approx 2.7724$

$\ln 18 \approx 2.8903$

$\ln 20 \approx 2.9956$

117. $\frac{27y^3}{8x^6}$

119. $\frac{x^2y^2}{x+y}$

121. 0, -5

123. $\pm\frac{1}{3}, \pm 2$ 125. $\pm\frac{1}{3}, \pm 5$

Section 3.4 (page 221)

Vocabulary Check (page 221)

1. solve

2. (a) $x = y$ (b) $x = y$ (c) x (d) x

3. extraneous

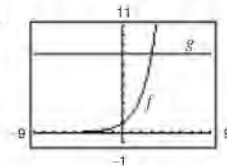
1. (a) Yes (b) No

3. (a) No (b) Yes (c) Yes, approximate

5. (a) Yes, approximate (b) No (c) Yes

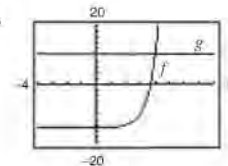
7. (a) Yes (b) Yes, approximate (c) No

9.



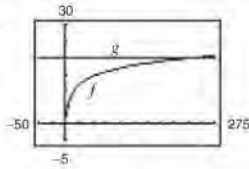
(3, 8); 3

11.



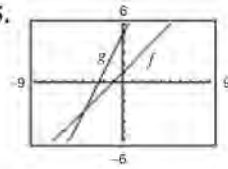
(4, 10); 4

13.



(243, 20); 243

15.



(-4, -3); -4

 17. 2 19. -4 21. -2 23. -4 25. $\log_{10} 36$

 27. 5 29. 5 31. e^{-7} 33. 5 35. 0.1

 37. $\frac{e^5 + 1}{2}$ 39. x^2 41. $5x + 2$

 43. $2x - 1$ 45. 0.944 47. 2 49. -1.498

51. 6.960 53. -277.951 55. 0.439 57. 0.511

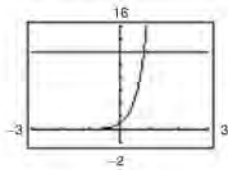
59. 0 61. 1.609 63. 184.444 65. -1, 2

67. 0.586, 3.414 69. 1.946 71. 183.258

73.

x	0.6	0.7	0.8	0.9	1.0
e^{3x}	6.05	8.17	11.02	14.88	20.09

(0.8, 0.9)

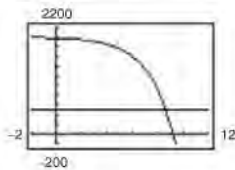


0.828

75.

x	5	6	7	8	9
$20(100 - e^{x/2})$	1756	1598	1338	908	200

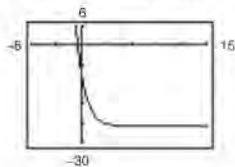
(8, 9)



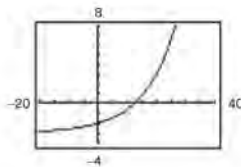
8.635

77. 21.330 79. 3.656

81.


 $x = -0.427$

83.


 $x = 12.207$

85. 0.050 87. 2.042 89. 4453.242 91. 1

93. 103 95. 17.945 97. 5.389

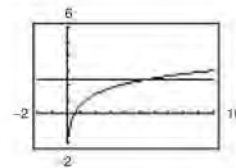
99. 1.718, -3.718 101. 2 103. No real solution

105. 180.384

107.

x	2	3	4	5	6
$\ln 2x$	1.39	1.79	2.08	2.30	2.48

(5, 6)

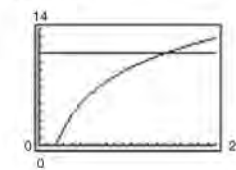


5.512

109.

x	12	13	14	15	16
$6 \log_3(0.5x)$	9.79	10.22	10.63	11.00	11.36

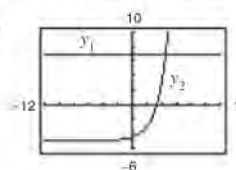
(14, 15)



14.988

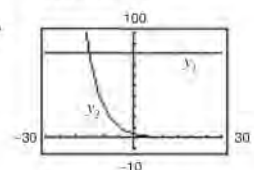
111. 1.469, 0.001 113. 2.928 115. 3.423

117.



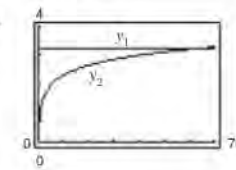
(4.585, 7)

119.



(-14.979, 80)

121.



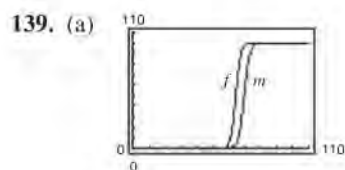
(663.142, 3.25)

 123. -1, 0 125. 1 127. $e^{-1/2} \approx 0.607$

 129. $e^{-1} \approx 0.368$ 131. (a) 9.24 years (b) 14.65 years

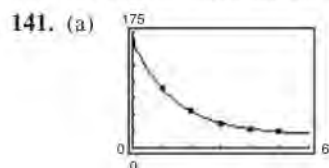
133. (a) 27.73 years (b) 43.94 years

135. (a) 1426 units (b) 1498 units 137. 2001



(b) $y = 100$ and $y = 0$; The percentages cannot exceed 100% or be less than 0%.

(c) Males: 69.71 inches; females: 64.51 inches



(b) $y = 20$. The object's temperature cannot cool below the room's temperature.

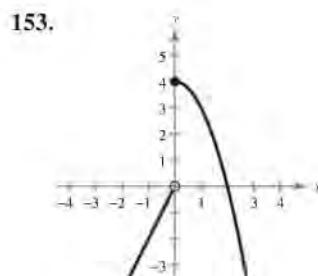
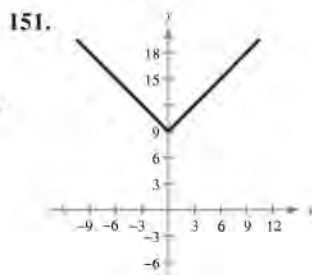
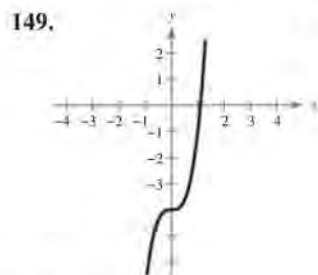
(c) 0.81 hour

143. False; $e^x = 0$ has no solutions.

145. (a) Isolate the exponential term by grouping, factoring, etc. Take the logarithms of both sides and solve for the variable.

(b) Isolate the logarithmic term by grouping, collapsing, etc. Exponentiate both sides and solve for the variable.

147. Yes. The investment will double every $\frac{\ln 2}{r}$ years.



Section 3.5 (page 232)

Vocabulary Check (page 232)

1. (a) iv (b) i (c) vi (d) iii
(e) vii (f) ii (g) v

2. Normally 3. Sigmoidal

4. Bell-shaped, mean

1. c 2. e 3. b 4. a 5. d 6. f

Initial Investment	Annual % Rate	Time to Double	Amount After 10 Years
--------------------	---------------	----------------	-----------------------

7. \$10,000 3.5% 19.8 years \$14,190.68

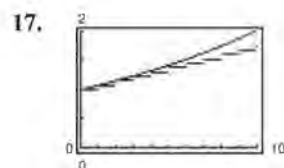
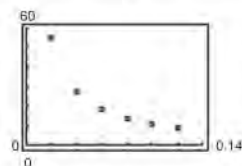
9. \$7500 3.30% 21 years \$10,432.26

11. \$5000 1.25% 55.45 years \$5,665.74

13. \$63,762.82 4.5% 15.40 years \$100,000.00

15.

r	2%	4%	6%	8%	10%	12%
t	54.93	27.47	18.31	13.73	10.99	9.16



Continuous compounding

Isotope	Half-Life (years)	Initial Quantity	Amount After 1000 Years
19. ^{226}Ra	1599	10 grams	6.48 grams
21. ^{14}C	5715	3 grams	2.66 grams

23. $y = e^{0.768x}$ 25. $y = 4e^{-0.2773x}$

27. (a) Australia: $y = 19.2e^{0.00848x}$; 24.76 million

Canada: $y = 31.3e^{0.00915x}$; 41.19 million

Phillipines: $y = 79.7e^{0.0185x}$; 138.83 million

South Africa: $y = 44.1e^{-0.00183x}$; 41.74 million

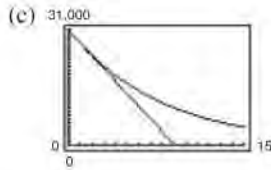
Turkey: $y = 65.7e^{0.01095x}$; 91.25 million

(b) b ; Population changes at a faster rate for a greater magnitude of b .

(c) b ; b is positive when the population is increasing and negative when the population is decreasing.

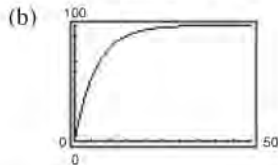
29. (a) 0.0295 (b) About 241,734 people 31. 95.8%

33. (a) $V = -3394t + 30,788$ (b) $V = 30,788e^{-0.1245t}$

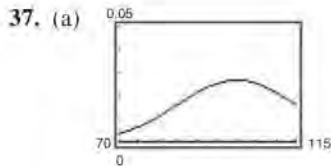


(d) Exponential model (e) Answers will vary.

35. (a) $S(t) = 100(1 - e^{-0.1625t})$

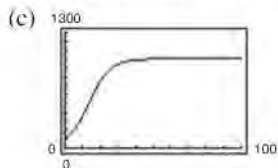


(c) 55,625 units



(b) 100

39. (a) ≈ 203 animals (b) ≈ 13 months



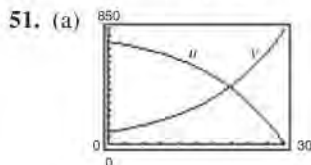
The horizontal asymptotes occur at $p = 1000$ and $p = 0$. The asymptote at $p = 1000$ means there will not be more than 1000 animals in the preserve.

41. (a) 1,258,925 (b) 39,810,717

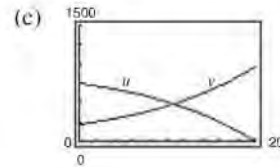
(c) 1,000,000,000

43. (a) 20 decibels (b) 70 decibels (c) 120 decibels

45. 97.49% 47. 4.64 49. 10,000,000 times



(b) Interest; $t \approx 20.7$ years



Interest; $t \approx 10.73$ years
Answers will vary.

53. 3:00 A.M.

55. False. The domain can be all real numbers.

57. True. Any x -value in the Gaussian model will give a positive y -value.

59. a; $(0, -3)$, $(\frac{9}{4}, 0)$ 60. b; $(0, 2)$, $(5, 0)$

61. d; $(0, 25)$, $(\frac{100}{9}, 0)$ 62. c; $(0, 4)$, $(2, 0)$

63. Falls to the left and rises to the right

65. Rises to the left, falls to the right

67. $2x^2 + 3 + \frac{3}{x - 4}$

69. Answers will vary.

Section 3.6 (page 242)

Vocabulary Check (page 242)

1. $y = ax + b$ 2. quadratic 3. $y = ax^b$

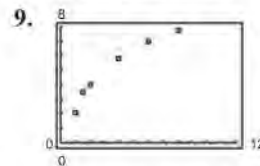
4. sum, squared differences 5. $y = ab^x$, ae^{cx}

1. Logarithmic model

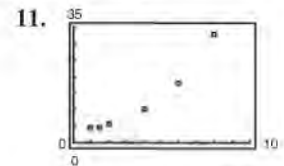
3. Quadratic model

5. Exponential model

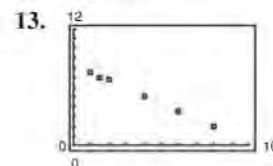
7. Quadratic model



Logarithmic model

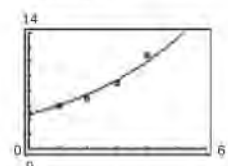


Exponential model

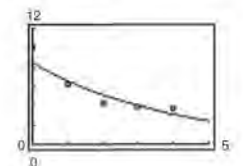


Linear model

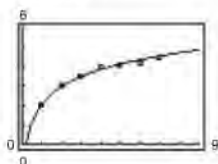
15. $y = 4.752(1.2607)^x$;
0.96773



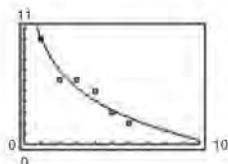
17. $y = 8.463(0.7775)^x$;
0.86639



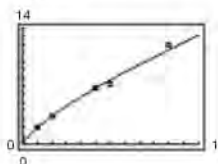
19. $y = 2.083 + 1.257 \ln x$; 0.98672



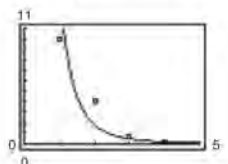
21. $y = 9.826 - 4.097 \ln x$; 0.93704



23. $y = 1.985x^{0.760}$; 0.99686

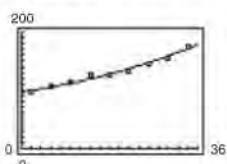


25. $y = 16.103x^{-3.174}$; 0.88161

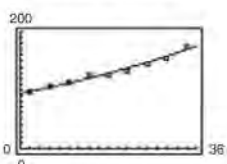


27. (a) Quadratic model: $y = 0.031t^2 + 1.13t + 97.1$
Exponential model: $y = 94.435(1.0174)^t$
Power model: $y = 77.837t^{0.1918}$

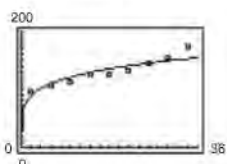
(b) Quadratic model:



Exponential model:



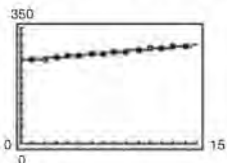
Power model:



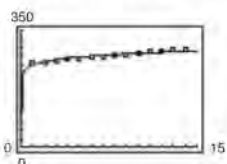
(c) Exponential model

(d) 2008: 181.89 million; 2012: 194.89 million

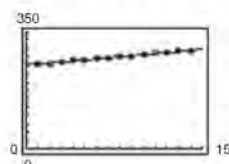
29. (a) Linear model: $P = 3.11t + 250.9$; 0.99942



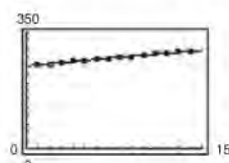
(b) Power model: $P = 246.52t^{0.0587}$; 0.90955



(c) Exponential model: $P = 251.57(1.0114)^t$; 0.99811



(d) Quadratic model: $P = -0.020t^2 + 3.41t + 250.1$; 0.99994



(e) Quadratic model; the coefficient of determination for the quadratic model is closest to 1, so the quadratic model best fits the data.

(f) Linear model:

Year	2005	2006	2007	2008	2009	2010
Population (in millions)	297.6	300.7	303.8	306.9	310.0	313.1

Power model:

Year	2005	2006	2007	2008	2009	2010
Population (in millions)	289.0	290.1	291.1	292.1	293.0	293.9

Exponential model:

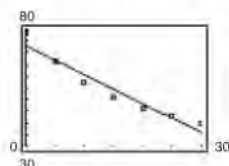
Year	2005	2006	2007	2008	2009	2010
Population (in millions)	298.2	301.6	305.0	308.5	312.0	315.6

Quadratic model:

Year	2005	2006	2007	2008	2009	2010
Population (in millions)	296.8	299.5	302.3	305.0	307.7	310.3

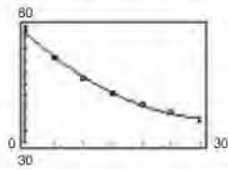
(g) and (h) Answers will vary.

31. (a) Linear model: $T = -1.24t + 73.0$



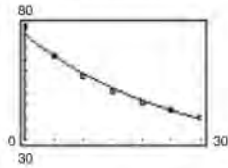
The data does not appear to be linear. Answers will vary.

- (b) Quadratic model: $T = 0.034t^2 - 2.26t + 77.3$



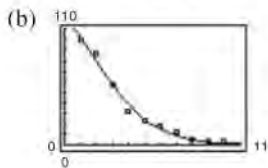
The data appears to be quadratic. When $t = 60$, the temperature of the water should decrease, not increase as shown in this model.

- (c) Exponential model: $T = 54.438(0.964)^t + 21$



- (d) Answers will vary.

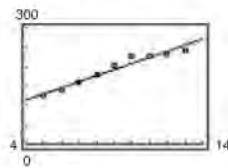
33. (a) $P = \frac{162.4}{1 + 0.34e^{0.5609x}}$



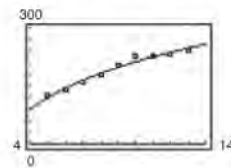
The model closely represents the actual data.

35. (a) Linear model: $y = 15.71t + 51.0$
 Logarithmic model: $y = 134.67 \ln t - 97.5$
 Quadratic model: $y = -1.292t^2 + 38.96t - 45$
 Exponential model: $y = 85.97(1.091)^t$
 Power model: $y = 37.274t^{0.7506}$

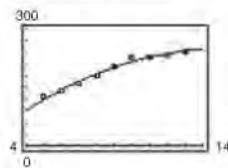
- (b) Linear model:



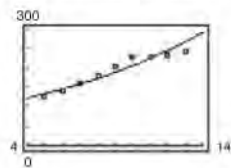
- Logarithmic model:



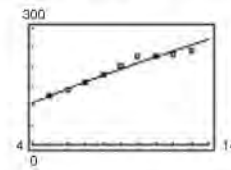
- Quadratic model:



- Exponential model:



- Power model:

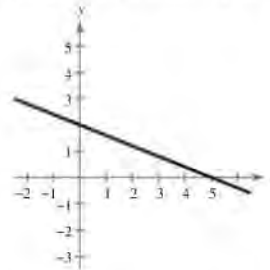


The quadratic model best fits the data.

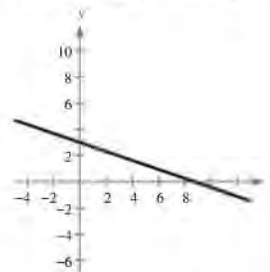
- (c) Linear model: 803.9, logarithmic model: 411.7, quadratic model: 289.8, exponential model: 1611.4, and power model: 667.1. The quadratic model best fits the data.
 (d) Linear model: 0.9485, logarithmic model: 0.9736, quadratic model: 0.9814, exponential model: 0.9274, and power model: 0.9720. The quadratic model best fits the data.
 (e) The quadratic model is the best-fitting model.

37. True

39. Slope: $-\frac{2}{5}$; y-intercept: (0, 2)



41. Slope: $-\frac{12}{35}$; y-intercept: (0, 3)



Review Exercises (page 247)

1. 10.3254 3. 0.0001 5. 2980.9580

7. 8.1662 9. c 10. d 11. b 12. a

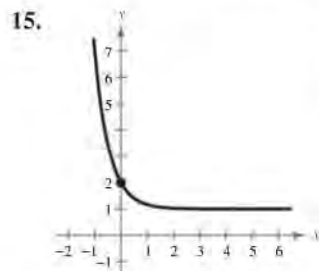
- 13.



Horizontal asymptote: $y = 0$

y-intercept: (0, 1)

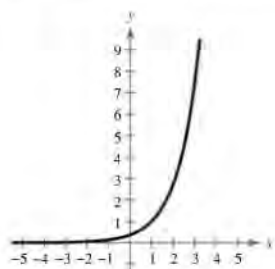
Increasing on $(-\infty, \infty)$



Horizontal asymptote: $y = 1$
 y -intercept: $(0, 2)$
 Decreasing on $(-\infty, \infty)$

17.

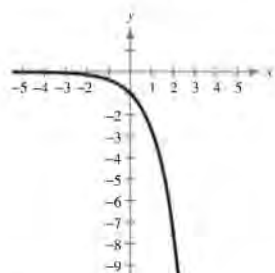
x	0	1	2	3	4
$h(x)$	0.37	1	2.72	7.39	20.09



Horizontal asymptote: $y = 0$

19.

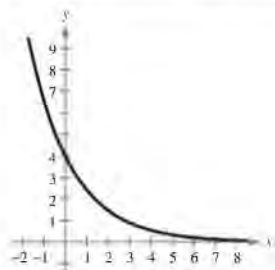
x	-2	-1	0	1	2
$h(x)$	-0.14	-0.37	-1	-2.72	-7.39



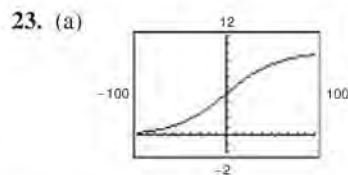
Horizontal asymptote: $y = 0$

21.

x	-1	0	1	2	3	4
$f(x)$	6.59	4	2.43	1.47	0.89	0.54



Horizontal asymptote: $y = 0$



(b) Horizontal asymptotes:
 $y = 0$ and $y = 10$

25.

t	1	10	20
A	\$10,832.87	\$22,255.41	\$49,530.32

t	30	40	50
A	\$110,231.76	\$245,325.30	\$545,981.50

27. (a) (b) \$14,625

(c) When it is first sold; Yes; Answers will vary.

29. $5^3 = 125$ 31. $64^{1/6} = 2$ 33. $e^4 = e^4$

35. $\log_4 64 = 3$ 37. $\log_{25} 125 = \frac{3}{2}$

39. $\log_{1/2} 8 = -3$ 41. $\ln 1096.6331 \dots = 7$

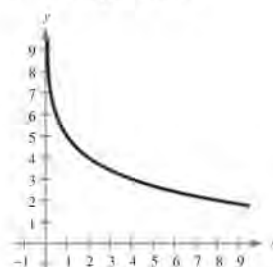
43. 3 45. -1

47. Domain: $(0, \infty)$

Vertical asymptote:

$x = 0$

x -intercept: $(32, 0)$

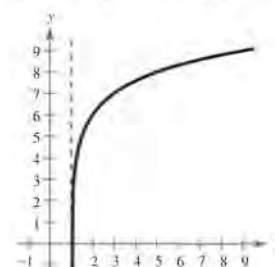


49. Domain: $(1, \infty)$

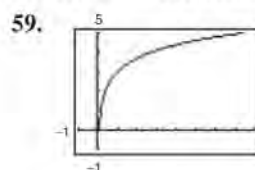
Vertical asymptote:

$x = 1$

x -intercept: $(1.016, 0)$



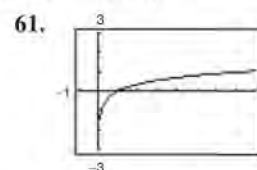
51. 3.068 53. 0.896 55. 3 57. $\frac{1}{9}$



Domain: $(0, \infty)$

Vertical asymptote: $x = 0$

x -intercept: $(0.05, 0)$

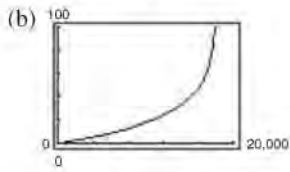


Domain: $(0, \infty)$

Vertical asymptote: $x = 0$

x -intercept: $(1, 0)$

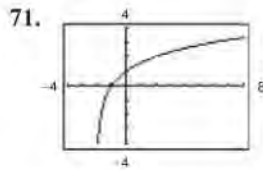
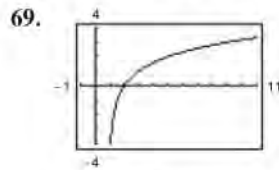
63. (a) $0 \leq h < 18,000$

Asymptote: $h = 18,000$

(c) The time required to increase its altitude further increases.

(d) 5.46 minutes

65. 1.585 67. 2.132

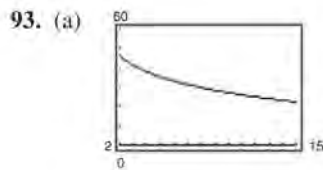


73. 1.13 75. 0.41 77. $\ln 5 - 2$ 79. $2 + \log_{10} 2$

81. $1 + 2 \log_5 x$ 83. $\log_{10} 5 + \frac{1}{2} \log_{10} y - 2 \log_{10} x$

85. $\ln(x+3) - \ln x - \ln y$ 87. $\log_2 5x$

89. $\ln \frac{\sqrt{2x-1}}{(x+1)^2}$ 91. $\ln \frac{3\sqrt[3]{4-x^2}}{x}$



(b)

h	4	6	8	10	12	14
s	38	33	30	27	25	23

(c) The decrease in productivity starts to level off.

95. 3 97. -3 99. -5 101. 2401 103. 9

105. e^{-4} 107. $e^2 + 1$ 109. -0.757 111. 4.459

113. 1.760 115. 3.916 117. 1,609, 0.693

119. 1213.650 121. 22.167 123. 53.598

125. No solution 127. 0.9 129. -1 131. 0.368

133. 15.15 years 135. e 136. b 137. f

138. d 139. a 140. c 141. $y = 2e^{0.1014x}$

143. $y = \frac{1}{2}e^{0.4605x}$ 145. $k \approx 0.0259$; 606,000

147. (a) 5.78% (b) \$10,595.03

149. (a) 7.64 weeks (b) 13.21 weeks

151. Logistic model 153. Logarithmic model

155. (a) Linear model: $S = 297.8t + 739$; 0.97653

Quadratic model: $S = 11.79t^2 + 38.5t + 2118$; 0.98112

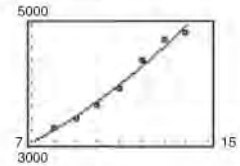
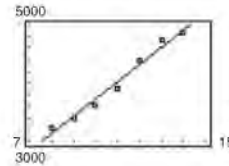
Exponential model: $S = 1751.5(1.077)^t$; 0.98225

Logarithmic model: $S = 3169.8 \ln t - 3532$; 0.95779

Power model: $S = 598.1t^{0.7950}$; 0.97118

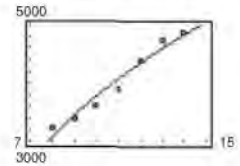
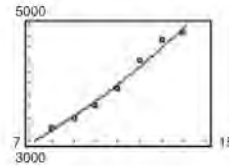
(b) Linear model:

Quadratic model:

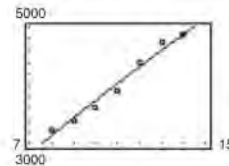


Exponential model:

Logarithmic model:



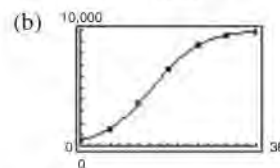
Power model:



(c) Exponential model; the coefficient of determination for the exponential model is closest to 1.

(d) \$7722 million (e) 2004

157. (a) $P = \frac{9999.887}{1 + 19.0e^{-0.2x}}$



(c) The model fits the data well. (d) 10,000 fish

159. True 161. False, $\ln(xy) = \ln x + \ln y$

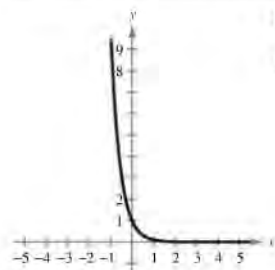
163. False. $x > 0$

165. Because $1 < \sqrt{2} < 2$, then $2^1 < 2^{\sqrt{2}} < 2^2$.

Chapter Test (page 252)

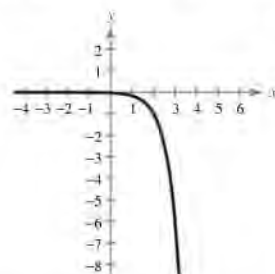
1.

x	-2	-1	0	1	2
$f(x)$	100	10	1	0.1	0.01

Horizontal asymptote: $y = 0$ y -intercept: $(0, 1)$

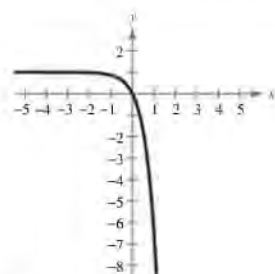
2.

x	0	2	3	4
$f(x)$	-0.03	-1	-6	-36

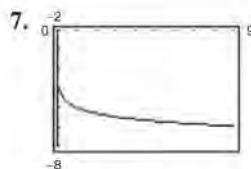
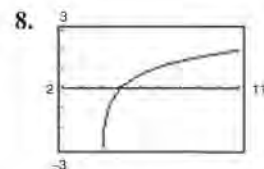
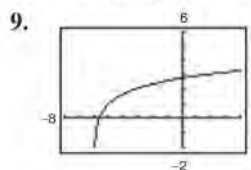
Horizontal asymptote: $y = 0$ y -intercept: $(0, -\frac{1}{36})$

3.

x	-2	-1	0	1	2
$f(x)$	0.9817	0.8647	0	-6.3891	-53.5982

Horizontal asymptote: $y = 1$ Intercept: $(0, 0)$

4. -0.89 5. 9.2 6. 0

Domain: $(0, \infty)$ Vertical asymptote:
 $x = 0$ Intercept: $(0, 0)$ Domain: $(4, \infty)$ Vertical asymptote:
 $x = 4$ x -intercept: $(5, 0)$ Domain: $(-6, \infty)$ Vertical asymptote: $x = -6$ x -intercept: $(-5.63, 0)$

10. 1.945 11. 0.115 12. 1.328

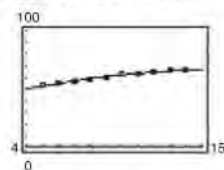
13. $\log_2 3 + 4 \log_2 a$ 14. $\ln 5 + \frac{1}{2} \ln x - \ln 6$ 15. $\ln x + \frac{1}{2} \ln(x+1) - \ln 2 - 4$ 16. $\log_3 13y$ 17. $\ln\left(\frac{x^4}{y^4}\right)$ 18. $\ln\left[\frac{x(2x-3)}{x+2}\right]$ 19. 4

20. 2.431 21. 343 22. 100,004 23. 1.321

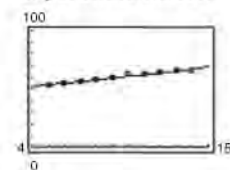
24. 1 25. 1.597 26. 1.649 27. 54.96%

28. (a) Quadratic model: $R = -0.092t^2 + 3.29t + 38.1$ Exponential model: $R = 46.99(1.026)^t$ Power model: $R = 36.00t^{0.233}$

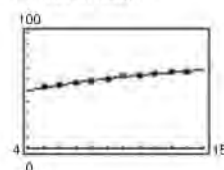
(b) Quadratic model:



Exponential model:



Power model:



(c) Quadratic model, 67.1

Cumulative Test for Chapters 1–3 (page 253)

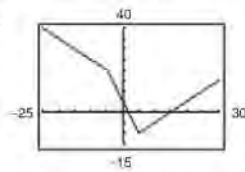
1. (a) $x + y - 3 = 0$
(b) Answers will vary. Sample answer: (0, 3), (1, 2), (2, 1)
2. (a) $2x + y = 0$
(b) Answers will vary. Sample answer: (0, 0), (1, -2), (2, -4)

3. (a) $7x + 3 = 0$
(b) Answers will vary. Sample answer: $(-\frac{3}{7}, 0)$, $(-\frac{3}{7}, 1)$, $(-\frac{3}{7}, -3)$

4. (a) $\frac{5}{3}$ (b) Undefined (c) $\frac{5 + 4s}{3 + 4s}$

5. (a) -32 (b) 4 (c) 20

6. No. It doesn't pass the Vertical Line Test.

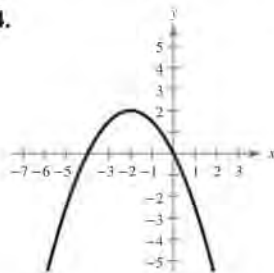
7.  Decreasing on $(-\infty, 5)$
Increasing on $(5, \infty)$

8. (a) Vertical shrink (b) Vertical shift
(c) Horizontal shift and reflection in the x -axis

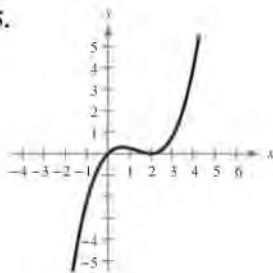
9. -53 10. $\frac{197}{16}$ 11. -79 12. 42

13. $h^{-1}(x) = \frac{x + 2}{5}$

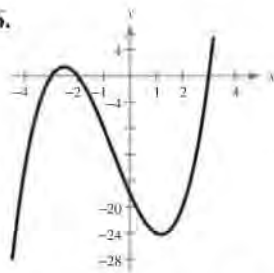
14.



15.



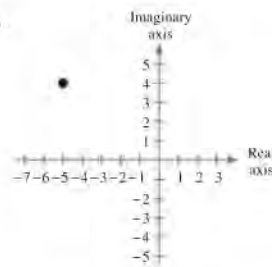
16.



17. $x = -2, \pm 2i$ 18. 1.424

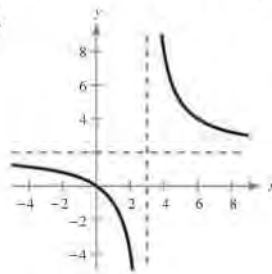
19. $4x + 2 - \frac{15}{x + 3}$ 20. $2x^2 + 7x + 48 + \frac{268}{x - 6}$

21.



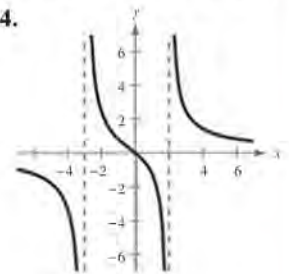
22. Answers will vary. Sample answer: $f(x) = x^4 + x^3 + 18x$

23.



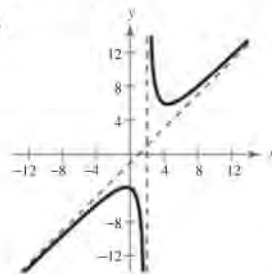
Asymptotes:
 $x = 3, y = 2$

24.



Asymptotes:
 $x = -3, x = 2, y = 0$

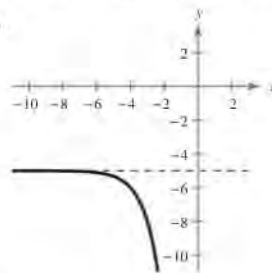
25.



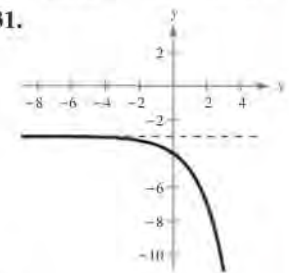
Asymptotes: $x = 2, y = x - 1$

26. 6.733 27. 8772.934 28. 0.162 29. 51.743

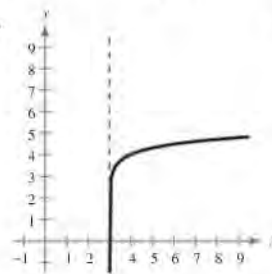
30.



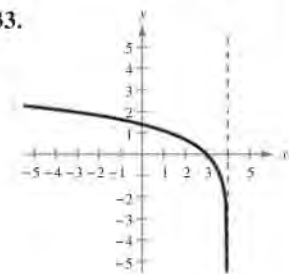
31.



32.



33.



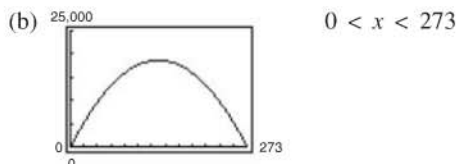
34. 1.892 35. 0.872 36. 0.585

37. $\ln(x+2) + \ln(x-2) - \ln(x^2+1)$

38. $\ln\left(\frac{x^2(x+1)}{x-1}\right)$ 39. 1.242 40. 6.585 41. 12.8

42. 152.018 43. 0, 1 44. No solution

45. (a) $A = x(273 - x)$



(c) 76.23 feet $\times$ 196.77 feet

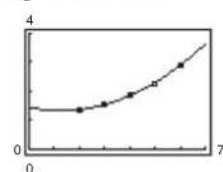
46. (a) Quadratic model:

$y = 0.0707x^2 - 0.183x + 1.45; 0.99871$

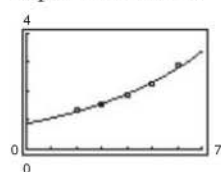
Exponential model: $y = 0.89(1.21)^x; 0.98862$

Power model: $y = 0.79(x)^{0.68}; 0.93130$

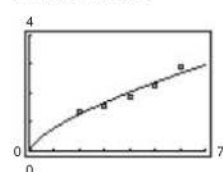
(b) Quadratic model:



Exponential model:



Power model:



(c) Quadratic model; the coefficient of determination for the quadratic model is closest to 1.

(d) \$4.51, \$6.69; Answers will vary.

Chapter 4

Section 4.1 (page 265)

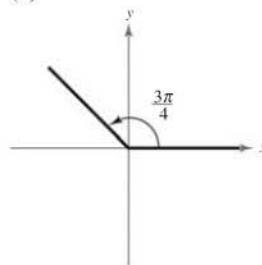
Vocabulary Check (page 265)

1. Trigonometry 2. angle
3. standard position 4. coterminal 5. radian
6. complementary 7. supplementary
8. degree 9. linear 10. angular

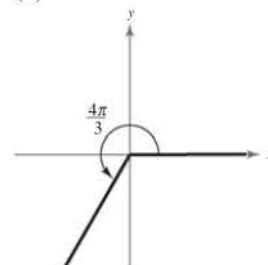
1. 2 3. (a) Quadrant IV (b) Quadrant II

5. (a) Quadrant IV (b) Quadrant III

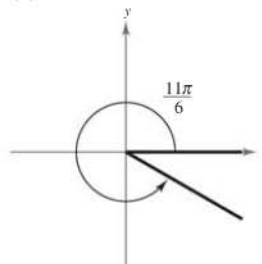
7. (a)



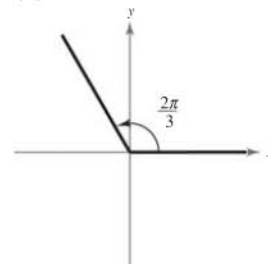
(b)



9. (a)



(b)



11. (a) $\frac{13\pi}{6}, -\frac{11\pi}{6}$ (b) $\frac{8\pi}{3}, -\frac{4\pi}{3}$

13. (a) $\frac{7\pi}{4}, -\frac{\pi}{4}$ (b) $\frac{28\pi}{15}, -\frac{32\pi}{15}$

15. Complement: $\frac{\pi}{6}$; supplement: $\frac{2\pi}{3}$

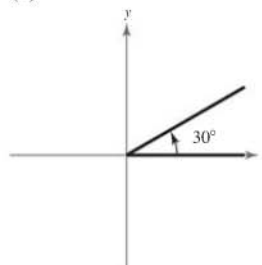
17. Complement: $\frac{\pi}{3}$; supplement: $\frac{5\pi}{6}$

19. Complement: 0.571; supplement: 2.142

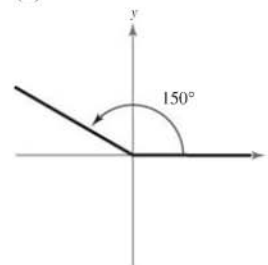
21. 210° 23. (a) Quadrant II (b) Quadrant IV

25. (a) Quadrant III (b) Quadrant I

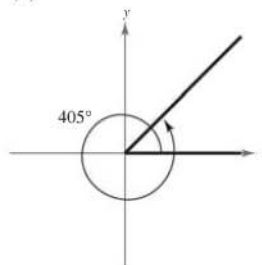
27. (a)



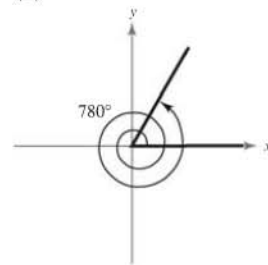
(b)



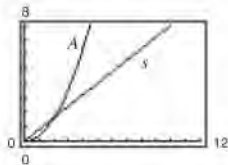
29. (a)



(b)

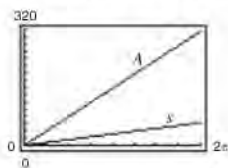


31. (a) $412^\circ, -308^\circ$ (b) $324^\circ, -396^\circ$
 33. (a) $660^\circ, -60^\circ$ (b) $590^\circ, -130^\circ$
 35. Complement: 66° ; supplement: 156°
 37. Complement: 3° ; supplement: 93°
 39. (a) $\frac{\pi}{6}$ (b) $\frac{5\pi}{6}$ 41. (a) $-\frac{\pi}{9}$ (b) $-\frac{4\pi}{3}$
 43. (a) 270° (b) -210° 45. (a) 420° (b) -39°
 47. 2.007 49. -3.776 51. -0.014 53. 25.714°
 55. 1170° 57. -114.592° 59. 64.75° 61. 85.308°
 63. -125.01° 65. $280^\circ 36'$ 67. $-345^\circ 7' 12''$
 69. $-20^\circ 20' 24''$ 71. $\frac{6}{5}$ rad 73. $\frac{32}{7}$ rad
 75. 0, $\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$ 77. $\frac{8}{15}$ rad
 79. $\frac{70}{29}$ rad 81. 14π inches 83. 18π meters
 85. 22.92 feet 87. 34.80 miles
 89. 1141.81 miles 91. $4^\circ 2' 33''$ 93. 23.87°
 95. (a) $540^\circ = 3\pi$ rad (b) $900^\circ = 5\pi$ rad
 (c) $1260^\circ = 7\pi$ rad
 97. (a) 80π radians per second (b) 25π feet per second
 99. (a) 35.70 miles per hour
 (b) 941.18 revolutions per minute
 101. False. A radian is larger: $1 \text{ rad} \approx 57.3^\circ$.
 103. True. The sum of the angles of a triangle must equal
 $180^\circ = \pi$ radians, and $\frac{2\pi}{3} + \frac{\pi}{4} + \frac{\pi}{12} = \pi$.
 105. The length of the intercepted arc is increasing.
 Answers will vary.
 107. $\frac{50\pi}{3}$ square meters
 109. (a) $A = 0.4r^2, r > 0; s = 0.8r, r > 0$



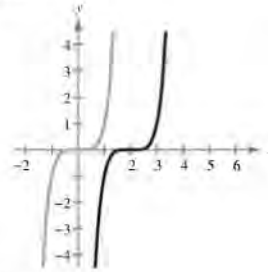
The area function changes more rapidly for $r > 1$ because it is quadratic and the arc length function is linear.

- (b) $A = 50\theta, 0 < \theta < 2\pi; s = 10\theta, 0 < \theta < 2\pi$

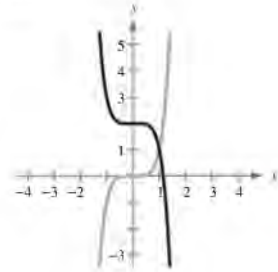


111. Answers will vary.

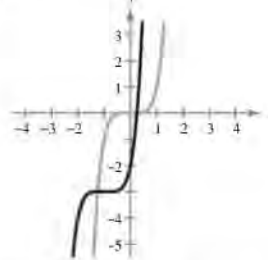
113.



115.



117.



Section 4.2 (page 274)

Vocabulary Check (page 274)

1. unit circle 2. periodic 3. odd, even

1. $\sin \theta = \frac{15}{17}$ 3. $\sin \theta = -\frac{5}{13}$
 $\cos \theta = -\frac{8}{17}$ $\cos \theta = \frac{12}{13}$
 $\tan \theta = -\frac{15}{8}$ $\tan \theta = -\frac{5}{12}$
 $\csc \theta = \frac{17}{15}$ $\csc \theta = -\frac{13}{5}$
 $\sec \theta = -\frac{17}{8}$ $\sec \theta = \frac{13}{12}$
 $\cot \theta = -\frac{8}{15}$ $\cot \theta = -\frac{12}{5}$
 5. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ 7. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ 9. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
 11. $(0, -1)$ 13. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ 15. $(0, 1)$
 17. $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ 19. $\sin \frac{7\pi}{6} = -\frac{1}{2}$
 $\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ $\cos \frac{7\pi}{6} = -\frac{\sqrt{3}}{2}$
 $\tan \frac{\pi}{4} = 1$ $\tan \frac{7\pi}{6} = \frac{\sqrt{3}}{3}$

21. $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$\cos \frac{2\pi}{3} = -\frac{1}{2}$

$\tan \frac{2\pi}{3} = -\sqrt{3}$

25. $\sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$

$\cos\left(-\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$

$\tan\left(-\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{3}$

29. $\sin\left(-\frac{3\pi}{2}\right) = 1$

$\cos\left(-\frac{3\pi}{2}\right) = 0$

$\tan\left(-\frac{3\pi}{2}\right)$ is undefined.

31. $\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}$

$\cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

$\tan \frac{3\pi}{4} = -1$

$\csc \frac{3\pi}{4} = \sqrt{2}$

$\sec \frac{3\pi}{4} = -\sqrt{2}$

$\cot \frac{3\pi}{4} = -1$

35. $\sin\left(-\frac{2\pi}{3}\right) = -\frac{\sqrt{3}}{2}$

$\cos\left(-\frac{2\pi}{3}\right) = -\frac{1}{2}$

$\tan\left(-\frac{2\pi}{3}\right) = \sqrt{3}$

$\csc\left(-\frac{2\pi}{3}\right) = -\frac{2\sqrt{3}}{3}$

$\sec\left(-\frac{2\pi}{3}\right) = -2$

$\cot\left(-\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{3}$

37. 0

39. $-\frac{1}{2}$

41. $\frac{\sqrt{3}}{2}$

43. $-\frac{\sqrt{2}}{2}$

45. (a) $-\frac{1}{3}$ (b) -3

47. (a) $-\frac{1}{5}$ (b) -5

49. (a) $\frac{4}{5}$ (b) $-\frac{4}{5}$

51. 0.6428

53. 0.8090

23. $\sin\left(-\frac{5\pi}{3}\right) = \frac{\sqrt{3}}{2}$

$\cos\left(-\frac{5\pi}{3}\right) = \frac{1}{2}$

$\tan\left(-\frac{5\pi}{3}\right) = \sqrt{3}$

27. $\sin\left(-\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$\cos\left(-\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$\tan\left(-\frac{7\pi}{4}\right) = 1$

55. 1.0378

57. -0.1288

59. 1.3940

61. -1.4486

63. -1.3386

65. -1.0025

67. -4.7439

69. (a) -0.9 (b) -0.4

71. (a) 0.25, 2.89 (b) 1.82, 4.46

73. 0.79 ampere

75. (a) 0.25 foot (b) 0.02 foot (c) -0.25 foot

77. False. $\sin(-t) = -\sin t$ means that the function is odd, not that the sine of a negative angle is a negative number.

79. False. The real number 0 corresponds to the point (1, 0).

81. (a) y-axis (b) $\sin t_1 = \sin(\pi - t_1)$

(c) $\cos(\pi - t_1) = -\cos t_1$

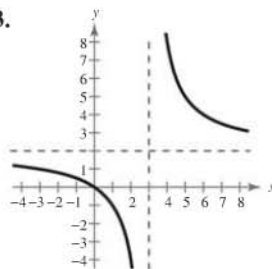
83. Answers will vary. 85. Answers will vary.

87. It is an odd function.

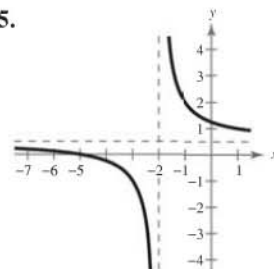
89. $f^{-1}(x) = \frac{2}{3}(x + 1)$

91. $f^{-1}(x) = \sqrt{x^2 + 4}, x \geq 0$

93.



95.



97. Domain: all real numbers x

Intercepts: $(-4, 0)$, $(1, 0)$, $(0, -4)$

No asymptotes

99. Domain: all real numbers x

Intercept: $(0, 5)$

Asymptote: $y = 2$

Section 4.3 (page 284)

Vocabulary Check (page 284)

1. (a) iii (b) vi (c) ii (d) v (e) i (f) iv

2. hypotenuse, opposite, adjacent

3. elevation, depression

1. $\sin \theta = \frac{3}{5}$

$\cos \theta = \frac{4}{5}$

$\tan \theta = \frac{3}{4}$

$\csc \theta = \frac{5}{3}$

$\sec \theta = \frac{5}{4}$

$\cot \theta = \frac{4}{3}$

3. $\sin \theta = \frac{8}{17}$

$\cos \theta = \frac{15}{17}$

$\tan \theta = \frac{8}{15}$

$\csc \theta = \frac{17}{8}$

$\sec \theta = \frac{17}{15}$

$\cot \theta = \frac{15}{8}$

5. $\sin \theta = \frac{3}{5}$

$\cos \theta = \frac{4}{5}$

$\tan \theta = \frac{3}{4}$

$\csc \theta = \frac{5}{3}$

$\sec \theta = \frac{5}{4}$

$\cot \theta = \frac{4}{3}$

The triangles are similar and corresponding sides are proportional.

9. $\cos \theta = \frac{\sqrt{11}}{6}$

$\tan \theta = \frac{5\sqrt{11}}{11}$

$\csc \theta = \frac{6}{5}$

$\sec \theta = \frac{6\sqrt{11}}{11}$

$\cot \theta = \frac{\sqrt{11}}{5}$

7. $\sin \theta = \frac{1}{3}$

$\cos \theta = \frac{2\sqrt{2}}{3}$

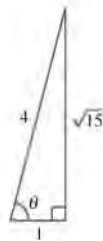
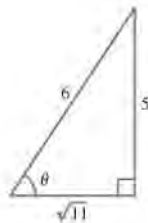
$\tan \theta = \frac{\sqrt{2}}{4}$

$\csc \theta = 3$

$\sec \theta = \frac{3\sqrt{2}}{4}$

$\cot \theta = 2\sqrt{2}$

The triangles are similar and corresponding sides are proportional.



11. $\sin \theta = \frac{\sqrt{15}}{4}$

$\cos \theta = \frac{1}{4}$

$\tan \theta = \sqrt{15}$

$\csc \theta = \frac{4\sqrt{15}}{15}$

$\cot \theta = \frac{\sqrt{15}}{15}$

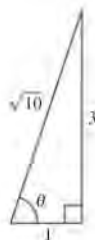
13. $\sin \theta = \frac{3\sqrt{10}}{10}$

$\cos \theta = \frac{\sqrt{10}}{10}$

$\csc \theta = \frac{\sqrt{10}}{3}$

$\sec \theta = \sqrt{10}$

$\cot \theta = \frac{1}{3}$



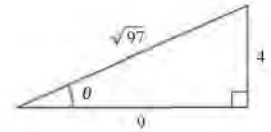
15. $\sin \theta = \frac{4\sqrt{97}}{97}$

$\cos \theta = \frac{9\sqrt{97}}{97}$

$\tan \theta = \frac{4}{9}$

$\csc \theta = \frac{\sqrt{97}}{4}$

$\sec \theta = \frac{\sqrt{97}}{9}$



17. $\frac{\pi}{6}, \frac{1}{2}$

19. $60^\circ, \sqrt{3}$

21. $60^\circ, \frac{\pi}{3}$

23. $30^\circ, \frac{\sqrt{3}}{2}$

25. $45^\circ, \frac{\pi}{4}$

27. $\csc \theta$

29. $\cot \theta$

31. $\cos \theta$

33. $\frac{\sin \theta}{\cos \theta}$

35. 1

37. $\cos \theta$

39. $\cot \theta$

41. $\csc \theta$

43. (a) $\sqrt{3}$

(b) $\frac{1}{2}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{\sqrt{3}}{3}$

45. (a) $\frac{1}{3}$

(b) $\frac{2\sqrt{2}}{3}$

(c) $\frac{\sqrt{2}}{4}$

(d) 3

47. (a) 4

(b) $\pm \frac{\sqrt{15}}{4}$

(c) $\pm \frac{\sqrt{15}}{15}$

(d) $\frac{1}{4}$

49–55. Answers will vary.

57. (a) 0.6561 (b) 0.0523

59. (a) 1.3499 (b) 1.3432

61. (a) 5.0273 (b) 0.4142

63. (a) $30^\circ = \frac{\pi}{6}$

(b) $30^\circ = \frac{\pi}{6}$

65. (a) $60^\circ = \frac{\pi}{3}$

(b) $45^\circ = \frac{\pi}{4}$

67. (a) $60^\circ = \frac{\pi}{3}$

(b) $45^\circ = \frac{\pi}{4}$

69. $y = 35\sqrt{3}, r = 70\sqrt{3}$

71. $x = 8, y = 8\sqrt{3}$

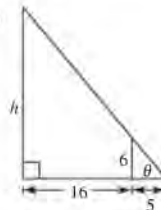
73. $x = 20, r = 20\sqrt{2}$

75. $x = 2\sqrt{5}, r = 2\sqrt{10}$

77. (a)

(b) $\tan \theta = \frac{h}{21}$

(c) $h = 25.2$ feet



79. 160 feet

81. (a) 45°

(b) $50\sqrt{2}$ feet

(c) $\frac{25\sqrt{2}}{3}$ feet per second; $\frac{25}{3}$ feet per second

83. $(x_1, y_1) = (28\sqrt{3}, 28)$
 $(x_2, y_2) = (28, 28\sqrt{3})$

85. True. $\csc x = \frac{1}{\sin x}$ 87. True. $1 + \cot^2 x = \csc^2 x$

89. (a)

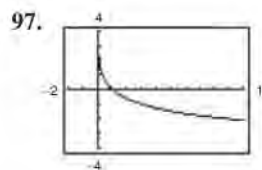
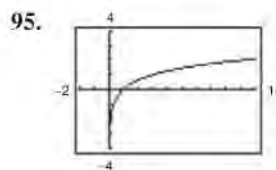
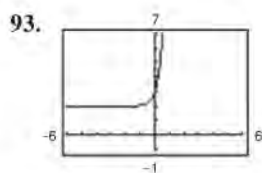
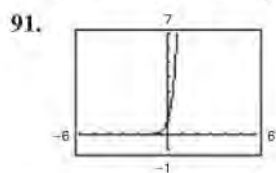
θ	0°	20°	40°	60°	80°
$\sin \theta$	0	0.3420	0.6428	0.8660	0.9848
$\cos \theta$	1	0.9397	0.7660	0.5	0.1736
$\tan \theta$	0	0.3640	0.8391	1.7321	5.6713

(b) Sine: increasing

Cosine: decreasing

Tangent: increasing

(c) Answers will vary.



Section 4.4 (page 294)

Vocabulary Check (page 294)

1. $\frac{y}{r}$ 2. $\csc \theta$ 3. $\frac{y}{x}$ 4. $\frac{r}{x}$
 5. $\cos \theta$ 6. $\cot \theta$ 7. reference

1. (a) $\sin \theta = \frac{3}{5}$

$\cos \theta = \frac{4}{5}$

$\tan \theta = \frac{3}{4}$

$\csc \theta = \frac{5}{3}$

$\sec \theta = \frac{5}{4}$

$\cot \theta = \frac{4}{3}$

(b) $\sin \theta = -\frac{15}{17}$

$\cos \theta = -\frac{8}{17}$

$\tan \theta = \frac{15}{8}$

$\csc \theta = -\frac{17}{15}$

$\sec \theta = -\frac{17}{8}$

$\cot \theta = \frac{8}{15}$

3. (a) $\sin \theta = -\frac{1}{2}$

$\cos \theta = -\frac{\sqrt{3}}{2}$

$\tan \theta = \frac{\sqrt{3}}{3}$

$\csc \theta = -2$

$\sec \theta = -\frac{2\sqrt{3}}{3}$

$\cot \theta = \sqrt{3}$

(b) $\sin \theta = \frac{\sqrt{2}}{2}$

$\cos \theta = -\frac{\sqrt{2}}{2}$

$\tan \theta = -1$

$\csc \theta = \sqrt{2}$

$\sec \theta = -\sqrt{2}$

$\cot \theta = -1$

5. $\sin \theta = \frac{24}{25}$

$\cos \theta = \frac{7}{25}$

$\tan \theta = \frac{24}{7}$

$\csc \theta = \frac{25}{24}$

$\sec \theta = \frac{25}{7}$

$\cot \theta = \frac{7}{24}$

9. $\sin \theta = \frac{5\sqrt{29}}{29}$

$\cos \theta = -\frac{2\sqrt{29}}{29}$

$\tan \theta = -\frac{5}{2}$

$\csc \theta = \frac{\sqrt{29}}{5}$

$\sec \theta = -\frac{\sqrt{29}}{2}$

$\cot \theta = -\frac{2}{5}$

7. $\sin \theta = -\frac{12}{13}$

$\cos \theta = \frac{5}{13}$

$\tan \theta = -\frac{12}{5}$

$\csc \theta = -\frac{13}{12}$

$\sec \theta = \frac{13}{5}$

$\cot \theta = -\frac{5}{12}$

11. $\sin \theta = \frac{4\sqrt{41}}{41}$

$\cos \theta = -\frac{5\sqrt{41}}{41}$

$\tan \theta = -\frac{4}{5}$

$\csc \theta = \frac{\sqrt{41}}{4}$

$\sec \theta = -\frac{\sqrt{41}}{5}$

$\cot \theta = -\frac{5}{4}$

13. Quadrant III

17. $\sin \theta = \frac{3}{5}$

$\cos \theta = -\frac{4}{5}$

$\tan \theta = -\frac{3}{4}$

$\csc \theta = \frac{5}{3}$

$\sec \theta = -\frac{5}{4}$

$\cot \theta = -\frac{4}{3}$

15. Quadrant I

19. $\sin \theta = -\frac{15}{17}$

$\cos \theta = \frac{8}{17}$

$\tan \theta = -\frac{15}{8}$

$\csc \theta = -\frac{17}{15}$

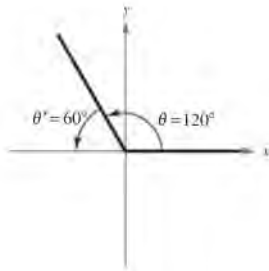
$\sec \theta = \frac{17}{8}$

$\cot \theta = -\frac{8}{15}$

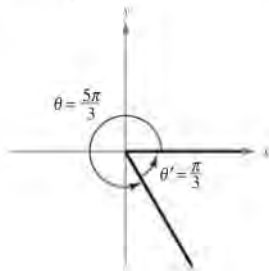
$$\begin{aligned}
 21. \sin \theta &= \frac{\sqrt{3}}{2} \\
 \cos \theta &= -\frac{1}{2} \\
 \tan \theta &= -\sqrt{3} \\
 \csc \theta &= \frac{2\sqrt{3}}{3} \\
 \sec \theta &= -2 \\
 \cot \theta &= -\frac{\sqrt{3}}{3}
 \end{aligned}$$

$$\begin{aligned}
 25. \sin \theta &= \frac{\sqrt{2}}{2} \\
 \cos \theta &= -\frac{\sqrt{2}}{2} \\
 \tan \theta &= -1 \\
 \csc \theta &= \sqrt{2} \\
 \sec \theta &= -\sqrt{2} \\
 \cot \theta &= -1
 \end{aligned}$$

$$\begin{aligned}
 29. -1 \quad 31. 0 \quad 33. 1 \quad 35. \text{Undefined} \\
 37. \theta' = 60^\circ
 \end{aligned}$$



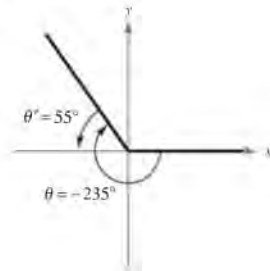
$$41. \theta' = \frac{\pi}{3}$$



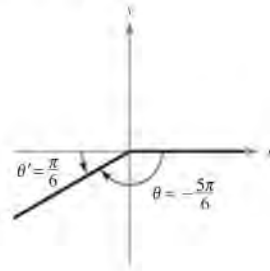
$$\begin{aligned}
 23. \sin \theta &= 0 \\
 \cos \theta &= -1 \\
 \tan \theta &= 0 \\
 \csc \theta &\text{ is undefined.} \\
 \sec \theta &= -1 \\
 \cot \theta &\text{ is undefined.}
 \end{aligned}$$

$$\begin{aligned}
 27. \sin \theta &= -\frac{2\sqrt{5}}{5} \\
 \cos \theta &= -\frac{\sqrt{5}}{5} \\
 \tan \theta &= 2 \\
 \csc \theta &= -\frac{\sqrt{5}}{2} \\
 \sec \theta &= -\sqrt{5} \\
 \cot \theta &= \frac{1}{2}
 \end{aligned}$$

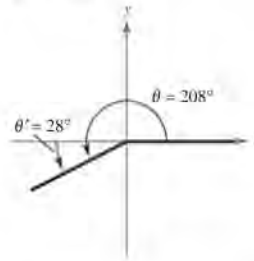
$$39. \theta' = 45^\circ$$



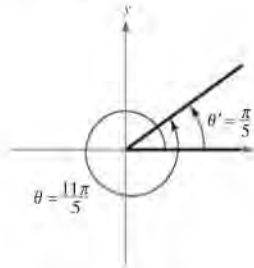
$$43. \theta' = \frac{\pi}{6}$$



$$45. \theta' = 28^\circ$$



$$49. \theta' = \frac{\pi}{5}$$



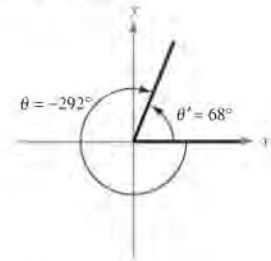
$$\begin{aligned}
 53. \sin 225^\circ &= -\frac{\sqrt{2}}{2} \\
 \cos 225^\circ &= -\frac{\sqrt{2}}{2} \\
 \tan 225^\circ &= 1
 \end{aligned}$$

$$\begin{aligned}
 57. \sin \frac{5\pi}{3} &= -\frac{\sqrt{3}}{2} \\
 \cos \frac{5\pi}{3} &= \frac{1}{2} \\
 \tan \frac{5\pi}{3} &= -\sqrt{3}
 \end{aligned}$$

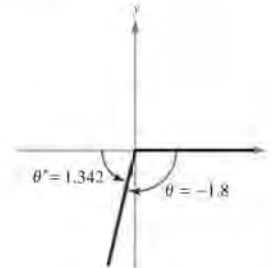
$$\begin{aligned}
 61. \sin \frac{11\pi}{4} &= \frac{\sqrt{2}}{2} \\
 \cos \frac{11\pi}{4} &= -\frac{\sqrt{2}}{2} \\
 \tan \frac{11\pi}{4} &= -1
 \end{aligned}$$

$$65. \frac{4}{5} \quad 67. -\sqrt{3} \quad 69. \frac{\sqrt{65}}{4}$$

$$47. \theta' = 68^\circ$$



$$51. \theta' = 1.342$$



$$\begin{aligned}
 55. \sin(-750^\circ) &= -\frac{1}{2} \\
 \cos(-750^\circ) &= \frac{\sqrt{3}}{2} \\
 \tan(-750^\circ) &= -\frac{\sqrt{3}}{3}
 \end{aligned}$$

$$\begin{aligned}
 59. \sin\left(-\frac{\pi}{6}\right) &= -\frac{1}{2} \\
 \cos\left(-\frac{\pi}{6}\right) &= \frac{\sqrt{3}}{2} \\
 \tan\left(-\frac{\pi}{6}\right) &= -\frac{\sqrt{3}}{3}
 \end{aligned}$$

$$\begin{aligned}
 63. \sin\left(-\frac{17\pi}{6}\right) &= -\frac{1}{2} \\
 \cos\left(-\frac{17\pi}{6}\right) &= -\frac{\sqrt{3}}{2} \\
 \tan\left(-\frac{17\pi}{6}\right) &= \frac{\sqrt{3}}{3}
 \end{aligned}$$

71. $\cos \theta = -\frac{\sqrt{21}}{5}$

73. $\sin \theta = \frac{4\sqrt{17}}{17}$

$\tan \theta = -\frac{2\sqrt{21}}{21}$

$\cos \theta = -\frac{\sqrt{17}}{17}$

$\csc \theta = \frac{5}{2}$

$\csc \theta = \frac{\sqrt{17}}{4}$

$\sec \theta = -\frac{5\sqrt{21}}{21}$

$\sec \theta = -\sqrt{17}$

$\cot \theta = -\frac{\sqrt{21}}{2}$

$\cot \theta = -\frac{1}{4}$

75. $\sin \theta = -\frac{2}{3}$

$\cos \theta = \frac{\sqrt{5}}{3}$

$\tan \theta = -\frac{2\sqrt{5}}{5}$

$\sec \theta = \frac{3\sqrt{5}}{5}$

$\cot \theta = -\frac{\sqrt{5}}{2}$

77. 0.1736

79. 2.1445

81. -0.3420

83. 5.7588

85. 0.8391

87. -2.9238

89. (a) $30^\circ = \frac{\pi}{6}$, $150^\circ = \frac{5\pi}{6}$

(b) $210^\circ = \frac{7\pi}{6}$, $330^\circ = \frac{11\pi}{6}$

91. (a) $60^\circ = \frac{\pi}{3}$, $120^\circ = \frac{2\pi}{3}$

(b) $135^\circ = \frac{3\pi}{4}$, $315^\circ = \frac{7\pi}{4}$

93. (a) $150^\circ = \frac{5\pi}{6}$, $210^\circ = \frac{7\pi}{6}$

(b) $120^\circ = \frac{2\pi}{3}$, $240^\circ = \frac{4\pi}{3}$

95. (a) $\frac{1+\sqrt{3}}{2}$

(b) $\frac{\sqrt{3}-1}{2}$

(c) $\frac{3}{4}$

(d) $\frac{\sqrt{3}}{4}$

(e) 1

(f) $\frac{\sqrt{3}}{2}$

97. (a) 0

(b) $\sqrt{2}$

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

(e) $-\sqrt{2}$

(f) $\frac{\sqrt{2}}{2}$

99. (a) $\frac{1-\sqrt{3}}{2}$

(b) $-\frac{1+\sqrt{3}}{2}$

(c) $\frac{3}{4}$

(d) $-\frac{\sqrt{3}}{4}$

(e) 1

(f) $-\frac{\sqrt{3}}{2}$

101. (a) $-\frac{1+\sqrt{3}}{2}$

(b) $\frac{1-\sqrt{3}}{2}$

(c) $\frac{3}{4}$

(d) $\frac{\sqrt{3}}{4}$

(e) -1

(f) $-\frac{\sqrt{3}}{2}$

103. (a) $-\frac{1+\sqrt{3}}{2}$

(b) $\frac{-1+\sqrt{3}}{2}$

(c) $\frac{1}{4}$

(d) $\frac{\sqrt{3}}{4}$

(e) $-\sqrt{3}$

(f) $-\frac{1}{2}$

105. (a) -1

(b) 1

(c) 0

(d) 0

(e) -2

(f) 0

107. (a) -1

(b) 1

(c) 0

(d) 0

(e) -2

(f) 0

109. (a) 29° F

(b) 70° F

(c) 31.75° F

111. (a) 12 miles

(b) 6 miles

(c) 6.93 miles

113. True. The angles have the same reference angle.

115. (a)

θ	0°	20°	40°
$\sin \theta$	0	0.3420	0.6428
$\sin(180^\circ - \theta)$	0	0.3420	0.6428

θ	60°	80°
$\sin \theta$	0.8660	0.9848
$\sin(180^\circ - \theta)$	0.8660	0.9848

(b) $\sin \theta = \sin(180^\circ - \theta)$

117. 7

119. 3.449, -1.449

121. 4.908, -5.908

123. -1.752

125. 0.002

Section 4.5 (page 304)

Vocabulary Check (page 304)

1. amplitude

2. one cycle

3. $\frac{2\pi}{b}$

4. phase shift

1. (a) $x = -2\pi, -\pi, 0, \pi, 2\pi$

(b) $y = 0$

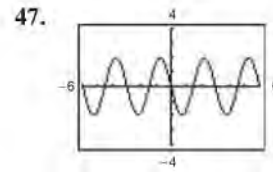
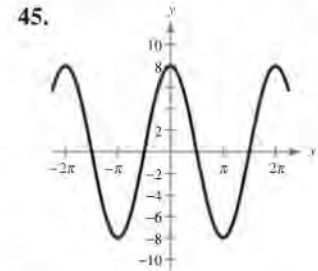
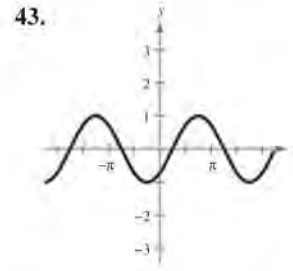
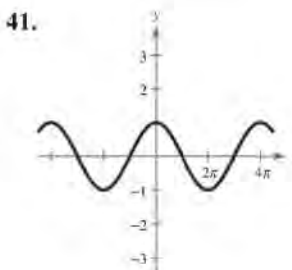
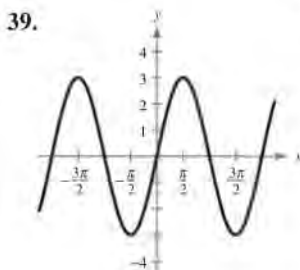
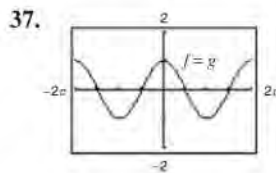
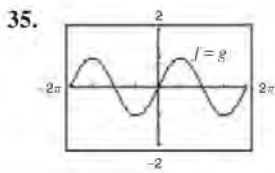
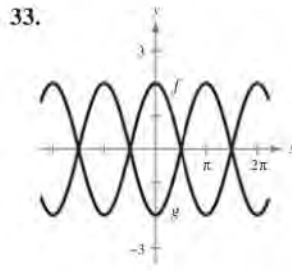
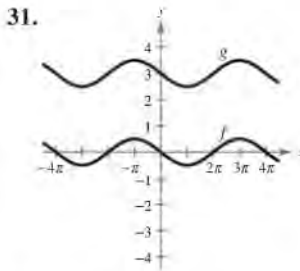
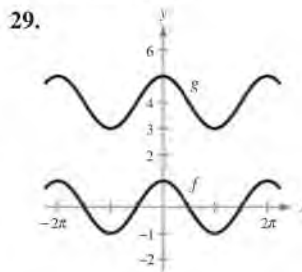
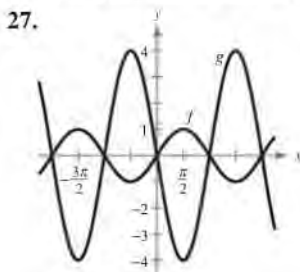
(c) Increasing: $\left(-2\pi, -\frac{3\pi}{2}\right), \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \left(\frac{3\pi}{2}, 2\pi\right)$

Decreasing: $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right), \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

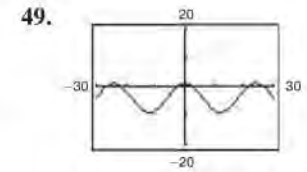
(d) Relative maxima: $\left(-\frac{3\pi}{2}, 1\right), \left(\frac{\pi}{2}, 1\right)$

Relative minima: $\left(-\frac{\pi}{2}, -1\right), \left(\frac{3\pi}{2}, -1\right)$

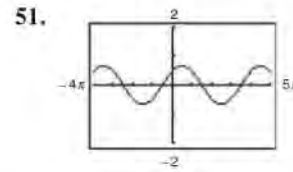
3. Period: π 5. Period: 4π 7. Period: 2
 Amplitude: 3 Amplitude: $\frac{5}{2}$ Amplitude: $\frac{2}{3}$
 9. Period: 2π 11. Period: 3π
 Amplitude: 2 Amplitude: $\frac{1}{4}$
 13. Period: $\frac{1}{2}$ 15. g is a shift of f π units to the right.
 Amplitude: $\frac{1}{3}$
 17. g is a reflection of f in the x -axis.
 19. g is a reflection of f in the x -axis and has five times the amplitude of f .
 21. g is a shift of f five units upward.
 23. g has twice the amplitude of f .
 25. g is a horizontal shift of f π units to the right.



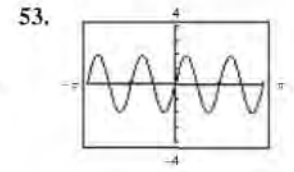
Amplitude: 2
Period: 3



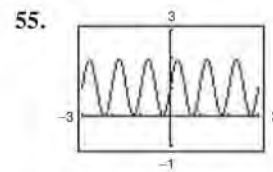
Amplitude: 5
Period: 24



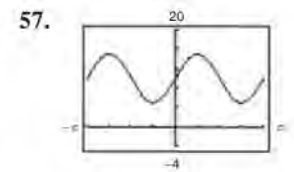
Amplitude: $\frac{2}{3}$
Period: 4π



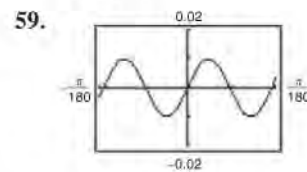
Amplitude: 2
Period: $\frac{\pi}{2}$



Amplitude: 1
Period: 1



Amplitude: 5
Period: π



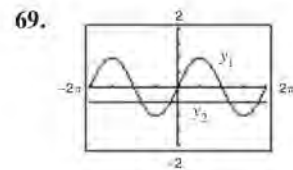
Amplitude: $\frac{1}{100}$
Period: $\frac{1}{60}$

61. $a = -4, d = 4$

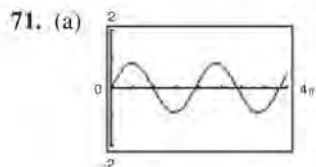
63. $a = -6, d = 1$

65. $a = -3, b = 2, c = 0$

67. $a = 1, b = 1, c = \frac{\pi}{4}$

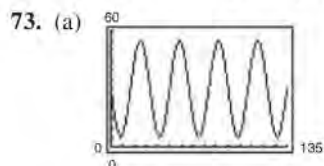


$x = -\frac{\pi}{6}, -\frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$



- (b) 6 seconds
(c) 10 cycles per minute

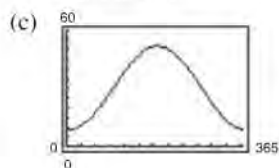
(d) The period of the model would decrease because the time for a respiratory cycle would decrease.



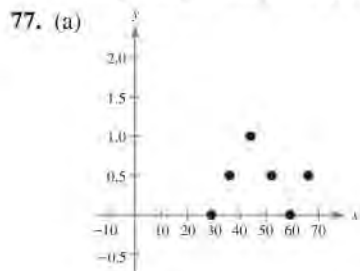
- (b) Minimum height: 5 feet
Maximum height: 55 feet

75. (a) 365 days. The cycle is 1 year.

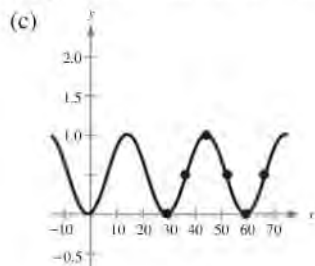
(b) 30.3 gallons per day. The average is the constant term of the model.



Consumption exceeds 40 gallons per day from the beginning of May through part of September.



(b) $y = 0.506 \sin(0.209x - 1.336) + 0.526$

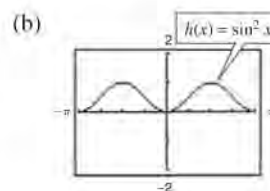
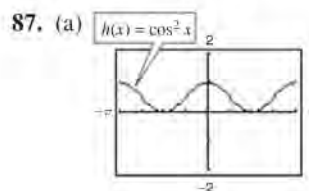


Answers will vary.

- (d) 30.06
(e) 27.09%

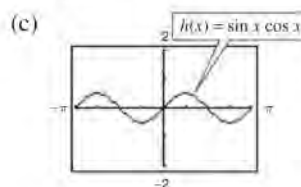
79. True, $2\pi \cdot \frac{10}{3} = \frac{20\pi}{3}$

81. True 83. e 85. c



Even

Even



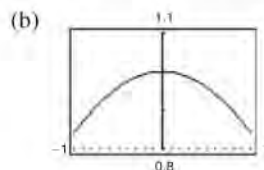
Odd

89. (a)

x	-1	-0.1	-0.01
$\frac{\sin x}{x}$	0.8415	0.9983	1.0000

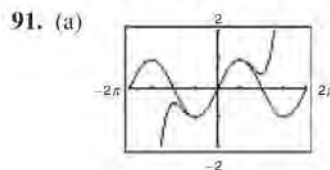
x	-0.001	0	0.001
$\frac{\sin x}{x}$	1.0000	Undefined	1.0000

x	0.01	0.1	1
$\frac{\sin x}{x}$	1.0000	0.9983	0.8415

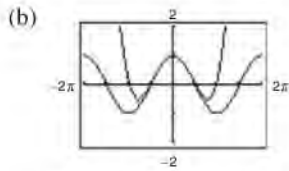


$f \rightarrow 1$ as $x \rightarrow 0$

(c) The ratio approaches 1 as x approaches 0.

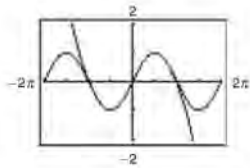


The polynomial function is a good approximation of the sine function when x is close to 0.

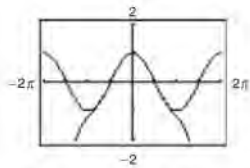


The polynomial function is a good approximation of the cosine function when x is close to 0.

(c) $\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$

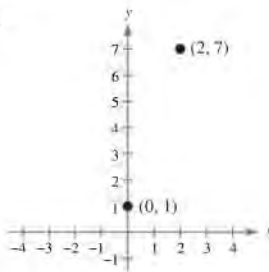


$\cos x \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$



The accuracy increased.

93.



$m = 3$

95. 487.014° 97. Answers will vary.

Section 4.6 (page 316)

Vocabulary Check (page 316)

1. vertical 2. reciprocal 3. damping

1. (a) $x = -2\pi, -\pi, 0, \pi, 2\pi$ (b) $y = 0$

(c) Increasing on $\left(-2\pi, -\frac{3\pi}{2}\right), \left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right), \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \frac{3\pi}{2}\right), \left(\frac{3\pi}{2}, 2\pi\right)$

(d) No relative extrema

(e) $x = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

3. (a) No x -intercepts

(b) $y = 0$

(c) Increasing on $\left(-2\pi, -\frac{3\pi}{2}\right), \left(-\frac{3\pi}{2}, -\pi\right), \left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right)$

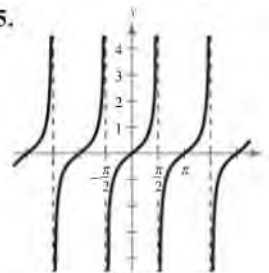
Decreasing on $\left(-\pi, -\frac{\pi}{2}\right), \left(-\frac{\pi}{2}, 0\right), \left(\pi, \frac{3\pi}{2}\right), \left(\frac{3\pi}{2}, 2\pi\right)$

(d) Relative minima: $(-2\pi, 1), (0, 1), (2\pi, 1)$

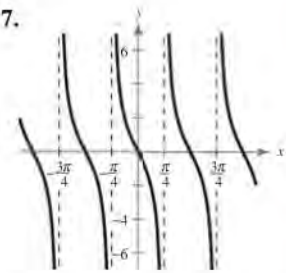
Relative maxima: $(-\pi, -1), (\pi, -1)$

(e) $x = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

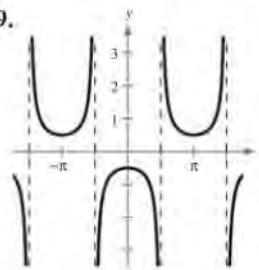
5.



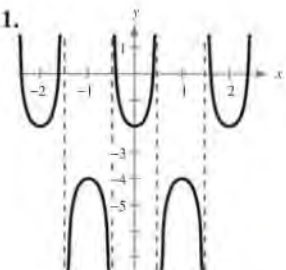
7.



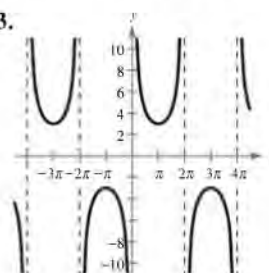
9.



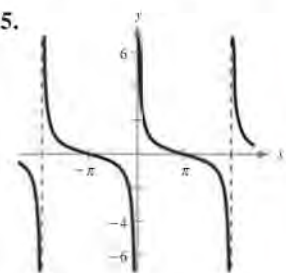
11.



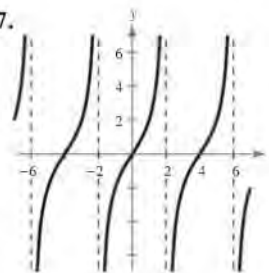
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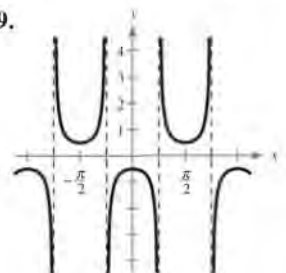
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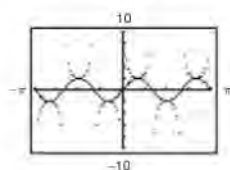
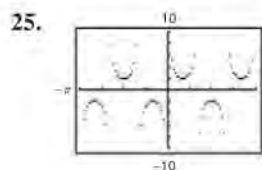
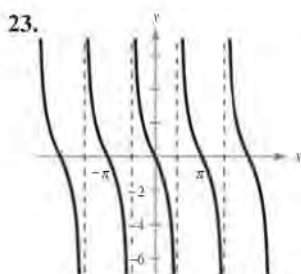
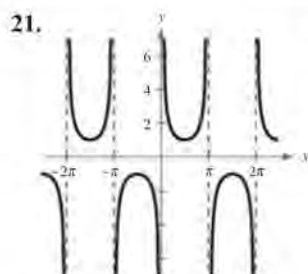


17.

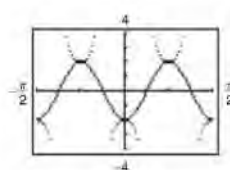
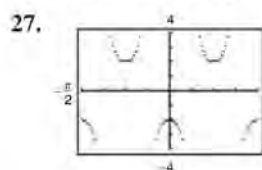


19.

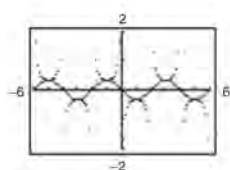
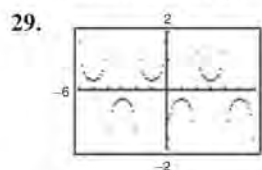




Answers will vary.



Answers will vary.

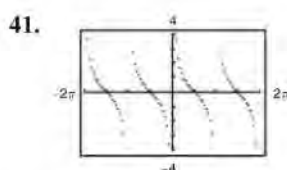
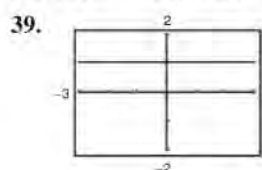


Answers will vary.

31. $-5.498, -2.356, 0.785, 3.927$

33. $-4.189, -2.094, 2.094, 4.189$

35. Even 37. Odd



Not equivalent;
 y_1 is undefined at $x = 0$.

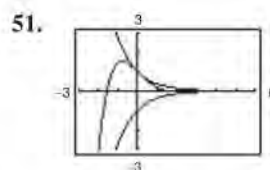
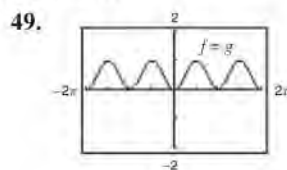
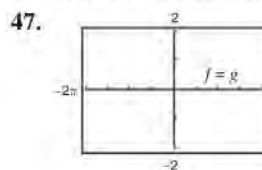
Equivalent

43. d; as x approaches 0, $f(x)$ approaches 0.

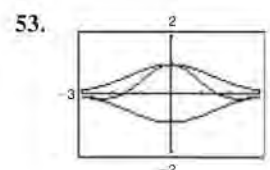
44. a; as x approaches 0, $f(x)$ approaches 0.

45. b; as x approaches 0, $g(x)$ approaches 0.

46. c; as x approaches 0, $g(x)$ approaches 0.



$f \rightarrow 0$ as $x \rightarrow \infty$



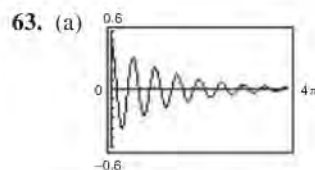
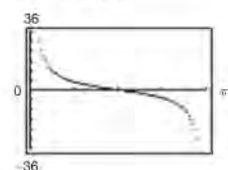
$h \rightarrow 0$ as $x \rightarrow \infty$

55. (a) $f \rightarrow -\infty$ (b) $f \rightarrow \infty$ (c) $f \rightarrow -\infty$ (d) $f \rightarrow \infty$

57. (a) $f \rightarrow \infty$ (b) $f \rightarrow -\infty$ (c) $f \rightarrow \infty$ (d) $f \rightarrow -\infty$

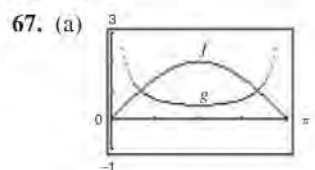
59. As the predator population increases, the number of prey decreases. When the number of prey is small, the number of predators decreases.

61. $d = 5 \cot x$



(b) Not periodic and damped; approaches 0 as t increases.

65. True



(b) $0.524 < x < 2.618$

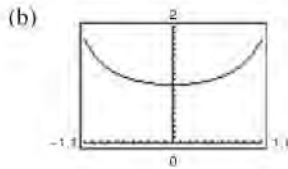
(c) f approaches 0 and g approaches ∞ , because g is the reciprocal of f .

69. (a)

x	-1	-0.1	-0.01
$\frac{\tan x}{x}$	1.5574	1.0033	1.0000

x	-0.001	0	0.001
$\frac{\tan x}{x}$	1.0000	Undefined	1.0000

x	0.01	0.1	1
$\frac{\tan x}{x}$	1.0000	1.0033	1.5574

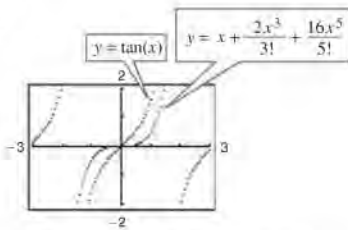


$$f \rightarrow 1 \text{ as } x \rightarrow 0$$

(c) The ratio approaches 1 as x approaches 0.

71. a

73.



The polynomial function is a good approximation of the tangent function when x is close to 0.

75. Distributive Property 77. Additive Identity Property

79. Not one-to-one

81. One-to-one. $f^{-1}(x) = \frac{x^2 + 14}{3}, x \geq 0$

Section 4.7 (page 327)

Vocabulary Check (page 327)

1. $y = \sin^{-1} x, -1 \leq x \leq 1$

2. $y = \arccos x, 0 \leq y \leq \pi$

3. $y = \tan^{-1} x, -\infty < x < \infty, -\frac{\pi}{2} < y < \frac{\pi}{2}$

1. (a) $\frac{\pi}{6}$ (b) 0 3. (a) $\frac{\pi}{2}$ (b) 0

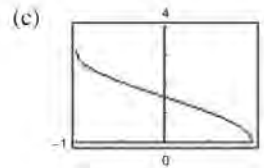
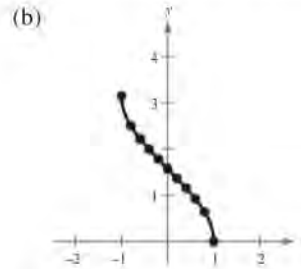
5. (a) $\frac{\pi}{6}$ (b) $-\frac{\pi}{4}$ 7. (a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$

9. (a) $\frac{\pi}{3}$ (b) $-\frac{\pi}{6}$

11. (a)

x	-1	-0.8	-0.6	-0.4	-0.2
y	3.142	2.498	2.214	1.982	1.772

x	0	0.2	0.4	0.6	0.8	1
y	1.571	1.369	1.159	0.927	0.644	0



They are the same.

(d) Intercepts: $(0, \frac{\pi}{2}), (1, 0)$

13. $(-\sqrt{3}, -\frac{\pi}{3}), (-\frac{\sqrt{3}}{3}, -\frac{\pi}{6}), (1, \frac{\pi}{4})$ 15. 0.72

17. -0.85 19. -1.41 21. $\theta = \arctan \frac{x}{8}$

23. $\theta = \arcsin \frac{x+2}{5}$

25. $\sqrt{4-x^2}; \theta = \arcsin \frac{x}{2}, \theta = \arccos \frac{\sqrt{4-x^2}}{2},$

$$\theta = \arctan \frac{x}{\sqrt{4-x^2}}$$

27. $\sqrt{x^2+2x+5}; \theta = \arcsin \frac{x+1}{\sqrt{x^2+2x+5}},$

$$\theta = \arccos \frac{2}{\sqrt{x^2+2x+5}}, \theta = \arctan \frac{x+1}{2}$$

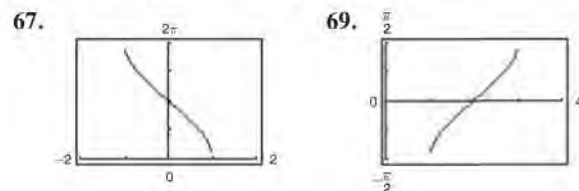
29. 0.7 31. -0.3 33. 0 35. $-\frac{\pi}{6}$ 37. $\frac{\pi}{2}$

39. $\frac{\pi}{2}$ 41. 0 43. $\frac{\sqrt{2}}{2}$ 45. $\frac{\pi}{3}$ 47. $\frac{4}{5}$ 49. $\frac{7}{25}$

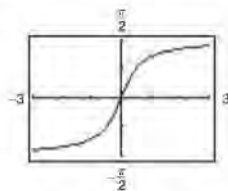
51. $\frac{\sqrt{34}}{5}$ 53. $\frac{\sqrt{5}}{3}$ 55. $\frac{1}{x}$ 57. $\sqrt{-x^2-4x-3}$

59. $\frac{\sqrt{25-x^2}}{x}$ 61. $\frac{\sqrt{x^2+7}}{x}$

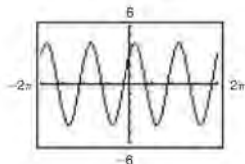
63. $\frac{14}{\sqrt{x^2+196}}$ 65. $\frac{|x-1|}{\sqrt{x^2-2x+10}}$



71.



73. $3\sqrt{2} \sin\left(2t + \frac{\pi}{4}\right)$



The two forms are equivalent.

75. $\frac{\pi}{2}$ 77. $\frac{\pi}{2}$ 79. π

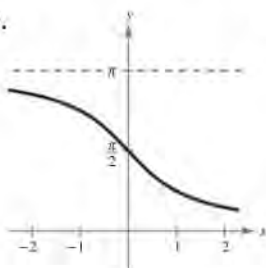
81. (a) $\theta = \arcsin \frac{10}{s}$ (b) 0.19 rad, 0.39 rad

83. (a) $\theta = \arctan \frac{s}{750}$ (b) 0.49 rad, 1.13 rad

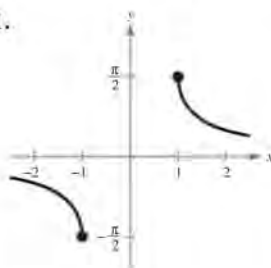
85. (a) $\theta = \arctan \frac{6}{x}$ (b) 0.54 rad, 1.11 rad

87. False. $5\pi/6$ is not in the range of the arcsine function.

89.



91.



93. $\frac{\pi}{4}$ 95. $\frac{5\pi}{6}$ 97. Proof 99. Proof

101. $\frac{\sqrt{2}}{2}$ 103. $\frac{\sqrt{3}}{3}$

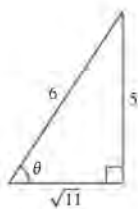
105. $\cos \theta = \frac{\sqrt{11}}{6}$

$$\tan \theta = \frac{5\sqrt{11}}{11}$$

$$\csc \theta = \frac{6}{5}$$

$$\sec \theta = \frac{6\sqrt{11}}{11}$$

$$\cot \theta = \frac{\sqrt{11}}{5}$$



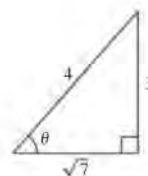
107. $\cos \theta = \frac{\sqrt{7}}{4}$

$$\tan \theta = \frac{3\sqrt{7}}{7}$$

$$\csc \theta = \frac{4}{3}$$

$$\sec \theta = \frac{4\sqrt{7}}{7}$$

$$\cot \theta = \frac{\sqrt{7}}{3}$$



Section 4.8 (page 337)

Vocabulary Check (page 337)

1. elevation, depression
2. bearing
3. harmonic motion

1. $B = 60^\circ$ 3. $A = 19^\circ$ 5. $A \approx 26.57^\circ$

$a \approx 5.77$ $a \approx 4.82$ $B \approx 63.43^\circ$

$c \approx 11.55$ $c \approx 14.81$ $c \approx 13.42$

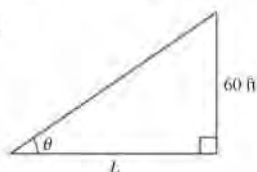
7. $A \approx 72.76^\circ$ 9. $B = 77^\circ 45'$

$B \approx 17.24^\circ$ $a \approx 91.34$

$a \approx 51.58$ $b \approx 420.70$

11. 5.12 inches 13. 8.21 feet

15. (a)



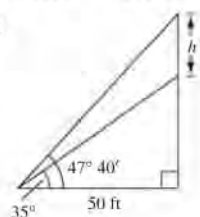
(b) $L = 60 \cot \theta$

θ	10°	20°	30°	40°	50°
L	340.28	164.85	103.92	71.51	50.35

(d) No. The cotangent is not a linear function.

17. 19.70 feet 19. 76.60 feet

21. (a)



(b) $h = 50(\tan 47^\circ 40' - \tan 35^\circ)$ (c) 19.87 feet

23. (a) $l = \sqrt{h^2 + 34h + 10,289}$

(b) $\theta = \arccos \frac{100}{l}$

(c) 53 feet

25. 38.29° 27. 75.97°

29. 5099 feet 31. 0.66 mile

33. 104.95 nautical miles south, 58.18 nautical miles west

35. (a) $N 58^\circ E$ (b) 68.82 meters

37. $N 56.31^\circ W$ 39. 1933.32 feet 41. 3.23 miles

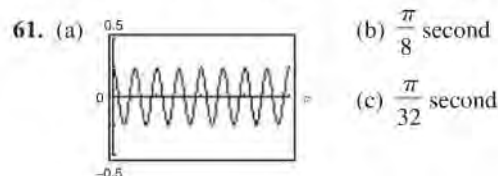
43. (a) 54.46° ; 20.47° (b) 56 feet

45. 78.69° 47. 35.26° 49. $y = \sqrt{3}r$

51. $d = 8 \sin \pi t$ 53. $d = 3 \cos\left(\frac{4\pi}{3}t\right)$

55. (a) 4 (b) 4 (c) 4 (d) $\frac{1}{16}$

57. (a) $\frac{1}{16}$ (b) 70 (c) 0 (d) $\frac{1}{140}$ 59. $\omega = 528\pi$



63. (a)

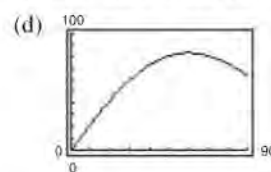
Base 1	Base 2	Altitude	Area
8	$8 + 16 \cos 10^\circ$	$8 \sin 10^\circ$	22.06
8	$8 + 16 \cos 20^\circ$	$8 \sin 20^\circ$	42.46
8	$8 + 16 \cos 30^\circ$	$8 \sin 30^\circ$	59.71
8	$8 + 16 \cos 40^\circ$	$8 \sin 40^\circ$	72.65
8	$8 + 16 \cos 50^\circ$	$8 \sin 50^\circ$	80.54
8	$8 + 16 \cos 60^\circ$	$8 \sin 60^\circ$	83.14
8	$8 + 16 \cos 70^\circ$	$8 \sin 70^\circ$	80.71

(b)

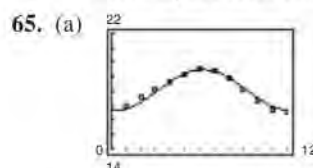
Base 1	Base 2	Altitude	Area
8	$8 + 16 \cos 56^\circ$	$8 \sin 56^\circ$	82.73
8	$8 + 16 \cos 58^\circ$	$8 \sin 58^\circ$	83.04
8	$8 + 16 \cos 59^\circ$	$8 \sin 59^\circ$	83.11
8	$8 + 16 \cos 60^\circ$	$8 \sin 60^\circ$	83.14
8	$8 + 16 \cos 61^\circ$	$8 \sin 61^\circ$	83.11
8	$8 + 16 \cos 62^\circ$	$8 \sin 62^\circ$	83.04

83.14 square feet

(c) $A = 64(1 + \cos \theta)(\sin \theta)$



83.14 square feet; They are the same.



(b) 12 months. Yes; one period is 1 year.

(c) 1.41 hours; maximum displacement of 1.41 hours from the average sunset time of 18.09

67. False. $a = \frac{22.56}{\tan 48.1^\circ}$ 69. $4x - y + 6 = 0$

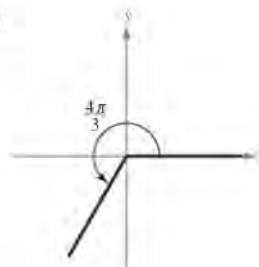
71. $4x + 5y - 22 = 0$ 73. All real numbers x

75. All real numbers x

Review Exercises (page 344)

1. 1 rad

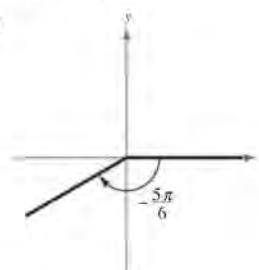
3. (a)



(b) Quadrant III

(c) $\frac{10\pi}{3}, -\frac{2\pi}{3}$

5. (a)



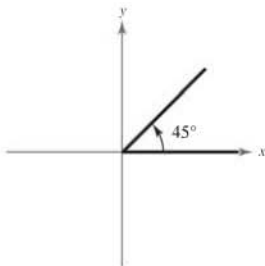
(b) Quadrant III

(c) $\frac{7\pi}{6}, -\frac{17\pi}{6}$

7. Complement: $\frac{3\pi}{8}$; supplement: $\frac{7\pi}{8}$

9. Complement: $\frac{\pi}{5}$; supplement: $\frac{7\pi}{10}$

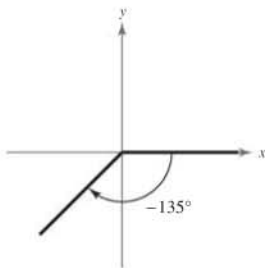
11. (a)



(b) Quadrant I

(c) $405^\circ, -315^\circ$

13. (a)



(b) Quadrant III

(c) $225^\circ, -495^\circ$

15. Complement: 85° ; supplement: 175°

17. Complement: none; supplement: 9° 19. 135.279°

21. 5.381° 23. $135^\circ 17' 24''$ 25. $-85^\circ 21' 36''$

27. 7.243 29. -1.257 31. 128.571°

33. -200.535° 35. $\frac{25}{12}$ rad 37. $\frac{46\pi}{3}$ meters

39. 6000π centimeters per minute

41. $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ 43. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

45. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ 47. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

49. $\sin \frac{7\pi}{6} = -\frac{1}{2}$

$\cos \frac{7\pi}{6} = -\frac{\sqrt{3}}{2}$

$\tan \frac{7\pi}{6} = \frac{\sqrt{3}}{3}$

$\csc \frac{7\pi}{6} = -2$

$\sec \frac{7\pi}{6} = -\frac{2\sqrt{3}}{3}$

$\cot \frac{7\pi}{6} = \sqrt{3}$

51. $\sin 2\pi = 0$

$\cos 2\pi = 1$

$\tan 2\pi = 0$

$\csc 2\pi$ is undefined.

$\sec 2\pi = 1$

$\cot 2\pi$ is undefined.

53. $\sin\left(-\frac{11\pi}{6}\right) = \frac{1}{2}$

$\cos\left(-\frac{11\pi}{6}\right) = \frac{\sqrt{3}}{2}$

$\tan\left(-\frac{11\pi}{6}\right) = \frac{\sqrt{3}}{3}$

$\csc\left(-\frac{11\pi}{6}\right) = 2$

$\sec\left(-\frac{11\pi}{6}\right) = \frac{2\sqrt{3}}{3}$

$\cot\left(-\frac{11\pi}{6}\right) = \sqrt{3}$

55. $\sin\left(-\frac{\pi}{2}\right) = -1$

$\cos\left(-\frac{\pi}{2}\right) = 0$

$\tan\left(-\frac{\pi}{2}\right)$ is undefined.

$\csc\left(-\frac{\pi}{2}\right) = -1$

$\sec\left(-\frac{\pi}{2}\right)$ is undefined.

$\cot\left(-\frac{\pi}{2}\right) = 0$

57. $\frac{\sqrt{2}}{2}$

59. $-\frac{1}{2}$

61. (a) $-\frac{3}{5}$ (b) $-\frac{5}{3}$

63. (a) $\frac{2}{3}$ (b) $\frac{3}{2}$

65. -0.8935 67. 0.5

69. $\sin \theta = \frac{\sqrt{65}}{9}$

71. $\sin \theta = \frac{5\sqrt{61}}{61}$

$\cos \theta = \frac{4}{9}$

$\cos \theta = \frac{6\sqrt{61}}{61}$

$\tan \theta = \frac{\sqrt{65}}{4}$

$\tan \theta = \frac{5}{6}$

$\csc \theta = \frac{9\sqrt{65}}{65}$

$\csc \theta = \frac{\sqrt{61}}{5}$

$\sec \theta = \frac{9}{4}$

$\sec \theta = \frac{\sqrt{61}}{6}$

$\cot \theta = \frac{4\sqrt{65}}{65}$

$\cot \theta = \frac{6}{5}$

73. Answers will vary.

75. (a) 0.1045 (b) 0.1045

77. (a) 0.7071

(b) 1.4142

79. 235 feet

81. $\sin \theta = \frac{4}{5}$

83. $\sin \theta = \frac{2\sqrt{53}}{53}$

$\cos \theta = \frac{3}{5}$

$\cos \theta = -\frac{7\sqrt{53}}{53}$

$\tan \theta = \frac{4}{3}$

$\tan \theta = -\frac{2}{7}$

$\csc \theta = \frac{5}{4}$

$\csc \theta = \frac{\sqrt{53}}{2}$

$\sec \theta = \frac{5}{3}$

$\sec \theta = -\frac{\sqrt{53}}{7}$

$\cot \theta = \frac{3}{4}$

$\cot \theta = -\frac{7}{2}$

$$85. \sin \theta = \frac{15\sqrt{481}}{481}$$

$$\cos \theta = \frac{16\sqrt{481}}{481}$$

$$\tan \theta = \frac{15}{16}$$

$$\csc \theta = \frac{\sqrt{481}}{15}$$

$$\sec \theta = \frac{\sqrt{481}}{16}$$

$$\cot \theta = \frac{16}{15}$$

$$89. \cos \theta = -\frac{\sqrt{55}}{8}$$

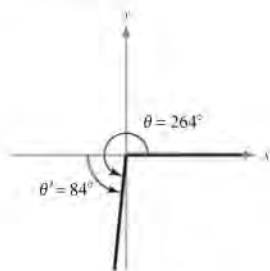
$$\tan \theta = -\frac{3\sqrt{55}}{55}$$

$$\csc \theta = \frac{8}{3}$$

$$\sec \theta = -\frac{8\sqrt{55}}{55}$$

$$\cot \theta = -\frac{\sqrt{55}}{3}$$

$$91. \theta' = 84^\circ$$



$$95. \sin 240^\circ = -\frac{\sqrt{3}}{2}$$

$$\cos 240^\circ = -\frac{1}{2}$$

$$\tan 240^\circ = \sqrt{3}$$

$$99. \sin\left(-\frac{9\pi}{4}\right) = -\frac{\sqrt{2}}{2}$$

$$\cos\left(-\frac{9\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$\tan\left(-\frac{9\pi}{4}\right) = -1$$

$$87. \sin \theta = -\frac{\sqrt{11}}{6}$$

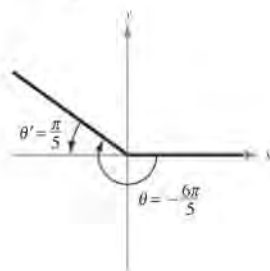
$$\cos \theta = \frac{5}{6}$$

$$\tan \theta = -\frac{\sqrt{11}}{5}$$

$$\csc \theta = -\frac{6\sqrt{11}}{11}$$

$$\cot \theta = -\frac{5\sqrt{11}}{11}$$

$$93. \theta' = \frac{\pi}{5}$$



$$97. \sin(-210^\circ) = \frac{1}{2}$$

$$\cos(-210^\circ) = -\frac{\sqrt{3}}{2}$$

$$\tan(-210^\circ) = -\frac{\sqrt{3}}{3}$$

$$101. \sin 4\pi = 0$$

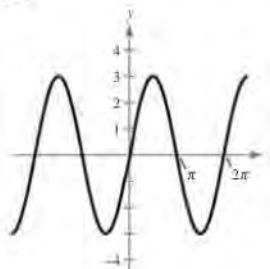
$$\cos 4\pi = 1$$

$$\tan 4\pi = 0$$

$$103. 0.6494$$

$$105. 3.2361$$

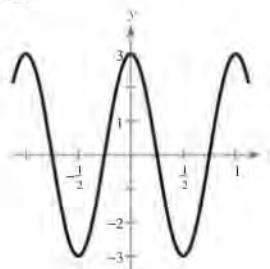
107.



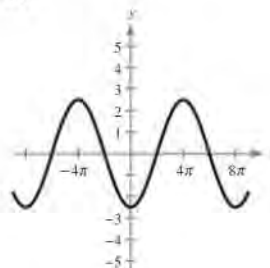
111. Period: 2; amplitude: 5

 113. Period: π ; amplitude: 3.4

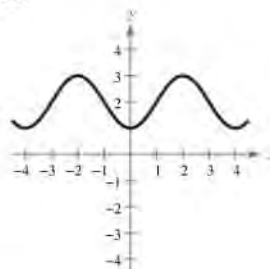
115.



119.

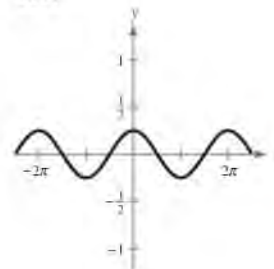


123.

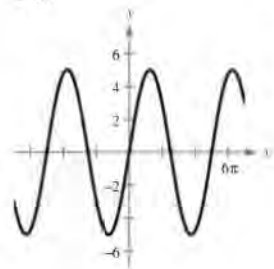


$$127. a = -2, b = 1, c = \frac{\pi}{4}$$

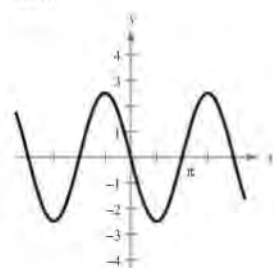
109.



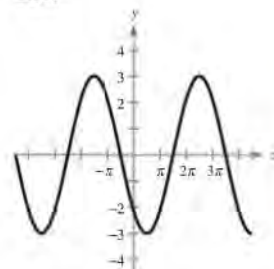
117.



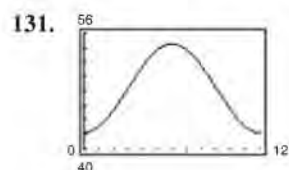
121.



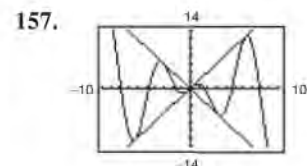
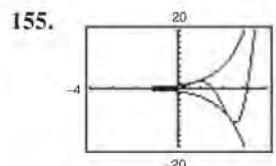
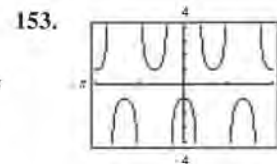
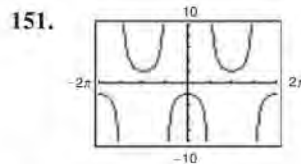
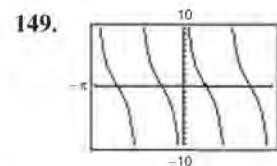
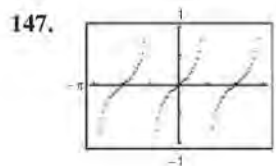
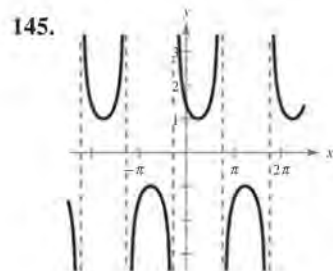
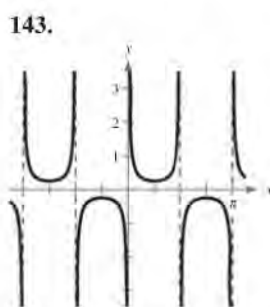
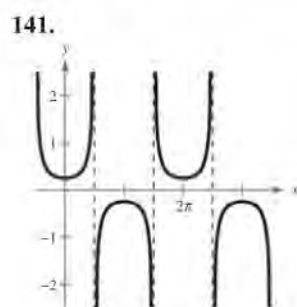
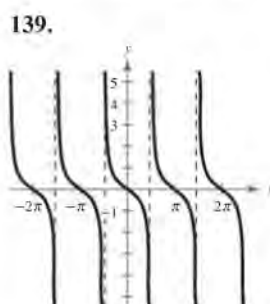
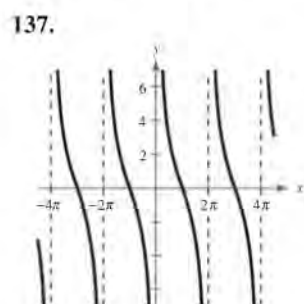
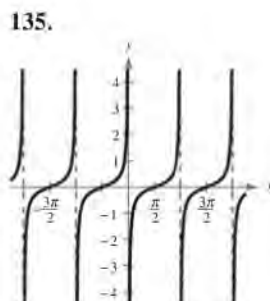
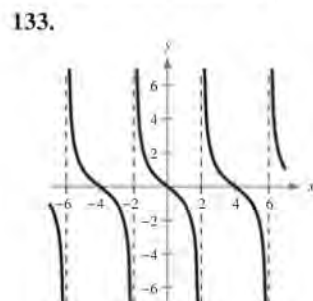
125.



$$129. a = -4, b = 2, c = \frac{\pi}{2}$$



Maximum sales: June
Minimum sales: December



As $x \rightarrow \infty$, f oscillates
between $-\infty$ and ∞ .

As $x \rightarrow \infty$, f oscillates
between $-2x$ and $2x$.

159. (a) $-\frac{\pi}{2}$ (b) Does not exist 161. (a) $\frac{\pi}{4}$ (b) $\frac{5\pi}{6}$

163. 1.14 165. -1.22 167. -1.49

169. 0.68 171. $\theta = \arcsin\left(\frac{x+3}{16}\right)$ 173. $\frac{1}{\sqrt{2x-x^2}}$

175. $\frac{2\sqrt{4-2x^2}}{4-x^2}$ 177. 0.071 kilometer

179. 9.47 miles 181. $y = 3 \cos\left(\frac{2\pi t}{15}\right)$

183. False. y is a function but is not one-to-one on $30^\circ \leq \theta \leq 150^\circ$.

185. (a)

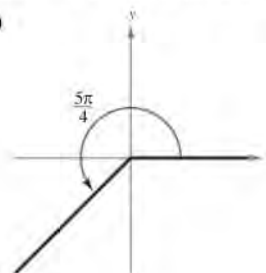
s	10	20	30
θ	0.0224	0.0894	0.1989

s	40	50	60
θ	0.3441	0.5105	0.6786

(b) θ is not a linear function of s .

Chapter Test (page 349)

1. (a)



(b) Answers will vary.
Sample answer:

$$\frac{13\pi}{4}, -\frac{3\pi}{4}$$

(c) 225°

2. 2400 radians per minute

3. $\sin \theta = \frac{4\sqrt{17}}{17}$

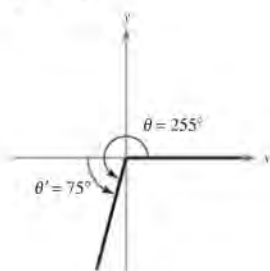
$$\cos \theta = -\frac{\sqrt{17}}{17}$$

$$\tan \theta = -4$$

$$\csc \theta = \frac{\sqrt{17}}{4}$$

$$\sec \theta = -\sqrt{17}$$

$$\cot \theta = -\frac{1}{4}$$

 5. $\theta' = 75^\circ$


6. Quadrant III

 7. $135^\circ, 225^\circ$

8. 1.33, 1.81

9. $\sin \theta = \frac{4}{5}$

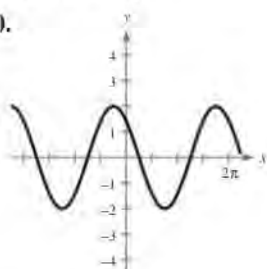
$$\tan \theta = -\frac{4}{3}$$

$$\csc \theta = \frac{5}{4}$$

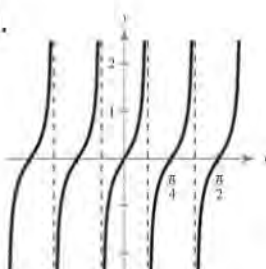
$$\sec \theta = -\frac{5}{3}$$

$$\cot \theta = -\frac{3}{4}$$

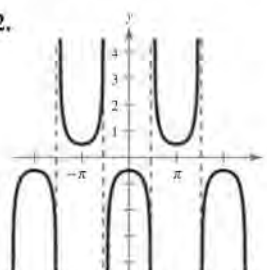
10.



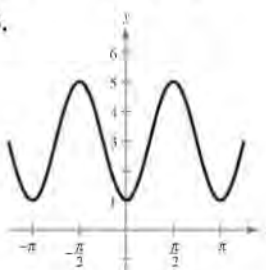
11.



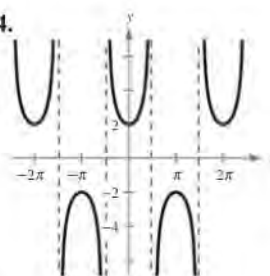
12.



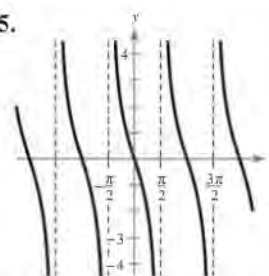
13.



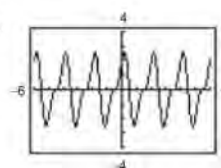
14.



15.

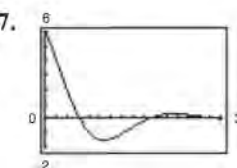


16.



Period: 2

17.

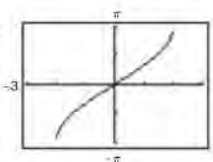


Not periodic

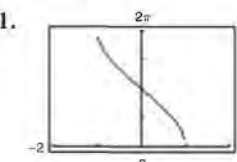
18. $a = -2, b = \frac{1}{2}, c = -\frac{\pi}{4}$

19. $\frac{\sqrt{5}}{2}$

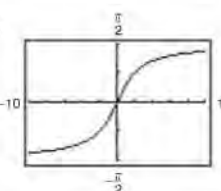
20.



21.



22.


 23. 214.51°

Chapter 5

Section 5.1 (page 357)

Vocabulary Check (page 357)

1. $\sec u$ 2. $\tan u$ 3. $\cot u$ 4. $\csc u$
5. $\tan^2 u$ 6. $\csc^2 u$ 7. $\sin u$ 8. $\sec u$
9. $-\tan u$ 10. $\cos u$

1. $\tan x = \frac{\sqrt{3}}{3}$

$$\csc x = 2$$

$$\sec x = \frac{2\sqrt{3}}{3}$$

$$\cot x = \sqrt{3}$$

3. $\cos \theta = \frac{\sqrt{2}}{2}$

$$\tan \theta = -1$$

$$\csc \theta = -\sqrt{2}$$

$$\cot \theta = -1$$

$$\begin{aligned} 5. \sin x &= -\frac{7}{25} \\ \cos x &= -\frac{24}{25} \\ \csc x &= -\frac{25}{7} \\ \cot x &= \frac{24}{7} \end{aligned}$$

$$\begin{aligned} 9. \sin x &= \frac{2}{3} \\ \cos x &= -\frac{\sqrt{5}}{3} \\ \csc x &= \frac{3}{2} \\ \sec x &= -\frac{3\sqrt{5}}{5} \\ \cot x &= -\frac{\sqrt{5}}{2} \end{aligned}$$

$$\begin{aligned} 13. \sin \theta &= 0 \\ \cos \theta &= -1 \\ \tan \theta &= 0 \\ \sec \theta &= -1 \\ \cot \theta &\text{ is undefined.} \end{aligned}$$

$$15. d \quad 16. a \quad 17. b \quad 18. f \quad 19. e$$

$$20. c \quad 21. b \quad 22. c \quad 23. f \quad 24. a$$

$$25. e \quad 26. d \quad 27. \cos x \quad 29. \cos^2 \phi$$

$$31. \sec x \quad 33. 1 \quad 35. \cot x \quad 37. 1 + \sin y$$

$$39\text{--}49. \text{ Answers will vary. } 51. \cos^2 x \quad 53. \cos x + 2$$

$$55. \sec^4 x \quad 57. \sin^2 x - \cos^2 x \quad 59. (\csc x - 1) \cot^2 x$$

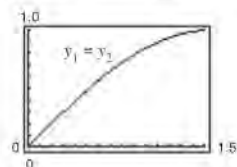
$$61. 1 + 2 \sin x \cos x \quad 63. \cot^2 x \quad 65. 2 \csc^2 x$$

$$67. -\cot x \quad 69. 1 + \cos y \quad 71. 3(\sec x + \tan x)$$

73.

x	0.2	0.4	0.6	0.8
y_1	0.1987	0.3894	0.5646	0.7174
y_2	0.1987	0.3894	0.5646	0.7174

x	1.0	1.2	1.4
y_1	0.8415	0.9320	0.9854
y_2	0.8415	0.9320	0.9854



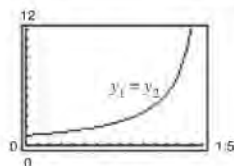
$$\begin{aligned} 7. \cos \phi &= -\frac{15}{17} \\ \tan \phi &= -\frac{8}{15} \\ \csc \phi &= \frac{17}{8} \\ \cot \phi &= -\frac{15}{8} \end{aligned}$$

$$\begin{aligned} 11. \sin \theta &= -\frac{2\sqrt{5}}{5} \\ \cos \theta &= -\frac{\sqrt{5}}{5} \\ \csc \theta &= -\frac{\sqrt{5}}{2} \\ \sec \theta &= -\sqrt{5} \\ \cot \theta &= \frac{1}{2} \end{aligned}$$

75.

x	0.2	0.4	0.6	0.8
y_1	1.2230	1.5085	1.8958	2.4650
y_2	1.2230	1.5085	1.8958	2.4650

x	1.0	1.2	1.4
y_1	3.4082	5.3319	11.6814
y_2	3.4082	5.3319	11.6814



$$77. \csc x \quad 79. \tan x \quad 81. 5 \cos \theta \quad 83. 3 \tan \theta$$

$$85. 3 \cos \theta \quad 87. 3 \sec \theta \quad 89. 3 \tan \theta \quad 91. \sqrt{2} \cos \theta$$

$$93. 0 \leq \theta \leq \pi \quad 95. 0 \leq \theta < \frac{\pi}{2}, \frac{3\pi}{2} < \theta < 2\pi$$

$$97. \ln|\cot \theta| \quad 99. \ln|(\cos x)(1 + \sin x)|$$

$$101. \text{ The identity is not true when } \theta = \frac{7\pi}{6}.$$

$$103. \text{ The identity is not true when } \theta = \frac{5\pi}{3}.$$

$$105. \text{ The identity is not true when } \theta = \frac{7\pi}{4}.$$

$$107. \text{ (a) and (b) Answers will vary.}$$

$$109. \text{ (a) and (b) Answers will vary.}$$

$$111. \text{ Answers will vary. } 113. \text{ True, for all } \theta \neq n\pi$$

$$115. 1, 1 \quad 117. \infty, 0$$

$$119. \cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

$$\tan \theta = \pm \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

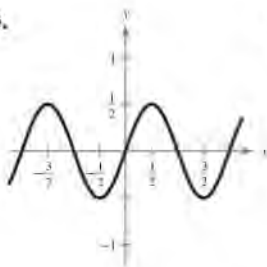
$$\sec \theta = \pm \frac{1}{\sqrt{1 - \sin^2 \theta}}$$

$$\cot \theta = \pm \frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta}$$

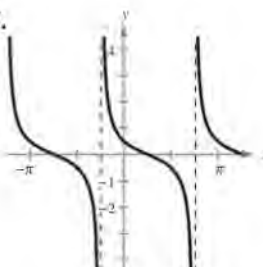
The sign depends on the choice of θ .

$$121. \text{ Answers will vary.}$$

123.



125.



Section 5.2 (page 365)

Vocabulary Check (page 365)

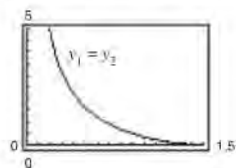
1. conditional 2. identity 3. $\cot u$ 4. $\sin u$
 5. $\tan u$ 6. $\cos u$ 7. $\cos^2 u$ 8. $\cot u$
 9. $-\sin u$ 10. $\sec u$

1–9. Answers will vary.

11.

x	0.2	0.4	0.6	0.8
y_1	4.8348	2.1785	1.2064	0.6767
y_2	4.8348	2.1785	1.2064	0.6767

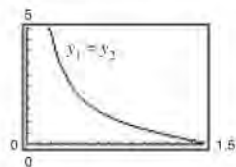
x	1.0	1.2	1.4
y_1	0.3469	0.1409	0.0293
y_2	0.3469	0.1409	0.0293



13.

x	0.2	0.4	0.6	0.8
y_1	4.8348	2.1785	1.2064	0.6767
y_2	4.8348	2.1785	1.2064	0.6767

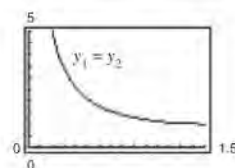
x	1.0	1.2	1.4
y_1	0.3469	0.1409	0.0293
y_2	0.3469	0.1409	0.0293



15.

x	0.2	0.4	0.6	0.8
y_1	5.0335	2.5679	1.7710	1.3940
y_2	5.0335	2.5679	1.7710	1.3940

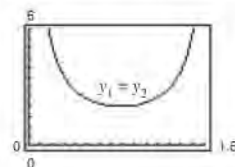
x	1.0	1.2	1.4
y_1	1.1884	1.0729	1.0148
y_2	1.1884	1.0729	1.0148



17.

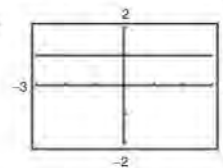
x	0.2	0.4	0.6	0.8
y_1	5.1359	2.7880	2.1458	2.0009
y_2	5.1359	2.7880	2.1458	2.0009

x	1.0	1.2	1.4
y_1	2.1995	2.9609	5.9704
y_2	2.1995	2.9609	5.9704

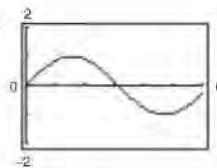

 19. $\cot(-x) = -\cot(x)$

21–49. Answers will vary.

51.


 $y = 1$

53.


 $y = \sin x$

55. Answers will vary. 57. Answers will vary.

59. 1 61. 2 63–69. Answers will vary.

 71. $\mu = \tan \theta$, $W \neq 0$ 73. True

 75. False. $\sin^2\left(\frac{\pi}{4}\right) + \cos^2\left(\frac{\pi}{4}\right) \neq 1 + \tan^2\left(\frac{\pi}{4}\right)$

77. (a) Answers will vary. (b) No. Division by zero.

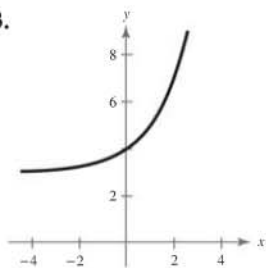
79. (a) Answers will vary. (b) Yes. Answers will vary.

81. $a \cos \theta$ 83. $a \sec \theta$ 85. $\sqrt{\tan^2 x} = |\tan x|$; $\frac{3\pi}{4}$

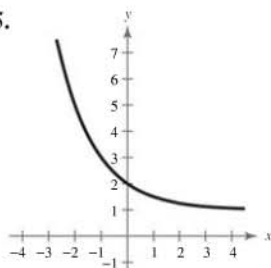
87. Answers will vary. 89. $f(x) = x^3 - x^2 + 64x - 64$

91. $f(x) = x^3 - 16x^2 + 85x - 148$

93.



95.



97. Quadrant II 99. Quadrant IV

Section 5.3 (page 376)

Vocabulary Check (page 376)

1. general 2. quadratic 3. extraneous

1–5. Answers will vary. 7. $30^\circ, 150^\circ$ 9. $120^\circ, 240^\circ$

11. $45^\circ, 225^\circ$ 13. $\frac{5\pi}{6}, \frac{7\pi}{6}$ 15. $\frac{3\pi}{4}, \frac{7\pi}{4}$

17. $\frac{5\pi}{6}, \frac{11\pi}{6}$ 19. $\frac{7\pi}{6}, \frac{11\pi}{6}$ 21. $\frac{\pi}{6}, \frac{7\pi}{6}$ 23. $\frac{3\pi}{4}, \frac{7\pi}{4}$

25. $\frac{2\pi}{3} + 2n\pi, \frac{4\pi}{3} + 2n\pi$ 27. $\frac{\pi}{6} + 2n\pi, \frac{11\pi}{6} + 2n\pi$

29. $\frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi$ 31. $\frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi$

33. $\frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi$ 35. $\frac{2\pi}{3}, \frac{5\pi}{3}$

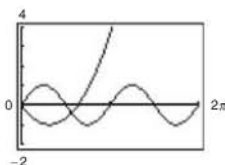
37. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ 39. $0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$

41. $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$ 43. No solution 45. $\frac{\pi}{3}, \frac{5\pi}{3}$

47. $2.0344, 5.1760, \frac{\pi}{4}, \frac{5\pi}{4}$ 49. $3.6652, 4.7124, 5.7596$

51. $0.8614, 5.4218$ 53. 1.5708 55. $0.5236, 2.6180$

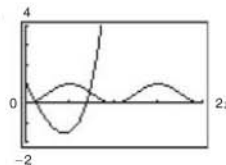
57. (a)



(b) $\sin 2x = x^2 - 2x$

(c) $(0, 0), (1.7757, -0.3984)$

59. (a)



(b) $\sin^2 x = e^x - 4x$

(c) $(0.3194, 0.0986), (2.2680, 0.5878)$

61. $2\pi + 4n\pi$ 63. $\frac{\pi}{8} + \frac{n\pi}{2}$ 65. $\frac{2\pi}{3} + n\pi, \frac{5\pi}{6} + n\pi$

67. $\frac{\pi}{8} + \frac{n\pi}{4}$ 69. $\frac{n\pi}{3}, \frac{\pi}{4} + n\pi$

71. $\frac{\pi}{2} + 4n\pi, \frac{7\pi}{2} + 4n\pi$ 73. $x = -1, 3$

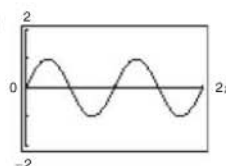
75. $x = \pm 2$ 77. $1.1071, 4.2487$ 79. $0.8603, 3.4256$

81. $0, 2.6779, 3.1416, 5.8195$

83. $0.3398, 0.8481, 2.2935, 2.8018$

85. $-1.154, 0.534$ 87. 1.110

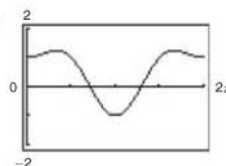
89. (a)


Maxima: $(0.7854, 1), (3.9270, 1)$

Minima: $(2.3562, -1), (5.4978, -1)$

(b) $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

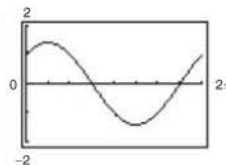
91. (a)


Maxima: $(1.0472, 1.25), (5.2360, 1.25)$

Minima: $(0, 1), (3.1416, -1), (6.2832, 1)$

(b) $0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$

93. (a)


Maximum: $(0.7854, 1.4142)$

Minimum: $(3.9270, -1.4142)$

(b) $\frac{\pi}{4}, \frac{5\pi}{4}$

95. 1

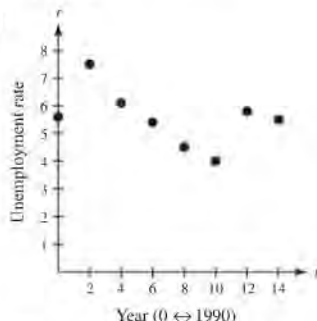
97. (a) All real numbers x except $x = 0$
 (b) y -axis symmetry; horizontal asymptote: $y = 1$
 (c) Oscillates
 (d) Infinite number of solutions
 (e) Yes, 0.6366

99. April, May, June, July

101. 0.04 second, 0.43 second, 0.83 second

 103. 36.87° , 53.13°

105. (a)



- (b) By graphing the curves, it can be seen that model (1), $r = 124 \sin(0.47t + 0.40) + 5.45$, best fits the data.
 (c) The constant terms gives the average unemployment rate of 5.45%.
 (d) 13.37 years
 (e) 2010

107. False

109. False. The range of the sine function does not include 3.4.

 111. 2.164 rad 113. -0.007 rad

115. 24.25 117. 2290.38 feet

Section 5.4 (page 384)

Vocabulary Check (page 384)

1. $\sin u \cos v - \cos u \sin v$
 2. $\cos u \cos v - \sin u \sin v$ 3. $\frac{\tan u + \tan v}{1 - \tan u \tan v}$
 4. $\sin u \cos v + \cos u \sin v$
 5. $\cos u \cos v + \sin u \sin v$ 6. $\frac{\tan u - \tan v}{1 + \tan u \tan v}$

1. (a) $-\frac{1}{2}$ (b) $-\frac{3}{2}$ 3. (a) $\frac{\sqrt{2} - \sqrt{6}}{4}$ (b) $\frac{1 + \sqrt{2}}{2}$

5. (a) $-\frac{\sqrt{2} + \sqrt{6}}{4}$ (b) $-\frac{\sqrt{2} + \sqrt{3}}{2}$

7. $\sin 105^\circ = \frac{\sqrt{2} + \sqrt{6}}{4}$

$\cos 105^\circ = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\tan 105^\circ = -2 - \sqrt{3}$

11. $\sin \frac{11\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{4}$

$\cos \frac{11\pi}{12} = -\frac{\sqrt{2} + \sqrt{6}}{4}$

$\tan \frac{11\pi}{12} = -2 + \sqrt{3}$

15. $\sin 75^\circ = \frac{\sqrt{2} + \sqrt{6}}{4}$

$\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$

$\tan 75^\circ = 2 + \sqrt{3}$

19. $\sin \frac{13\pi}{12} = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\cos \frac{13\pi}{12} = -\frac{\sqrt{2} + \sqrt{6}}{4}$

$\tan \frac{13\pi}{12} = 2 - \sqrt{3}$

21. $\sin\left(-\frac{7\pi}{12}\right) = -\frac{\sqrt{2} + \sqrt{6}}{4}$

$\cos\left(-\frac{7\pi}{12}\right) = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\tan\left(-\frac{7\pi}{12}\right) = 2 + \sqrt{3}$

9. $\sin 195^\circ = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\cos 195^\circ = -\frac{\sqrt{2} + \sqrt{6}}{4}$

$\tan 195^\circ = 2 - \sqrt{3}$

13. $\sin\left(-\frac{\pi}{12}\right) = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\cos\left(-\frac{\pi}{12}\right) = \frac{\sqrt{2} + \sqrt{6}}{4}$

$\tan\left(-\frac{\pi}{12}\right) = \sqrt{3} - 2$

17. $\sin(-225^\circ) = \frac{\sqrt{2}}{2}$

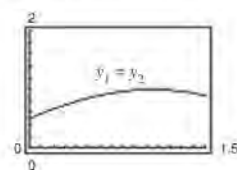
$\cos(-225^\circ) = -\frac{\sqrt{2}}{2}$

$\tan(-225^\circ) = -1$

23. $\cos 80^\circ$ 25. $\tan 239^\circ$ 27. $\sin 2.3$ 29. $\cos \frac{16\pi}{63}$

x	0.2	0.4	0.6	0.8
y_1	0.6621	0.7978	0.9017	0.9696
y_2	0.6621	0.7978	0.9017	0.9696

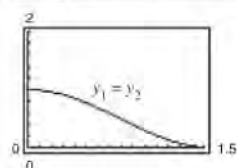
x	1.0	1.2	1.4
y_1	0.9989	0.9883	0.9384
y_2	0.9989	0.9883	0.9384



33.

x	0.2	0.4	0.6	0.8
y_1	0.9605	0.8484	0.6812	0.4854
y_2	0.9605	0.8484	0.6812	0.4854

x	1.0	1.2	1.4
y_1	0.2919	0.1313	0.0289
y_2	0.2919	0.1313	0.0289



35. $-\frac{63}{65}$ 37. $-\frac{63}{16}$ 39. $\frac{36}{85}$ 41. $\frac{13}{85}$ 43. 1

45. $\frac{2x^2 - \sqrt{1-x^2}}{\sqrt{4x^2+1}}$ 47. 1 49. -1 51. $-\frac{1}{2}$

53. $\frac{\sqrt{3}}{3}$ 55. 0 57. 0 59. $\frac{33}{65}$ 61. $\frac{24}{25}$

63–69. Answers will vary. 71. $\frac{\pi}{2}$ 73. $0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

75. 0.7854, 5.4978 77. 0, 3.1416

79. Answers will vary.

81. False. $\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$

83. Answers will vary. 85. Answers will vary.

87. (a) $\sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right)$ (b) $\sqrt{2} \cos\left(\theta - \frac{\pi}{4}\right)$

89. (a) $13 \sin(3\theta + 0.3948)$ (b) $13 \cos(3\theta - 1.1760)$

91. $2 \cos \theta$ 93. Answers will vary.

95. $u + v = w$. Answers will vary. 97. (0, 19), (38, 0)

99. (0, 4), (2, 0), (7, 0) 101. $\frac{\pi}{6}$ 103. $\frac{\pi}{2}$

Section 5.5 (page 394)

Vocabulary Check (page 394)

1. $2 \sin u \cos u$ 2. $\frac{1 + \cos 2u}{2}$ 3. $\cos 2u$

4. $\tan \frac{u}{2}$ 5. $\frac{2 \tan u}{1 - \tan^2 u}$

6. $\frac{1}{2}[\cos(u - v) + \cos(u + v)]$ 7. $\sin^2 u$

8. $\cos \frac{u}{2}$ 9. $\frac{1}{2}[\sin(u + v) + \sin(u - v)]$

10. $2 \sin\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right)$

1. (a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{7}{25}$ (d) $\frac{24}{25}$

(e) $\frac{24}{7}$ (f) $\frac{25}{7}$ (g) $\frac{25}{24}$ (h) $\frac{7}{24}$

3. 0, 1.0472, 3.1416, 5.2360; $0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

5. 0.2618, 1.3090, 3.4034, 4.4506; $\frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

7. 0, 2.0944, 4.1888; $0, \frac{2\pi}{3}, \frac{4\pi}{3}$

9. 0, 1.5708, 3.1416, 4.7124; $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

11. 1.5708, 3.6652, 5.7596; $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

13. $\sin 2u = \frac{24}{25}$

$\cos 2u = \frac{7}{25}$

$\tan 2u = \frac{24}{7}$

15. $\sin 2u = \frac{4}{5}$

$\cos 2u = \frac{3}{5}$

$\tan 2u = \frac{4}{3}$

17. $\sin 2u = -\frac{4\sqrt{21}}{25}$

$\cos 2u = -\frac{17}{25}$

$\tan 2u = \frac{4\sqrt{21}}{17}$

19. $4 \sin 2x$ 21. $6 \cos 2x$

23. $\frac{1}{8}(3 + 4 \cos 2x + \cos 4x)$ 25. $\frac{1}{8}(1 - \cos 4x)$

27. $\frac{1}{32}(2 + \cos 2x - 2 \cos 4x - \cos 6x)$

29. $\frac{1}{2}(1 - \cos 4x)$ 31. $\frac{1}{2}(1 + \cos x)$

33. $\frac{1}{8}(1 - \cos 8x)$ 35. $\frac{1}{8}(3 - 4 \cos x + \cos 2x)$

37. (a) $\frac{4\sqrt{17}}{17}$ (b) $\frac{\sqrt{17}}{17}$ (c) $\frac{1}{4}$ (d) $\frac{\sqrt{17}}{4}$

(e) $\sqrt{17}$ (f) 4 (g) $\frac{8}{17}$ (h) $\frac{2\sqrt{17}}{17}$

39. $\sin 15^\circ = \frac{\sqrt{2 - \sqrt{3}}}{2}$

$\cos 15^\circ = \frac{\sqrt{2 + \sqrt{3}}}{2}$

$\tan 15^\circ = 2 - \sqrt{3}$

43. $\sin \frac{\pi}{8} = \frac{\sqrt{2 - \sqrt{2}}}{2}$

$\cos \frac{\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$

$\tan \frac{\pi}{8} = \sqrt{2} - 1$

41. $\sin 112^\circ 30' = \frac{\sqrt{2 + \sqrt{2}}}{2}$

$\cos 112^\circ 30' = -\frac{\sqrt{2 - \sqrt{2}}}{2}$

$\tan 112^\circ 30' = -1 - \sqrt{2}$

45. $\sin \frac{3\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$

$\cos \frac{3\pi}{8} = \frac{\sqrt{2 - \sqrt{2}}}{2}$

$\tan \frac{3\pi}{8} = \sqrt{2} + 1$

47. $\sin \frac{u}{2} = \frac{5\sqrt{26}}{26}$

$\cos \frac{u}{2} = \frac{\sqrt{26}}{26}$

$\tan \frac{u}{2} = 5$

51. $\sin \frac{u}{2} = \frac{3\sqrt{10}}{10}$

$\cos \frac{u}{2} = -\frac{\sqrt{10}}{10}$

$\tan \frac{u}{2} = -3$

53. $|\sin 3x|$

55. $-|\tan 4x|$

57. $\frac{\pi}{3}, \frac{5\pi}{3}$

59. $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

61. $3\left(\sin \frac{2\pi}{3} + \sin 0\right)$

63. $\frac{1}{2}(\sin 8\theta + \sin 2\theta)$

65. $5(\cos 60^\circ + \cos 90^\circ)$

67. $\frac{5}{2}(\cos 8\beta + \cos 2\beta)$

69. $\frac{1}{2}(\cos 2y - \cos 2x)$

71. $\frac{1}{2}(\sin 2\theta + \sin 2\pi)$

73. $2 \cos 3\theta \sin 2\theta$

75. $2 \cos 4x \cos 2x$

77. $2 \cos \alpha \sin \beta$

79. $-2 \sin \theta \sin \frac{\pi}{2}$

81. $\frac{\sqrt{2}}{2}$

83. $\frac{\sqrt{6}}{2}$

85. $0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$

87. $\frac{\pi}{6}, \frac{5\pi}{6}$

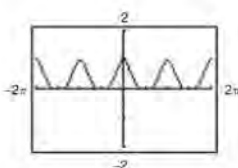
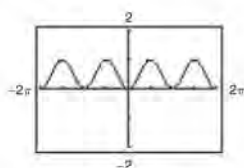
89. $\frac{25}{169}$

91. $\frac{4}{13}$

93–109. Answers will vary.

111. $\frac{1 - \cos 2x}{2}$

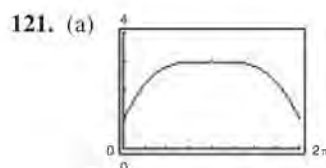
113. $\frac{3 + 4 \cos 2x + \cos 4x}{8}$



115. $2x\sqrt{1-x^2}$

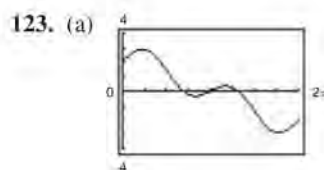
117. $1 - 2x^2$

119. $\frac{1-x^2}{1+x^2}$



Maximum: (3.1416, 3)

(b) π



Minima: (2.6078, -0.3486), (5.5839, -2.8642)

Maxima: (0.6993, 2.8642), (3.6754, 0.3486)

(b) 0.6993, 2.6078, 3.6754, 5.5839

125. (a) $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$

(b) $x = \arccos \frac{1 \pm \sqrt{33}}{8}, 2\pi - \arccos \frac{1 \pm \sqrt{33}}{8}$

127. (a) $r = \frac{1}{16}v_0^2 \sin \theta \cos \theta$

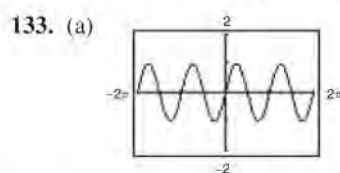
(b) 198.90 feet

(c) 98.73 feet per second

(d) $\theta = 45^\circ$; Answers will vary.

129. $x = 2r(1 - \cos \theta)$

131. False. $\sin \frac{x}{2} = -\sqrt{\frac{1 - \cos x}{2}}$ for $\pi \leq \frac{x}{2} \leq 2\pi$.



(b) $y = \sin 2x$

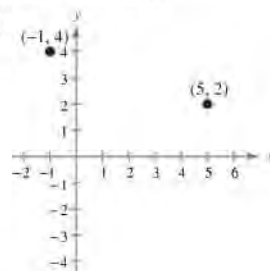
(c) Answers will vary.

135. Answers will vary.

137. (a) $(-1, 4)$

(b) $2\sqrt{10}$

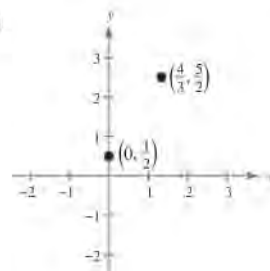
(c) (2, 3)



139. (a) $\left(\frac{4}{3}, \frac{5}{2}\right)$

(b) $\frac{2\sqrt{13}}{3}$

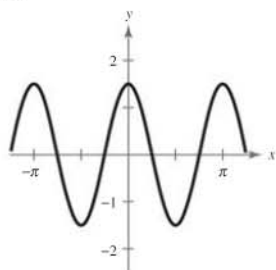
(c) $\left(\frac{2}{3}, \frac{3}{2}\right)$



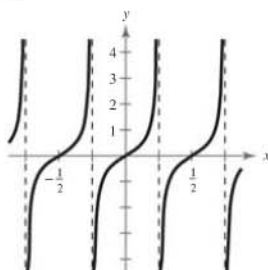
141. (a) Complement: 35° ; supplement: 125° (b) Complement: none; supplement: 18° 143. (a) Complement: $\frac{4\pi}{9}$; supplement: $\frac{17\pi}{18}$ (b) Complement: $\frac{\pi}{20}$; supplement: $\frac{11\pi}{20}$

145. 0.4667 rad

147.



149.



Review Exercises (page 400)

1. $\sec x$ 3. $\cos x$ 5. $|\sin x|$ 7. $\sec x$ 9. $\sec x$

11. $\tan x = \frac{4}{3}$

13. $\cos x = \frac{\sqrt{2}}{2}$

$\csc x = \frac{5}{4}$

$\tan x = -1$

$\csc x = -\sqrt{2}$

$\sec x = \frac{5}{3}$

$\sec x = \sqrt{2}$

$\cot x = \frac{3}{4}$

$\cot x = -1$

15. $\cos^2 x$ 17. $1 + \cot x$ 19. 1 21. $\csc x$

23–35. Answers will vary.

37. $\frac{\pi}{6} + 2n\pi, \frac{5\pi}{6} + 2n\pi$ 39. $\frac{\pi}{3} + 2n\pi, \frac{2\pi}{3} + 2n\pi$

41. $\frac{\pi}{6} + n\pi$ 43. $\frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi$

45. $\frac{\pi}{6} + n\pi, \frac{5\pi}{6} + n\pi$ 47. $n\pi$ 49. $0, \frac{2\pi}{3}, \frac{4\pi}{3}$

51. $0, \frac{\pi}{2}, \pi$ 53. $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$

55. $0, \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$ 57. $\frac{\pi}{2}, \frac{3\pi}{2}$

59. $\frac{\pi}{12} + n\pi, \frac{5\pi}{12} + n\pi$ 61. $\frac{\pi}{12} + \frac{n\pi}{6}$ 63. $0, \pi$

65. $\arctan 2, \arctan(-5) + \pi, \arctan 2 + \pi, \arctan(-5) + 2\pi$

67. $\sin 285^\circ = -\frac{\sqrt{2} + \sqrt{6}}{4}$

$\cos 285^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$

$\tan 285^\circ = -2 - \sqrt{3}$

69. $\sin \frac{31\pi}{12} = \frac{\sqrt{2} + \sqrt{6}}{4}$

$\cos \frac{31\pi}{12} = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\tan \frac{31\pi}{12} = -2 - \sqrt{3}$

71. $\sin 180^\circ$ 73. $\tan 35^\circ$ 75. $-\frac{117}{125}$

77. $\frac{3}{4}$ 79. $-\frac{44}{125}$ 81. 0 83. 0

85–89. Answers will vary. 91. $\frac{\pi}{4}, \frac{7\pi}{4}$

93. $\sin 2u = \frac{20\sqrt{6}}{49}$

95. $\sin 2u = -\frac{36}{85}$

$\cos 2u = -\frac{1}{49}$

$\cos 2u = \frac{77}{85}$

$\tan 2u = -20\sqrt{6}$

$\tan 2u = -\frac{36}{77}$

97 and 99. Answers will vary.

101. $15^\circ, 75^\circ$ 103. $\frac{1}{32}(10 - 15 \cos 2x + 6 \cos 4x - \cos 6x)$

105. $\frac{1}{8}(3 + 4 \cos 4x + \cos 8x)$

107. $\sin 105^\circ = \frac{\sqrt{2} + \sqrt{3}}{2}$

$\cos 105^\circ = -\frac{\sqrt{2} - \sqrt{3}}{2}$

$\tan 105^\circ = -2 - \sqrt{3}$

109. $\sin \frac{7\pi}{8} = \frac{\sqrt{2} - \sqrt{2}}{2}$

111. $\sin \frac{u}{2} = \frac{\sqrt{10}}{10}$

$\cos \frac{7\pi}{8} = -\frac{\sqrt{2} + \sqrt{2}}{2}$

$\cos \frac{u}{2} = \frac{3\sqrt{10}}{10}$

$\tan \frac{7\pi}{8} = 1 - \sqrt{2}$

$\tan \frac{u}{2} = \frac{1}{3}$

113. $\sin \frac{u}{2} = \frac{3\sqrt{14}}{14}$

$\cos \frac{u}{2} = \frac{\sqrt{70}}{14}$

$\tan \frac{u}{2} = \frac{3\sqrt{5}}{5}$

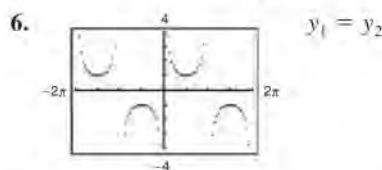
115. $-|\cos 4x|$ 117. $V = \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ cubic meters

119. $3\left(\sin \frac{\pi}{2} + \sin 0\right)$ 121. $\frac{1}{2}(\cos \alpha - \cos 9\alpha)$
 123. $2 \cos\left(\frac{9\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)$ 125. $2 \cos x \sin \frac{\pi}{4}$
 127. $y = \frac{1}{2} \sqrt{10} \sin\left(8t - \arctan \frac{1}{3}\right)$ 129. $\frac{\sqrt{10}}{2}$ feet
 131. False. $\cos \frac{\theta}{2} > 0$ 133. True
 135. Answers will vary. 137. $y_3 = y_2 + 1$

Chapter Test (page 403)

1. $\sin \theta = \frac{-3\sqrt{13}}{13}$ 2. 1 3. 1
 $\cos \theta = \frac{-2\sqrt{13}}{13}$
 $\csc \theta = \frac{-\sqrt{13}}{3}$
 $\sec \theta = \frac{-\sqrt{13}}{2}$
 $\cot \theta = \frac{2}{3}$

4. $\csc \theta \sec \theta$ 5. $0, \frac{\pi}{2} < \theta \leq \pi, \frac{3\pi}{2} < \theta < 2\pi$



- 7–12. Answers will vary. 13. $-2 - \sqrt{3}$

14. $\frac{1}{16} \left[\frac{10 - 15 \cos 2x + 6 \cos 4x - \cos 6x}{1 + \cos 2x} \right]$

15. $\tan 2\theta$ 16. $2(\sin 6\theta + \sin 2\theta)$

17. $-2 \cos \frac{7\theta}{2} \sin \frac{\theta}{2}$ 18. $0, \frac{3\pi}{4}, \pi, \frac{7\pi}{4}$

19. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ 20. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

21. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ 22. $-2.938, -2.663, 1.170$

23. $\sin 2u = \frac{4}{5}$ 24. 76.52°
 $\cos 2u = -\frac{3}{5}$
 $\tan 2u = -\frac{4}{3}$

Chapter 6

Section 6.1 (page 414)

Vocabulary Check (page 414)

1. oblique 2. $\frac{b}{\sin B}$
 3. (a) Two; any; AAS; ASA
 (b) Two; an opposite; SSA
 4. $\frac{1}{2}bc \sin A; \frac{1}{2}ab \sin C; \frac{1}{2}ac \sin B$

1. $C = 95^\circ$, $b \approx 24.59$ inches, $c \approx 28.29$ inches
 3. $A = 40^\circ$, $a \approx 15.69$ centimeters, $b \approx 6.32$ centimeters
 5. $C = 74^\circ 15'$, $a \approx 6.41$ kilometers, $c \approx 6.26$ kilometers
 7. $B \approx 21.55^\circ$, $C \approx 122.45^\circ$, $c \approx 11.49$
 9. $B = 60.9^\circ$, $b \approx 19.32$, $c \approx 6.36$
 11. $B \approx 18^\circ 13'$, $C \approx 51^\circ 32'$, $c \approx 40.05$
 13. $B \approx 48.74^\circ$, $C \approx 21.26^\circ$, $c \approx 48.23$
 15. No solution
 17. Two solutions
 $B \approx 72.21^\circ$, $C \approx 49.79^\circ$, $c \approx 10.27$
 $B \approx 107.79^\circ$, $C \approx 14.21^\circ$, $c \approx 3.30$
 19. 28.19 square units 21. 1782.32 square units
 23. 2888.57 square units
 25. (a)

(b) $\frac{16}{\sin 70^\circ} = \frac{h}{\sin 34^\circ}$ (c) 9.52 meters

27. 240.03°

29. (a) (b) 4385.71 feet
 (c) 3061.80 feet

31. 15.53 kilometers from Colt Station; 42.43 kilometers from Pine Knob

A214 Answers to Odd-Numbered Exercises and Tests

33. 16.08°

35. (a) $\alpha \approx 5.36^\circ$

(b) $\beta = \arcsin\left(\frac{d \sin \theta}{58.36}\right)$

(c) $d = \sin(84.64 - \theta) \left[\frac{58.36}{\sin \theta} \right]$

(d)

θ	10°	20°	30°	40°	50°	60°
d	324.08	154.19	95.19	63.80	43.30	28.10

37. False. The triangle can't be solved if only three angles are known.

39. Yes, the Law of Sines can be used to solve a right triangle provided that at least one side and one angle are given or two sides are given. Answers will vary.

41. $C = 83^\circ$, $b \approx 17.83$, $c \approx 22.46$

43. $\tan \theta = -\frac{12}{5}$; $\csc \theta = -\frac{13}{12}$; $\sec \theta = \frac{13}{5}$; $\cot \theta = -\frac{5}{12}$

45. $3(\sin 11\theta + \sin 5\theta)$ 47. $\frac{3}{2}\left(\sin \frac{11\pi}{6} + \sin \frac{3\pi}{2}\right)$

Section 6.2 (page 421)

Vocabulary Check (page 421)

1. $c^2 = a^2 + b^2 - 2ab \cos C$ 2. Heron's Area

3. $\frac{1}{2}bh$, $\sqrt{s(s-a)(s-b)(s-c)}$

1. $A \approx 40.80^\circ$, $B \approx 60.61^\circ$, $C \approx 78.59^\circ$

3. $A \approx 49.51^\circ$, $B \approx 55.40^\circ$, $C \approx 75.09^\circ$

5. $A \approx 31.40^\circ$, $C \approx 128.60^\circ$, $b \approx 6.56$ millimeters

7. $A \approx 53^\circ 45'$, $C \approx 75^\circ 45'$, $b \approx 9.95$ feet

9. $A \approx 26.38^\circ$, $B \approx 36.34^\circ$, $C \approx 117.28^\circ$

11. $B \approx 29.44^\circ$, $C \approx 100.56^\circ$, $a \approx 23.38$

13. $A \approx 36.87^\circ$, $B \approx 53.13^\circ$, $C = 90^\circ$

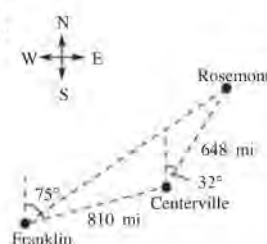
15. $A \approx 103.52^\circ$, $B \approx 38.24^\circ$, $C \approx 38.24^\circ$

17. $A \approx 154^\circ 14'$, $C \approx 17^\circ 31'$, $b \approx 8.58$

19. $A \approx 37^\circ 6' 7''$, $C \approx 67^\circ 33' 53''$, $b \approx 9.94$

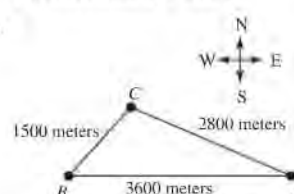
	a	b	c	d	θ	ϕ
21.	4	8	11.64	4.96	30°	150°
23.	10	14	20	13.86	68.20°	111.80°
25.	15	16.96	25	20	77.22°	102.78°
27.	104.57 square inches		29.	19.81		
31.	0.27 square foot		33.	15.52		35. 35.19

37.



1357.85 miles, 236.01°

39.



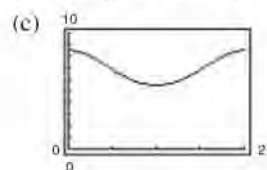
N 43.03° E, S 66.95° E

41. 43.27 miles 43. $PQ \approx 9.43$, $QS \approx 5$, $RS \approx 12.81$

45. 18,617.66 square feet

47. (a) $49 = 2.25 + x^2 - 3x \cos \theta$

(b) $x = \frac{1}{2}(3 \cos \theta + \sqrt{9 \cos^2 \theta + 187})$



(d) 6 inches

49. False. A triangle cannot be formed with sides of lengths 10 feet, 16 feet, and 5 feet.

51. False. $s = \frac{a+b+c}{2}$ 53 and 55. Proof.

57. Answers will vary. Yes; Answers will vary.

59. $-\frac{\pi}{2}$ 61. $\frac{\pi}{3}$

Section 6.3 (page 433)

Vocabulary Check (page 433)

1. directed line segment 2. initial, terminal

3. magnitude 4. vector

5. standard position 6. unit vector

7. multiplication, addition 8. resultant

9. linear combination, horizontal, vertical

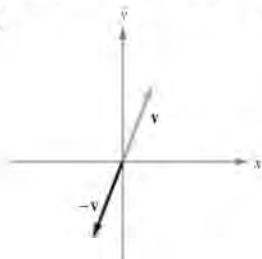
1. Answers will vary.

3. $\langle 4, 3 \rangle$, $\|v\| = 5$ 5. $\langle -3, 2 \rangle$, $\|v\| = \sqrt{13}$

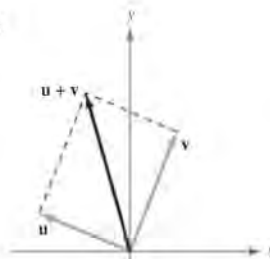
7. $\langle 0, 5 \rangle$, $\|v\| = 5$ 9. $\langle \frac{3}{5}, -\frac{3}{5} \rangle$, $\|v\| = \frac{3\sqrt{2}}{5}$

11. $\langle \frac{7}{6}, \frac{9}{5} \rangle$, $\|v\| = \frac{\sqrt{4141}}{30}$

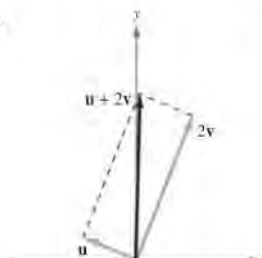
13.



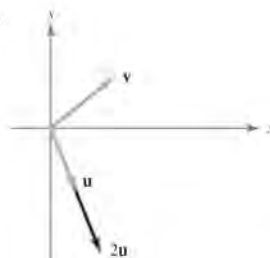
15.



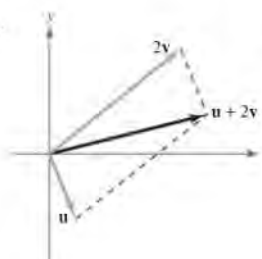
17.



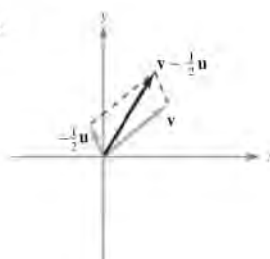
19.



21.



23.



25. (a) $\langle 11, 3 \rangle$ (b) $\langle -3, 1 \rangle$ (c) $\langle -13, 1 \rangle$ (d) $\langle 23, 9 \rangle$

27. (a) $\langle -4, -4 \rangle$ (b) $\langle -8, -12 \rangle$ (c) $\langle -18, -28 \rangle$

(d) $\langle -22, -28 \rangle$

29. (a) $3\mathbf{i} - 2\mathbf{j}$ (b) $-\mathbf{i} + 4\mathbf{j}$ (c) $-4\mathbf{i} + 11\mathbf{j}$

(d) $6\mathbf{i} + \mathbf{j}$

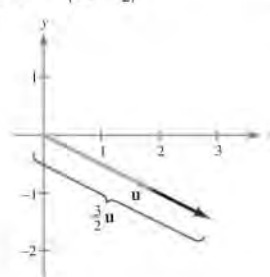
31. $\mathbf{u} + \mathbf{v}$ 33. $\mathbf{w} - \mathbf{v}$ 35. $\langle 1, 0 \rangle$

37. $\langle -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \rangle$ 39. $\langle -\frac{24}{25}, -\frac{7}{25} \rangle$ 41. $\frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$

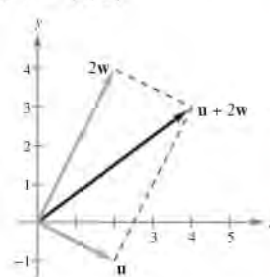
43. $\mathbf{j}$ 45. $\frac{40\sqrt{61}}{61}\mathbf{i} + \frac{48\sqrt{61}}{61}\mathbf{j}$ 47. $\frac{21}{5}\mathbf{i} + \frac{28}{5}\mathbf{j}$

49. $-8\mathbf{i}$ 51. $7\mathbf{i} + 4\mathbf{j}$ 53. $3\mathbf{i} + 8\mathbf{j}$

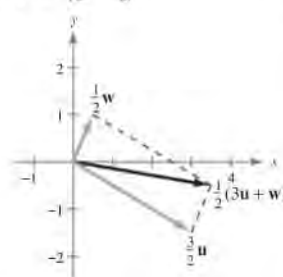
55. $\mathbf{v} = \langle 3, -\frac{3}{2} \rangle$



57. $\mathbf{v} = \langle 4, 3 \rangle$



59. $\mathbf{v} = \langle \frac{7}{2}, -\frac{1}{2} \rangle$



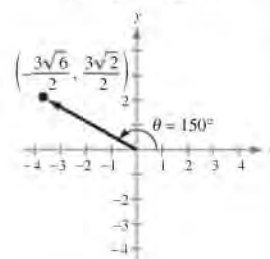
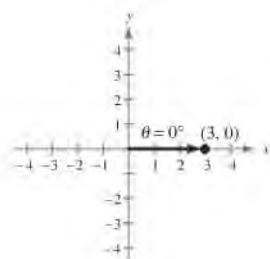
61. $\|v\| = 5$, $\theta = 30^\circ$

63. $\|v\| = 6\sqrt{2}$, $\theta = 315^\circ$

65. $\|v\| = \sqrt{29}$, $\theta = 111.80^\circ$

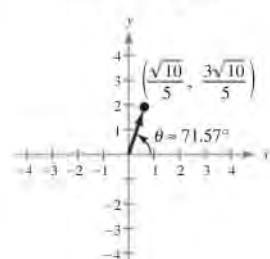
67. $\mathbf{v} = \langle 3, 0 \rangle$

69. $\mathbf{v} = \langle -\frac{3\sqrt{6}}{2}, \frac{3\sqrt{2}}{2} \rangle$



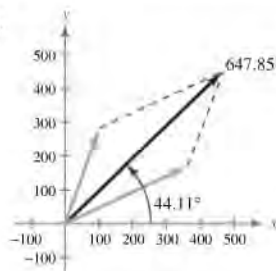
71. $\mathbf{v} = \langle \frac{\sqrt{10}}{5}, \frac{3\sqrt{10}}{5} \rangle$

73. $\langle \frac{5}{2}, \frac{10 + 5\sqrt{3}}{2} \rangle$



75. $\langle 10\sqrt{2} - 25\sqrt{3}, 25 + 10\sqrt{2} \rangle$ 77. 90°

79.

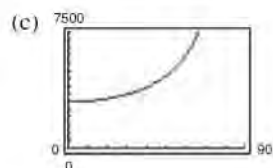


$$\|v\| \approx 647.85, \theta \approx 44.11^\circ$$

81. 62.72° 83. Horizontal component: ≈ 53.62 feet per secondVertical component: ≈ 45.00 feet per second85. $T_{AC} \approx 3611.11$ pounds, $T_{BC} \approx 2169.51$ pounds87. (a) $T = 3000 \sec \theta$; Domain: $0^\circ \leq \theta < 90^\circ$

θ	10°	20°	30°
T	3046.28	3192.53	3464.10

θ	40°	50°	60°
T	3916.22	4667.17	6000



(d) The component in the direction of the motion of the barge decreases.

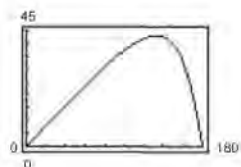
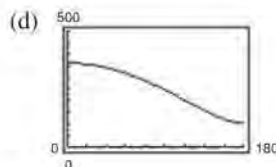
89. N 26.67° E, 130.35 kilometers per hour91. (a) 12.10° , 357.85 newtons

$$(b) M = 10\sqrt{660 \cos \theta + 709}$$

$$\alpha = \arctan \frac{15 \sin \theta}{15 \cos \theta + 22}$$

θ	0°	30°	60°	90°
M	370°	357.85°	322.34°	266.27°
α	0°	12.10°	23.77°	34.29°

θ	120°	150°	180°
M	194.68°	117.23°	70°
α	41.86°	39.78°	0°

(e) For increasing θ , the two vectors tend to work against each other, resulting in a decrease in the magnitude of the resultant.93. True 95. True. $a = b = 0$ 97. True

99. True 101. True 103. False

105. (a) 0° (b) 180° (c) No. The magnitude is equal to the sum when the angle between the vectors is 0° .107. Proof. 109. $\langle 1, 3 \rangle$ or $\langle -1, -3 \rangle$ 111. $12x^3y^7, x \neq 0, y \neq 0$ 113. $48xy^2, x \neq 0$ 115. 7.14×10^5 117. $7 \cos \theta$ 119. $10 \csc \theta$ 121. $\frac{\pi}{2} + n\pi, \pi + 2n\pi$ 123. $\frac{\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi$

Section 6.4 (page 445)

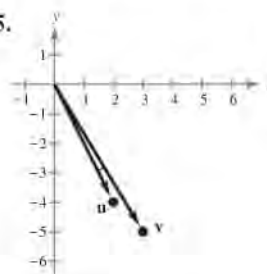
Vocabulary Check (page 445)

1. dot product 2. $\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$ 3. orthogonal4. $\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\right)\mathbf{v}$ 5. $\|\text{proj}_{PQ} \mathbf{F}\| \|\overrightarrow{PQ}\|, \mathbf{F} \cdot \overrightarrow{PQ}$

1. 0 3. 14 5. 8, scalar 7. 4, scalar

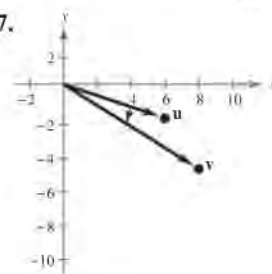
9. $\langle -114, -114 \rangle$, vector 11. 13 13. $5\sqrt{41}$ 15. 4 17. 90° 19. 70.56° 21. 90° 23. $\frac{5\pi}{12}$

25.



$$\theta \approx 4.40^\circ$$

27.



$$\theta \approx 13.57^\circ$$

29. $26.57^\circ, 63.43^\circ, 90^\circ$ 31. $-162\sqrt{2}$ 33. Parallel

35. Neither 37. Orthogonal 39. 3

41. 10 43. 0

45. $\frac{16}{17}\langle 4, 1 \rangle$, $\mathbf{u} = \left\langle \frac{64}{17}, \frac{16}{17} \right\rangle + \left\langle -\frac{13}{17}, \frac{52}{17} \right\rangle$

47. $\frac{45}{229}\langle 2, 15 \rangle$, $\mathbf{u} = \left\langle \frac{90}{229}, \frac{675}{229} \right\rangle + \left\langle -\frac{90}{229}, \frac{12}{229} \right\rangle$ 49. $\mathbf{u}$

51. $\mathbf{0}$ 53. $\langle 3, -1 \rangle$, $\langle -3, 1 \rangle$ 55. $-\frac{3}{4}\mathbf{i} - \frac{1}{2}\mathbf{j}$, $\frac{3}{4}\mathbf{i} + \frac{1}{2}\mathbf{j}$

57. 32

59. (a) 37,289; It is the total dollar value of the picture frames produced by the company.

 (b) Multiply $\mathbf{v}$ by 1.02.

61. (a) Force = $30,000 \sin d^\circ$

d	0°	1°	2°	3°
Force	0	523.57	1046.98	1570.08

d	4°	5°	6°	7°
Force	2092.69	2614.67	3135.85	3656.08

d	8°	9°	10°
Force	4175.19	4693.03	5209.45

(c) 29,885.84 pounds

63. (a) $W = 15,691 \frac{\sqrt{3}}{2} d$

(b)

d	0	200	400	800
Work	0	2,717,760.92	5,435,521.84	10,871,043.69

65. 21,650.64 foot-pounds 67. 725.05 foot-pounds

69. True. The zero vector is orthogonal to every vector.

 71. Orthogonal. $\mathbf{u} \cdot \mathbf{v} = 0$

 73. (a) $\mathbf{u}$ and $\mathbf{v}$ are parallel. (b) $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

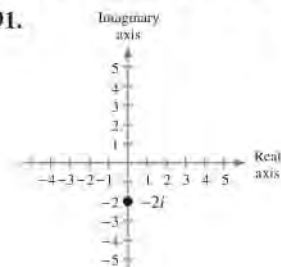
75 and 77. Proof

 79. g is a horizontal shift of f four units to the right.

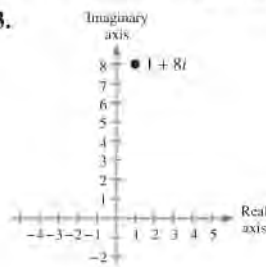
 81. g is a vertical shift of f six units upward.

83. $-1 + 2i$ 85. $15 + 12i$ 87. 10 89. $\frac{47}{26} - \frac{27}{26}i$

91.



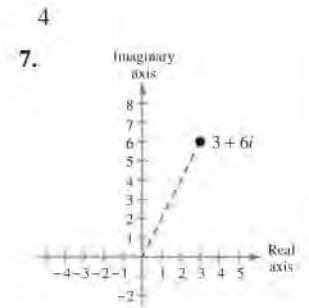
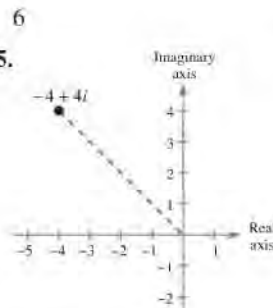
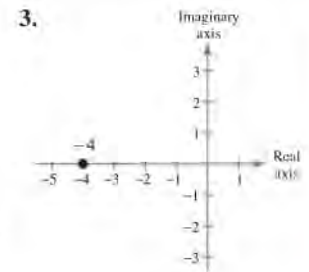
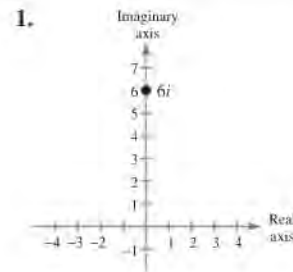
93.



Section 6.5 (page 456)

Vocabulary Check (page 456)

- absolute value
- trigonometric form, modulus, argument
- DeMoivre's
- n th root



$4\sqrt{2}$

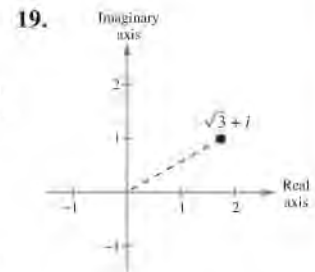
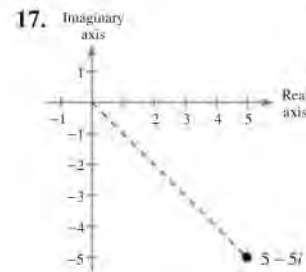
$3\sqrt{5}$

9. $3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$

11. $2(\cos \pi + i \sin \pi)$

13. $2\sqrt{2}\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$

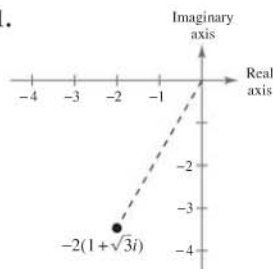
15. $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$



$5\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$

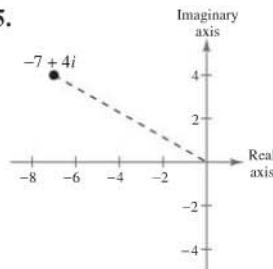
$2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$

21.



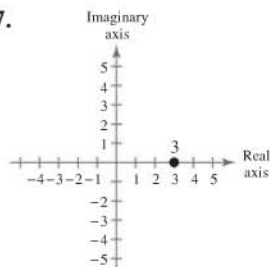
$$4\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$$

25.



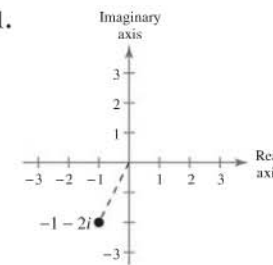
$$\sqrt{65}(\cos 2.622 + i \sin 2.622)$$

27.



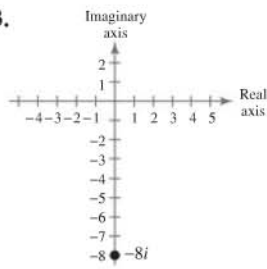
$$3(\cos 0 + i \sin 0)$$

31.



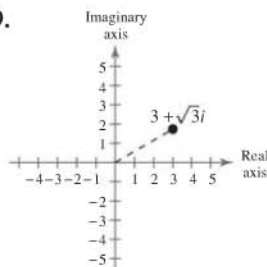
$$\sqrt{5}(\cos 4.249 + i \sin 4.249)$$

23.



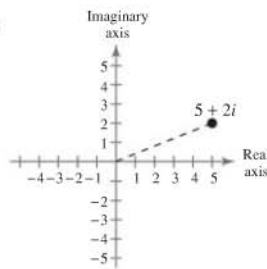
$$8\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$$

29.



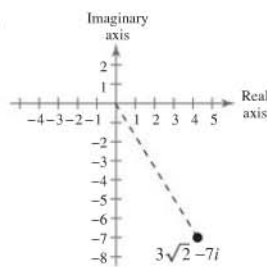
$$2\sqrt{3}\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$$

33.



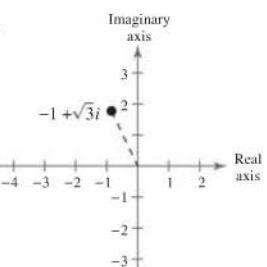
$$\sqrt{29}(\cos 0.381 + i \sin 0.381)$$

35.



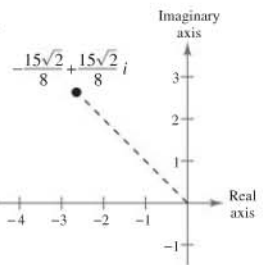
$$\sqrt{67}(\cos 5.257 + i \sin 5.257)$$

37.



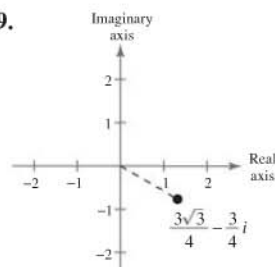
$$-1 + \sqrt{3}i$$

41.



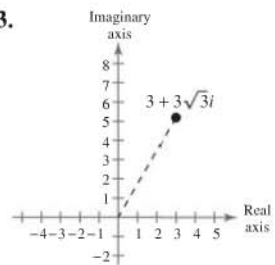
$$-\frac{15\sqrt{2}}{8} + \frac{15\sqrt{2}}{8}i$$

39.

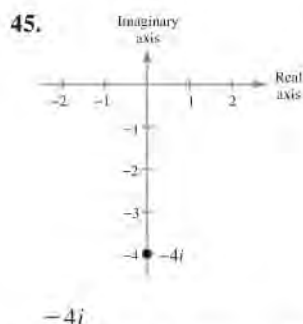


$$\frac{3\sqrt{3}}{4} - \frac{3}{4}i$$

43.

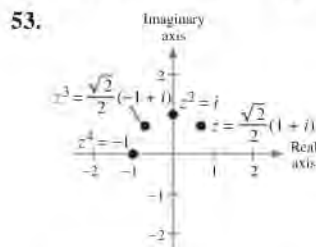


$$3 + 3\sqrt{3}i$$



$$-4i$$

49. $4.6985 + 1.7101i$



The absolute value of each is 1.

55. $12\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$

57. $\frac{10}{9}(\cos 200^\circ + i \sin 200^\circ)$

59. $\frac{11}{50}(\cos 130^\circ + i \sin 130^\circ)$

61. $\cos 30^\circ + i \sin 30^\circ$

63. $\frac{1}{2}(\cos 80^\circ + i \sin 80^\circ)$

65. $6(\cos 312^\circ + i \sin 312^\circ)$

67. (a) $2\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$
 $\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

(b) and (c) 4

69. (a) $2\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$
 $\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$

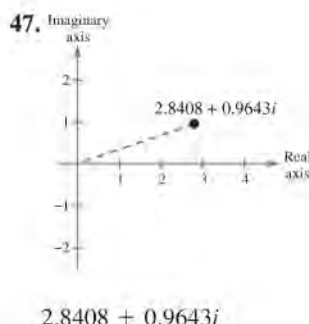
(b) and (c) 4

71. (a) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$
 $\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

(b) and (c) $2 - 2i$

73. (a) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$
 $2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$

(b) and (c) $-2 - 2\sqrt{3}i$



$$2.8408 + 0.9643i$$

75. (a) $2(\cos 0 + i \sin 0)$

$$\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$$

(b) and (c) $2 - 2i$

77. (a) $3\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$
 $2\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$

(b) and (c) $\frac{3 - 3\sqrt{3}}{4} + \frac{3 + \sqrt{3}}{4}i$

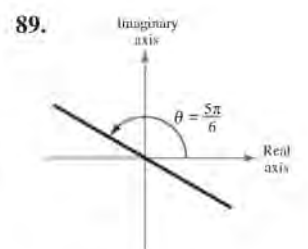
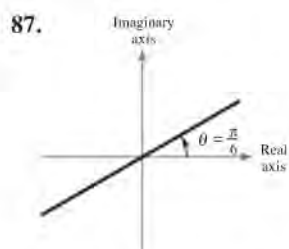
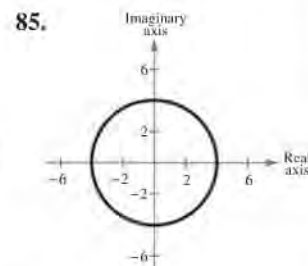
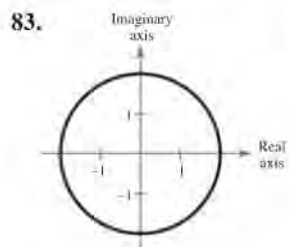
79. (a) $5(\cos 0 + i \sin 0)$

$$2\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$$

(b) and (c) $\frac{5}{4} - \frac{5}{4}i$

81. (a) $4\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$
 $\sqrt{2}\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$

(b) and (c) $2 - 2i$



91. $-2 + 2i$

93. $-32i$

95. $-32\sqrt{3} + 32i$

97. $\frac{125}{2} + \frac{125\sqrt{3}}{2}i$

99. i

101. $4.5386 - 15.3428i$

103. 256

105. $-597 - 122i$

107. $2048 + 2048\sqrt{3}i$

109. $\frac{9\sqrt{2}}{2} + \frac{9\sqrt{2}}{2}i$

111. Answers will vary.

113. (a) $2(\cos 30^\circ + i \sin 30^\circ)$; $2(\cos 150^\circ + i \sin 150^\circ)$;
 $2(\cos 270^\circ + i \sin 270^\circ)$

(b) and (c) 8i

115. (a) $\cos 0^\circ + i \sin 0^\circ$; $\cos 120^\circ + i \sin 120^\circ$;
 $\cos 240^\circ + i \sin 240^\circ$

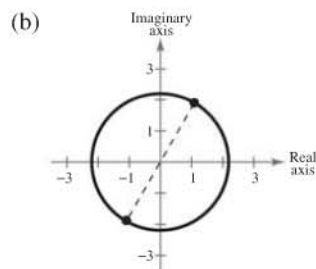
(b) and (c) 1

117. $1 + i, -1 - i$ 119. $-\frac{\sqrt{6}}{2} + \frac{\sqrt{6}}{2}i, \frac{\sqrt{6}}{2} - \frac{\sqrt{6}}{2}i$

121. $-1.5538 + 0.6436i, 1.5538 - 0.6436i$

123. $\frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i, -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i$

125. (a) $\sqrt{5}(\cos 60^\circ + i \sin 60^\circ)$
 $\sqrt{5}(\cos 240^\circ + i \sin 240^\circ)$



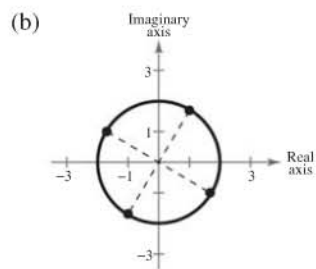
(c) $\frac{\sqrt{5}}{2} + \frac{\sqrt{15}}{2}i, -\frac{\sqrt{5}}{2} - \frac{\sqrt{15}}{2}i$

127. (a) $2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$

$2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$

$2\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$

$2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$

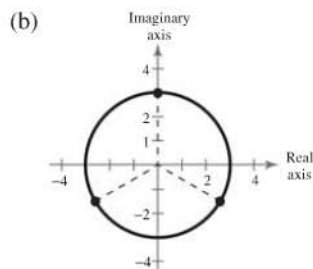


(c) $1 + \sqrt{3}i, -\sqrt{3} + i, -1 - \sqrt{3}i, \sqrt{3} - i$

129. (a) $3\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$

$3\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$

$3\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$

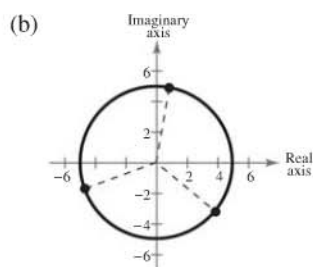


(c) $3i, -\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \frac{3\sqrt{3}}{2} - \frac{3}{2}i$

131. (a) $5\left(\cos \frac{4\pi}{9} + i \sin \frac{4\pi}{9}\right)$

$5\left(\cos \frac{10\pi}{9} + i \sin \frac{10\pi}{9}\right)$

$5\left(\cos \frac{16\pi}{9} + i \sin \frac{16\pi}{9}\right)$

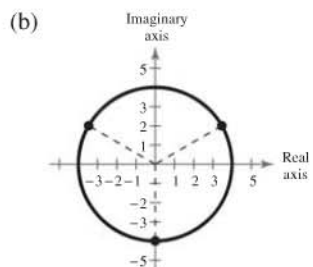


(c) $0.8682 + 4.9240i, -4.6985 - 1.7101i$
 $3.8302 - 3.2139i$

133. (a) $4\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$

$4\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$

$4\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$



(c) $2\sqrt{3} + 2i, -2\sqrt{3} + 2i, -4i$

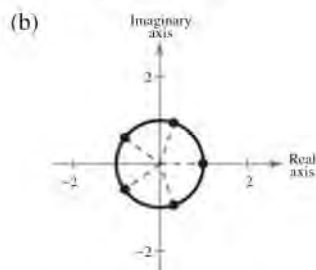
135. (a) $\cos 0 + i \sin 0$

$$\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

$$\cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$$

$$\cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

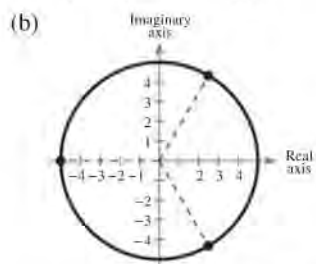


(c) $1, 0.3090 + 0.9511i, -0.8090 + 0.5878i,$
 $-0.8090 - 0.5878i, 0.3090 - 0.9511i$

137. (a) $5\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$

$$5(\cos \pi + i \sin \pi)$$

$$5\left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}\right)$$



(c) $\frac{5}{2} + \frac{5\sqrt{3}}{2}i, -5, \frac{5}{2} - \frac{5\sqrt{3}}{2}i$

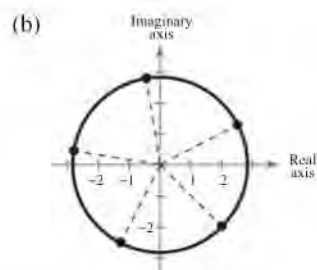
139. (a) $2\sqrt{2}\left(\cos \frac{3\pi}{20} + i \sin \frac{3\pi}{20}\right)$

$$2\sqrt{2}\left(\cos \frac{11\pi}{20} + i \sin \frac{11\pi}{20}\right)$$

$$2\sqrt{2}\left(\cos \frac{19\pi}{20} + i \sin \frac{19\pi}{20}\right)$$

$$2\sqrt{2}\left(\cos \frac{27\pi}{20} + i \sin \frac{27\pi}{20}\right)$$

$$2\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$$



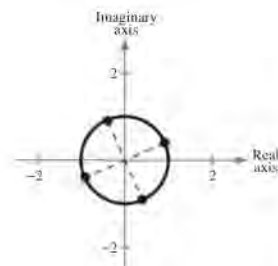
(c) $2.5201 + 1.2841i, -0.4425 + 2.7936i,$
 $-2.7936 + 0.4425i, -1.2841 - 2.5201i,$
 $2 - 2i$

141. $\cos \frac{\pi}{8} + i \sin \frac{\pi}{8}$

$$\cos \frac{5\pi}{8} + i \sin \frac{5\pi}{8}$$

$$\cos \frac{9\pi}{8} + i \sin \frac{9\pi}{8}$$

$$\cos \frac{13\pi}{8} + i \sin \frac{13\pi}{8}$$



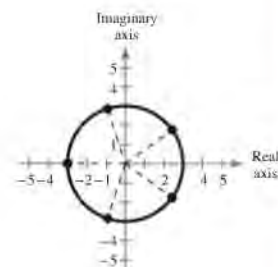
143. $3\left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}\right)$

$$3\left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}\right)$$

$$3(\cos \pi + i \sin \pi)$$

$$3\left(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}\right)$$

$$3\left(\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}\right)$$

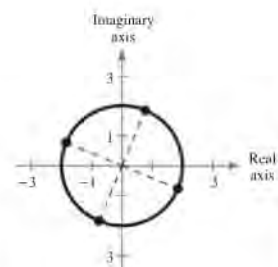


145. $2\left(\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8}\right)$

$$2\left(\cos \frac{7\pi}{8} + i \sin \frac{7\pi}{8}\right)$$

$$2\left(\cos \frac{11\pi}{8} + i \sin \frac{11\pi}{8}\right)$$

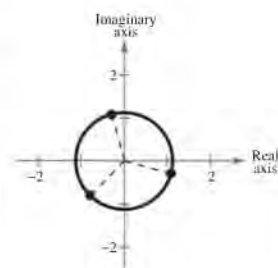
$$2\left(\cos \frac{15\pi}{8} + i \sin \frac{15\pi}{8}\right)$$



147. $\sqrt[5]{2}\left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12}\right)$

$$\sqrt[5]{2}\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$$

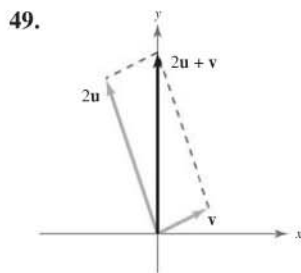
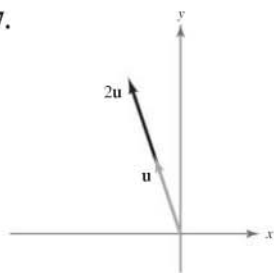
$$\sqrt[5]{2}\left(\cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12}\right)$$



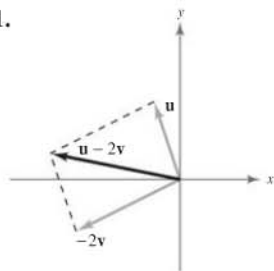
149. $34 + 38i$ 151. $\frac{3}{2} - \frac{1}{2}i$ 153. $\frac{39}{34} + \frac{3}{34}i$
 155. True 157. True 159. Answers will vary.
 161. Answers will vary.
 163. Maximum displacement: 16; $t = 2$
 165. Maximum displacement: $\frac{1}{8}$; $t = \frac{1}{24}$
 167. $\frac{\pi}{2}, \frac{3\pi}{2}$ 169. $\frac{5\pi}{6}, \frac{7\pi}{6}$

Review Exercises (page 461)

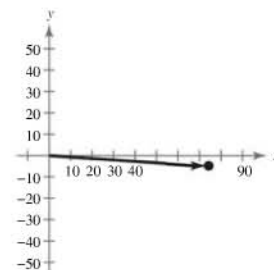
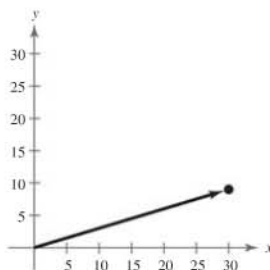
1. $C = 98^\circ, b \approx 23.13, c \approx 29.90$
 3. $A = 50^\circ, a \approx 19.83, b \approx 10.94$
 5. $C = 74^\circ 15', a \approx 5.84, c \approx 6.48$
 7. No solution
 9. $A \approx 34.23^\circ, C \approx 30.77^\circ, c \approx 8.18$
 11. Two solutions
 $A \approx 60.52^\circ, B \approx 69.48^\circ, b \approx 26.90$
 $A \approx 119.48^\circ, B \approx 10.52^\circ, b \approx 5.23$
 13. 9.08 15. 221.34
 17. 15.29 meters 19. 31.01 feet
 21. $A \approx 82.82^\circ, B \approx 41.41^\circ, C \approx 55.77^\circ$
 23. $A \approx 15.29^\circ, B \approx 20.59^\circ, C \approx 144.11^\circ$
 25. $A \approx 13.19^\circ, B \approx 20.98^\circ, C \approx 145.83^\circ$
 27. $A \approx 86.38^\circ, B \approx 28.62^\circ, c \approx 22.70$
 29. $A = 35^\circ, C = 35^\circ, b \approx 6.55$
 31. $A \approx 41^\circ 41', C \approx 82^\circ 49', b \approx 15.37$
 33. 4.29 feet, 12.63 feet 35. 1135.45 miles
 37. 9.80 square units 39. 511.71 square units
 41. $\langle 7, -5 \rangle$ 43. $\langle 7, -7 \rangle$ 45. $\langle -4, 4\sqrt{3} \rangle$
 47.



51.



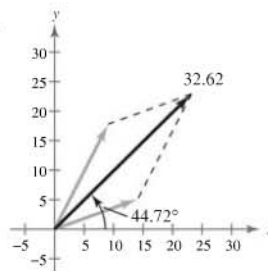
53. (a) $\langle -4, 3 \rangle$ (b) $\langle 2, -9 \rangle$ (c) $\langle -3, -9 \rangle$
 (d) $\langle -11, -3 \rangle$
 55. (a) $\langle -1, 6 \rangle$ (b) $\langle -9, -2 \rangle$ (c) $\langle -15, 6 \rangle$
 (d) $\langle -17, 18 \rangle$
 57. (a) $7\mathbf{i} + 2\mathbf{j}$ (b) $-3\mathbf{i} - 4\mathbf{j}$ (c) $6\mathbf{i} - 3\mathbf{j}$
 (d) $20\mathbf{i} + \mathbf{j}$
 59. (a) $3\mathbf{i} + 6\mathbf{j}$ (b) $5\mathbf{i} - 6\mathbf{j}$ (c) $12\mathbf{i}$ (d) $18\mathbf{i} + 12\mathbf{j}$
 61. $\langle 30, 9 \rangle$ 63. $\langle 74, -5 \rangle$



65. $\langle 0, -1 \rangle$ 67. $\frac{\sqrt{29}}{29} \langle 5, -2 \rangle$ 69. $9\mathbf{i} - 8\mathbf{j}$

71. $10\sqrt{2} \left[\left(\cos \frac{3\pi}{4} \right) \mathbf{i} + \left(\sin \frac{3\pi}{4} \right) \mathbf{j} \right]$

73.

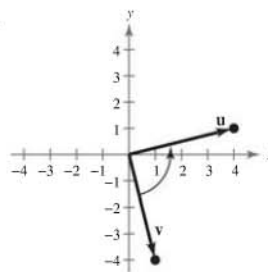


Magnitude: 32.62

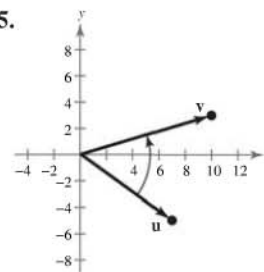
Direction: 44.72°

75. $60^\circ, 76.79$ pounds 77. 180 pounds
 79. 422.30 miles per hour, 130.41° 81. -20 83. 7
 85. 25 87. -40 89. 2.802 91. $\frac{11\pi}{12}$

93.

 90°

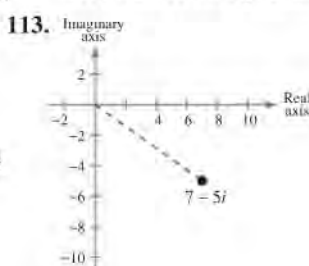
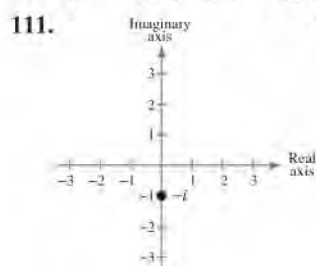
95.

 52.2°

97. Parallel 99. Neither 101. $\frac{1}{2}$ 103. -1

105. $\frac{13}{17}\langle -4, -1 \rangle, \langle -\frac{52}{17}, -\frac{13}{17} \rangle + \langle -\frac{16}{17}, \frac{64}{17} \rangle$

107. $\frac{5}{2}\langle -1, 1 \rangle, \langle -\frac{5}{2}, \frac{5}{2} \rangle + \langle \frac{9}{2}, \frac{9}{2} \rangle$ 109. 72,000 foot-pounds



1

 $\sqrt{74}$

115. $2\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$

117. $2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$ 119. $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$

121. $10\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ 123. $4(\cos 240^\circ + i \sin 240^\circ)$

125. (a) $2\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$
 $3\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

(b) and (c) 12

127. (a) $\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$
 $2\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

(b) and (c) $2 - 2i$

129. (a) $3\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$
 $2\sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

(b) and (c) $-\frac{3}{2}i$

131. $\frac{625}{2} + \frac{625\sqrt{3}}{2}i$ 133. $2035 - 828i$

135. $\pm(0.3660 + 1.3660i)$ 137. $-1 + i, 1 - i$

139. $\pm(0.6436 - 1.5538i)$

141. (a) (b)

$3\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

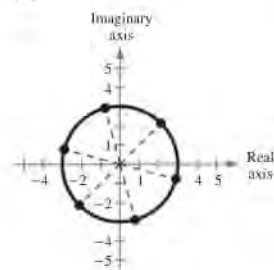
$3\left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12}\right)$

$3\left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12}\right)$

$3\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$

$3\left(\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12}\right)$

$3\left(\cos \frac{23\pi}{12} + i \sin \frac{23\pi}{12}\right)$

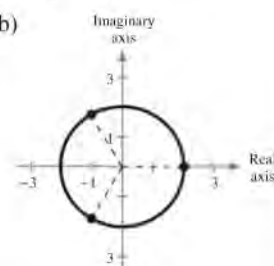


(c) $2.1213 + 2.1213i, -0.7765 + 2.8978i,$
 $-2.8978 + 0.7765i, -2.1213 - 2.1213i,$
 $0.7765 - 2.8978i, 2.8978 - 0.7765i$

143. (a) $2(\cos 0 + i \sin 0)$ (b)

$2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$

$2\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$



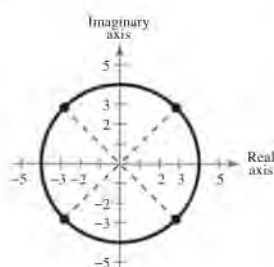
(c) $2, -1 + \sqrt{3}i, -1 - \sqrt{3}i$

145. $4\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$

$4\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$

$4\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$

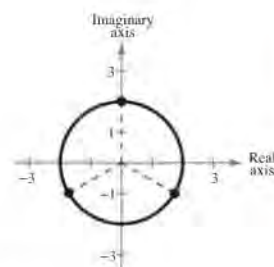
$4\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$



147. $2\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$

$2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$

$2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$

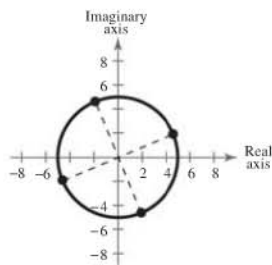


149. True 151. Direction and magnitude

153. $z_1 z_2 = -4, \frac{z_1}{z_2} = -\frac{1}{4} z_1^2$

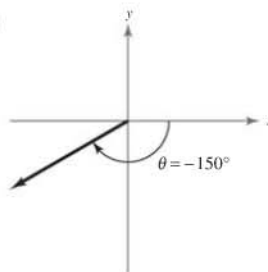
Chapter Test (page 465)

- $C = 46^\circ, a \approx 13.07, b \approx 22.03$
- $A \approx 22.33^\circ, B \approx 49.46^\circ, C \approx 108.21^\circ$
- $B \approx 40.11^\circ, C = 104.89^\circ, a \approx 7.12$
- Two solutions
 $B \approx 41.10^\circ, C \approx 113.90^\circ, c \approx 38.94$
 $B \approx 138.90^\circ, C \approx 16.10^\circ, c \approx 11.81$
- No solution
- $B \approx 14.79^\circ, C \approx 15.21^\circ, c \approx 4.93$
- 675 feet
- 2337 square meters
- $\mathbf{w} = \langle 12, 13 \rangle, \|\mathbf{w}\| \approx \sqrt{313}$
- (a) $\langle -4, -20 \rangle$ (b) $\langle 6, 20 \rangle$ (c) $\langle 2, -12 \rangle$
- (a) $\langle -4, -23 \rangle$ (b) $\langle 1, 27 \rangle$ (c) $\langle -9, -5 \rangle$
- (a) $13\mathbf{i} + 17\mathbf{j}$ (b) $-17\mathbf{i} - 28\mathbf{j}$ (c) $-\mathbf{i} - 14\mathbf{j}$
- (a) $-\mathbf{j}$ (b) $5\mathbf{i} + 9\mathbf{j}$ (c) $11\mathbf{i} + 17\mathbf{j}$
- $\frac{\sqrt{65}}{65} \langle 7, 4 \rangle$
- $\left\langle \frac{18\sqrt{34}}{17}, -\frac{30\sqrt{34}}{17} \right\rangle$
- $\theta \approx 14.87^\circ, 250.15$ pounds
- 3
- 105.95°
- No, because $\mathbf{u} \cdot \mathbf{v} = 24$, not 0.
- $\left\langle \frac{185}{26}, \frac{37}{26} \right\rangle, \mathbf{u} = \left\langle \frac{185}{26}, \frac{37}{26} \right\rangle + \left\langle -\frac{29}{26}, \frac{145}{26} \right\rangle$
- $z = 2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$
- $-50 - 50\sqrt{3}i$
- $-\frac{6561}{2} + \frac{6561\sqrt{3}}{2}i$
- 5832i
- $4 \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$
 $4 \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right)$
 $4 \left(\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right)$
 $4 \left(\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right)$
- $5 \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right)$
 $5 \left(\cos \frac{5\pi}{8} + i \sin \frac{5\pi}{8} \right)$
 $5 \left(\cos \frac{9\pi}{8} + i \sin \frac{9\pi}{8} \right)$
 $5 \left(\cos \frac{13\pi}{8} + i \sin \frac{13\pi}{8} \right)$



Cumulative Test for Chapters 4–6 (page 466)

1. (a)

(b) 210° (c) $-\frac{5\pi}{6}$ (d) 30°

(e) $\sin(-150^\circ) = -\frac{1}{2}$

$$\cos(-150^\circ) = -\frac{\sqrt{3}}{2}$$

$$\tan(-150^\circ) = \frac{\sqrt{3}}{3}$$

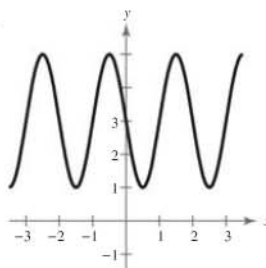
$$\csc(-150^\circ) = -2$$

$$\sec(-150^\circ) = -\frac{2\sqrt{3}}{3}$$

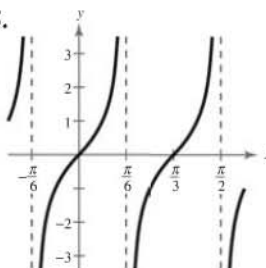
$$\cot(-150^\circ) = \sqrt{3}$$

2. 146.1° 3. $\cos \theta = -\frac{5}{13}$

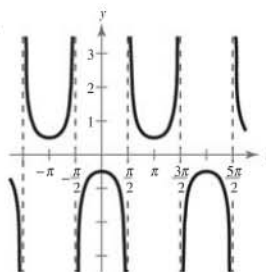
4.



5.



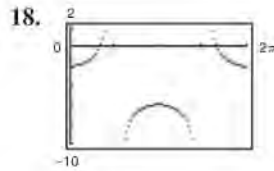
6.

7. $a = -3, b = \pi, c = 0$ 8. $\frac{3}{5}$ 9. $-\frac{\sqrt{3}}{3}$ 10. $\frac{2x}{\sqrt{4x^2 + 1}}$ 11. $2 \tan \theta$

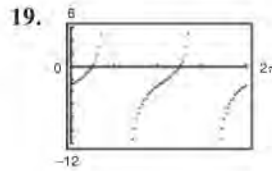
12–14. Answers will vary.

15. $\frac{3\pi}{2} + 2n\pi$ 16. $\frac{\pi}{6} + n\pi, \frac{5\pi}{6} + n\pi$

17. 1.7646, 4.5186



$$\frac{\pi}{3}, \frac{5\pi}{3}$$



$$\frac{\pi}{4}, \frac{5\pi}{4}$$

20. $\frac{16}{63}$ 21. $\frac{4}{3}$ 22. $\frac{2\sqrt{5}}{5}$

23. $2 \cos 6x \cos 2x$ 24–27. Answers will vary.

28. $B \approx 14.89^\circ$ 29. $B \approx 52.82^\circ$

$C \approx 119.11^\circ$ $C \approx 95.18^\circ$

$b \approx 17.00$ $a \approx 5.32$

30. $B = 55^\circ$ 31. $A \approx 26.07^\circ$

$b \approx 20.14$ $B \approx 33.33^\circ$

$c \approx 24.13$ $C \approx 120.60^\circ$

32. 131.71 square inches 33. 94.10 square inches

34. $3\mathbf{i} + 5\mathbf{j}$ 35. $\frac{\sqrt{5}}{5}\mathbf{i} - \frac{2\sqrt{5}}{5}\mathbf{j}$ 36. -5 37. 1

38. $\langle -\frac{1}{13}, -\frac{5}{13} \rangle$; $\mathbf{u} = \langle \frac{108}{13}, -\frac{21}{13} \rangle + \langle -\frac{1}{13}, -\frac{5}{13} \rangle$

39. $3\sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$ 40. $-4\sqrt{3} + 4i$

41. $-12\sqrt{3} + 12i$

42. $1.4553 + 0.3436i, -1.4553 - 0.3436i$

43. $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

44. $3(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5})$
 $3(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5})$

$3(\cos \pi + i \sin \pi)$

$3(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5})$

$3(\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5})$

45. 5 feet 46. $d = 4 \sin \frac{\pi}{4}t$

47. 32.63° , 543.88 kilometers per hour

48. 80.28°

Chapter 7

Section 7.1 (page 481)

Vocabulary Check (page 481)

1. system, equations
2. solution
3. method, substitution
4. point, intersection
5. break-even point

1. (a) No (b) No (c) No (d) Yes

3. (a) No (b) Yes (c) No (d) No

5. (2, 2) 7. (2, 6), (-1, 3)

9. (0, 2), $(\sqrt{3}, 2 - 3\sqrt{3})$, $(-\sqrt{3}, 2 + 3\sqrt{3})$

11. (4, 4) 13. (5, 5) 15. $(\frac{1}{2}, 3)$ 17. (1, 1)

19. $(\frac{20}{3}, \frac{40}{3})$ 21. No solution 23. (-2, 0), (3, 5)

25. No real solution 27. (0, 0), (1, 1), (-1, -1)

29. (4, 3) 31. $(\frac{5}{2}, \frac{3}{2})$ 33. No real solution

35. (3, 6), (-3, 0) 37. (4, -0.5) 39. (8, 3), (3, -2)

41. $(\pm 1.540, 2.372)$ 43. (0, 1) 45. (2.318, 2.841)

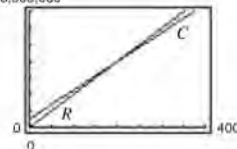
47. (2.25, 5.5) 49. (0, -13), $(\pm 12, 5)$ 51. (1, 2)

53. (-2, 0), $(\frac{29}{10}, \frac{21}{10})$ 55. No real solution

57. (0.25, 1.5) 59. (0.287, 1.751)

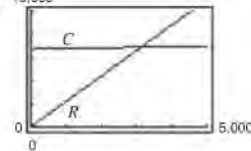
61. (0, 1), (1, 0) 63. $(-4, -\frac{1}{4})$, $(\frac{1}{2}, 2)$

65. 3,500,000



192 units; \$1,910,400

67. 15,000



3133 units; \$10,308

69. (a)

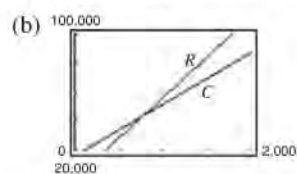
Week	Animated	Horror
1	336	42
2	312	60
3	288	78
4	264	96
5	240	114
6	216	132
7	192	150
8	168	168
9	144	186
10	120	204
11	96	222
12	72	240

(b) and (c) $x = 8$

(d) The answers are the same.

(e) During week 8 the same number of animated and horror films were rented.

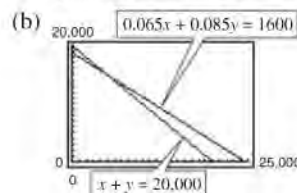
71. (a) $C = 35.45x + 16,000$
 $R = 55.95x$



780 units

73. Sales greater than \$11,667

75. (a)
$$\begin{cases} x + y = 20,000 \\ 0.065x + 0.085y = 1,600 \end{cases}$$



More invested at 6.5% means less invested at 8.5% and less interest.

(c) \$5000

77. (a)

Year	Missouri	Tennessee
1990	5104	4875
1994	5294	5181
1998	5483	5487
2002	5673	5793
2006	5862	6099
2010	6052	6405

(b) From 1998 to 2010

(c) (d) (7.87, 5477.01)

(7.87, 5477.01)

(e) During 1997 the populations of Missouri and Tennessee were equal.

79. 6 meters $\times$ 9 meters 81. 8 miles $\times$ 12 miles

83. False. You can solve for either variable before back-substituting.

85. For a linear system, the result will be a contradictory equation such as $0 = N$, where N is a nonzero real number. For a nonlinear system, there may be an equation with imaginary roots.

87. (a) $y = x + 1$ (b) $y = 0$ (c) $y = -2$

89.
$$\begin{cases} y = x - 3 \\ 2y = x - 4 \end{cases}$$
 91. $2x + 7y - 45 = 0$

93. $y - 3 = 0$ 95. $30x - 17y - 18 = 0$

97. Domain: All real numbers x except $x = 6$

Asymptotes: $y = 0, x = 6$

99. Domain: All real numbers x except $x = \pm 4$

Asymptotes: $y = 1, x = \pm 4$

101. Domain: All real numbers x

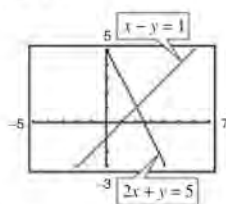
Asymptote: $y = 0$

Section 7.2 (page 491)

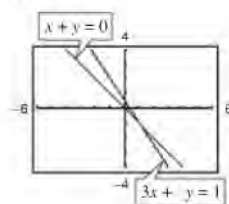
Vocabulary Check (page 491)

- method, elimination
- equivalent
- consistent, inconsistent

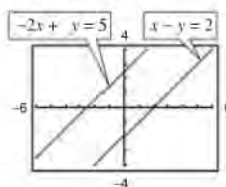
1. (2, 1)



3. (1, -1)



5. Inconsistent


 7. $(\frac{5}{2}, \frac{3}{4})$ 9. (3, 4) 11. (4, -1) 13. $(\frac{12}{7}, \frac{18}{7})$

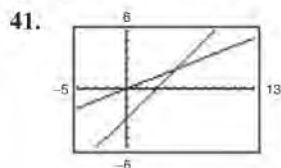
15. Inconsistent 17. b; One solution, consistent

 19. c; One solution, consistent 21. $(-\frac{6}{35}, \frac{43}{35})$

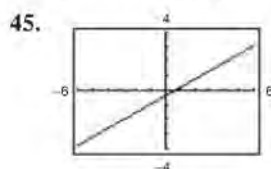
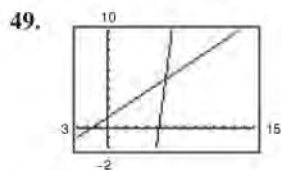
 23. Inconsistent 25. All points on $6x + 8y - 1 = 0$

27. (5, -2) 29. (7, 1)

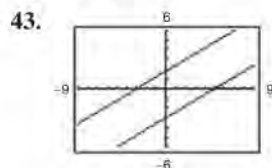
 31. All points on $5x - 6y - 3 = 0$ 33. (101, 96)

 35. $(\frac{90}{31}, -\frac{67}{31})$ 37. (-1, 1) 39. $(1, \frac{1}{2})$


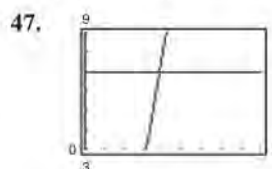
Consistent; (5, 2)


 Consistent; all points on $8x - 14y = 5$


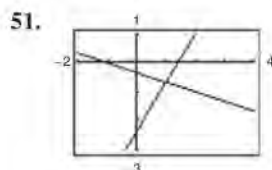
(6, 5)



Inconsistent

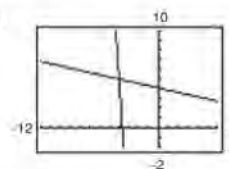


(3.833, 7)



(1, -0.667)

53.



(-4, 5)

 55. (4, 1) 57. $(-\frac{5}{3}, -\frac{11}{3})$

 59. (6, -3) 61. $(\frac{43}{6}, \frac{25}{6})$

 63. $\begin{cases} 3x + \frac{1}{2}y = 4 \\ x + 3y = 24 \end{cases}$ 65. $\begin{cases} 2x + 2y = 11 \\ x - 4y = -7 \end{cases}$

67. (80, 10) 69. (2,000,000, 100)

71. Plane: 550 miles per hour; wind: 50 miles per hour

 73. (a) $\begin{cases} 5.00A + 3.50C = 5087.50 \\ A + C = 1175 \end{cases}$

 (b) $A = 650, C = 525$; Answers will vary.

 (c) $A = 650, C = 525$

75. 9 oranges, 7 grapefruits

77. 185 movies, 125 video game cartridges

 79. $y = 0.97x + 2.1$ 81. $y = -2.5x + 5.5$

 83. (a) and (b) $y = 14x + 19$

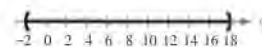
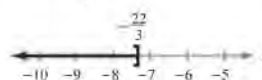
(c) (d) 41.4 bushels per acre

85. True. A linear system can have only one solution, no solution, or infinitely many solutions.

87. (39,600, 398). The solution is outside the viewing window.

89. No. Two lines can only intersect once, never, or infinitely many times.

 91. $k = -4$ 93. $u = 1; v = -\tan x$

 95. $x \leq -\frac{22}{3}$ 97. $-2 < x < 18$

 99. $-5 < x < \frac{7}{2}$

 101. $\ln 6x$

 103. $\log_9 \frac{12}{x}$

 105. $\ln \frac{x^2}{x+2}$

107. Answers will vary.

Section 7.3 (page 505)

Vocabulary Check (page 505)

1. row-echelon 2. ordered triple 3. Gaussian
 4. independent, dependent 5. nonsquare
 6. three-dimensional
 7. partial fraction decomposition

1. (a) No (b) Yes (c) No (d) No
 3. (a) No (b) No (c) Yes (d) No
 5. $(2, -2, 2)$ 7. $(3, 10, 2)$ 9. $(\frac{1}{2}, -2, 2)$

$$11. \begin{cases} x - 2y + 3z = 5 \\ y - 2z = 9 \\ 2x - 3z = 0 \end{cases}$$

It removed the x -term from Equation 2.

$$13. \begin{cases} x - 2y + z = 1 \\ 3y + z = -2 \\ 3x - y - 4z = 1 \end{cases}$$

It removed the x -term from Equation 2.

15. $(1, 2, 3)$ 17. $(-4, 8, 5)$ 19. $(2, -3, -2)$

21. Inconsistent 23. $(1, -\frac{3}{2}, \frac{1}{2})$

25. $(-3a + 10, 5a - 7, a)$ 27. $(1, -1, 2)$

29. $(-a + 3, a + 1, a)$ 31. Inconsistent

33. Inconsistent 35. $(-1, 1, 0)$

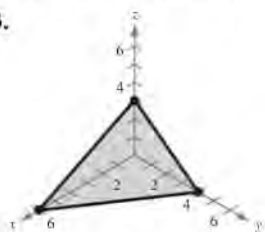
37. $(2a, 21a - 1, 8a)$ 39. $(-\frac{3}{2}a + \frac{1}{2}, -\frac{2}{3}a + 1, a)$

41. $(-5a + 3, -a - 5, a)$ 43. $(\frac{1}{2}, 0, \frac{3}{2})$

45. $(1, -2, -1)$ 47. $(1, -1, 2)$

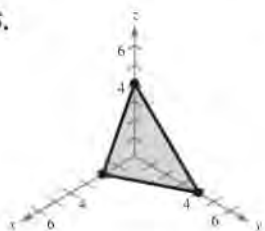
$$49. \begin{cases} 3x + y - z = 9 \\ x + 2y - z = 0 \\ -x + y + 3z = 1 \end{cases} \quad 51. \begin{cases} x + 6y + 4z = 7 \\ 2x - 2y - 4z = 0 \\ -x + y + z = -\frac{7}{4} \end{cases}$$

53.



- $(6, 0, 0), (0, 4, 0),$
 $(0, 0, 3), (4, 0, 1)$

55.



- $(2, 0, 0), (0, 4, 0),$
 $(0, 0, 4), (0, 2, 2)$

$$57. \frac{A}{x} + \frac{B}{x-14} \quad 59. \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-10}$$

$$61. \frac{A}{x-5} + \frac{B}{(x-5)^2} + \frac{C}{(x-5)^3}$$

$$63. \frac{1}{2} \left(\frac{1}{x-1} - \frac{1}{x+1} \right) \quad 65. \frac{1}{x} - \frac{1}{x+1}$$

$$67. \frac{1}{x} - \frac{2}{2x+1} \quad 69. \frac{3}{2x-1} - \frac{2}{x+1}$$

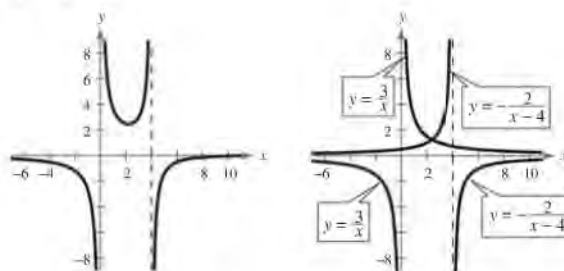
$$71. -\frac{3}{x} - \frac{1}{x+2} + \frac{5}{x-2} \quad 73. \frac{3}{x} - \frac{1}{x^2} + \frac{1}{x+1}$$

$$75. \frac{3}{x} - \frac{3}{x-3} + \frac{2}{(x-3)^2}$$

$$77. 2x - 7 + \frac{17}{x+2} + \frac{1}{x+1}$$

$$79. x + 3 + \frac{6}{x-1} + \frac{4}{(x-1)^2} + \frac{1}{(x-1)^3}$$

$$81. \frac{3}{x} - \frac{2}{x-4}$$



The vertical asymptotes are the same.

$$83. s = -16t^2 + 144 \quad 85. s = -16t^2 - 32t + 500$$

$$87. y = \frac{1}{2}x^2 - 2x \quad 89. y = x^2 - 6x + 8$$

$$91. x^2 + y^2 - 4x = 0 \quad 93. x^2 + y^2 + 6x - 8y = 0$$

$$95. \$366,666.67 \text{ at } 8\%, \$316,666.67 \text{ at } 9\%, \text{ and } \$91,666.67 \text{ at } 10\%$$

$$97. \$156,250 + 0.75g \text{ in certificates of deposit}$$

$$\$125,000 \text{ in municipal bonds}$$

$$\$218,750 - 1.75g \text{ in blue-chip stocks}$$

$$g \text{ in growth stocks}$$

$$99. 24 \text{ two-point field goals, } 6 \text{ three-point field goals, } 18 \text{ free throws}$$

$$101. \text{ Four touchdowns, four extra-point kicks, one field goal}$$

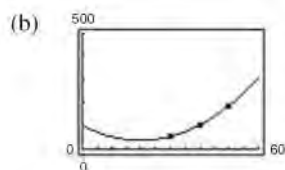
$$103. I_1 = 1, I_2 = 2, I_3 = 1$$

$$105. \text{ Four par-3 holes, } 10 \text{ par-4 holes, four par-5 holes}$$

$$107. y = -\frac{5}{24}x^2 - \frac{3}{10}x + \frac{41}{6} \quad 109. y = x^2 - x$$

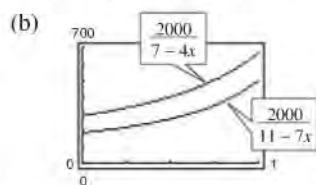
111. (a)
$$\begin{cases} 900a + 30b + c = 55 \\ 1600a + 40b + c = 105 \\ 2500a + 50b + c = 188 \end{cases}$$

$$y = 0.165x^2 - 6.55x + 103$$



(c) 453 feet

113. (a) $\frac{2000}{7-4x} - \frac{2000}{11-7x}, 0 \leq x \leq 1$



115. False. The leading coefficients are not all 1.

117. False. $\frac{A}{x+10} + \frac{B}{x-10} + \frac{C}{(x-10)^2}$

119. $\frac{1}{2a} \left(\frac{1}{a+x} + \frac{1}{a-x} \right)$ 121. $\frac{1}{a} \left(\frac{1}{y} + \frac{1}{a-y} \right)$

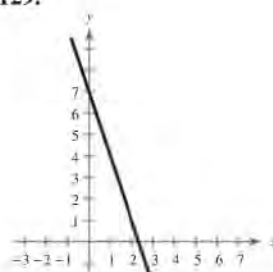
123. No. There are two arithmetic errors. The constant in the second equation should be -11 and the coefficient of z in the third equation should be 2.

125. $x = 5, y = 5, \lambda = -5$

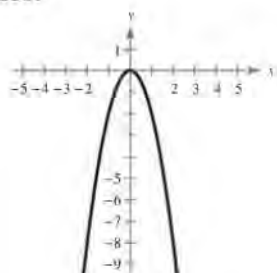
127. $x = \frac{\sqrt{2}}{2}, y = \frac{1}{2}, \lambda = 1$ 129.

$x = -\frac{\sqrt{2}}{2}, y = \frac{1}{2}, \lambda = 1$

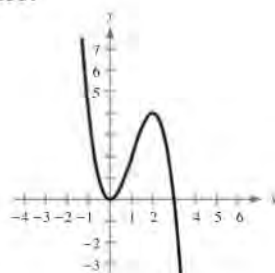
$x = 0, y = 0, \lambda = 0$



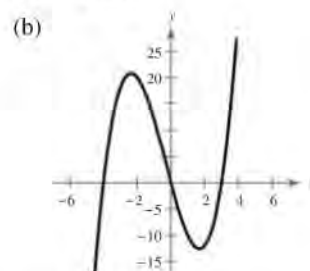
131.



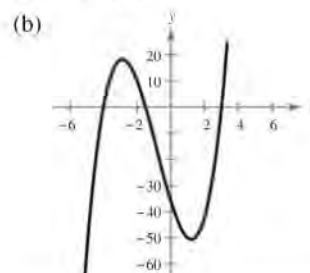
133.



135. (a) $-4, 0, 3$

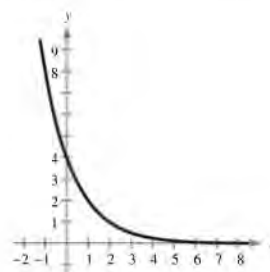


137. (a) $-4, -\frac{3}{2}, 3$



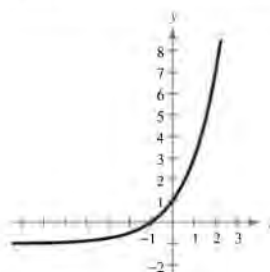
139.

x	-2	-1	0	1	2
y	16	8	4	2	1



141.

x	-2	-1	0	1	2
y	-0.5	0	1	3	7



143. Answers will vary.

Section 7.4 (page 521)

Vocabulary Check (page 521)

1. matrix 2. square
3. row matrix, column matrix 4. augmented matrix
5. coefficient matrix 6. row-equivalent
7. reduced row-echelon form
8. Gauss-Jordan elimination

1. 1×2 3. 3×1 5. 2×2

$$7. \begin{bmatrix} 6 & -7 & \vdots & 11 \\ -2 & 5 & \vdots & -1 \end{bmatrix}$$

$$9. \begin{bmatrix} 1 & 10 & -2 & \vdots & 2 \\ 5 & -3 & 4 & \vdots & 0 \\ 2 & 1 & 0 & \vdots & 6 \end{bmatrix}$$

$$11. \begin{cases} 3x + 4y = 9 \\ x - y = -3 \end{cases} \quad 13. \begin{cases} 9x + 12y + 3z = 0 \\ -2x + 18y + 5z = 10 \\ x + 7y - 8z = -4 \end{cases}$$

$$15. \begin{bmatrix} 1 & 4 & 3 \\ 0 & 2 & -1 \end{bmatrix}$$

$$17. \begin{bmatrix} 1 & 1 & 4 & -1 \\ 0 & 5 & -2 & 6 \\ 0 & 3 & 20 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 4 & -1 \\ 0 & 1 & -\frac{2}{5} & \frac{6}{5} \\ 0 & 3 & 20 & 4 \end{bmatrix}$$

19. Add -3 times R_2 to R_1 .
21. Interchange R_1 and R_2 .

23. Reduced row-echelon form

25. Not in row-echelon form 27. Not in row-echelon form

$$29. (a) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 3 & 1 & -1 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & -5 & -10 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 0 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(e) \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is in reduced row-echelon form.

$$31. (a) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 3 & 1 & -1 \end{bmatrix}$$

$$(b) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & -5 & -10 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(e) \begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is in reduced row-echelon form.

$$33. \begin{bmatrix} 1 & 3 & \frac{3}{2} & 5 \\ 0 & 1 & \frac{3}{14} & 0 \\ 0 & 0 & 1 & -\frac{35}{12} \end{bmatrix} \quad 35. \begin{bmatrix} 1 & -1 & -1 & 1 \\ 0 & 1 & 6 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$37. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad 39. \begin{bmatrix} 1 & 0 & -\frac{3}{7} & -\frac{8}{7} \\ 0 & 1 & -\frac{12}{7} & \frac{10}{7} \end{bmatrix}$$

$$41. \begin{cases} x - 2y = 4 \\ y = -3 \end{cases} \quad 43. \begin{cases} x - y + 2z = 4 \\ y - z = 2 \\ z = -2 \end{cases}$$

45. $(7, -5)$ 47. $(-4, -8, 2)$ 49. $(3, 2)$
51. Inconsistent 53. $(3, -2, 5, 0)$ 55. $(4, -3, 2)$

$$57. (2a + 1, 3a + 2, a) \quad 59. (7, -3, 4)$$

$$61. (0, 2 - 4a, a) \quad 63. (-5a, a, 3)$$

$$65. \text{Yes; } (-1, 1, -3) \quad 67. \text{No} \quad 69. y = x^2 + 2x + 5$$

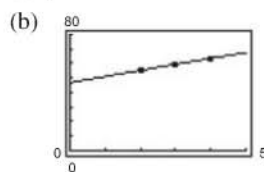
$$71. y = 2x^2 - x + 1 \quad 73. f(x) = -9x^2 - 5x + 11$$

$$75. f(x) = x^3 - 2x^2 - 4x + 1$$

$$77. \$150,000 \text{ at } 7\%, \$750,000 \text{ at } 8\%, \$600,000 \text{ at } 10\%$$

$$79. I_1 = \frac{13}{10}, I_2 = \frac{11}{5}, I_3 = \frac{9}{10}$$

$$81. (a) y = -0.0400t^2 + 4.350t + 46.83$$



$$(c) 2005: \$67.58; 2010: \$86.33; 2015: \$103.08$$

(d) Answers will vary.

$$83. \begin{cases} 5.5g + 4.25s + 3.75m = 480 \\ -g + s + m = 0 \\ g + s + m = 100 \end{cases}$$

50 pounds of glossy, 35 pounds of semi-gloss, 15 pounds of matte

$$85. (a) x_1 = s, x_2 = t, x_3 = 600 - s,$$

$$x_4 = s - t, x_5 = 500 - t, x_6 = s, x_7 = t$$

$$(b) x_1 = 0, x_2 = 0, x_3 = 600, x_4 = 0, x_5 = 500,$$

$$x_6 = 0, x_7 = 0$$

$$(c) x_1 = 0, x_2 = -500, x_3 = 600, x_4 = 500,$$

$$x_5 = 1000, x_6 = 0, x_7 = -500$$

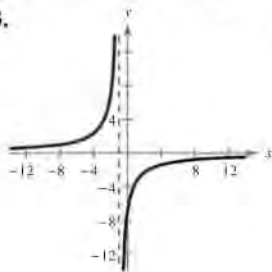
87. False. It is a 2×4 matrix.

$$89. \begin{cases} x + y + 7z = -1 \\ x + 2y + 11z = 0 \\ 2x + y + 10z = -3 \end{cases}$$

(Answer is not unique.)

91. Gauss-Jordan elimination was not performed on the last column of the matrix.

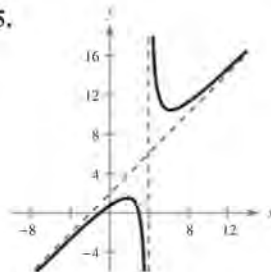
93.



Asymptotes:

$$x = -1, y = 0$$

95.



Asymptotes:

$$x = 4, y = x + 2$$

Section 7.5 (page 536)

Vocabulary Check (page 536)

1. equal 2. scalars 3. zero, O 4. identity
 5. (a) iii (b) i (c) iv (d) v (e) ii
 6. (a) ii (b) iv (c) i (d) iii

1. $x = -4, y = 22$ 3. $x = -1, y = 4, z = 6$

5. (a) $\begin{bmatrix} 3 & -2 \\ 1 & 7 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 0 \\ 3 & -9 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & -3 \\ 6 & -3 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & -1 \\ 8 & -19 \end{bmatrix}$

7. (a) $\begin{bmatrix} 9 & 5 \\ 1 & -2 \\ -3 & 15 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & -7 \\ 3 & 8 \\ -5 & -5 \end{bmatrix}$

(c) $\begin{bmatrix} 24 & -3 \\ 6 & 9 \\ -12 & 15 \end{bmatrix}$ (d) $\begin{bmatrix} 22 & -15 \\ 8 & 19 \\ -14 & -5 \end{bmatrix}$

9. (a) $\begin{bmatrix} 5 & 5 & -2 & 4 & 4 \\ -5 & 10 & 0 & -4 & -7 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 5 & 0 & 2 & 4 \\ 7 & -6 & -4 & 2 & 7 \end{bmatrix}$

(c) $\begin{bmatrix} 12 & 15 & -3 & 9 & 12 \\ 3 & 6 & -6 & -3 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 10 & 15 & -1 & 7 & 12 \\ 15 & -10 & -10 & 3 & 14 \end{bmatrix}$

11. (a) Not possible (b) Not possible

(c) $\begin{bmatrix} 18 & 0 & 9 \\ -3 & -12 & 0 \end{bmatrix}$ (d) Not possible

13. $\begin{bmatrix} -8 & -7 \\ 15 & -1 \end{bmatrix}$

15. $\begin{bmatrix} -24 & -4 & 12 \\ -12 & 32 & 12 \end{bmatrix}$

17. $\begin{bmatrix} -1 & 5 \\ 1 & -2 \end{bmatrix}$

19. $\begin{bmatrix} -14.646 & 21.306 \\ -41.546 & -69.137 \\ 78.117 & -32.064 \end{bmatrix}$

21. $\begin{bmatrix} -6 & -9 \\ -1 & 0 \\ 17 & -10 \end{bmatrix}$

23. $\begin{bmatrix} 3 & 3 \\ -\frac{1}{2} & 0 \\ -\frac{13}{2} & \frac{11}{2} \end{bmatrix}$

25. Not possible 27. $\begin{bmatrix} -2 & 51 \\ -8 & 33 \\ 0 & 27 \end{bmatrix}$

29. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{7}{2} \end{bmatrix}$

31. $\begin{bmatrix} -15 & -5 & -25 & -45 \\ -18 & -6 & -30 & -54 \end{bmatrix}$

33. (a) $\begin{bmatrix} 0 & 15 \\ 8 & 11 \end{bmatrix}$

(b) $\begin{bmatrix} -3 & 2 \\ 39 & 14 \end{bmatrix}$

(c) $\begin{bmatrix} 11 & 6 \\ 15 & 14 \end{bmatrix}$

35. (a) $\begin{bmatrix} 0 & -10 \\ 10 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & -10 \\ 10 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 8 & -6 \\ 6 & 8 \end{bmatrix}$

37. (a) $\begin{bmatrix} 7 & 7 & 14 \\ 8 & 8 & 16 \\ -1 & -1 & -2 \end{bmatrix}$

(b) $[13]$

(c) Not possible

39. $\begin{bmatrix} 70 & -17 & 73 \\ 32 & 11 & 6 \\ 16 & -38 & 70 \end{bmatrix}$

41. $\begin{bmatrix} 151 & 25 & 48 \\ 516 & 279 & 387 \\ 47 & -20 & 87 \end{bmatrix}$

43. Not possible

45. $\begin{bmatrix} 5 & 8 \\ -4 & -16 \end{bmatrix}$

47. $\begin{bmatrix} -4 & 10 \\ 3 & 14 \end{bmatrix}$

49. (a) No (b) Yes (c) No (d) No

51. (a) No (b) Yes (c) No (d) No

53. (a) $\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ (b) $\begin{bmatrix} 4 \\ 8 \end{bmatrix}$

55. (a) $\begin{bmatrix} -2 & -3 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 \\ -36 \end{bmatrix}$ (b) $\begin{bmatrix} -7 \\ 6 \end{bmatrix}$

57. (a) $\begin{bmatrix} 1 & -2 & 3 \\ -1 & 3 & -1 \\ 2 & -5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -6 \\ 17 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

59. (a) $\begin{bmatrix} 1 & -5 & 2 \\ -3 & 1 & -1 \\ 0 & -2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -20 \\ 8 \\ -16 \end{bmatrix}$ (b) $\begin{bmatrix} -1 \\ 3 \\ -2 \end{bmatrix}$

61. (a) $\begin{bmatrix} 7 & -2 & 5 \\ -6 & 13 & -8 \\ 16 & 11 & -3 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & -2 & 5 \\ -6 & 13 & -8 \\ 16 & 11 & -3 \end{bmatrix}$

The answers are the same.

$$63. (a) \begin{bmatrix} 26 & 11 & 0 \\ 11 & 20 & -3 \\ 11 & 14 & 0 \end{bmatrix} \quad (b) \begin{bmatrix} 26 & 11 & 0 \\ 11 & 20 & -3 \\ 11 & 14 & 0 \end{bmatrix}$$

The answers are the same.

$$65. (a) \begin{bmatrix} 25 & -34 & 28 \\ -53 & 34 & -7 \\ -76 & 30 & 21 \end{bmatrix} \quad (b) \begin{bmatrix} 25 & -34 & 28 \\ -53 & 34 & -7 \\ -76 & 30 & 21 \end{bmatrix}$$

The answers are the same.

$$67. (a) \text{ and } (b) \begin{bmatrix} -1 & 10 & -4 \\ -5 & -1 & 0 \end{bmatrix}$$

69. Not possible, undefined 71. Not possible, undefined

$$73. (a) \text{ and } (b) \begin{bmatrix} -6 & -12 & 12 \\ 6 & -6 & 0 \end{bmatrix}$$

$$75. \begin{bmatrix} -4 & 0 \\ 8 & 2 \end{bmatrix} \quad 77. \begin{bmatrix} 84 & 60 & 30 \\ 42 & 120 & 84 \end{bmatrix}$$

$$79. [\$1037.50 \quad \$1400.00 \quad \$1012.50]$$

The entries represent the total profit made at each outlet.

$$81. \begin{bmatrix} \$15,770 & \$18,300 \\ \$26,500 & \$29,250 \\ \$21,260 & \$24,150 \end{bmatrix}$$

The entries are the total wholesale and retail prices of the inventory at each outlet.

$$83. \begin{bmatrix} 0.40 & 0.15 & 0.15 \\ 0.28 & 0.53 & 0.17 \\ 0.32 & 0.32 & 0.68 \end{bmatrix}$$

P^2 represents the changes in party affiliations after two elections.

85. True. To add two matrices, you add corresponding entries.

87. Not possible 89. Not possible 91. 2×2

93. 2×3 95. Answers will vary.

97. Answers will vary.

$$99. AC = BC = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}, A \neq B$$

$$101. (a) A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, A^3 = \begin{bmatrix} -i & 0 \\ 0 & -i \end{bmatrix},$$

$$A^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The entries on the main diagonal are i^2 in A^2 , i^3 in A^3 , and i^4 in A^4 .

$$(b) B^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

B^2 is the identity matrix.

$$103. (a) A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

(Answers are not unique.)

(b) A^2 and B^3 are zero matrices.

$$(c) A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

A^4 is the zero matrix.

(d) A^n is the zero matrix.

$$105. \ln\left(\frac{64}{\sqrt[3]{x^2+3}}\right) \quad 107. \ln\left(\frac{\sqrt{x}(x+5)}{\sqrt{x-8}}\right)$$

Section 7.6 (page 547)

Vocabulary Check (page 547)

1. square 2. inverse 3. nonsingular, singular

1–9. Answers will vary.

$$11. \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \quad 13. \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix} \quad 15. \text{ Does not exist}$$

$$17. \begin{bmatrix} 1 & 1 & -1 \\ -3 & 2 & -1 \\ 3 & -3 & 2 \end{bmatrix} \quad 19. \text{ Does not exist}$$

$$21. \text{ Does not exist} \quad 23. \begin{bmatrix} -12 & -5 & -9 \\ -4 & -2 & -4 \\ -8 & -4 & -6 \end{bmatrix}$$

$$25. \frac{5}{11} \begin{bmatrix} 0 & -4 & 2 \\ -22 & 11 & 11 \\ 22 & -6 & -8 \end{bmatrix} \quad 27. \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$$29. \begin{bmatrix} \frac{1}{4} & \frac{1}{8} \\ -\frac{1}{4} & -\frac{5}{8} \end{bmatrix} \quad 31. \frac{1}{59} \begin{bmatrix} 16 & 15 \\ -4 & 70 \end{bmatrix} \quad 33. \begin{bmatrix} \frac{5}{13} & -\frac{3}{13} \\ \frac{1}{13} & \frac{2}{13} \end{bmatrix}$$

$$35. \begin{bmatrix} -1 & 0 \\ -\frac{3}{2} & -\frac{1}{2} \end{bmatrix} \quad 37. k = 0 \quad 39. k = \frac{3}{5} \quad 41. (5, 0)$$

$$43. (-8, -6) \quad 45. (3, 8, -11) \quad 47. (2, 1, 0, 0)$$

$$49. (2, -2)$$

51. Not possible, because A is not invertible.

$$53. (-4, -8) \quad 55. (-1, 3, 2)$$

$$57. (0.3125t + 0.8125, 1.8175t + 0.6875, t)$$

$$59. (5, 0, -2, 3)$$

61. (a)
- $(-3, 2)$
- (b)
- $(-1, 1)$

(c) $B = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$; they correspond to the same point.

(d) $A^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(e) $A^{-1}B = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$; $A^{-1}B$ represents X .

63. (a)
- $(2, -4)$
- (b)
- $(-1, 0)$

(c) $B = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$; they correspond to the same point.

(d) $A^{-1} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$

(e) $A^{-1}B = \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix}$; $A^{-1}B$ represents X .

65. \$10,000 in AAA-rated bonds, \$5,000 in A-rated bonds, and \$10,000 in B-rated bonds

67. \$20,000 in AAA-rated bonds, \$15,000 in A-rated bonds, and \$30,000 in B-rated bonds

- 69.
- $I_1 = -3$
- amperes,
- $I_2 = 8$
- amperes,
- $I_3 = 5$
- amperes

71. 0 muffins, 300 bones, 200 cookies

73. 100 muffins, 300 bones, 150 cookies

75. (a)
$$\begin{cases} 2f + 2.5h + 3s = 26 \\ f + h + s = 10 \\ h - s = 0 \end{cases}$$

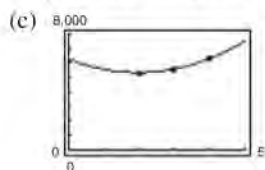
(b) $\begin{bmatrix} 2 & 2.5 & 3 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} f \\ h \\ s \end{bmatrix} = \begin{bmatrix} 26 \\ 10 \\ 0 \end{bmatrix}$

- (c) 2 pounds of French vanilla, 4 pounds of hazelnut, 4 pounds of Swiss chocolate

77. (a)
$$\begin{cases} 4a + 2b + c = 5343 \\ 9a + 3b + c = 5589 \\ 16a + 4b + c = 6309 \end{cases}$$

(b) $A^{-1} = \begin{bmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ -\frac{7}{2} & 6 & -\frac{5}{2} \\ 6 & -8 & 3 \end{bmatrix}$

$$y = 237t^2 - 939t + 6273$$



- (d) 2005: 7,503,000 snowboarders

2010: 20,583,000 snowboarders

2015: 45,513,000 snowboarders

- (e) Answers will vary.

79. True.
- $AA^{-1} = I = A^{-1}A$
81. Answers will vary.

83. $\frac{9}{2x+6}$, $x \neq 0$ 85. $\frac{x^2+2x-13}{x(x-2)}$, $x \neq \pm 3$

87. $\ln 3 \approx 1.099$ 89. $\frac{e^{12/7}}{3} \approx 1.851$

91. Answers will vary.

Section 7.7 (page 556)

Vocabulary Check (page 556)

1. determinant 2. minor 3. cofactor
-
4. expanding by cofactors 5. triangular
-
6. diagonal

1. 4 3. 16 5. 28 7. -24 9. 0

11. -9 13. -0.002

15. (a)
- $M_{11} = -5$
- ,
- $M_{12} = 2$
- ,
- $M_{21} = 4$
- ,
- $M_{22} = 3$

- (b)
- $C_{11} = -5$
- ,
- $C_{12} = -2$
- ,
- $C_{21} = -4$
- ,
- $C_{22} = 3$

17. (a)
- $M_{11} = 10$
- ,
- $M_{12} = -43$
- ,
- $M_{13} = 2$
- ,
- $M_{21} = -30$
- ,
-
- $M_{22} = 17$
- ,
- $M_{23} = -6$
- ,
- $M_{31} = 54$
- ,
- $M_{32} = -53$
- ,
-
- $M_{33} = -34$

- (b)
- $C_{11} = 10$
- ,
- $C_{12} = 43$
- ,
- $C_{13} = 2$
- ,
- $C_{21} = 30$
- ,
- $C_{22} = 17$
- ,
-
- $C_{23} = 6$
- ,
- $C_{31} = 54$
- ,
- $C_{32} = 53$
- ,
- $C_{33} = -34$

19. (a) -75 (b) -75 21. (a) 170 (b) 170

23. -58 25. -168 27. 412 29. -60

31. -168 33. -336 35. 410

37. (a) -3 (b) -2 (c)
- $\begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix}$
- (d) 6

39. (a) 2 (b) -6 (c)
- $\begin{bmatrix} 1 & 4 & 3 \\ -1 & 0 & 3 \\ 0 & 2 & 0 \end{bmatrix}$
- (d) -12

41. (a) -25 (b) -220

(c) $\begin{bmatrix} -7 & -16 & -1 & -28 \\ -4 & -14 & -11 & 8 \\ 13 & 4 & 4 & -4 \\ -2 & 3 & 2 & 2 \end{bmatrix}$ (d) 5500

- 43-47. Answers will vary. 49.
- $x = \pm 2$
- 51.
- $x = \pm \frac{3}{2}$

- 53.
- $x = 1 \pm \sqrt{2}$
- 55.
- $x = -4, -1$
- 57.
- $x = 1, \frac{1}{2}$

- 59.
- $x = 3$
- 61.
- $8uv - 1$
- 63.
- e^{5x}
- 65.
- $1 - \ln x$

67. True

69. Answers will vary.

Sample answer: $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -4 & 0 \\ 3 & 5 \end{bmatrix}$

$|A + B| = -30$, $|A| + |B| = -10$

71. (a) 6 (b) $\begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{6} \end{bmatrix}$ (c) $\frac{1}{6}$ (d) They are reciprocals.

73. (a) 2 (b) $\begin{bmatrix} -4 & -5 & 1.5 \\ -1 & -1 & 0.5 \\ -1 & -1 & 0 \end{bmatrix}$

(c) $\frac{1}{2}$ (d) They are reciprocals.

75. (a) Columns 2 and 3 are interchanged.

(b) Rows 1 and 3 are interchanged.

77. (a) 5 is factored from the first row.

(b) 4 and 3 are factored from the second and third columns, respectively.

79. $(x - 2)(x - 1)$ 81. $(2y - 3)^2$ 83. $(2, -4)$

Section 7.8 (page 567)

Vocabulary Check (page 567)

1. collinear 2. Cramer's Rule 3. cryptogram
4. uncoded, coded

1. $\frac{5}{2}$ 3. $\frac{33}{8}$ 5. 24 7. $x = 0, -\frac{16}{5}$ 9. Collinear

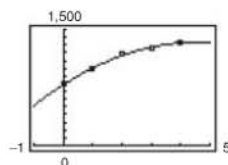
11. Not collinear 13. $x = 3$ 15. $(-3, -2)$

17. Not possible 19. $(\frac{32}{7}, \frac{30}{7})$ 21. $(-1, 3, 2)$

23. (a) and (b) $(0, -\frac{1}{2}, \frac{1}{2})$ 25. (a) and (b) $(1, -1, 2)$

27. (a) $y = -26.36t^2 + 241.5t + 784$

(b)



(c) Answers will vary.

29. Uncoded: $[3 \ 1 \ 12], [12 \ 0 \ 13], [5 \ 0 \ 20], [15 \ 13 \ 15], [18 \ 18 \ 15], [23 \ 0 \ 0]$

Encoded: $-68 \ 21 \ 35 \ -66 \ 14 \ 39 \ -115 \ 35 \ 60$
 $-62 \ 15 \ 32 \ -54 \ 12 \ 27 \ 23 \ -23 \ 0$

31. 38 63 51 -1 -14 -32 58 119 133 44 88 95

33. HAPPY NEW YEAR

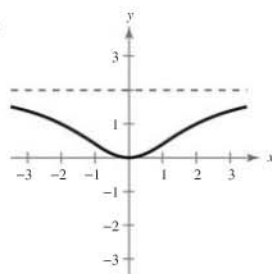
35. YANKEES WIN WORLD SERIES

37. True. Cramer's Rule divides by the determinant.

39. Answers will vary.

41. $x + 4y - 19 = 0$ 43. $2x - 7y - 27 = 0$

45.



Asymptote: $y = 2$

Review Exercises (page 570)

1. $(1, 1)$ 3. $(5, 4)$ 5. $(0, 0), (2, 8), (-2, 8)$

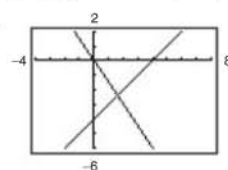
7. $(2, -0.5)$ 9. $(0, 0), (4, -4)$

11. $(-1, 2), (0.67, 2.56)$ 13. $(4, 4)$ 15. $(-1, 1)$

17. 4762 units 19. 96 meters $\times$ 144 meters 21. $(\frac{5}{2}, 3)$

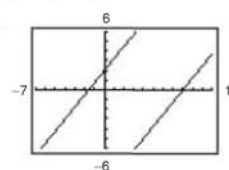
23. $(-\frac{1}{2}, \frac{4}{5})$ 25. $(0, 0)$ 27. $(\frac{14}{5} + \frac{8}{5}a, a)$

29.



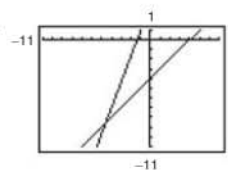
Consistent; $(1.6, -2.4)$

31.



Inconsistent

33.



Consistent; $(-4.6, -8.6)$

35. $(\frac{500,000}{7}, \frac{159}{7})$

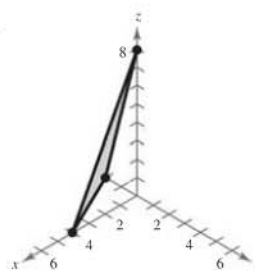
37. 218.75 miles per hour; 193.75 miles per hour

39. $(4, 3, -2)$ 41. $(\frac{38}{17}, \frac{40}{17}, -\frac{63}{17})$

43. $(3a + 4, 2a + 5, a)$ 45. $(-\frac{19}{6}, \frac{17}{12}, \frac{1}{3})$

47. $(a - 4, a - 3, a)$

49.



Sample answer: $(0, 0, 8), (0, -2, 0), (4, 0, 0), (1, -1, 2)$

$$51. (a) \frac{3}{x+2} - \frac{4}{x+4} \quad 53. \frac{1}{2} \left(\frac{3}{x-1} - \frac{x-3}{x^2+1} \right)$$

$$55. \frac{2x-1}{x^2+1} + \frac{-1}{x+2} \quad 57. y = 2x^2 + x - 5$$

59. Spray X: 10 gallons; spray Y: 5 gallons;
spray Z: 12 gallons

$$61. 3 \times 1 \quad 63. 1 \times 1$$

$$65. \begin{bmatrix} 3 & -10 & \vdots & 15 \\ 5 & 4 & \vdots & 22 \end{bmatrix}$$

$$67. \begin{bmatrix} 8 & -7 & 4 & \vdots & 12 \\ 3 & -5 & 2 & \vdots & 20 \\ 5 & 3 & -3 & \vdots & 26 \end{bmatrix}$$

$$69. \begin{cases} 5x + y + 7z = -9 \\ 4x + 2y = 10 \\ 9x + 4y + 2z = 3 \end{cases}$$

$$71. \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad 73. \begin{bmatrix} 1 & 0 & 3 & -2 \\ 0 & 1 & 4 & -3 \end{bmatrix}$$

$$75. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad 77. (10, -12) \quad 79. (-0.2, 0.7)$$

$$81. \left(\frac{1}{2}, -\frac{1}{3}, 1 \right) \quad 83. (3a + 1, -a, a) \quad 85. (2, -3, 3)$$

$$87. \left(1, 2, \frac{1}{2} \right) \quad 89. (3, 0, -4) \quad 91. (2, 6, -10, -3)$$

$$93. x = 12, y = -7 \quad 95. x = 1, y = 11$$

$$97. (a) \begin{bmatrix} 17 & -17 \\ 13 & 2 \end{bmatrix} \quad (b) \begin{bmatrix} -3 & 23 \\ -15 & 8 \end{bmatrix}$$

$$(c) \begin{bmatrix} 28 & 12 \\ -4 & 20 \end{bmatrix} \quad (d) \begin{bmatrix} 37 & -57 \\ 41 & -4 \end{bmatrix}$$

$$99. (a) \begin{bmatrix} 6 & 5 & 8 \\ 1 & 7 & 8 \\ 5 & 1 & 4 \end{bmatrix} \quad (b) \begin{bmatrix} 6 & -5 & 6 \\ 9 & -9 & -4 \\ 1 & 3 & 2 \end{bmatrix}$$

$$(c) \begin{bmatrix} 24 & 0 & 28 \\ 20 & -4 & 8 \\ 12 & 8 & 12 \end{bmatrix} \quad (d) \begin{bmatrix} 6 & 15 & 10 \\ -7 & 23 & 20 \\ 9 & -1 & 6 \end{bmatrix}$$

$$101. \begin{bmatrix} -13 & -8 & 18 \\ 0 & 11 & -19 \end{bmatrix} \quad 103. \begin{bmatrix} 9 & -7 \\ -9 & 4 \end{bmatrix}$$

$$105. \begin{bmatrix} 32 & -\frac{17}{2} & -\frac{3}{2} \\ 6 & 46 & 33 \end{bmatrix} \quad 107. \begin{bmatrix} -14 & -4 \\ 7 & -17 \\ -17 & -2 \end{bmatrix}$$

$$109. \frac{1}{3} \begin{bmatrix} 9 & 2 \\ -4 & 11 \\ 10 & 0 \end{bmatrix} \quad 111. \begin{bmatrix} 14 & -2 & 8 \\ 14 & -10 & 40 \\ 36 & -12 & 48 \end{bmatrix}$$

$$113. \begin{bmatrix} 11 & -6 \\ 18 & 39 \end{bmatrix} \quad 115. \begin{bmatrix} 14 & -22 & 22 \\ 19 & -41 & 80 \\ 42 & -66 & 66 \end{bmatrix}$$

$$117. \begin{bmatrix} 1 & 17 \\ 12 & 36 \end{bmatrix}$$

$$119. (a) \begin{bmatrix} 438.20 & 47.40 \\ 612.10 & 66.20 \\ 717.28 & 77.20 \end{bmatrix}$$

The dairy mart's sales and profits on milk for Friday, Saturday, and Sunday

(b) \$190.80

121. Answers will vary.

$$123. \begin{bmatrix} 4 & -5 \\ 5 & -6 \end{bmatrix} \quad 125. \begin{bmatrix} 13 & 6 & -4 \\ -12 & -5 & 3 \\ 5 & 2 & -1 \end{bmatrix}$$

$$127. \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ \frac{1}{10} & -\frac{1}{15} \end{bmatrix} \quad 129. \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & -1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$131. \begin{bmatrix} 1 & -1 \\ 4 & -\frac{7}{2} \end{bmatrix} \quad 133. \begin{bmatrix} -\frac{1}{2} & \frac{1}{4} \\ \frac{1}{20} & \frac{1}{40} \end{bmatrix} \quad 135. \left(1, -\frac{2}{5} \right)$$

$$137. (2, -1, -2) \quad 139. (1, -2, 1, 0) \quad 141. (-3, 1)$$

$$143. (1, 1, -2) \quad 145. -42 \quad 147. 550$$

$$149. (a) M_{11} = 4, M_{12} = 7, M_{21} = -1, M_{22} = 2$$

$$(b) C_{11} = 4, C_{12} = -7, C_{21} = 1, C_{22} = 2$$

$$151. (a) M_{11} = 30, M_{12} = -12, M_{13} = -21, M_{21} = 20,$$

$$M_{22} = 19, M_{23} = 22, M_{31} = 5, M_{32} = -2,$$

$$M_{33} = 19$$

$$(b) C_{11} = 30, C_{12} = 12, C_{13} = -21, C_{21} = -20,$$

$$C_{22} = 19, C_{23} = -22, C_{31} = 5, C_{32} = 2, C_{33} = 19$$

$$153. 130 \quad 155. -3 \quad 157. 279 \quad 159. -96$$

$$161. 16 \quad 163. 1.75 \quad 165. \frac{13}{2} \quad 167. 48$$

$$169. \text{Not collinear} \quad 171. (1, 2) \quad 173. (4, 7)$$

$$175. (-1, 4, 5) \quad 177. (0, -2.4, -2.6)$$

$$179. (a) \text{ and } (b) \left(\frac{53}{33}, -\frac{17}{33}, \frac{61}{66} \right)$$

181. Uncoded:

$$[9 \ 0 \ 8], [1 \ 22 \ 5], [0 \ 1 \ 0], [4 \ 18 \ 5], [1 \ 13 \ 0]$$

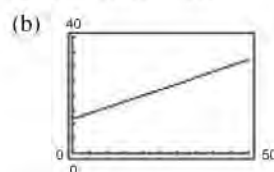
$$\text{Encoded: } -30, -2, 24, 38, 8, -51, 3, 0, -3, 32,$$

$$2, -39, 41, -2, -39$$

183. I WILL BE BACK

185. THAT IS MY FINAL ANSWER

$$187. (a) y = 0.41t + 11.74$$



(c) and (d) $t \approx 20.15$ or 2010

189. True

191. An $n \times n$ matrix A has an inverse A^{-1} if $\det(A) \neq 0$.

Chapter Test (page 576)

1. $(4, -2)$ 2. $(0, -1), (1, 0), (2, 1)$

3. $(8, 5), (2, -1)$ 4. $(\frac{28}{9}, -\frac{31}{9})$ 5. $(-\frac{2}{3}, -\frac{1}{2}, 1)$

6. $(1, 0, -2)$ 7. $y = -\frac{1}{2}x^2 + x + 6$

8. $\frac{5}{x-1} + \frac{3}{(x-1)^2}$ 9. $\frac{1}{x} + \frac{2}{x^2} - \frac{1}{x^2+1}$

10. $(-2a + 1.5, 2a + 1, a)$ 11. $(5, 2, -6)$

12. (a) $\begin{bmatrix} 1 & 0 & 4 \\ -7 & -6 & -1 \\ 0 & 4 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 15 & 12 & 12 \\ -12 & -12 & 0 \\ 3 & 6 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 7 & 4 & 12 \\ -18 & -16 & -2 \\ 1 & 10 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 36 & 20 & 4 \\ -28 & -24 & -4 \\ 10 & 8 & 2 \end{bmatrix}$

13. $\begin{bmatrix} -\frac{4}{3} & -\frac{5}{3} & 1 \\ -\frac{4}{3} & -\frac{8}{3} & 1 \\ \frac{1}{3} & \frac{2}{3} & 0 \end{bmatrix}, (-2, 3, -1)$ 14. 67 15. -2

16. 30 17. $(1, -\frac{1}{2})$

18. $x_1 = 700 - s - t, x_2 = 300 - s - t,$

$x_3 = s, x_4 = 100 - t, x_5 = t$

Chapter 8

Section 8.1 (page 587)

Vocabulary Check (page 587)

1. infinite sequence 2. terms 3. finite

4. recursively 5. factorial

6. summation notation

7. index, upper limit, lower limit 8. series

9. n th partial sum

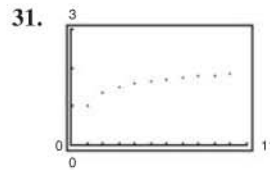
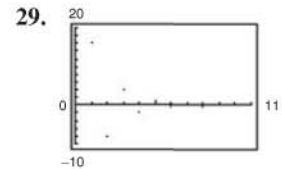
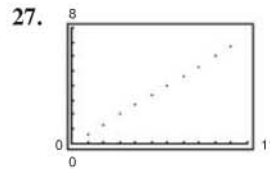
1. 7, 9, 11, 13, 15 3. 2, 4, 8, 16, 32

5. $-\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{32}$ 7. $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}$

9. $\frac{1}{2}, \frac{2}{5}, \frac{3}{10}, \frac{4}{17}, \frac{5}{26}$ 11. 0, 1, 0, $\frac{1}{2}$, 0 13. $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \frac{31}{32}$

15. $1, \frac{1}{2^{3/2}}, \frac{1}{3^{3/2}}, \frac{1}{4^{3/2}} = \frac{1}{8}, \frac{1}{5^{3/2}}$ 17. $-1, \frac{1}{4}, -\frac{1}{9}, \frac{1}{16}, -\frac{1}{25}$

19. 3, 15, 35, 63, 99 21. -73 23. $\frac{100}{101}$ 25. $\frac{64}{65}$



33. 9, 15, 21, 27, 33, 39, 45, 51, 57, 63

35. $3, \frac{5}{2}, \frac{7}{3}, \frac{9}{4}, \frac{11}{5}, \frac{13}{6}, \frac{15}{7}, \frac{17}{8}, \frac{19}{9}, \frac{21}{10}$

37. 0, 2, 0, 2, 0, 2, 0, 2, 0, 2

39. c 40. b 41. d 42. a

43. $a_n = 3n - 2$ 45. $a_n = n^2 - 1$ 47. $a_n = \frac{n+1}{n+2}$

49. $a_n = \frac{(-1)^{n+1}}{2^n}$ 51. $a_n = 1 + \frac{1}{n}$ 53. $a_n = \frac{1}{n!}$

55. $a_n = (-1)^n + 2(1)^n = (-1)^n + 2$

57. 28, 24, 20, 16, 12 59. 3, 4, 6, 10, 18

61. 6, 8, 10, 12, 14; $a_n = 2n + 4$

63. 81, 27, 9, 3, 1; $a_n = \frac{243}{3^n}$ 65. 1, 1, $\frac{1}{2}, \frac{1}{6}, \frac{1}{24}$

67. $1, \frac{1}{3}, \frac{2}{5}, \frac{6}{7}, \frac{8}{3}$ 69. $1, \frac{1}{2}, \frac{1}{24}, \frac{1}{720}, \frac{1}{40,320}$ 71. $\frac{1}{12}$

73. 495 75. $n + 1$ 77. $\frac{1}{2n(2n+1)}$ 79. 35

81. 40 83. 30 85. $\frac{9}{5}$ 87. 238 89. 30

91. 81 93. $\frac{47}{60}$ 95. $\sum_{i=1}^9 \frac{1}{3i} \approx 0.94299$

97. $\sum_{i=1}^8 \left[2\left(\frac{i}{8}\right) + 3 \right] = 33$ 99. $\sum_{i=1}^6 (-1)^{i+1} 3^i = -546$

101. $\sum_{i=1}^{20} \frac{(-1)^{i+1}}{i^2} \approx 0.821$ 103. $\sum_{i=1}^5 \frac{2^i - 1}{2^{i+1}} \approx 2.0156$

105. $\frac{75}{16}$ 107. $-\frac{3}{2}$ 109. (a) $\frac{3333}{5000}$ (b) $\frac{2}{3}$

111. (a) $\frac{1111}{10,000}$ (b) $\frac{1}{9}$

113. (a) $A_1 = \$5037.50, A_2 = \$5075.28,$

$A_3 = \$5113.35, A_4 = \$5151.70,$

$A_5 = \$5190.33, A_6 = \$5229.26,$

$A_7 = \$5268.48, A_8 = \5307.99

(b) \$6741.74

115. (a) $p_0 = 5500, p_n = (0.75)p_{n-1} + 500$

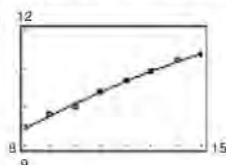
(b) 4625; 3969; 3477

(c) 2000 trout; Answers will vary. Sample answer: As time passes, the population of trout decreases at a decreasing rate. Because the population is growing smaller and still declines 25%, each time 25% is taken from a smaller number, there is a smaller decline in the number of trout.

117. (a) $a_0 = 50, a_n = a_{n-1}(1.005) + 50$

(b) \$616.78 (c) \$2832.26

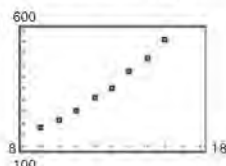
119. (a)



(b) 2010: \$12.25; 2015: \$12.64

(c) Answers will vary. (d) 2008

121. (a)



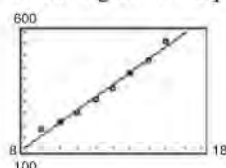
(b) Linear regression sequence:

$R_n = 54.58n - 336.3; 0.98656$

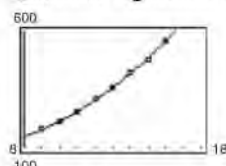
Quadratic regression sequence:

$R_n = 3.088n^2 - 22.62n + 130.0; 0.99919$

(c) Linear regression sequence:



Quadratic regression sequence:



(d) Quadratic regression sequence; the coefficient of determination for the quadratic regression sequence is closer to 1.

(e) 2010: \$912,850,000; 2015: \$1,494,560,000

(f) 2010

123. True 125. 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144;

$$1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \frac{21}{13}, \frac{34}{21}, \frac{55}{34}, \frac{89}{55}$$

127. 1, 1, 2, 3, 5

129. $a_{n+1} = \frac{1}{2}a_n + \frac{(1 + \sqrt{5})^n + (1 - \sqrt{5})^n}{2^{n+1}}$

$$a_{n+2} = \frac{3}{2}a_n + \frac{(1 + \sqrt{5})^n + (1 - \sqrt{5})^n}{2^{n+1}}$$

131. $x, \frac{x^2}{2}, \frac{x^3}{6}, \frac{x^4}{24}, \frac{x^5}{120}$ 133. $-\frac{x^3}{3}, \frac{x^5}{5}, -\frac{x^7}{7}, \frac{x^9}{9}, -\frac{x^{11}}{11}$

135. $-\frac{x^2}{2}, \frac{x^4}{24}, -\frac{x^6}{720}, \frac{x^8}{40,320}, -\frac{x^{10}}{3,628,800}$

137. $-x, \frac{x^2}{2}, -\frac{x^3}{6}, \frac{x^4}{24}, -\frac{x^5}{120}$

139. $x + 1, -\frac{(x+1)^2}{2}, \frac{(x+1)^3}{6}, -\frac{(x+1)^4}{24}, \frac{(x+1)^5}{120}$

141. $\frac{1}{4}, \frac{1}{12}, \frac{1}{24}, \frac{1}{40}, \frac{1}{60}, \frac{1}{2}, -\frac{1}{2n+2}$

143. $\frac{1}{6}, \frac{1}{12}, \frac{1}{20}, \frac{1}{30}, \frac{1}{42}, \frac{1}{2}, -\frac{1}{n+2}$

145. 0, $\ln 2$, $\ln 3$, $\ln 4$, $\ln 5$; $\ln(n!)$

147. (a) $\begin{bmatrix} 8 & 1 \\ -3 & 7 \end{bmatrix}$ (b) $\begin{bmatrix} -22 & -7 \\ 3 & -18 \end{bmatrix}$

(c) $\begin{bmatrix} 18 & 9 \\ 18 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 6 \\ 27 & 18 \end{bmatrix}$

149. (a) $\begin{bmatrix} -3 & -7 & 4 \\ 4 & 4 & 1 \\ 1 & 4 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & 17 & -14 \\ -12 & -13 & -9 \\ -3 & -15 & -10 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 7 & -16 \\ 4 & 42 & 45 \\ 1 & 23 & 48 \end{bmatrix}$ (d) $\begin{bmatrix} 16 & 31 & 42 \\ 10 & 47 & 31 \\ 13 & 22 & 25 \end{bmatrix}$

Section 8.2 (page 598)

Vocabulary Check (page 598)

1. arithmetic, common 2. $a_n = dn + c$ 3. n th partial sum1. Arithmetic sequence, $d = -2$ 3. Arithmetic sequence, $d = -\frac{1}{2}$ 5. Arithmetic sequence, $d = 8$ 7. Arithmetic sequence, $d = 0.6$

9. 21, 34, 47, 60, 73

Arithmetic sequence, $d = 13$

11. $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}$

Not an arithmetic sequence

13. 143, 136, 129, 122, 115

Arithmetic sequence, $d = -7$

15. 1, 5, 1, 5, 1

Not an arithmetic sequence

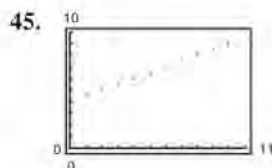
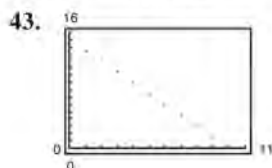
17. $a_n = -2 + 3n$ 19. $a_n = 108 - 8n$ 21. $a_n = \frac{13}{2} - \frac{5}{2}n$ 23. $a_n = \frac{10}{3}n + \frac{5}{3}$ 25. $a_n = 103 - 3n$ 27. 5, 11, 17, 23, 29

29. -10, -22, -34, -46, -58 31. -2, 2, 6, 10, 14

33. 22.45, 20.725, 19, 17.275, 15.55

35. 15, 19, 23, 27, 31; $d = 4$; $a_n = 11 + 4n$ 37. $\frac{3}{5}, \frac{1}{2}, \frac{2}{5}, \frac{3}{10}, \frac{1}{5}$; $d = -\frac{1}{10}$; $a_n = -\frac{1}{10}n + \frac{7}{10}$

39. 59 41. 18.6



47. -1, 3, 7, 11, 15, 19, 23, 27, 31, 35

49. 19.25, 18.5, 17.75, 17, 16.25, 15.5, 14.75, 14, 13.25, 12.5

51. 1.55, 1.6, 1.65, 1.7, 1.75, 1.8, 1.85, 1.9, 1.95, 2

53. 110 55. -25 57. 2550 59. -4585

61. 620 63. 41 65. 4000 67. 1275

69. 25,250 71. 355 73. 129,250 75. 440

77. 2575 79. 14,268 81. 405 bricks

83. \$150,000

85. (a) $a_n = 0.91n + 5.7$

(b)

Year	1997	1998	1999	2000
Sales (in billions of dollars)	12.1	13.0	13.9	14.8

Year	2001	2002	2003	2004
Sales (in billions of dollars)	15.7	16.6	17.5	18.4

The model fits the data.

(c) \$122.0 billion

(d) \$180.3 billion; Answers will vary.

87. True. Given a_1 and a_2 , you know that $d = a_2 - a_1$. Hence, $a_n = a_1 + (n - 1)d$.89. $x, 3x, 5x, 7x, 9x, 11x, 13x, 15x, 17x, 19x$ 91. 4

93. (a) -7, -4, -1, 2, 5, 8, 11

$$a_1 = -7, a_{n+1} = a_n + 3$$

(b) 17, 23, 29, 35, 41, 47, 53, 59

$$a_1 = 17, a_{n+1} = a_n + 6$$

(c) Not possible

(d) 4, 7.5, 11, 14.5, 18, 21.5, 25, 28.5

$$a_1 = 4, a_{n+1} = a_n + 3.5$$

(e) Not possible

Answers will vary.

95. 20,100 97. 2601 99. (1, 5, -1)

101. 15 square units 103. Answers will vary.

Section 8.3 (page 607)

Vocabulary Check (page 607)

1. geometric, common 2. $a_n = a_1 r^{n-1}$

3. $S_n = \sum_{i=1}^n a_1 r^{i-1} = a_1 \left(\frac{1-r^n}{1-r} \right)$

4. geometric series 5. $S = \sum_{i=0}^{\infty} a_1 r^i = \frac{a_1}{1-r}$

1. Geometric sequence, $r = 3$

3. Not a geometric sequence

5. Geometric sequence, $r = -\frac{1}{2}$ 7. Geometric sequence, $r = 2$

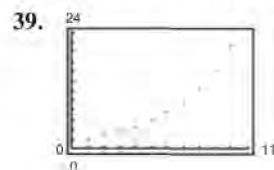
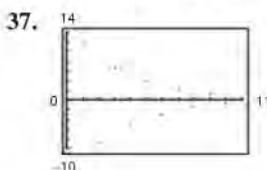
9. Not a geometric sequence 11. 6, 18, 54, 162, 486

13. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$ 15. 5, $-\frac{1}{2}, \frac{1}{20}, -\frac{1}{200}, \frac{1}{2000}$ 17. 1, e, e^2, e^3, e^4 19. 64, 32, 16, 8, 4; $r = \frac{1}{2}$; 21. 9, 18, 36, 72, 144; $r = 2$;

$$a_n = 64\left(\frac{1}{2}\right)^{n-1} \quad a_n = 9(2)^{n-1}$$

23. 6, -9, $\frac{27}{2}, -\frac{81}{4}, \frac{243}{8}$; $r = -\frac{3}{2}$;

$$a_n = 6\left(-\frac{3}{2}\right)^{n-1}$$

25. $\frac{1}{128}$ 27. $-\frac{2}{3^{10}}$ 29. $500(1.02)^{13}$ 31. $-\frac{2}{9}$ 33. $a_n = 7(3)^{n-1}$; 45,927 35. $a_n = 5(6)^{n-1}$; 50,388,480

41. 8, 4, 6, 5

43.	n	S_n	45. 511	47. 43
	1	16		
	2	24		
	3	28		
	4	30		
	5	31		
	6	31.5		
	7	31.75		
	8	31.875		
	9	31.9375		
	10	31.96875		

49. 29,921.31 51. 6.4 53. 2092.60

55. $\sum_{n=1}^7 5(3)^{n-1}$ 57. $\sum_{n=1}^7 2\left(-\frac{1}{4}\right)^{n-1}$ 59. 50 61. $\frac{10}{3}$

63. Series does not have a finite sum because $\left|\frac{7}{3}\right| > 1$.

65. $\frac{1000}{89}$ 67. $-\frac{30}{19}$ 69. 32 71. $\frac{9}{4}$

73. $\frac{4}{11}$ 75. $\frac{113}{90}$

77. (a) \$1343.92 (b) \$1346.86 (c) \$1348.35
(d) \$1349.35 (e) \$1349.84

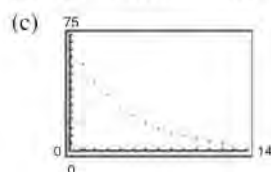
79. \$6480.83 81. Answers will vary.

83. (a) \$26,198.27 (b) \$26,263.88

85. (a) \$153,237.86 (b) \$153,657.02

87. 126 square inches

89. (a) $T_n = 70(0.8)^n$ (b) 18.4°F; 4.8°F



3.5 hours

91. \$1600 93. \$1250 95. \$2181.82

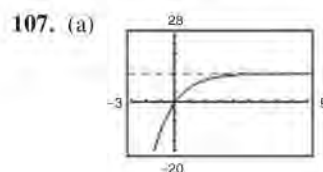
97. (a) Option 1. Answers will vary. Sample answer: You make a cumulative amount of \$157,689.86 from option 1 and \$169,131.31 from option 2.

(b) Option 2. Answers will vary. Sample answer: You make about \$33,114.39 from option 1 and \$35,179.05 from option 2 the year prior to re-evaluation.

99. (a) 3208.53 feet; 2406.40 feet; 5614.93 feet total
(b) 5950 feet

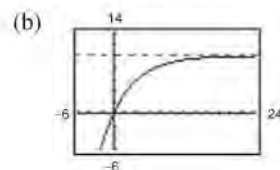
101. False. A sequence is geometric if the ratios of consecutive terms are the same.

103. $3, \frac{3x}{2}, \frac{3x^2}{4}, \frac{3x^3}{8}, \frac{3x^4}{16}$ 105. $100e^{8x}$



Horizontal asymptote: $y = 12$

Corresponds to the sum of the series



Horizontal asymptote: $y = 10$

Corresponds to the sum of the series

109. Divide the second term by the first to obtain the common ratio. The n th term is the first term times the common ratio raised to the $(n - 1)$ th power.

111. 45.65 miles per hour 113. -102

115. Answers will vary.

Section 8.4 (page 617)

Vocabulary Check (page 617)

1. mathematical induction
2. first
3. arithmetic
4. second

1. $\frac{5}{(k+1)(k+2)}$ 3. $\frac{2^{k+1}}{(k+2)!}$

5. $1 + 6 + 11 + \cdots + (5k - 4) + (5k + 1)$

7-19. Answers will vary. 21. 1,625,625 23. 572

25-41. Answers will vary.

43. 0, 3, 6, 9, 12

First differences: 3, 3, 3, 3

Second differences: 0, 0, 0

Linear

45. 3, 1, -2, -6, -11

First differences: -2, -3, -4, -5

Second differences: -1, -1, -1

Quadratic

47. 0, 1, 3, 6, 10

First differences: 1, 2, 3, 4

Second differences: 1, 1, 1

Quadratic

49. 2, 4, 6, 8, 10

First differences: 2, 2, 2, 2

Second differences: 0, 0, 0

Linear

51. $a_n = n^2 - 3n + 5$, $n \geq 1$ 53. $a_n = \frac{1}{2}n^2 + n - 3$

55. (a) $a_n = 3(4)^{n-1}$

(b) $a_n = \frac{\sqrt{3}}{4} \left[1 + \frac{1}{3} \sum_{k=2}^n \left(\frac{4}{9} \right)^{k-2} \right]$ (c) $3 \left(\frac{4}{3} \right)^{n-1}$

57. False. Not necessarily

59. False. It has $n - 2$ second differences.

61. $4x^4 - 4x^2 + 1$ 63. $-64x^3 + 240x^2 - 300x + 125$

65. $7\sqrt{3}i$ 67. $40(1 + \sqrt[3]{2})$

Section 8.5 (page 624)

Vocabulary Check (page 624)

1. binomial coefficients

2. Binomial Theorem, Pascal's Triangle

3. ${}_nC_r$ or $\binom{n}{r}$ 4. expanding, binomial

1. 21 3. 1 5. 15,504 7. 14 9. 4950

11. 749,398 13. 4950 15. 31,125

17. $x^4 + 8x^3 + 24x^2 + 32x + 16$

19. $a^3 + 9a^2 + 27a + 27$

21. $y^4 - 8y^3 + 24y^2 - 32y + 16$

23. $x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$

25. $729r^6 + 2916r^5s + 4860r^4s^2 + 4320r^3s^3 + 2160r^2s^4 + 576rs^5 + 64s^6$

27. $x^5 - 5x^4y + 10x^3y^2 - 10x^2y^3 + 5xy^4 - y^5$

29. $1 - 12x + 48x^2 - 64x^3$

31. $x^8 + 8x^6 + 24x^4 + 32x^2 + 16$

33. $x^{10} - 25x^8 + 250x^6 - 1250x^4 + 3125x^2 - 3125$

35. $x^8 + 4x^6y^2 + 6x^4y^4 + 4x^2y^6 + y^8$

37. $x^{18} - 6x^{15}y + 15x^{12}y^2 - 20x^9y^3 + 15x^6y^4 - 6x^3y^5 + y^6$

39. $\frac{1}{x^5} + \frac{5y}{x^4} + \frac{10y^2}{x^3} + \frac{10y^3}{x^2} + \frac{5y^4}{x} + y^5$

41. $\frac{16}{x^4} - \frac{32y}{x^3} + \frac{24y^2}{x^2} - \frac{8y^3}{x} + y^4$

43. $-512x^4 + 576x^3 - 240x^2 + 44x - 3$

45. $2x^4 - 24x^3 + 113x^2 - 246x + 207$

47. $-4x^6 - 24x^5 - 60x^4 - 83x^3 - 42x^2 - 60x + 20$

49. $61,440x^7$ 51. $360x^3y^2$ 53. $1,259,712x^2y^7$

55. $32,476,950,000x^4y^8$ 57. 3,247,695 59. 180

61. $-489,888$ 63. 210 65. 21 67. 6

69. $81t^4 - 216t^3v + 216t^2v^2 - 96tv^3 + 16v^4$

71. $32x^5 - 240x^4y + 720x^3y^2 - 1080x^2y^3 + 810xy^4 - 243y^5$

73. $x^2 + 20x^{3/2} + 150x + 500x^{1/2} + 625$

75. $x^2 - 3x^{4/3}y^{1/3} + 3x^{2/3}y^{2/3} - y$

77. $3x^2 + 3xh + h^2$, $h \neq 0$

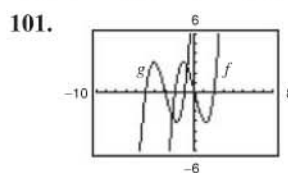
79. $6x^5 + 15x^4h + 20x^3h^2 + 15x^2h^3 + 6xh^4 + h^5$, $h \neq 0$

81. $\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{1}{\sqrt{x+h} + \sqrt{x}}$, $h \neq 0$

83. -4 85. $161 + 240i$ 87. $2035 + 828i$

89. $-115 + 236i$ 91. $-23 + 208\sqrt{3}i$

93. 1 95. $-\frac{1}{8}$ 97. 1.172 99. 510,568.785

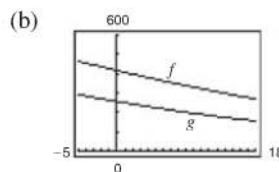
 g is shifted three units to the left of f .

$g(x) = x^3 + 9x^2 + 23x + 15$

103. 105. 0.273 107. 0.171

 $p(x)$ is the expansion of $f(x)$.

109. (a) $g(t) = 0.064t^2 - 6.74t + 256.1$, $-15 \leq t \leq 3$

111. False. The correct term is $126,720x^4y^8$.

113. The first and last numbers in each row are 1. Every other number in each row is formed by adding the two numbers immediately above the number.

115. The terms of the expansion of $(x - y)^n$ alternate between positive and negative.

117 and 119. Answers will vary.

121. $g(x)$ is shifted eight units up from $f(x)$.123. $g(x)$ is the reflection of $f(x)$ in the y -axis.

125. $\begin{bmatrix} 4 & -5 \\ 5 & -6 \end{bmatrix}$

Section 8.6 (page 634)

Vocabulary Check (page 634)

1. Fundamental Counting Principle
2. permutation
3. ${}_nP_r = \frac{n!}{(n-r)!}$
4. distinguishable permutations
5. combinations

1. 6
3. 5
5. 3
7. 7
9. 120
11. 1024
13. (a) 900 (b) 648
15. 16,000,000
17. (a) 35,152 (b) 3902
19. (a) 100,000 (b) 20,000
21. (a) 720 (b) 48
23. 24
25. 336
27. 120
29. 27,907,200
31. 197,149,680
33. 120
35. 362,880
37. 11,880
39. 50,653
41. ABCD, ABDC, ACBD, ACDB, ADBC, ADCB, BACD, BADC, CABD, CADB, DABC, DACB, BCAD, BDAC, CBAD, CDAB, DBAC, DCAB, BCDA, BDCA, CBDA, CDBA, DBCA, DCBA
43. 420
45. 2520
47. 10
49. 4
51. 1
53. 4845
55. 850,668
57. AB, AC, AD, AE, AF, BC, BD, BE, BF, CD, CE, CF, DE, DF, EF
59. $4.418694268 \times 10^{16}$
61. 13,983,816
63. 36
65. 3744
67. (a) 495 (b) 210
69. 292,600
71. 5
73. 20
75. $n = 5$ or $n = 6$
77. $n = 10$
79. $n = 3$
81. $n = 2$
83. False.
85. For some calculators the answer is too large.
87. They are equal.
- 89 and 91. Answers will vary.
93. 8.303
95. 35
97. $(-2, -8)$
99. $(-1, 1)$

Section 8.7 (page 645)

Vocabulary Check (page 645)

1. experiment, outcomes
2. sample space
3. probability
4. impossible, certain
5. mutually exclusive
6. independent
7. complement
8. (a) iii (b) i (c) iv (d) ii

1. $\{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$
3. $\{ABC, ACB, BAC, BCA, CAB, CBA\}$
5. $\{(A, B), (A, C), (A, D), (A, E), (B, C), (B, D), (B, E), (C, D), (C, E), (D, E)\}$
7. $\frac{3}{8}$
9. $\frac{7}{8}$
11. $\frac{3}{13}$
13. $\frac{4}{13}$
15. $\frac{5}{36}$
17. $\frac{11}{12}$
19. $\frac{1}{5}$
21. $\frac{2}{5}$
23. 0.25
25. $\frac{1}{3}$
27. 0.88
29. $\frac{7}{20}$
31. (a) 1.22 million (b) 0.41 (c) 0.24 (d) 0.26
33. (a) 37.59 million (b) 0.01 (c) 0.012
35. (a) 0.45 (b) 0.45 (c) 0.23
37. (a) $\frac{672}{1254}$ (b) $\frac{582}{1254}$ (c) $\frac{548}{1254}$
39. $P(\{\text{Taylor wins}\}) = \frac{1}{2}$
 $P(\{\text{Moore wins}\}) = P(\{\text{Perez wins}\}) = \frac{1}{4}$
41. (a) $\frac{21}{1292} \approx 0.016$ (b) $\frac{225}{646} \approx 0.348$ (c) $\frac{49}{323} \approx 0.152$
43. (a) $\frac{1}{120}$ (b) $\frac{1}{24}$
45. (a) 0.008875 (b) $\frac{3}{8788}$
47. (a) 6.84×10^{-7} (b) 6.84×10^{-6}
49. (a) $\frac{5}{13}$ (b) $\frac{1}{2}$ (c) $\frac{4}{13}$
51. (a) $\frac{14}{55}$ (b) $\frac{12}{55}$ (c) $\frac{54}{55}$
53. 0.1024
55. (a) 0.9702 (b) 0.9998 (c) 0.0002
57. (a) $\frac{1}{15,625}$ (b) $\frac{4096}{15,625}$ (c) $\frac{11,529}{15,625}$
59. (a) $\frac{\pi}{4}$ (b) Answers will vary.
61. True
63. (a) As you consider successive people with distinct birthdays, the probabilities must decrease to take into account the birth dates already used. Because the birth dates of people are independent events, multiply the respective probabilities of distinct birthdays.
 (b) $\frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365}$
 (c) Answers will vary.
 (d) Q_n is the probability that the birthdays are *not* distinct, which is equivalent to at least two people having the same birthday.
 (e)

n	10	15	20	23	30	40	50
P_n	0.88	0.75	0.59	0.49	0.29	0.11	0.03
Q_n	0.12	0.25	0.41	0.51	0.71	0.89	0.97

- (f) 23
65. $x = \frac{11}{2}$
67. $x = -10$
69. $x = \ln 28 \approx 3.332$
71. $x = \frac{1}{6}e^4 \approx 9.100$
73. 60
75. 6,652,800
77. 15
79. 165

Review Exercises (page 651)

1. $\frac{2}{3}, \frac{4}{5}, \frac{8}{9}, \frac{16}{17}, \frac{32}{33}$ 3. $-1, \frac{1}{2}, -\frac{1}{6}, \frac{1}{24}, -\frac{1}{120}$ 5. $a_n = 5n$

7. $a_n = \frac{2}{2n-1}$ 9. 9, 5, 1, -3, -7 11. $\frac{1}{380}$

13. $(n+1)(n)$ 15. 30 17. $\frac{205}{24}$ 19. 51,005,000

21. 43,078 23. $\sum_{k=1}^{20} \frac{1}{2k} = 1.799$

25. $\sum_{k=1}^9 \frac{k}{k+1} = 7.071$ 27. (a) $\frac{1111}{2000}$ (b) $\frac{5}{9}$

29. (a) $\frac{15}{8}$ (b) 2

31. (a) 2512.50, 2525.06, 2537.69, 2550.38, 2563.13,
2575.94, 2588.82, 2601.77
(b) \$3051.99

33. Arithmetic sequence, $d = -2$

35. Arithmetic sequence, $d = \frac{1}{2}$

37. 3, 7, 11, 15, 19 39. 1, 4, 7, 10, 13

41. 35, 32, 29, 26, 23; $d = -3$;
 $a_n = 38 - 3n$

43. 9, 16, 23, 30, 37; $d = 7$;
 $a_n = 2 + 7n$

45. $a_n = 103 - 3n$; 1430 47. 80 49. 88

51. 25,250 53. (a) \$43,000 (b) \$192,500

55. Geometric sequence, $r = 2$

57. Geometric sequence, $r = -\frac{1}{3}$

59. 4, -1, $\frac{1}{4}$, $-\frac{1}{16}$, $\frac{1}{64}$ 61. 9, 6, 4, $\frac{8}{3}$, $\frac{16}{9}$ or 9, -6, 4, $-\frac{8}{3}$, $\frac{16}{9}$

63. 120, 40, $\frac{40}{3}$, $\frac{40}{9}$, $\frac{40}{27}$; $r = \frac{1}{3}$;
 $a_n = 120\left(\frac{1}{3}\right)^{n-1}$

65. 25, -15, 9, $-\frac{27}{5}$, $\frac{81}{25}$; $r = -\frac{3}{5}$;
 $a_n = 25\left(-\frac{3}{5}\right)^{n-1}$

67. $a_n = 16\left(-\frac{1}{2}\right)^{n-1}$; 10.67

69. $a_n = 100(1.05)^{n-1}$; 3306.60 71. 127 73. 3277

75. 1301.01 77. 24.85 79. 32 81. 12

83. (a) $a_t = 120,000(0.7)^t$ (b) \$20,168.40

85 and 87. Answers will vary. 89. 465 91. 4648

93. 5, 10, 15, 20, 25

First differences: 5, 5, 5, 5

Second differences: 0, 0, 0

Linear model

95. 16, 15, 14, 13, 12

First differences: -1, -1, -1, -1

Second differences: 0, 0, 0

Linear model

97. 45 99. 126 101. 20 103. 70

105. $x^4 + 20x^3 + 150x^2 + 500x + 625$

107. $a^5 - 20a^4b + 160a^3b^2 - 640a^2b^3 + 1280ab^4 - 1024b^5$

109. $1241 + 2520i$ 111. 10

113. (a) 216 (b) 108 (c) 36 115. 45

117. 239,500,800 119. 4950 121. 999,000

123. 5040 125. 3,628,800 127. 15,504

129. $n = 3$ 131. $\frac{1}{9}$

133. (a) 0.416 (b) 0.8 (c) 0.074 135. 0.0475

137. True. $\frac{(n+2)!}{n!} = \frac{(n+2)(n+1)n!}{n!} = (n+2)(n+1)$

139. (a) Each term is obtained by adding the same constant (common difference) to the preceding term.

(b) Each term is obtained by multiplying the same constant (common ratio) by the preceding term.

141. (a) Arithmetic. There is a constant difference between consecutive terms.

(b) Geometric. Each term is a constant multiple of the preceding term. In this case the common ratio is greater than 1.

143. Each term of the sequence is defined using a previous term or terms.

145. If n is even, the expressions are the same. If n is odd, the expressions are negatives of each other.

Chapter Test (page 655)

1. $1, -\frac{2}{3}, \frac{4}{9}, -\frac{8}{27}, \frac{16}{81}$ 2. 12, 16, 20, 24, 28

3. $-x, \frac{x^2}{2}, -\frac{x^3}{3}, \frac{x^4}{4}, -\frac{x^5}{5}$

4. $-\frac{x^3}{6}, -\frac{x^5}{120}, -\frac{x^7}{5040}, -\frac{x^9}{362,880}, -\frac{x^{11}}{39,916,800}$

5. 7920 6. $\frac{1}{n+1}$ 7. $2n$ 8. $a_n = n^2 + 1$

9. $a_n = 5100 - 100n$ 10. $a_n = 4\left(\frac{1}{2}\right)^{n-1}$

11. $\sum_{n=1}^{12} \frac{2}{3n+1}$ 12. $\sum_{n=1}^{\infty} 2\left(\frac{1}{4}\right)^{n-1}$ 13. 189

14. 28.80 15. $\frac{25}{7}$ 16. Answers will vary.

17. $16a^4 - 160a^3b + 600a^2b^2 - 1000ab^3 + 625b^4$

18. 84 19. 1140 20. 72 21. 328,440

22. $n = 3$ 23. 26,000 24. 12,650 25. $\frac{3}{26}$

26. $\frac{1}{462}$ 27. (a) $\frac{1}{4}$ (b) $\frac{121}{3600}$ (c) $\frac{1}{60}$ 28. 0.25

Chapter 9

Section 9.1 (page 667)

Vocabulary Check (page 667)

1. conic section 2. locus 3. circle, center
 4. parabola, directrix, focus 5. vertex
 6. axis 7. tangent

1. $x^2 + y^2 = 18$ 3. $(x - 3) + (y - 7)^2 = 53$

5. $(x + 3)^2 + (y + 1)^2 = 7$

7. Center: (0, 0)

Radius: 7

11. Center: (1, 0)

Radius: $\sqrt{15}$

15. $x^2 + y^2 = \frac{3}{4}$

Center: (0, 0)

Radius: $\frac{\sqrt{3}}{2}$

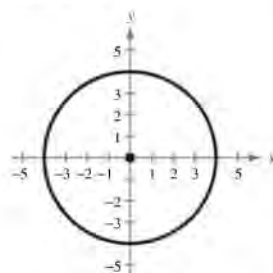
17. $(x - 1)^2 + (y + 3)^2 = 1$

Center: (1, -3)

Radius: 1

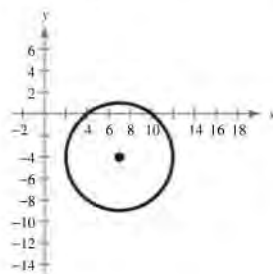
21. Center: (0, 0)

Radius: 4



25. Center: (7, -4)

Radius: 5



9. Center: (-2, 7)

Radius: 4

13. $x^2 + y^2 = 4$

Center: (0, 0)

Radius: 2

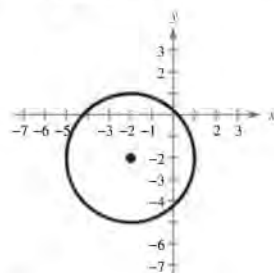
19. $(x + \frac{3}{2})^2 + (y - 3)^2 = 1$

Center: $(-\frac{3}{2}, 3)$

Radius: 1

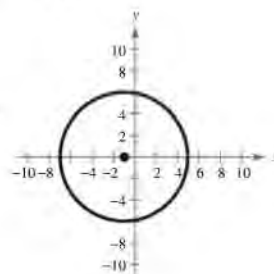
23. Center: (-2, -2)

Radius: 3



27. Center: (-1, 0)

Radius: 6



29. x-intercept: (2, 0)

y-intercepts:

$(0, -3 \pm \sqrt{5})$

33. x-intercept:

$(6 \pm \sqrt{7}, 0)$

y-intercept: none

37. e

38. b

39. d

40. f

41. a

42. c

43. $x^2 = \frac{3}{2}y$

45. $x^2 = -6y$

47. $y^2 = -8x$

49. $x^2 = 4y$

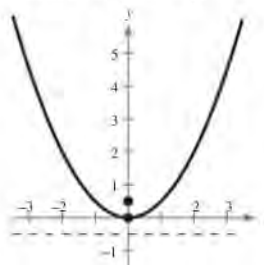
51. $y^2 = -8x$

53. $y^2 = 9x$

55. Vertex: (0, 0)

Focus: $(0, \frac{1}{2})$

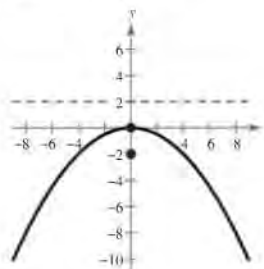
Directrix: $y = -\frac{1}{2}$



59. Vertex: (0, 0)

Focus: (0, -2)

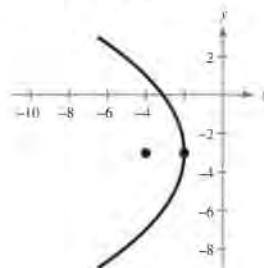
Directrix: $y = 2$



63. Vertex: (-2, -3)

Focus: (-4, -3)

Directrix: $x = 0$



31. x-intercepts:

$(1 \pm 2\sqrt{2}, 0)$

y-intercepts:

$(0, 9), (0, -3)$

35. (a) $x^2 + y^2 = 6561$

(b) Yes

(c) 6 miles

36. f

38. b

40. f

42. c

44. $x^2 = \frac{3}{2}y$

46. $x^2 = -6y$

48. $y^2 = -8x$

50. $x^2 = 4y$

52. $y^2 = -8x$

54. $y^2 = 9x$

56. Vertex: (0, 0)

Focus: $(-\frac{3}{2}, 0)$

Directrix: $x = \frac{3}{2}$

58. Vertex: (0, 0)

Focus: (0, -2)

Directrix: $y = 2$

60. Vertex: (-1, -3)

Focus: (-1, -5)

Directrix: $y = -1$

62. Vertex: (-2, -3)

Focus: (-4, -3)

Directrix: $x = 0$

64. Vertex: $(-\frac{3}{2}, 2)$

Focus: $(-\frac{3}{2}, 3)$

Directrix: $y = 1$

66. Vertex: (0, 0)

Focus: (0, 2)

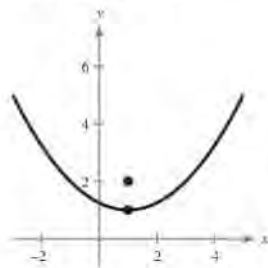
Directrix: $y = -2$

68. Vertex: (0, 0)

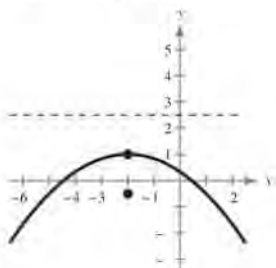
Focus: (0, 2)

Directrix: $y = -2$

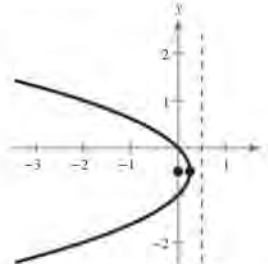
67. Vertex: $(1, 1)$
Focus: $(1, 2)$
Directrix: $y = 0$



69. Vertex: $(-2, 1)$
Focus: $(-2, -\frac{1}{2})$
Directrix: $y = \frac{5}{2}$

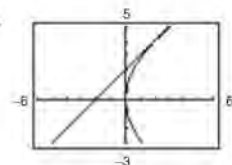


71. Vertex: $(\frac{1}{4}, -\frac{1}{2})$
Focus: $(0, -\frac{1}{2})$
Directrix: $x = \frac{1}{2}$



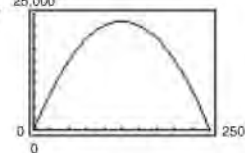
73. $(x-3)^2 = -(y-1)$ 75. $y^2 = 2(x+2)$
77. $(y-2)^2 = -8(x-5)$ 79. $x^2 = 8(y-4)$
81. $(y-2)^2 = 8x$

83. $(2, 4)$



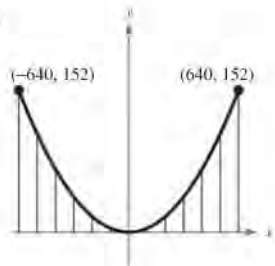
85. $4x - y - 8 = 0$; $(2, 0)$ 87. $4x - y + 2 = 0$; $(-\frac{1}{2}, 0)$

89. $x = 125$ televisions



91. (a) $y^2 = 6x$ (b) 2.67 inches

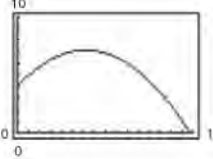
93. (a) (b) $x^2 = \frac{51,200}{19}y$



(c)

x	0	200	400	500	600
y	0	14,844	59,375	92,773	133,59

95. $y^2 = 640x$

97. (a)  (b) Highest point: 7.125 feet
Distance: 15.687 feet

99. $y = \frac{3}{4}x - \frac{25}{4}$ 101. $y = \frac{\sqrt{2}}{2}x - 3\sqrt{2}$

103. False. $x^2 + (y+5)^2 = 25$ represents a circle with its center at $(0, -5)$ and a radius of 5.

105. False. A circle is a conic section. 107. True

109. The resulting surface has the property that all incoming rays parallel to the axis are reflected through the focus of the parabola. Graphical representations will vary.

111. $y = \sqrt{6(x+1)} + 3$

113. Minimum: $(0.67, 0.22)$; maximum: $(-0.67, 3.78)$

115. Minimum: $(-0.79, 0.81)$

Section 9.2 (page 677)

Vocabulary Check (page 677)

1. ellipse 2. major axis, center
3. minor axis 4. eccentricity

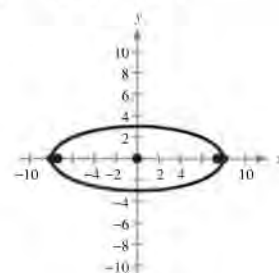
1. b 2. c 3. d 4. f 5. a 6. e

7. Center: $(0, 0)$

Vertices: $(\pm 8, 0)$

Foci: $(\pm \sqrt{55}, 0)$

Eccentricity: $\frac{\sqrt{55}}{8}$

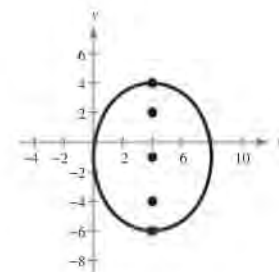


9. Center: $(4, -1)$

Vertices: $(4, 4), (4, -6)$

Foci: $(4, 2), (4, -4)$

Eccentricity: $\frac{3}{5}$



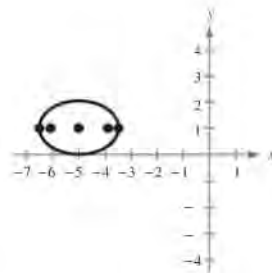
11. Center:
- $(-5, 1)$

Vertices:

$$\left(-\frac{7}{2}, 1\right), \left(-\frac{13}{2}, 1\right)$$

$$\text{Foci: } \left(-5 \pm \frac{\sqrt{5}}{2}, 1\right)$$

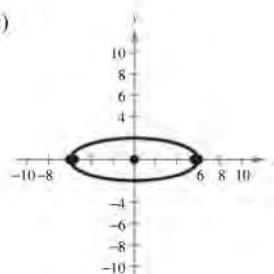
$$\text{Eccentricity: } \frac{\sqrt{5}}{2}$$



13. (a)
- $\frac{x^2}{36} + \frac{y^2}{4} = 1$

(b) Center: $(0, 0)$ Vertices: $(\pm 6, 0)$ Foci: $(\pm 4\sqrt{2}, 0)$

$$\text{Eccentricity: } \frac{2\sqrt{2}}{3}$$

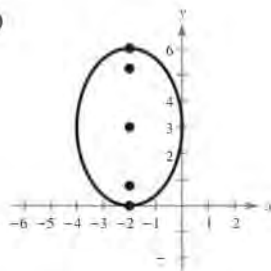


15. (a)
- $\frac{(x+2)^2}{4} + \frac{(y-3)^2}{9} = 1$

(b) Center: $(-2, 3)$ Vertices: $(-2, 6), (-2, 0)$ Foci: $(-2, 3 \pm \sqrt{5})$

$$\text{Eccentricity: } \frac{\sqrt{5}}{3}$$

(c)



17. (a)
- $\frac{(x+\frac{3}{2})^2}{4} + \frac{(y-\frac{5}{2})^2}{12} = 1$

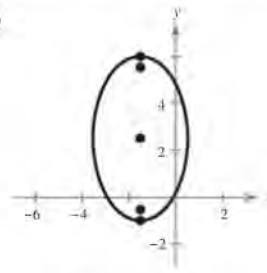
(b) Center: $(-\frac{3}{2}, \frac{5}{2})$

$$\text{Vertices: } \left(-\frac{3}{2}, \frac{5 \pm 4\sqrt{3}}{2}\right)$$

$$\text{Foci: } \left(-\frac{3}{2}, \frac{5}{2} \pm 2\sqrt{2}\right)$$

$$\text{Eccentricity: } \frac{\sqrt{6}}{3}$$

(c)



19. (a)
- $\frac{(x-1)^2}{\frac{25}{16}} + (y+1)^2 = 1$

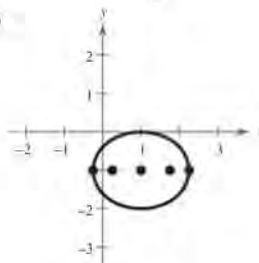
(b) Center: $(1, -1)$

$$\text{Vertices: } \left(\frac{9}{4}, -1\right), \left(-\frac{1}{4}, -1\right)$$

$$\text{Foci: } \left(\frac{7}{4}, -1\right), \left(\frac{1}{4}, -1\right)$$

$$\text{Eccentricity: } \frac{3}{5}$$

(c)



21. (a)
- $\frac{(x-\frac{1}{2})^2}{5} + \frac{(y+1)^2}{3} = 1$

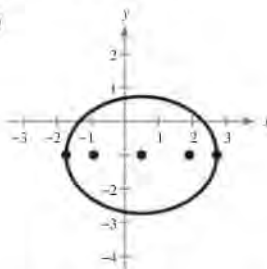
(b) Center: $(\frac{1}{2}, -1)$

$$\text{Vertices: } \left(\frac{1}{2} \pm \sqrt{5}, -1\right)$$

$$\text{Foci: } \left(\frac{1}{2} \pm \sqrt{2}, -1\right)$$

$$\text{Eccentricity: } \frac{\sqrt{10}}{5}$$

(c)



- 23.
- $\frac{x^2}{4} + \frac{y^2}{16} = 1$
- 25.
- $\frac{x^2}{9} + \frac{y^2}{5} = 1$
- 27.
- $\frac{x^2}{16} + \frac{y^2}{7} = 1$

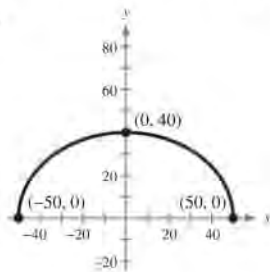
29. $\frac{x^2}{400/21} + \frac{y^2}{25} = 1$ 31. $\frac{(x-2)^2}{1} + \frac{(y-3)^2}{9} = 1$

33. $\frac{(x-4)^2}{16} + \frac{(y-2)^2}{1} = 1$ 35. $\frac{x^2}{308} + \frac{(y-4)^2}{324} = 1$

37. $\frac{(x-3)^2}{9} + \frac{(y-5)^2}{16} = 1$ 39. $\frac{x^2}{16} + \frac{(y-4)^2}{12} = 1$

41. $\frac{\sqrt{5}}{3}$ 43. $\frac{2\sqrt{2}}{3}$ 45. $\frac{x^2}{2} + \frac{y^2}{9} = 1$

47. (a)

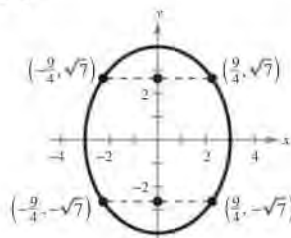


(b) $\frac{x^2}{2500} + \frac{y^2}{1600} = 1$ (c) 17.4 feet

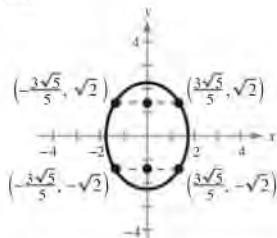
49. $(\pm\sqrt{5}, 0)$; 6 feet 51. 40 units

53. $\frac{x^2}{4.88} + \frac{y^2}{1.39} = 1$ 55. Answers will vary.

57.



59.



61. True

63. (a) 2a

(b) The sum of the distances from the two fixed points is constant.

65. $\frac{(x-6)^2}{324} + \frac{(y-2)^2}{308} = 1$ 67. Arithmetic

69. Geometric 71. 1093 73. 15.0990

Section 9.3 (page 687)

Vocabulary Check (page 687)

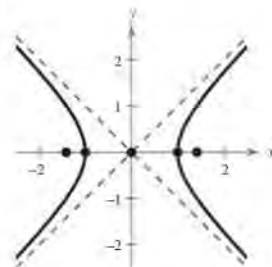
1. hyperbola 2. branches
3. transverse axis, center 4. asymptotes
5. $Ax^2 + Cy^2 + Dx + Ey + F = 0$

1. b 2. c 3. a 4. d

5. Center: (0, 0)

Vertices: $(\pm 1, 0)$

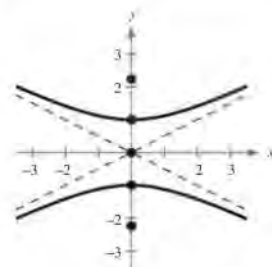
Foci: $(\pm\sqrt{2}, 0)$

Asymptotes: $y = \pm x$


7. Center: (0, 0)

Vertices: $(0, \pm 1)$

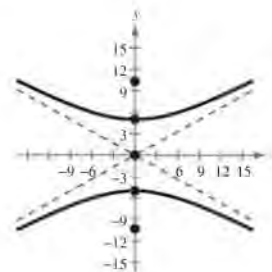
Foci: $(0, \pm\sqrt{5})$

Asymptotes: $y = \pm\frac{1}{2}x$


9. Center: (0, 0)

Vertices: $(0, \pm 5)$

Foci: $(0, \pm\sqrt{106})$

Asymptotes: $y = \pm\frac{5}{9}x$


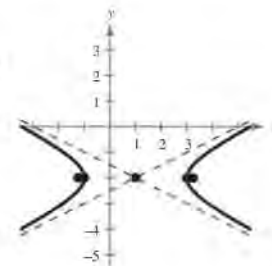
11. Center: (1, -2)

Vertices: $(3, -2), (-1, -2)$

Foci: $(1 \pm \sqrt{5}, -2)$

Asymptotes:

$$y = -2 \pm \frac{1}{2}(x - 1)$$



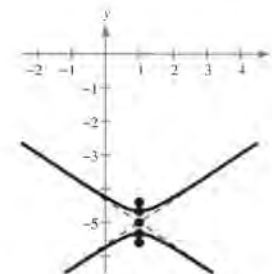
13. Center: (1, -5)

Vertices: $(1, -5 \pm \frac{1}{3})$

Foci: $(1, -5 \pm \frac{\sqrt{13}}{6})$

Asymptotes:

$$y = -5 \pm \frac{2}{3}(x - 1)$$

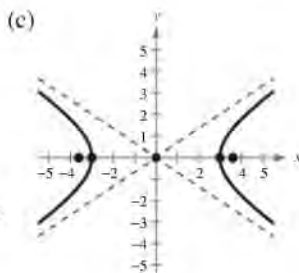


15. (a) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

 (b) Center: $(0, 0)$

 Vertices: $(\pm 3, 0)$

 Foci: $(\pm \sqrt{13}, 0)$

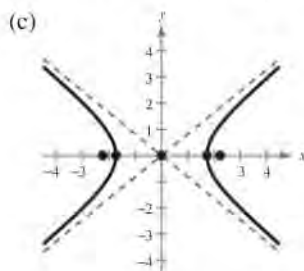
 Asymptotes: $y = \pm \frac{2}{3}x$


17. (a) $\frac{x^2}{3} - \frac{y^2}{2} = 1$

 (b) Center: $(0, 0)$

 Vertices: $(\pm \sqrt{3}, 0)$

 Foci: $(\pm \sqrt{5}, 0)$

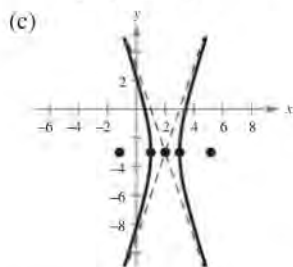
 Asymptotes: $y = \pm \frac{\sqrt{6}}{3}x$


19. (a) $(x - 2)^2 - \frac{(y + 3)^2}{9} = 1$

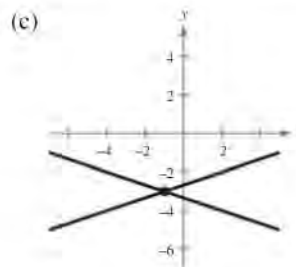
 (b) Center: $(2, -3)$

 Vertices: $(3, -3), (1, -3)$

 Foci: $(2 \pm \sqrt{10}, -3)$

 Asymptotes: $y = -3 \pm 3(x - 2)$


21. (a) $(x + 1)^2 - 9(y + 3)^2 = 0$

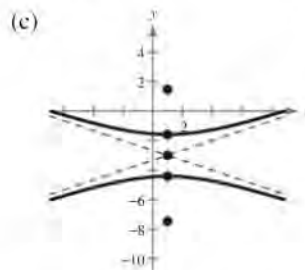
 (b) It is a degenerate conic. The graph of this equation is two lines intersecting at $(-1, -3)$.


23. (a) $\frac{(y + 3)^2}{2} - \frac{(x - 1)^2}{18} = 1$

 (b) Center: $(1, -3)$

 Vertices: $(1, -3 \pm \sqrt{2})$

 Foci: $(1, -3 \pm 2\sqrt{5})$

 Asymptotes: $y = -3 \pm \frac{1}{3}(x - 1)$


25. $\frac{y^2}{4} - \frac{x^2}{12} = 1$

27. $\frac{x^2}{1} - \frac{y^2}{25} = 1$

29. $\frac{17y^2}{1024} - \frac{17x^2}{64} = 1$

31. $\frac{(x - 4)^2}{4} - \frac{y^2}{12} = 1$

33. $\frac{(y - 5)^2}{16} - \frac{(x - 4)^2}{9} = 1$

35. $\frac{y^2}{9} - \frac{4(x - 2)^2}{9} = 1$

37. $\frac{(y - 2)^2}{4} - \frac{x^2}{4} = 1$

39. $\frac{(x - 2)^2}{1} - \frac{(y - 2)^2}{1} = 1$

41. $\frac{(x - 3)^2}{9} - \frac{(y - 2)^2}{4} = 1$

43. $\frac{x^2}{98,010,000} - \frac{y^2}{13,503,600} = 1$

45. (a) $x^2 - \frac{y^2}{27} = 1$

(b) 1.89 feet = 22.68 inches

47. $(12\sqrt{5} - 12, 0) \approx (14.83, 0)$

49. Ellipse

51. Hyperbola

53. Parabola

55. Circle

57. Parabola

 59. True. For a hyperbola, $c^2 = a^2 + b^2$. The larger the ratio of b to a , the larger the eccentricity of the hyperbola, $e = c/a$.

 61. False. If $D = E$ or $D = -E$, the graph is two intersecting lines. For example, the graph of $x^2 - y^2 - 2x + 2y = 0$ is two intersecting lines.

63. Proof 65. Answers will vary. 67. Proof

69. $x^3 + x^2 + 2x - 6$ 71. $x^2 - 2x + 1 + \frac{2}{x+2}$

73. $x(x+4)(x-4)$ 75. $2x(x-6)^2$

77. $2(2x+3)(4x^2-6x+9)$

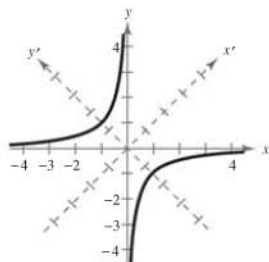
Section 9.4 (page 697)

Vocabulary Check (page 697)

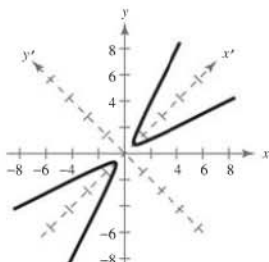
1. rotation, axes
2. invariant under rotation
3. discriminant

1. (3, 0)

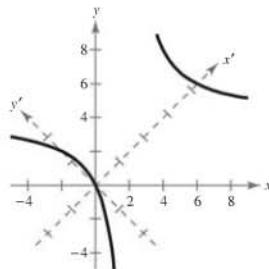
3. $\frac{(y')^2}{2} - \frac{(x')^2}{2} = 1$



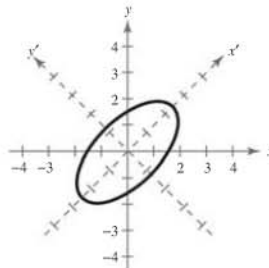
5. $(x')^2 - \frac{(y')^2}{1/3} = 1$



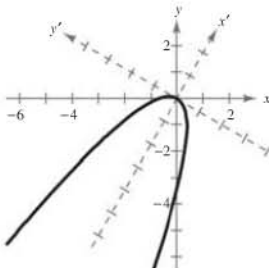
7. $\frac{(x' - 3\sqrt{2})^2}{16} - \frac{(y' - \sqrt{2})^2}{16} = 1$



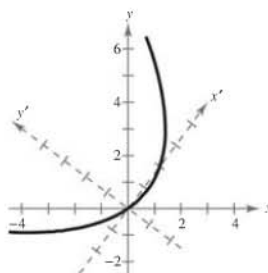
9. $\frac{(x')^2}{6} + \frac{(y')^2}{3/2} = 1$



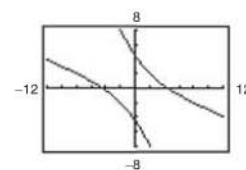
11. $x' = -(y')^2$



13. $y' = \frac{1}{6}(x')^2 - \frac{1}{3}x'$

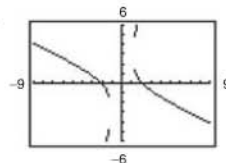


15.

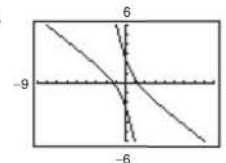


$\theta = 45^\circ$

17.



19.



$\theta = 26.57^\circ$

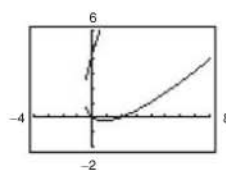
$\theta = 31.72^\circ$

21. e 22. b 23. f 24. a 25. d 26. c

27. (a) Parabola

(b) $y = \frac{24x + 40 \pm \sqrt{3000x + 1600}}{18}$

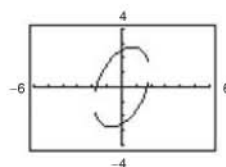
(c)



29. (a) Ellipse or circle

(b) $y = \frac{8x \pm \sqrt{-356x^2 + 1260}}{14}$

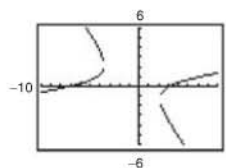
(c)



31. (a) Hyperbola

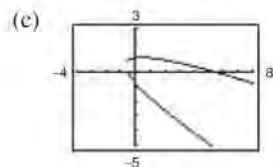
(b) $y = \frac{6x \pm \sqrt{56x^2 + 80x - 440}}{-10}$

(c)

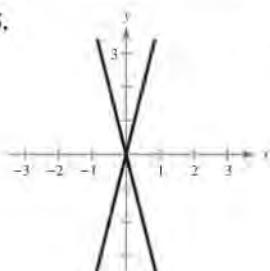


33. (a) Parabola

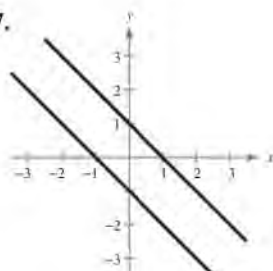
$$(b) y = \frac{-4x + 1 \pm \sqrt{72x + 49}}{8}$$



35.



37.


 39. $(1, \sqrt{3}), (1, -\sqrt{3})$

 41. $(-8, 12)$

 43. $(0, 8), (12, 8)$

 45. $(0, 4)$

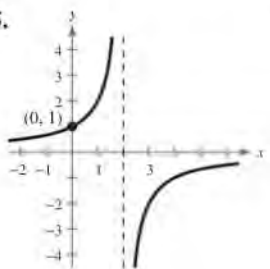
 47. $(\sqrt{3}, -2\sqrt{3}), (-\sqrt{3}, 2\sqrt{3})$

 49. $(8, 0)$

 51. $(-3, 0), (0, \frac{3}{2})$

53. True. The discriminant will be greater than zero.

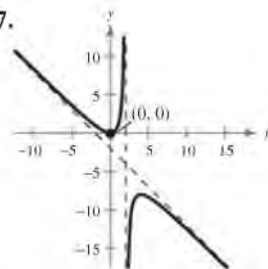
55.


 Intercept: $(0, 1)$

Asymptotes:

$$x = 2, y = 0$$

57.


 Intercept: $(0, 0)$

Asymptotes:

$$t = 2, y = -t - 2$$

 59. (a) $\begin{bmatrix} -15 & 9 \\ 25 & 7 \end{bmatrix}$

 (b) $\begin{bmatrix} 12 & 30 \\ 3 & -20 \end{bmatrix}$

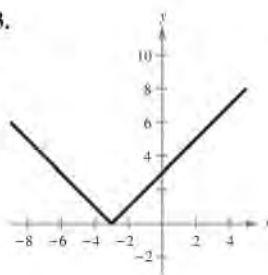
 (c) $\begin{bmatrix} -5 & -18 \\ 12 & 19 \end{bmatrix}$

 61. (a) $[45]$

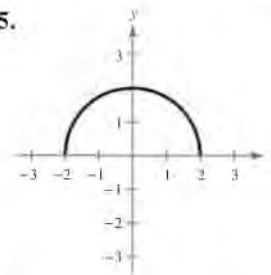
 (b) $\begin{bmatrix} 12 & -6 & 15 \\ -16 & 8 & -20 \\ 20 & -10 & 25 \end{bmatrix}$

(c) Not possible

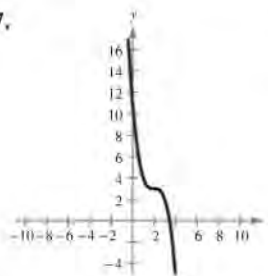
63.



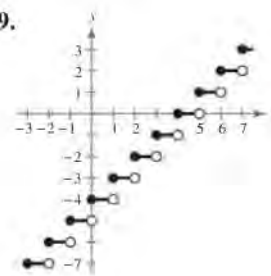
65.



67.



69.



71. 45.11

73. 48.60

Section 9.5 (page 704)

Vocabulary Check (page 704)

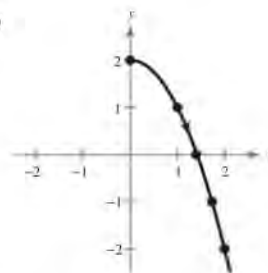
1. plane curve, parametric equations, parameter
2. orientation
3. eliminating, parameter

1. c 2. d 3. b 4. a 5. f 6. e

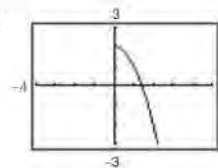
7. (a)

t	0	1	2	3	4
x	0	1	$\sqrt{2}$	$\sqrt{3}$	2
y	2	1	0	-1	-2

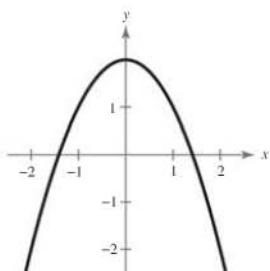
(b)



(c)



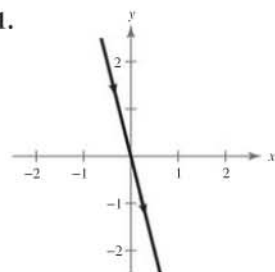
(d) $y = 2 - x^2$



The graph is an entire parabola rather than just the right half.

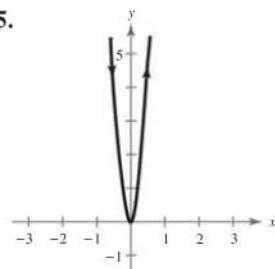
9. b

11.



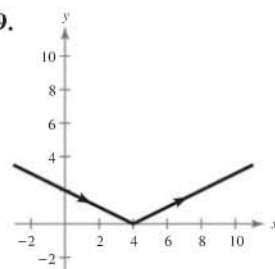
$y = -4x$

15.



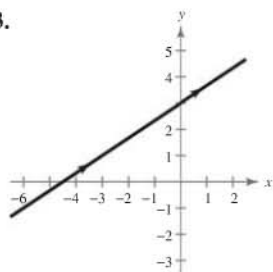
$y = 16x^2$

19.



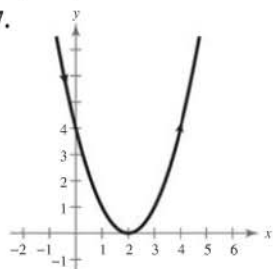
$y = \frac{1}{2}|x - 4|$

13.



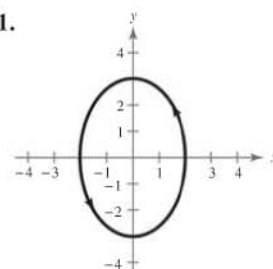
$y = \frac{2}{3}x + 3$

17.



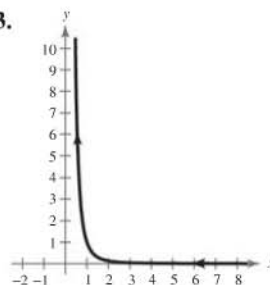
$y = (x - 2)^2$

21.



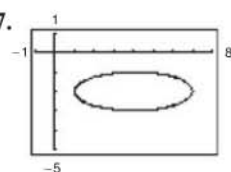
$\frac{x^2}{4} + \frac{y^2}{9} = 1$

23.

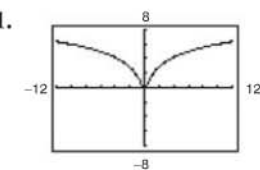


$y = x^{-3}, x > 0$

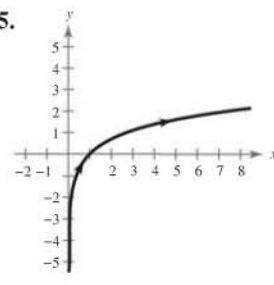
27.



31.

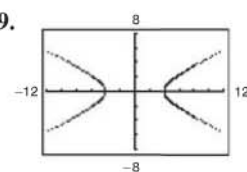


25.



$y = \ln x$

29.



33. Each curve represents a portion of the line $y = 2x + 1$.

Domain

Orientation

(a) $(-\infty, \infty)$

Left to right

(b) $[-1, 1]$

Depends on θ

(c) $(0, \infty)$

Right to left

(d) $(0, \infty)$

Left to right

35. $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

37. $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$

39. $x = 1 + 5t, y = 4 - 7t$

41. $x = 5 \cos \theta$

$y = 3 \sin \theta$

43. $x = t, y = 5t - 3$

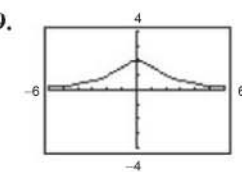
$x = \frac{1}{5}t, y = t - 3$

45. $x = t, y = \frac{1}{t}$
 $x = t^3, y = \frac{1}{t^3}$

47. $x = t, y = 6t^2 - 5$

$x = 2t, y = 24t^2 - 5$

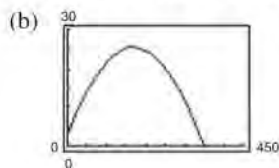
49.



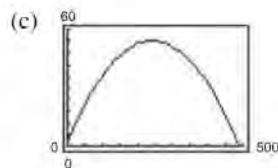
51. b 52. c 53. d 54. a

 55. (a) $x = (146.67 \cos \theta)t$

$$y = 3 + (146.67 \sin \theta)t - 16t^2$$



No



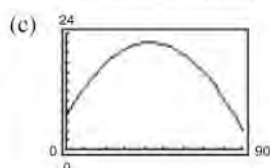
Yes

 (d) About 19.38°

 57. (a) $x = (v_0 \cos 35^\circ)t$

$$y = 7 + (v_0 \sin 35^\circ)t - 16t^2$$

(b) 54.09 feet per second



22.04 feet

(d) 2.03 seconds

 59. True. Both sets of parametric equations correspond to $y = x^2 + 1$.

 61. False. $x = t^2$, $y = t$ does not correspond to y as a function of x .

63. Answers will vary. Sample answer:

$$x = \cos \theta$$

$$y = -2 \sin \theta$$

65. Even 67. Neither

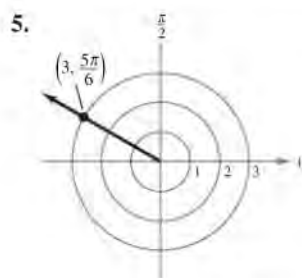
Section 9.6 (page 711)

Vocabulary Check (page 711)

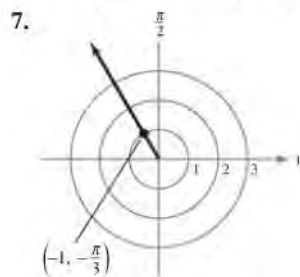
1. pole 2. directed distance, directed angle

3. polar

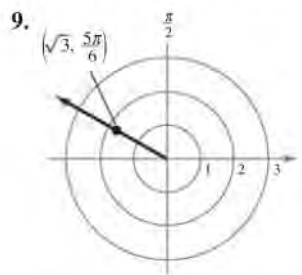
1. $(0, 4)$ 3. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$



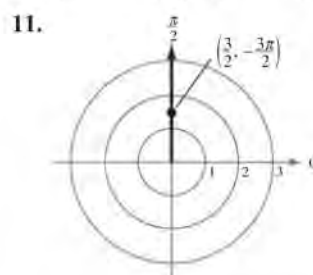
$$\left(3, -\frac{7\pi}{6}\right), \left(-3, \frac{11\pi}{6}\right), \left(-3, -\frac{\pi}{6}\right)$$



$$\left(-1, \frac{5\pi}{3}\right), \left(1, \frac{2\pi}{3}\right), \left(1, -\frac{4\pi}{3}\right)$$

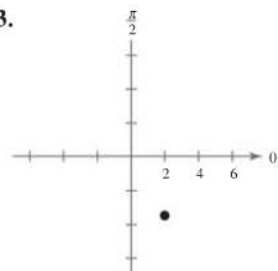


$$\left(-\sqrt{3}, \frac{11\pi}{6}\right), \left(\sqrt{3}, -\frac{7\pi}{6}\right), \left(-\sqrt{3}, -\frac{\pi}{6}\right)$$



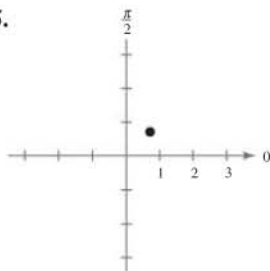
$$\left(\frac{3}{2}, \frac{\pi}{2}\right), \left(-\frac{3}{2}, \frac{3\pi}{2}\right), \left(-\frac{3}{2}, -\frac{\pi}{2}\right)$$

13.



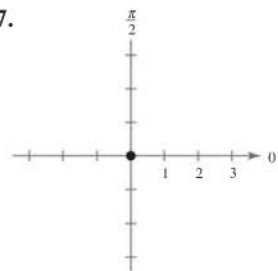
$$(2, -2\sqrt{3})$$

15.



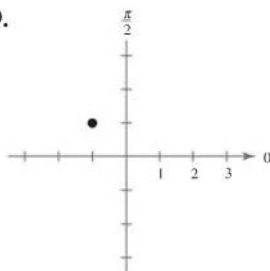
$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

17.



$$(0, 0)$$

19.



$$(-1.004, 0.996)$$

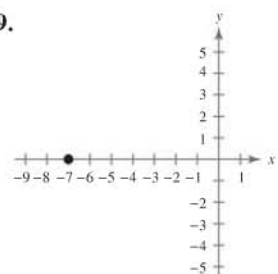
21. (1.53, 1.29)

23. (-1.20, -4.34)

25. (-0.02, 2.50)

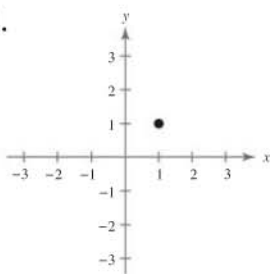
27. (-3.60, 1.97)

29.



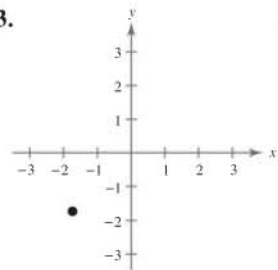
$$(7, \pi), (-7, 0)$$

31.



$$\left(\sqrt{2}, \frac{\pi}{4}\right), \left(-\sqrt{2}, \frac{5\pi}{4}\right)$$

33.



$$\left(\sqrt{6}, \frac{5\pi}{4}\right), \left(-\sqrt{6}, \frac{\pi}{4}\right) \quad (10.82, 0.98), (-10.82, 4.12)$$

37. (3.61, -0.59)

39. (2.65, 0.86)

41. (2.83, 0.49)

43. $r = 3$

45. $r = 4 \csc \theta$

47. $r = 8 \sec \theta$

$$49. r = -\frac{2}{3 \cos \theta - 6 \sin \theta}$$

$$51. r^2 = 8 \csc 2\theta$$

$$53. r^2 = 9 \cos 2\theta$$

$$55. r = 6 \cos \theta$$

$$57. r = 2a \cos \theta$$

$$59. r = \tan^2 \theta \sec \theta$$

$$61. x^2 + y^2 = 6y$$

$$63. y = \sqrt{3}x$$

$$65. y = -\frac{\sqrt{3}}{3}x$$

$$67. x = 0$$

$$69. x^2 + y^2 = 16$$

$$71. y = -3$$

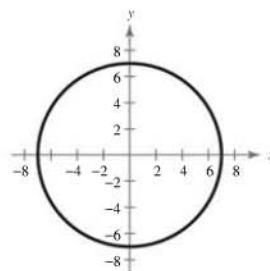
$$73. (x^2 + y^2)^3 = x^2$$

$$75. (x^2 + y^2)^2 = 6x^2y - 2y^3$$

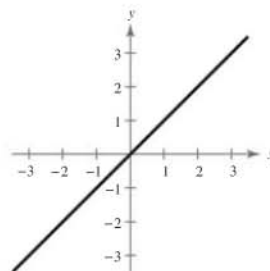
$$77. y^2 = 2x + 1$$

$$79. 4x^2 - 5y^2 = 36y + 36$$

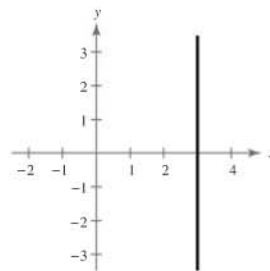
81. The graph is a circle centered at the origin with a radius of 7; $x^2 + y^2 = 49$.



83. The graph consists of all points on the line that makes an angle of $\pi/4$ with the positive x -axis; $x - y = 0$.



85. The graph is a vertical line through $(3, 0)$; $x - 3 = 0$.



87. True. Because r is a directed distance, (r, θ) can be represented by $(-r, \theta \pm (2n + 1)\pi)$, so $|r| = |-r|$.

89. (a) Answers will vary.

(b) The points lie on a line passing through the pole.

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2} = |r_1 - r_2|$$

(c) $d = \sqrt{r_1^2 + r_2^2}$ (Pythagorean Theorem)

Answers will vary.

(d) Answers will vary. The Distance Formula should give the same result in both cases.

91. $A \approx 30.68^\circ$ 93. $a \approx 16.16$ 95. $A \approx 119.09^\circ$
 $B \approx 48.23^\circ$ $b \approx 19.44$ $B \approx 25.91^\circ$
 $C \approx 101.09^\circ$ $B \approx 86^\circ$ $c \approx 5.25$

97. (2, 3) 99. (0, 0, 0) 101. (2, -3, 3)

103. Not collinear 105. Collinear

Section 9.7 (page 720)

Vocabulary Check (page 720)

1. $\theta = \frac{\pi}{2}$ 2. polar axis 3. convex limaçon
 4. circle 5. lemniscate 6. cardioid

1. Rose curve 3. Lemniscate 5. Rose curve

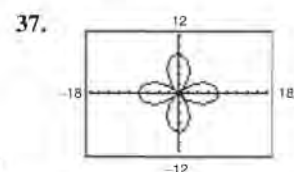
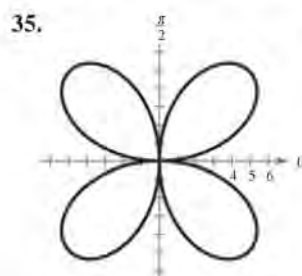
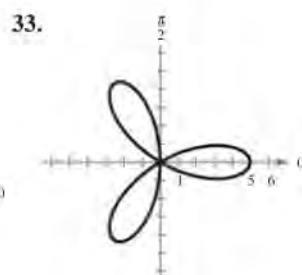
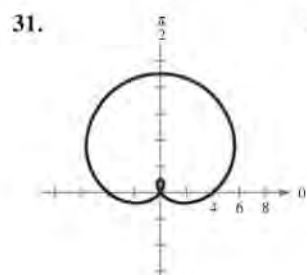
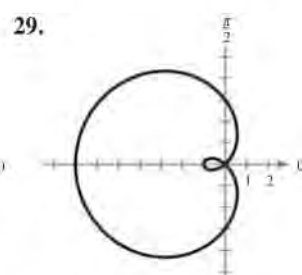
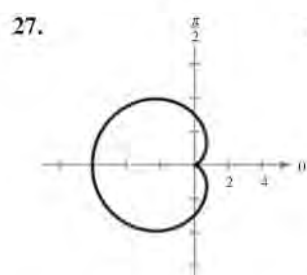
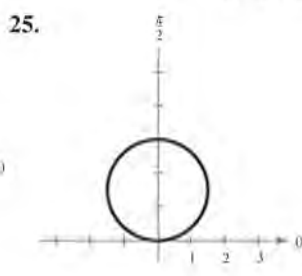
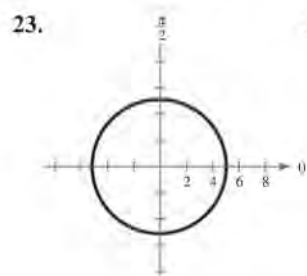
7. a 9. c 11. Polar axis 13. $\theta = \frac{\pi}{2}$

15. $\theta = \frac{\pi}{2}$ 17. Pole

19. Maximum: $|r| = 20$ 21. Maximum: $|r| = 4$

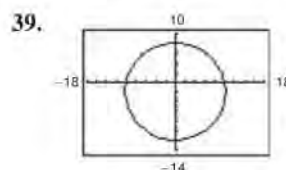
Zero of r : $\theta = \frac{\pi}{2}$

Zeros of r : $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$



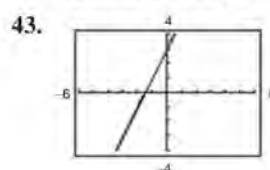
$$0 \leq \theta < 2\pi$$

Answers will vary.

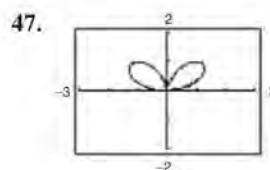


$$0 \leq \theta < 2\pi$$

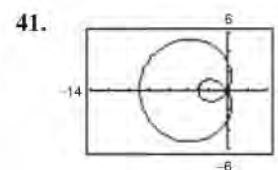
Answers will vary.



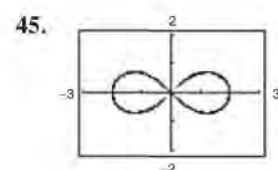
Answers will vary.



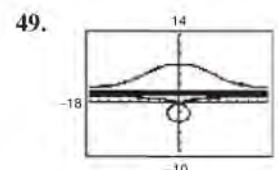
Answers will vary.



Answers will vary.

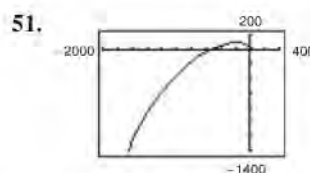


Answers will vary.

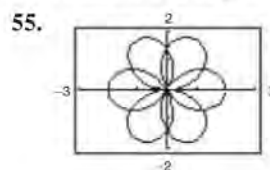


$$0 \leq \theta < 2\pi$$

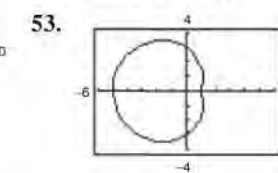
Answers will vary.



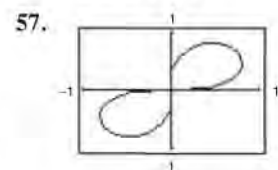
Answers will vary.



$$0 \leq \theta < 4\pi$$

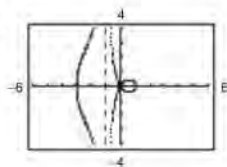


$$0 \leq \theta < 2\pi$$

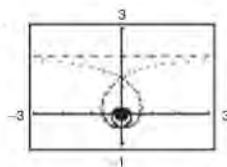


$$0 \leq \theta < \frac{\pi}{2}$$

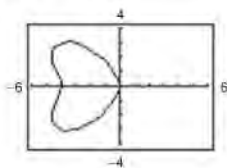
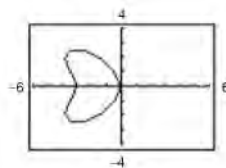
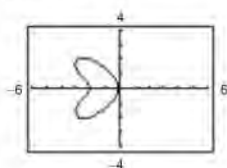
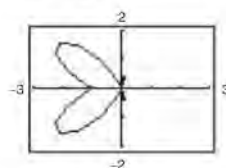
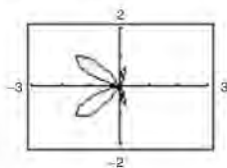
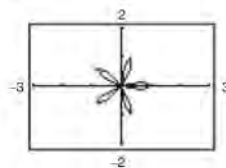
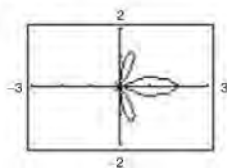
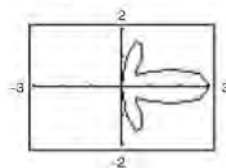
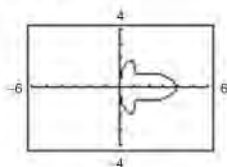
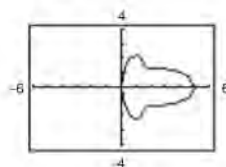
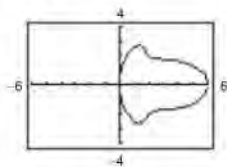
59.



61.



63. True

65. $n = -5$

 $n = -4$

 $n = -3$

 $n = -2$

 $n = -1$

 $n = 0$

 $n = 1$

 $n = 2$

 $n = 3$

 $n = 4$

 $n = 5$


Negative values of n produce the heart-shaped curves; positive values of n produce the bell-shaped curves.

67. (a), (b), and (c) Answers will vary.

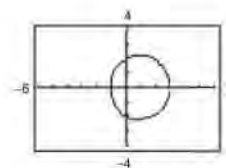
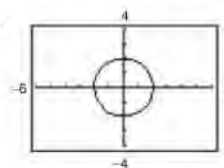
69. (a) $r = 4 \sin\left(\theta - \frac{\pi}{6}\right) \cos\left(\theta - \frac{\pi}{6}\right)$

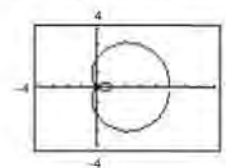
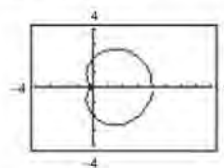
(b) $r = -4 \sin \theta \cos \theta$

(c) $r = 4 \sin\left(\theta - \frac{2\pi}{3}\right) \cos\left(\theta - \frac{2\pi}{3}\right)$

(d) $r = 4 \sin \theta \cos \theta$

71.


 $k = 0$; circle

 $k = 1$; convex limaçon

 $k = 2$; cardioid

 $k = 3$; limaçon with inner loop

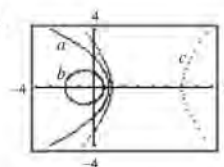
Section 9.8 (page 726)

Vocabulary Check (page 726)

1. conic 2. eccentricity, e

3. (a) i (b) iii (c) ii

1.

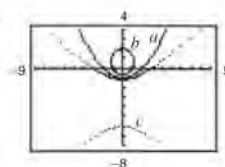


(a) Parabola

(b) Ellipse

(c) Hyperbola

3.



(a) Parabola

(b) Ellipse

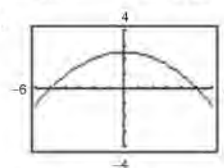
(c) Hyperbola

5. b 6. c 7. f 8. e 9. d 10. a

11. Parabola 13. Ellipse 15. Ellipse

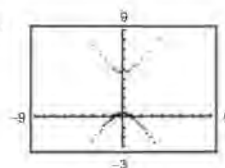
17. Ellipse 19. Hyperbola

21.

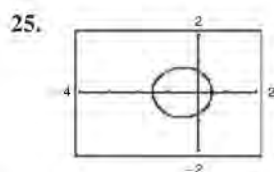


Parabola

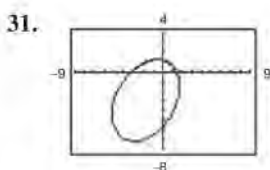
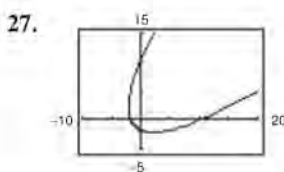
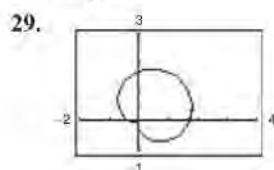
23.



Hyperbola



Ellipse



33. $r = \frac{1}{1 - \cos \theta}$

35. $r = \frac{1}{2 + \sin \theta}$

37. $r = \frac{2}{1 + 2 \cos \theta}$

39. $r = \frac{2}{1 - \sin \theta}$

41. $r = \frac{10}{1 - \cos \theta}$

43. $r = \frac{10}{3 + 2 \cos \theta}$

45. $r = \frac{20}{3 - 2 \cos \theta}$

47. $r = \frac{8}{3 + 5 \sin \theta}$

49. Answers will vary.

51. $r = \frac{9.2930 \times 10^7}{1 - 0.0167 \cos \theta}$

Perihelion: $\approx 9.1404 \times 10^7$ miles

Aphelion: $\approx 9.4508 \times 10^7$ miles

53. $r = \frac{7.7659 \times 10^8}{1 - 0.0484 \cos \theta}$

Perihelion: $\approx 7.4073 \times 10^8$ kilometers

Aphelion: $\approx 8.1609 \times 10^8$ kilometers

55. (a) $r_{\text{Neptune}} = \frac{4.4977 \times 10^9}{1 - 0.0086 \cos \theta}$

$r_{\text{Pluto}} = \frac{5.5404 \times 10^9}{1 - 0.2488 \cos \theta}$

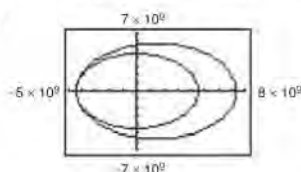
(b) Neptune: Perihelion $\approx 4.4593 \times 10^9$ kilometers

Aphelion $\approx 4.5367 \times 10^9$ kilometers

Pluto: Perihelion $\approx 4.4366 \times 10^9$ kilometers

Aphelion $\approx 7.3754 \times 10^9$ kilometers

(c)



(d) Yes; because on average, Pluto is farther from the sun than Neptune.

(e) Using a graphing utility, it would appear that the orbits intersect. No, Pluto and Neptune will never collide because the orbits do not intersect in three-dimensional space.

57. False. The equation can be rewritten as

$$r = \frac{-4/3}{1 + \sin \theta}$$

Because ep is negative, p must be negative and since p represents the distance between the pole and the directrix, the directrix has to be below the pole.

59. Answers will vary. 61. $r^2 = \frac{24,336}{169 - 25 \cos^2 \theta}$

63. $r^2 = \frac{400}{25 - 9 \cos^2 \theta}$ 65. $r^2 = \frac{144}{25 \sin^2 \theta - 16}$

67. (a) Ellipse

(b) $r = \frac{4}{1 + 0.4 \cos \theta}$ is reflected about the line $\theta = \frac{\pi}{2}$.

$r = \frac{4}{1 - 0.4 \sin \theta}$ is rotated 90° counterclockwise.

69. Answers will vary. 71. $\frac{\pi}{6} + n\pi$

73. $\frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi$ 75. $\frac{\pi}{2} + n\pi$

77. $\frac{\sqrt{2}}{10}$ 79. $\frac{\sqrt{2}}{10}$ 81. 220 83. 720

Review Exercises (page 730)

1. $x^2 + y^2 = 25$ 3. $(x - 2)^2 + (y - 4)^2 = 13$

5. $x^2 + y^2 = 36$ 7. $(x - \frac{1}{2})^2 + (y + \frac{3}{4})^2 = 1$

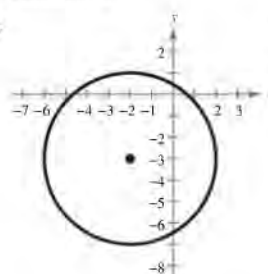
Center: $(0, 0)$

Center: $(\frac{1}{2}, -\frac{3}{4})$

Radius: 6

Radius: 1

9.



Center: $(-2, -3)$

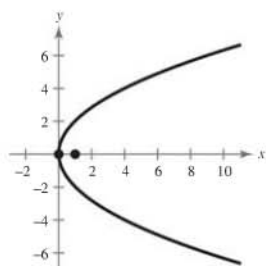
Radius: 4

11. $(3 \pm \sqrt{6}, 0)$

13. Vertex: $(0, 0)$

Focus: $(1, 0)$

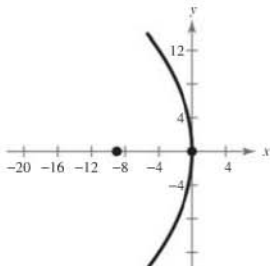
Directrix: $x = -1$



15. Vertex: $(0, 0)$

Focus: $(-9, 0)$

Directrix: $x = 9$



17. $y^2 = -24x$

19. $(y - 2)^2 = 12x$

21. $2x + y - 2 = 0$; $(1, 0)$

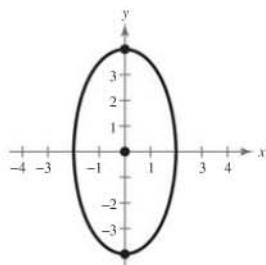
23. $8\sqrt{6}$ meters

25. Center: $(0, 0)$

Vertices: $(0, \pm 4)$

Foci: $(0, \pm 2\sqrt{3})$

Eccentricity: $\frac{\sqrt{3}}{2}$

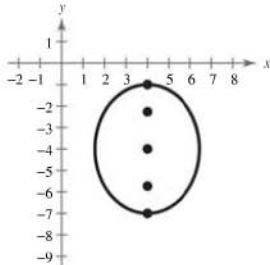


27. Center: $(4, -4)$

Vertices: $(4, -1), (4, -7)$

Foci: $(4, -4 \pm \sqrt{3})$

Eccentricity: $\frac{\sqrt{3}}{3}$



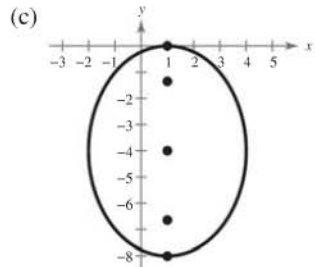
29. (a) $\frac{(x-1)^2}{9} + \frac{(y+4)^2}{16} = 1$

(b) Center: $(1, -4)$

Vertices: $(1, 0), (1, -8)$

Foci: $(1, -4 \pm \sqrt{7})$

Eccentricity: $\frac{\sqrt{7}}{4}$



31. (a) $\frac{(x+2)^2}{1/3} + \frac{(y-7)^2}{1/8} = 1$

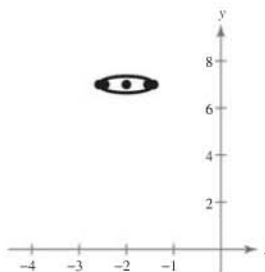
(b) Center: $(-2, 7)$

Vertices: $(-2 \pm \frac{\sqrt{3}}{3}, 7)$

Foci: $(-2 \pm \frac{\sqrt{30}}{12}, 7)$

Eccentricity: $\frac{\sqrt{30}}{4}$

(c)



33. $\frac{x^2}{25} + \frac{y^2}{9} = 1$

35. $\frac{(x-2)^2}{25} + \frac{y^2}{21} = 1$

37. The foci should be placed 3 feet on either side of the center at the same height as the pillars.

39. $e \approx 0.0543$

41. (a) $\frac{y^2}{4} - \frac{x^2}{5} = 1$

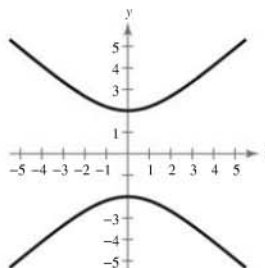
(b) Center: $(0, 0)$

Vertices: $(0, \pm 2)$

Foci: $(0, \pm 3)$

Eccentricity: $\frac{3}{2}$

(c)



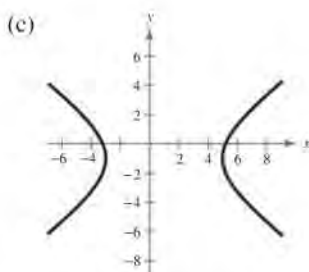
43. (a) $\frac{(x-1)^2}{16} - \frac{(y+1)^2}{9} = 1$

(b) Center: $(1, -1)$

Vertices: $(5, -1), (-3, -1)$

Foci: $(6, -1), (-4, -1)$

Eccentricity: $\frac{5}{4}$



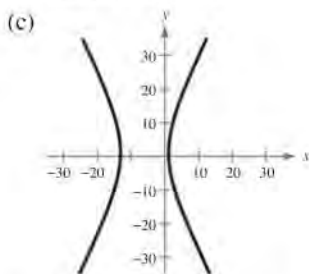
45. (a) $\frac{(x+6)^2}{\frac{101}{2}} - \frac{(y-1)^2}{202} = 1$

(b) Center: $(-6, 1)$

Vertices: $\left(-6 \pm \frac{\sqrt{202}}{2}, 1\right)$

Foci: $\left(-6 \pm \frac{\sqrt{1010}}{2}, 1\right)$

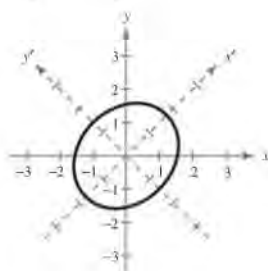
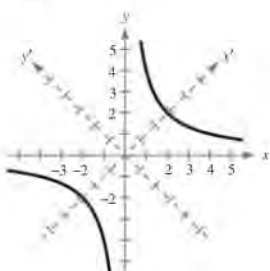
Eccentricity: $\sqrt{5}$



47. $\frac{x^2}{16} - \frac{y^2}{20} = 1$ 49. $\frac{(x-4)^2}{16/5} - \frac{y^2}{64/5} = 1$

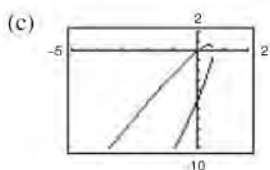
51. ≈ 72 miles 53. Ellipse 55. Hyperbola

57. $\frac{(x')^2}{8} - \frac{(y')^2}{8} = 1$ 59. $\frac{(x')^2}{3} + \frac{(y')^2}{2} = 1$



61. (a) Parabola

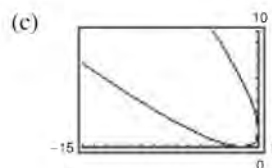
(b) $y = \frac{8x - 5 \pm \sqrt{(5-8x)^2 - 4(16x^2 - 10x)}}{2}$



63. (a) Parabola

(b)

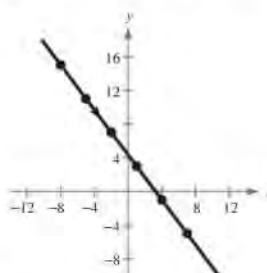
$$y = \frac{-(2x - 2\sqrt{2}) \pm \sqrt{(2x - 2\sqrt{2})^2 - 4(x^2 + 2\sqrt{2}x + 2)}}{2}$$



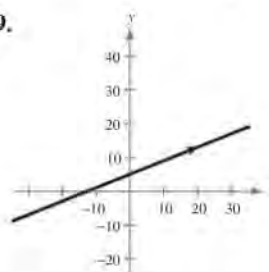
65. $(-10, 12)$

67.

t	-2	-1	0	1	2	3
x	-8	-5	-2	1	4	7
y	15	11	7	3	-1	-5

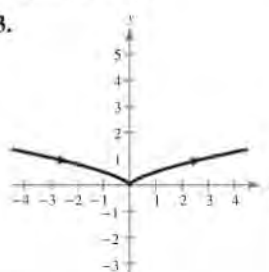


69.



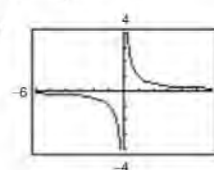
$$y = \frac{2}{5}x + \frac{27}{5}$$

73.

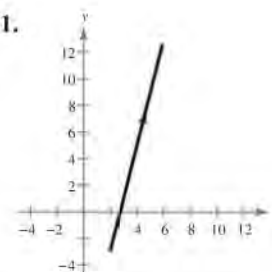


$$y = \frac{1}{2}x^{2/3}$$

77.

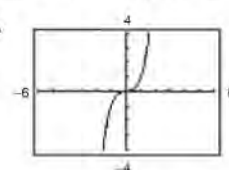


71.

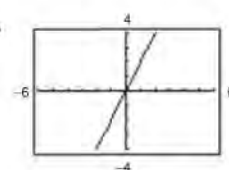


$$y = 4x - 11, x \geq 2$$

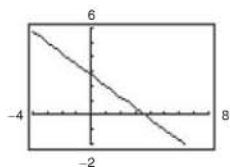
75.



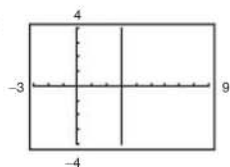
79.



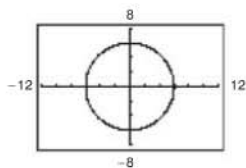
81.



83.



85.



87. $x = t, y = 6t + 2$

$x = 2t, y = 12t + 2$

89. $x = t, y = t^2 + 2$

$x = \frac{1}{2}t, y = \frac{1}{4}t^2 + 2$

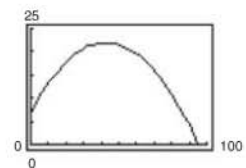
91. $x = t, y = 5$

93. $x = -1 + 11t$

$y = 6 - 6t$

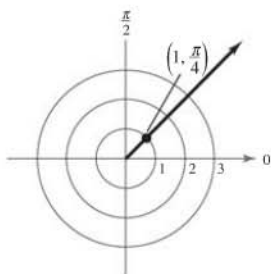
95. 54.22 feet per second

97.



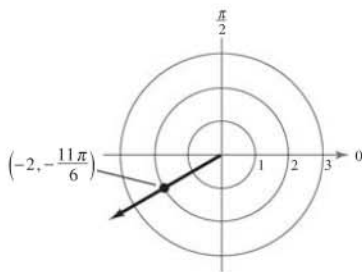
21.91 feet

99.



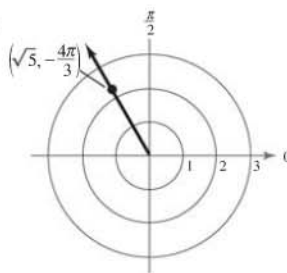
$\left(1, -\frac{7\pi}{4}\right), \left(-1, \frac{5\pi}{4}\right), \left(-1, -\frac{3\pi}{4}\right)$

101.



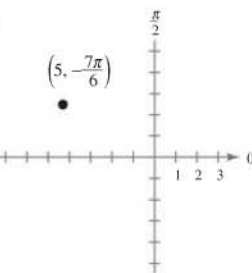
$\left(-2, \frac{\pi}{6}\right), \left(2, \frac{7\pi}{6}\right), \left(2, -\frac{5\pi}{6}\right)$

103.

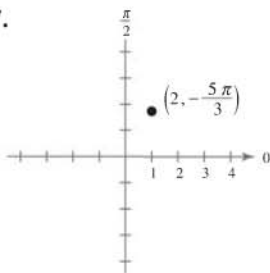


$\left(\sqrt{5}, -\frac{2\pi}{3}\right), \left(-\sqrt{5}, \frac{\pi}{3}\right), \left(-\sqrt{5}, -\frac{5\pi}{3}\right)$

105.



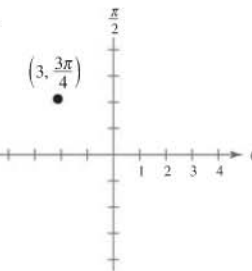
107.



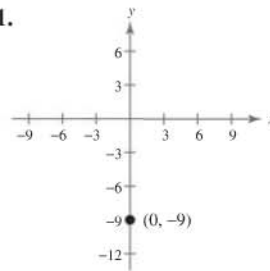
$\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$

$(1, \sqrt{3})$

109.



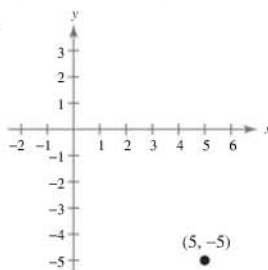
111.



$\left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$

$\left(-9, \frac{\pi}{2}\right), \left(9, \frac{3\pi}{2}\right)$

113.



$\left(-5\sqrt{2}, \frac{3\pi}{4}\right), \left(5\sqrt{2}, \frac{7\pi}{4}\right)$

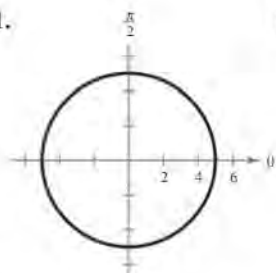
115. $r = 3$ 117. $r = 4 \cos \theta$ 119. $r^2 = 5 \sec \theta \csc \theta$

121. $r^2 = \frac{1}{1 + 3 \cos^2 \theta}$ 123. $x^2 + y^2 = 25$

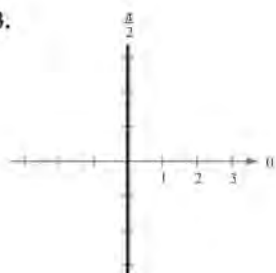
125. $x^2 + y^2 = 3x$ 127. $(x^2 + y^2)^2 - x^2 + y^2 = 0$

129. $y = -\frac{\sqrt{3}}{3}x$

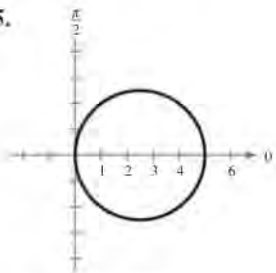
131.



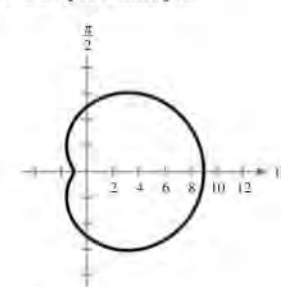
133.



135.



137. Dimpled limaçon

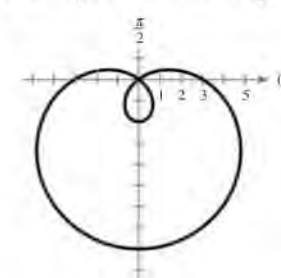


Symmetry: Polar axis

 Maximum: $|r| = 9$

 Zeros of r : None

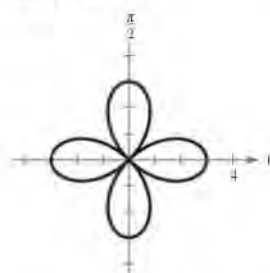
139. Limaçon with inner loop


 Symmetry: The line $\theta = \frac{\pi}{2}$

 Maximum: $|r| = 8$

 Zeros of r : $\theta \approx 0.64, 2.50$

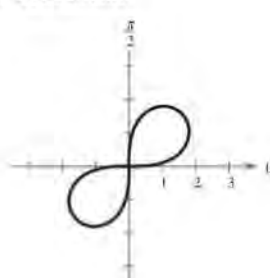
141. Rose curve


 Symmetry: Pole, polar axis, and the line $\theta = \frac{\pi}{2}$

 Maximum: $|r| = 3$

 Zeros of r : $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

143. Lemniscate

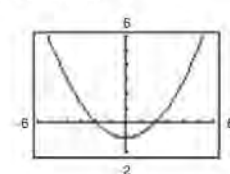


Symmetry: Pole

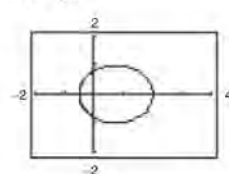
 Maximum: $|r| = \sqrt{5}$

 Zeros of r : $\theta = 0, \frac{\pi}{2}$

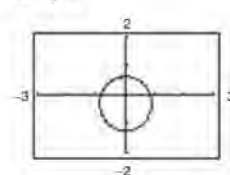
145. Parabola



147. Ellipse



149. Ellipse



151. $r = \frac{4}{1 - \cos \theta}$

153. $r = \frac{5}{3 - 2 \cos \theta}$

155. $r = \frac{1.512}{1 - 0.093 \cos \theta}$

Perihelion: 1.383 astronomical units

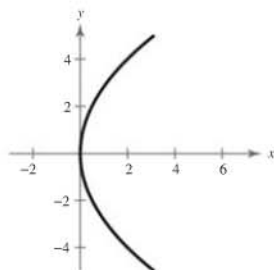
Aphelion: 1.667 astronomical units

157. False. The equation of a hyperbola is a second-degree equation.

159. (a) vertical translation (b) horizontal translation
(c) reflection in the y -axis (d) vertical shrink
161. 5; The ellipse gets closer and closer to circular and approaches a circle of radius 5.
163. (a) The time it takes to make one revolution is halved.
(b) The length of the major axis is increased by two units.

Chapter Test (page 734)

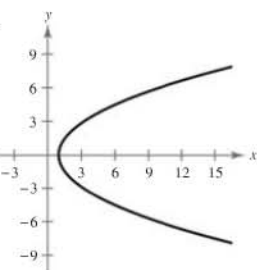
1.



Vertex: $(0, 0)$

Focus: $(2, 0)$

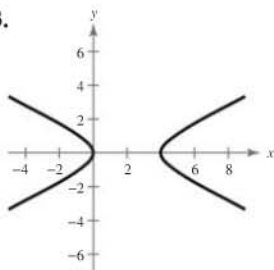
2.



Vertex: $(1, 0)$

Focus: $(2, 0)$

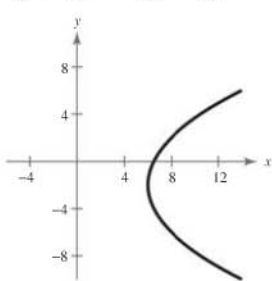
3.



Vertices: $(0, 0), (4, 0)$

Foci: $(2 \pm \sqrt{5}, 0)$

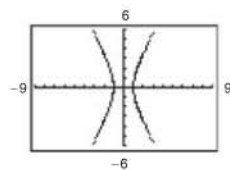
4. $(y + 2)^2 = 8(x - 6)$



5. $\frac{(x + 6)^2}{16} + \frac{(y - 3)^2}{49} = 1$

6. $\frac{y^2}{9} - \frac{x^2}{4} = 1$

7.

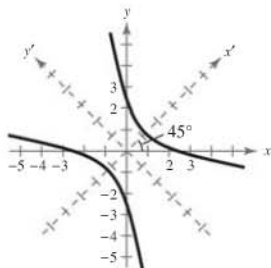


Answers will vary.

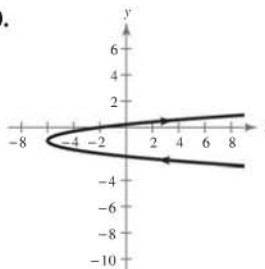
8. (a) 45°

9. No real solution

(b)

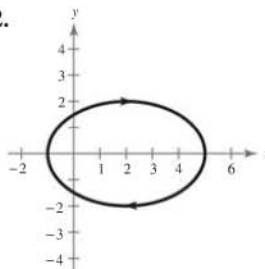


10.



$$(y + 1)^2 = \frac{1}{4}(x + 6)$$

12.



$$\frac{(x - 2)^2}{9} + \frac{y^2}{4} = 1$$

14. $x = 4 - t^2, y = t$

$$x = 4 - 4t^2, y = 2t$$

16. $(-7, 7\sqrt{3})$

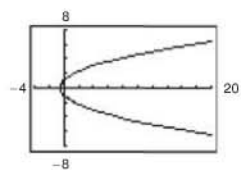
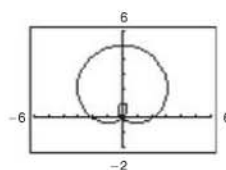
17. $(2\sqrt{2}, \frac{7\pi}{4}); (2\sqrt{2}, -\frac{\pi}{4}), (-2\sqrt{2}, \frac{3\pi}{4})$

18. $r = 12 \sin \theta$

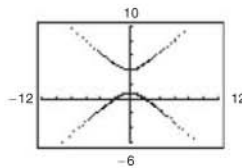
19. $x^2 + (y - 1)^2 = 1$

20. Limaçon with inner loop

21. Parabola



22. Hyperbola



23. $r = \frac{4}{4 + \sin \theta}$

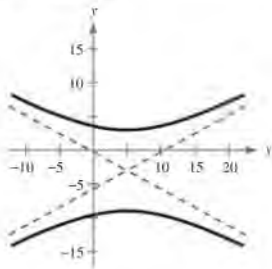
24. $r = \frac{10}{4 + 5 \sin \theta}$

25. Maximum: $|r| = 8$

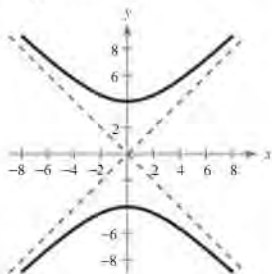
Zeros of r : $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$

Cumulative Test for Chapters 7–9
(page 735)

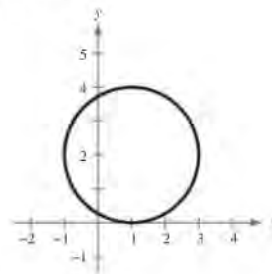
1. $(4, -3)$ 2. $(8, 4), (2, -2)$
 3. $(\frac{3}{5}, -4, -\frac{1}{5})$ 4. $(1, -4, -4)$
 5. $\begin{bmatrix} -7 & -10 & -16 \\ -6 & 18 & 9 \\ -12 & 16 & 7 \end{bmatrix}$ 6. $\begin{bmatrix} -18 & 15 & -14 \\ 28 & 11 & 34 \\ -20 & 52 & -1 \end{bmatrix}$
 7. $\begin{bmatrix} 3 & -31 & 2 \\ 22 & 18 & 6 \\ 52 & -40 & 14 \end{bmatrix}$ 8. $\begin{bmatrix} 5 & 36 & 31 \\ -36 & 12 & -36 \\ 16 & 0 & 18 \end{bmatrix}$
 9. (a) $\begin{bmatrix} -175 & 37 & -13 \\ 95 & -20 & 7 \\ 14 & -3 & 1 \end{bmatrix}$ (b) 1 10. 22
 11. (a) $\frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, -\frac{1}{11}, \frac{1}{13}$ (b) 3, 6, 12, 24, 48 12. 135
 13. $\frac{47}{52}$ 14. ≈ 34.48 15. ≈ 66.67 16. $\frac{15}{8}$
 17. $-\frac{5}{51}$ 18. $\frac{8}{3}$ 19. Answers will vary.
 20. $x^4 + 12x^3 + 54x^2 + 108x + 81$
 21. $32x^5 + 80x^4y^2 + 80x^3y^4 + 40x^2y^6 + 10xy^8 + y^{10}$
 22. $x^6 - 12x^5y + 60x^4y^2 - 160x^3y^3 + 240x^2y^4$
 $- 192xy^5 + 64y^6$
 23. $6561a^8 - 69,984a^7b + 326,592a^6b^2 - 870,912a^5b^3$
 $+ 1,451,520a^4b^4 - 1,548,288a^3b^5 + 1,032,192a^2b^6$
 $- 393,216ab^7 + 65,536b^8$
 24. 30 25. 120 26. 453,600 27. 151,200
 28. Hyperbola 29. Ellipse



30. Hyperbola

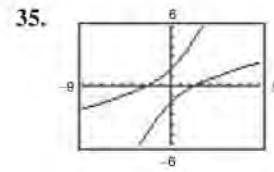


31. Circle



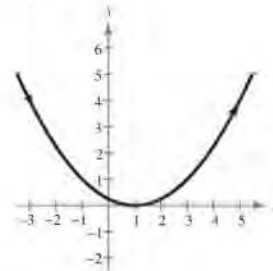
32. $(x - 2)^2 = -\frac{4}{3}(y - 3)$

33. $\frac{(x - 1)^2}{25} + \frac{(y - 4)^2}{4} = 1$ 34. $\frac{(y + 4)^2}{4} - \frac{x^2}{16/3} = 1$



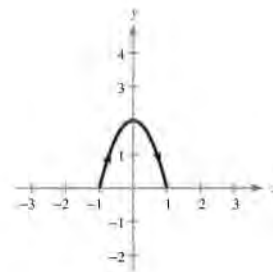
$\theta \approx 37.98^\circ$

36. (a) and (b)



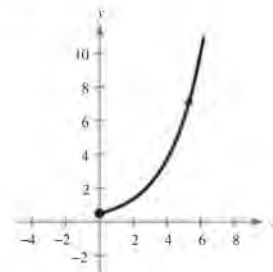
(c) $y = \frac{x^2 - 2x + 1}{4}$

37. (a) and (b)



(c) $y = 2 - 2x^2, -1 \leq x \leq 1$

38. (a) and (b)



(c) $y = 0.5e^{0.5x}, x \geq 0$

39. $x = t, y = 3t - 2$
 $x = 2t, y = 6t - 2$

40. $x = \pm \sqrt{16 + t^2}, y = t$
 $x = \pm 4\sqrt{1 + t^2}, y = 4t$

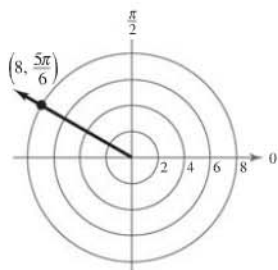
41. $x = t, y = \frac{2}{t}$

$x = \frac{1}{t}, y = 2t$

42. $x = t, y = \frac{e^{2t}}{e^{2t} + 1}$

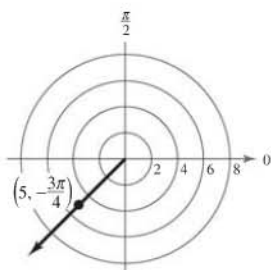
$x = 2t, y = \frac{e^{4t}}{e^{4t} + 1}$

43.



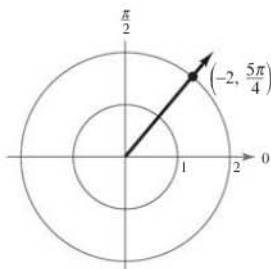
$(8, -\frac{7\pi}{6}), (-8, -\frac{\pi}{6}), (-8, \frac{11\pi}{6})$

44.



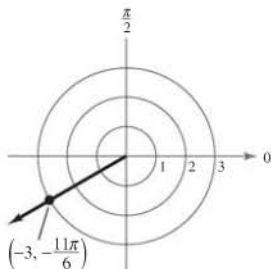
$(5, \frac{5\pi}{4}), (-5, -\frac{7\pi}{4}), (-5, \frac{\pi}{4})$

45.



$(-2, -\frac{3\pi}{4}), (2, -\frac{7\pi}{4}), (2, \frac{\pi}{4})$

46.



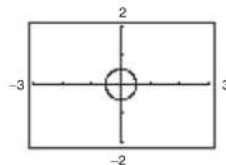
$(-3, \frac{\pi}{6}), (3, -\frac{5\pi}{6}), (3, \frac{7\pi}{6})$

47. $r = -\frac{1}{4 \sin \theta + 4 \cos \theta}$

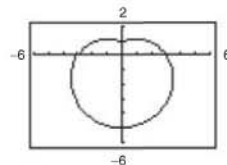
48. $(x - 1)^2 + y^2 = 1$

49. $\frac{(x + \frac{10}{9})^2}{\frac{64}{81}} - \frac{y^2}{\frac{4}{9}} = 1$

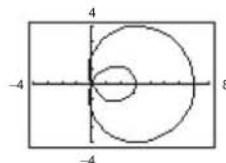
50. Circle



51. Dimpled limaçon



52. Limaçon with inner loop



53. \$701,303.32

54. $\frac{1}{4}$ 55. $24\sqrt{2}$ meters

Chapter 10

Section 10.1 (page 747)

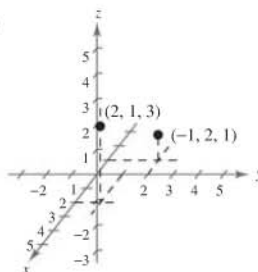
Vocabulary Check (page 747)

1. three-dimensional
2. xy -plane, xz -plane, yz -plane
3. octants
4. Distance Formula
5. $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2})$
6. sphere
7. surface, space
8. trace

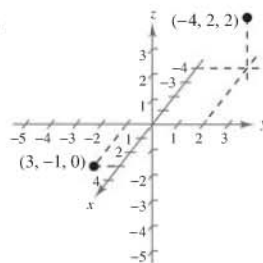
1. $A: (-1, 4, 3)$ $B: (1, 3, -2)$ $C: (-3, 0, -2)$

3. $A: (-2, -1, 4)$ $B: (3, -2, 0)$ $C: (-2, 2, -3)$

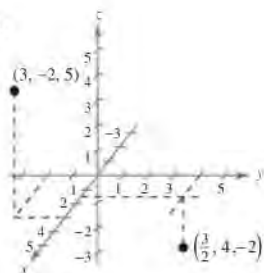
5.



7.



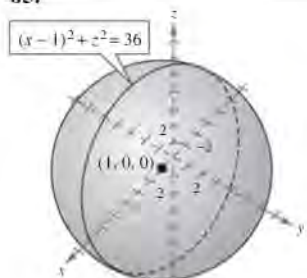
9.

11. $(-3, 3, 4)$ 13. $(10, 0, 0)$ 15. Octant IV

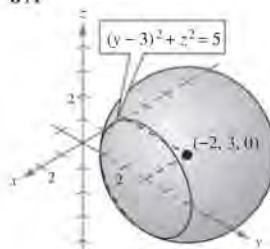
17. Octants I, II, III, and IV 19. Octants II, IV, VI, and VIII

21. $\sqrt{189}$ units 23. $\sqrt{114}$ units 25. $\sqrt{110}$ units27. $2\sqrt{5}$, 3, $\sqrt{29}$ 29. 3, 6, $3\sqrt{5}$ 31. 6, 6, $2\sqrt{10}$; isosceles triangle33. 6, 6, $2\sqrt{10}$; isosceles triangle35. $(0, -1, 7)$ 37. $(1, 0, \frac{11}{2})$ 39. $(\frac{5}{2}, 2, 6)$ 41. $(x-3)^2 + (y-2)^2 + (z-4)^2 = 16$ 43. $(x+1)^2 + (y-2)^2 + z^2 = 3$ 45. $x^2 + (y-4)^2 + (z-3)^2 = 9$ 47. $(x+3)^2 + (y-7)^2 + (z-5)^2 = 25$ 49. $(x-\frac{3}{2})^2 + y^2 + (z-3)^2 = \frac{45}{4}$ 51. Center: $(\frac{5}{2}, 0, 0)$; radius: $\frac{5}{2}$ 53. Center: $(2, -1, 0)$; radius: $\sqrt{5}$ 55. Center: $(2, -1, 3)$; radius: 257. Center: $(-2, 0, 4)$; radius: 159. Center: $(1, \frac{1}{3}, 4)$; radius: 361. Center: $(1, -2, 0)$; radius: $\frac{\sqrt{21}}{2}$ 63. Center: $(\frac{1}{3}, -1, 0)$; radius: 1

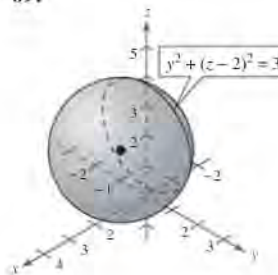
65.



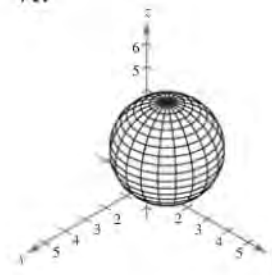
67.



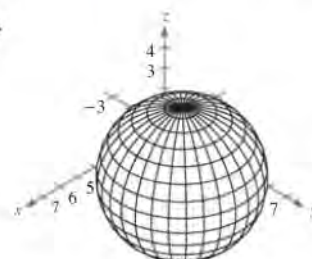
69.



71.



73.

75. $(3, 3, 3)$ 77. $x^2 + y^2 + z^2 = \frac{165^2}{4}$ 79. False. z is the directed distance from the xy -plane to P .

81. 0; 0; 0 83. A point or a circle

85. $(x_2, y_2, z_2) = (2x_m - x_1, 2y_m - y_1, 2z_m - z_1)$ 87. $v = -\frac{3 \pm \sqrt{17}}{2}$ 89. $x = \frac{5 \pm \sqrt{5}}{2}$ 91. $y = -\frac{1 \pm \sqrt{10}}{2}$ 93. $3\sqrt{2}, 315^\circ$ 95. $\sqrt{41}, 51.34^\circ$ 97. -7

99. 1, 2, 6, 15, 31

First differences: 1, 4, 9, 16

Second differences: 3, 5, 7

Neither

101. -1, 2, 5, 8, 11

First differences: 3, 3, 3, 3

Second differences: 0, 0, 0

Linear

103. $(x+5)^2 + (y-1)^2 = 49$ 105. $(y-1)^2 = -12(x-4)$ 107. $\frac{(x-3)^2}{9} + \frac{(y-3)^2}{4} = 1$ 109. $\frac{(x-6)^2}{4} - \frac{y^2}{32} = 1$

Section 10.2 (page 755)

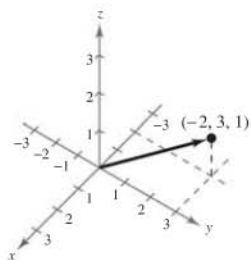
Vocabulary Check (page 755)

1. zero 2. $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$

3. component form 4. orthogonal 5. parallel

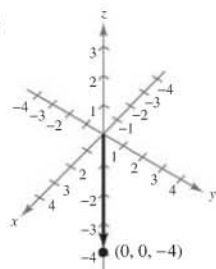
1. (a) $\langle -2, 3, 1 \rangle$

(b)



3. (a) $\langle 0, 0, -4 \rangle$

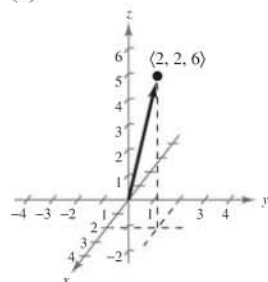
(b)



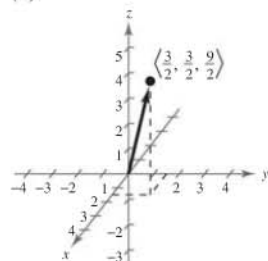
5. (a) $\langle 7, -5, 5 \rangle$ (b) $3\sqrt{11}$ (c) $\frac{\sqrt{11}}{33} \langle 7, -5, 5 \rangle$

7. (a) $\langle 2, 2, 0 \rangle$ (b) $2\sqrt{2}$ (c) $\frac{\sqrt{2}}{2} \langle 1, 1, 0 \rangle$

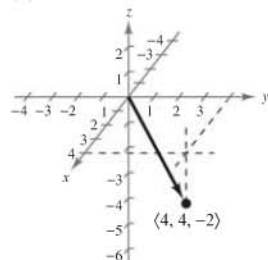
9. (a)



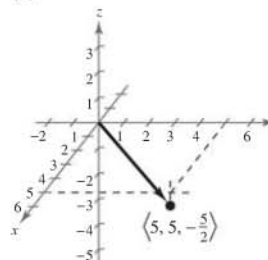
(c)



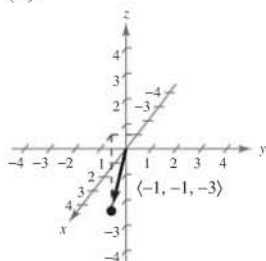
11. (a)



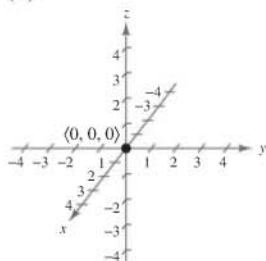
(c)



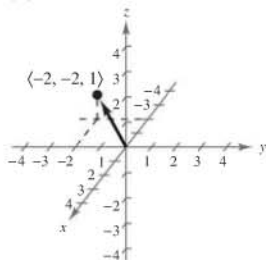
(b)



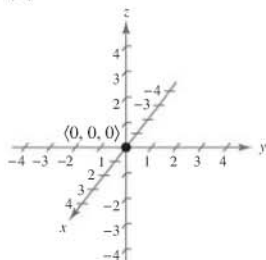
(d)



(b)



(d)



13. $\mathbf{z} = \langle -3, 7, 6 \rangle$ 15. $\mathbf{z} = \langle \frac{1}{2}, 6, \frac{3}{2} \rangle$

17. $\mathbf{z} = \langle -\frac{5}{2}, 12, \frac{15}{2} \rangle$ 19. $\mathbf{z} = \langle \frac{11}{2}, -\frac{5}{4}, -6 \rangle$

21. $9\sqrt{2}$ 23. $\sqrt{21}$ 25. $\sqrt{21}$ 27. $\sqrt{74}$

29. $\sqrt{34}$ 31. (a) $\frac{1}{13}(5\mathbf{i} - 12\mathbf{k})$ (b) $-\frac{1}{13}(5\mathbf{i} - 12\mathbf{k})$

33. (a) $\frac{\sqrt{74}}{74}(8\mathbf{i} + 3\mathbf{j} - \mathbf{k})$ (b) $-\frac{\sqrt{74}}{74}(8\mathbf{i} + 3\mathbf{j} - \mathbf{k})$

35. $\langle -26, 0, 48 \rangle$ 37. ≈ 8.73 39. -4 41. 0

43. $\approx 124.45^\circ$ 45. $\approx 109.92^\circ$ 47. Parallel

49. Orthogonal 51. Neither 53. Orthogonal

55. Not collinear 57. Collinear

59. Right triangle. Answers will vary.

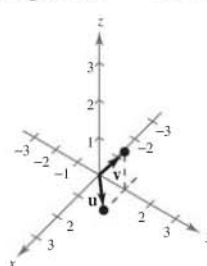
61. Acute triangle. Answers will vary. 63. $(3, 1, 7)$

65. $(6, \frac{5}{2}, -\frac{7}{4})$ 67. $\pm \frac{3\sqrt{14}}{14}$

69. $\langle 0, 2\sqrt{2}, 2\sqrt{2} \rangle$ or $\langle 0, 2\sqrt{2}, -2\sqrt{2} \rangle$

71. 10.91 pounds 73. True

75. (a)



(b) Answers will vary.

(c) $a = b = 1$

(d) Answers will vary.

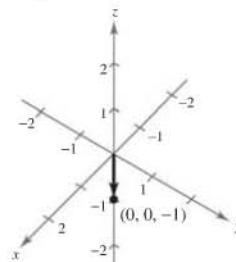
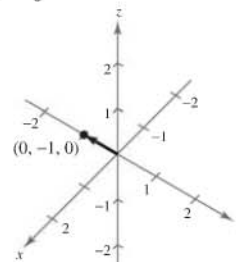
77. The angle between $\mathbf{u}$ and $\mathbf{v}$ is an obtuse angle.

Section 10.3 (page 762)

Vocabulary Check (page 762)

1. cross product 2. $\mathbf{0}$ 3. $\|\mathbf{u}\| \|\mathbf{v}\| \sin \theta$

4. triple scalar product

1. $-\mathbf{k}$

3. $-\mathbf{j}$


5. $\langle 1, 1, 1 \rangle$ 7. $\langle 3, -3, -3 \rangle$ 9. $\langle 0, 42, 0 \rangle$
 11. $-7\mathbf{i} + 13\mathbf{j} + 16\mathbf{k}$ 13. $-17\mathbf{i} + \mathbf{j} + 2\mathbf{k}$
 15. $-\frac{7}{6}\mathbf{i} - \frac{7}{8}\mathbf{j}$ 17. $-18\mathbf{i} - 6\mathbf{j}$ 19. $-\mathbf{i} - 2\mathbf{j} - \mathbf{k}$
 21. $\langle 10, -2, -4 \rangle$ 23. $-6\mathbf{i} - 15\mathbf{j} - 6\mathbf{k}$
 25. $-\frac{1}{4}\mathbf{i} - \frac{7}{10}\mathbf{j} - 2\mathbf{k}$ 27. $\frac{\sqrt{166}}{166}\langle 9, 6, -7 \rangle$
 29. $\frac{\sqrt{19}}{19}(\mathbf{i} - 3\mathbf{j} + 3\mathbf{k})$ 31. $\frac{\sqrt{7602}}{7602}(-71\mathbf{i} - 44\mathbf{j} + 25\mathbf{k})$
 33. $\frac{\sqrt{2}}{2}(\mathbf{i} - \mathbf{j})$ 35. 1 37. $\sqrt{806}$ 39. 14
 41. (a) Answers will vary.
 (b) $6\sqrt{10}$
 (c) The parallelogram is not a rectangle.
 43. $\frac{3\sqrt{13}}{2}$ 45. $\frac{1}{2}\sqrt{4290}$ 47. -16 49. 2
 51. 2 53. 12 55. 84
 57. (a) $T(p) = \frac{p}{2} \cos 40^\circ$

(b)

p	15	20	25	30	35	40	45
T	5.75	7.66	9.58	11.49	13.41	15.32	17.24

59. True 61 and 63. Proofs

65. $-\frac{1}{2}$ 67. $-\frac{1}{2}$

Section 10.4 (page 771)

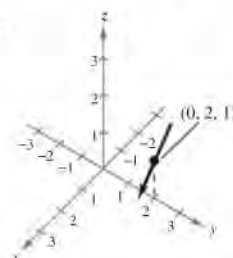
Vocabulary Check (page 771)

1. direction, $\frac{\overrightarrow{PQ}}{t}$ 2. parametric equations
 3. symmetric equations 4. normal
 5. $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

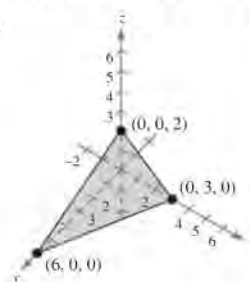
1. (a) $x = t, y = 2t, z = 3t$
 (b) $x = \frac{y}{2} = \frac{z}{3}$
 3. (a) $x = -4 + 3t, y = 1 + 8t, z = -6t$
 (b) $\frac{x+4}{3} = \frac{y-1}{8} = \frac{z}{-6}$
 5. (a) $x = 2 + 2t, y = -3 - 3t, z = 5 + t$
 (b) $\frac{x-2}{2} = \frac{y+3}{-3} = z - 5$

7. (a) $x = 2 - t, y = 4t, z = 2 - 5t$
 (b) $\frac{x-2}{-1} = \frac{y}{4} = \frac{z-2}{-5}$
 9. (a) $x = -3 + 4t, y = 8 - 10t, z = 15 + t$
 (b) $\frac{x+3}{4} = \frac{y-8}{-10} = z - 15$
 11. (a) $x = 3 - 4t, y = 1, z = 2 + 3t$
 (b) Not possible.
 13. (a) $x = -\frac{1}{2} + 3t, y = 2 - 5t, z = \frac{1}{2} - t$
 (b) $\frac{x+\frac{1}{2}}{3} = \frac{y-2}{-5} = \frac{z-\frac{1}{2}}{-1}$

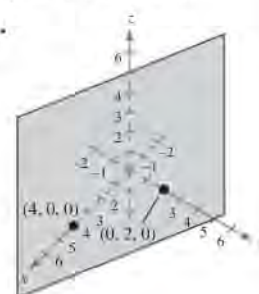
15.


 17. $x - 2 = 0$

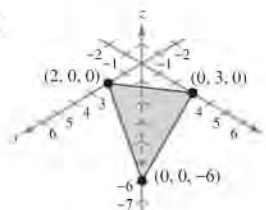
19. $-2x + y - 2z + 10 = 0$ 21. $-x - 2y + z + 2 = 0$
 23. $-3x - 9y + 7z = 0$ 25. $6x - 2y - z - 8 = 0$
 27. $y - 5 = 0$ 29. $y - z + 2 = 0$
 31. $7x + y - 11z - 5 = 0$ 33. $x = 2, y = 3, z = 4 + t$
 35. $x = 2 + 3t, y = 3 + 2t, z = 4 - t$
 37. $x = 5 + 2t, y = -3 - t, z = -4 + 3t$
 39. $x = 2 - t, y = 1 + t, z = 2 + t$
 41. Orthogonal 43. Orthogonal
 45. (a) 60.67° (b) $x = 2 - t, y = 8t, z = 7t$
 47. (a) 77.83° (b) $x = 1 + 6t, y = t, z = 1 + 7t$
 49.



51.



53.



55. $\frac{8}{9}$ 57. $\frac{2\sqrt{6}}{3}$ 59. 88.45°

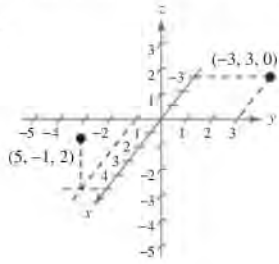
61. False. Lines that do not intersect and are not in the same plane may not be parallel.

63. Parallel. $\langle 10, -18, 20 \rangle$ is a scalar multiple of $\langle -15, 27, -30 \rangle$.

65. $x^2 + y^2 = 100$ 67. $x^2 + y^2 - 3x = 0$

69. $r = 7$ 71. $r = 5 \csc \theta$

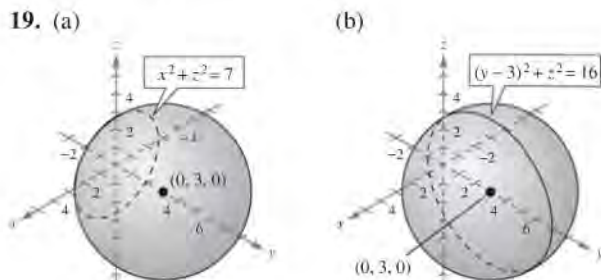
Review Exercises (page 774)

1.  3. $(-5, 4, 0)$ 5. $\sqrt{41}$

7. $\sqrt{29}, \sqrt{38}, \sqrt{67}$ 9. $(0, -1, 0)$ 11. $(1, 2, -9)$

13. $(x-2)^2 + (y-3)^2 + (z-5)^2 = 1$

15. $(x-1)^2 + (y-5)^2 + (z-2)^2 = 36$

17. Center: $(2, 3, 0)$; radius: 3


21. (a) $\langle 1, 4, -4 \rangle$ (b) $\sqrt{33}$ (c) $\frac{\sqrt{33}}{33} \langle 1, 4, -4 \rangle$

23. (a) $\langle -10, 6, 7 \rangle$ (b) $\sqrt{185}$ (c) $\frac{\sqrt{185}}{185} \langle -10, 6, 7 \rangle$

25. -9 27. 1 29. 90° 31. 90°

33. Orthogonal 35. Parallel 37. Not collinear

39. Collinear

41. A: 159.10 pounds of tension

B: 115.58 pounds of tension

C: 115.58 pounds of tension

43. $\langle -10, 0, -10 \rangle$

45. $\frac{\sqrt{7602}}{7602} (-71\mathbf{i} - 44\mathbf{j} + 25\mathbf{k})$

47. Answers will vary; $2\sqrt{43}$ 49. 75

51. (a) $x = 3 + 6t, y = 11t, z = 2 + 4t$

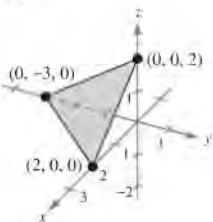
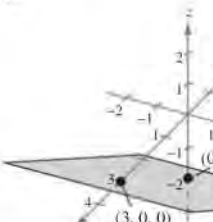
(b) $\frac{x-3}{6} = \frac{y}{11} = \frac{z-2}{4}$

53. (a) $x = -1 + 4t, y = 3 + 3t, z = 5 - 6t$

(b) $\frac{x+1}{4} = \frac{y-3}{3} = \frac{z-5}{-6}$

55. (a) $x = -2t, y = \frac{5}{2}t, z = t$ (b) $\frac{x}{-2} = \frac{y}{5/2} = z$

57. $-2x - 12y + 5z = 0$ 59. $z - 2 = 0$

61.  63. 

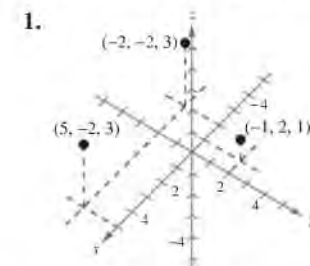
65. $\frac{\sqrt{110}}{110}$ 67. $\frac{\sqrt{110}}{55}$ 69. False. $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$

71 and 73. Answers will vary.

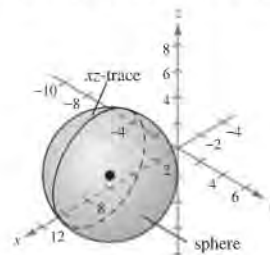
75. $\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2)\mathbf{i} - (u_1v_3 - u_3v_1)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k}$

77. The magnitude of the cross product will increase by a factor of 4.

Chapter Test (page 776)


2. No. Answers will vary. 3. $(7, 1, 2)$

4. $(x-7)^2 + (y-1)^2 + (z-2)^2 = 19$



5. $\langle 2, 5, -10 \rangle$; $\sqrt{129}$ 6. $\langle -3, -5, 8 \rangle$; $7\sqrt{2}$

7. $\mathbf{u} = \langle -2, 6, -6 \rangle$, $\mathbf{v} = \langle -12, 5, -5 \rangle$

8. (a) $\sqrt{194}$ (b) 84 (c) $\langle 0, 62, 62 \rangle$ 9. 46.23°

10. Answers will vary. Sample answer:

(a) $x = 8 - 2t$, $y = -2 + 6t$, $z = 5 - 6t$

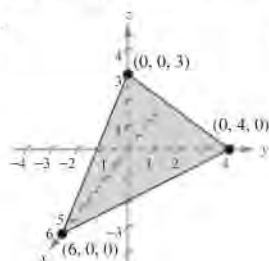
(b) $\frac{x-8}{-2} = \frac{y+2}{6} = \frac{z-5}{-6}$

11. Neither 12. Orthogonal

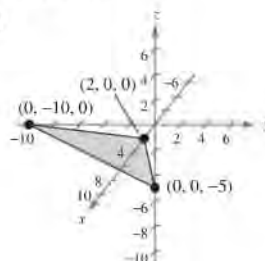
13. Answers will vary; $2\sqrt{230}$

14. $27x + 4y + 32z + 33 = 0$ 15. 200

16.



17.



18. $\frac{4\sqrt{14}}{7}$

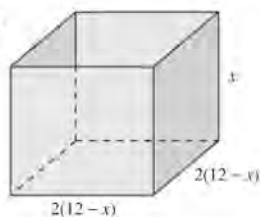
Chapter 11

Section 11.1 (page 788)

Vocabulary Check (page 788)

1. limit 2. oscillates 3. direct substitution

1. (a)

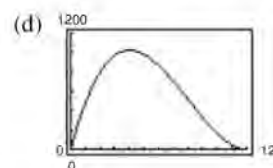


(b) Answers will vary.

x	3	3.5	3.9	4
V	972	1011.5	1023.5	1024

x	4.1	4.5	5
V	1023.5	1012.5	980

$$\lim_{x \rightarrow 4} V = 1024$$



x	1.9	1.99	1.999	2
$f(x)$	13.5	13.95	13.995	14

x	2.001	2.01	2.1
$f(x)$	14.005	14.05	14.5

14; Yes

x	2.9	2.99	2.999	3
$f(x)$	0.1695	0.1669	0.1667	Error

x	3.001	3.01	3.1
$f(x)$	0.1666	0.1664	0.1639

0.1667; No

x	-0.1	-0.01	-0.001	0
$f(x)$	1.9867	1.99987	1.9999987	Error

x	0.001	0.01	0.1
$f(x)$	1.9999987	1.99987	1.9867

2; No

x	-0.1	-0.01	-0.001	0
$f(x)$	1.8127	1.9801	1.9980	Error

x	0.001	0.01	0.1
$f(x)$	2.0020	2.0201	2.2140

2; No

x	0.9	0.99	0.999	1
$f(x)$	0.2564	0.2506	0.2501	Error

x	1.001	1.01	1.1
$f(x)$	0.2499	0.2494	0.2439

0.25

13.

x	-0.1	-0.01	-0.001	0
$f(x)$	0.2247	0.2237	0.2236	Error

x	0.001	0.01	0.1
$f(x)$	0.2236	0.2235	0.2225

0.2236

15.

x	-4.1	-4.01	-4.001	-4
$f(x)$	0.4762	0.4975	0.4998	Error

x	-3.999	-3.99	-3.9
$f(x)$	0.5003	0.5025	0.5263

0.5

17.

x	-0.1	-0.01	-0.001	0
$f(x)$	0.9983	0.99998	0.9999998	Error

x	0.001	0.01	0.1
$f(x)$	0.9999998	0.99998	0.9983

1

19.

x	-0.1	-0.01	-0.001	0
$f(x)$	-0.0997	-0.0100	-0.0010	Error

x	0.001	0.01	0.1
$f(x)$	0.0010	0.0100	0.0997

0

21.

x	-0.1	-0.01	-0.001	0
$f(x)$	0.9063	0.9901	0.9990	Error

x	0.001	0.01	0.1
$f(x)$	1.0010	1.0101	1.1070

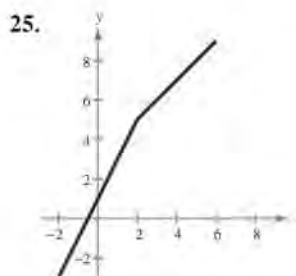
1

23.

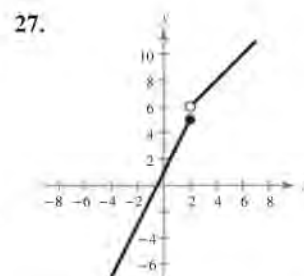
x	0.9	0.99	0.999	1
$f(x)$	2.2314	2.0203	2.0020	Error

x	1.001	1.01	1.1
$f(x)$	1.9980	1.9803	1.8232

2



$$\lim_{x \rightarrow 2} f(x) = 5$$



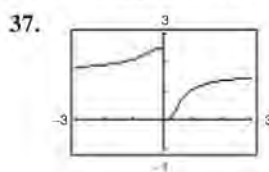
Limit does not exist.

29. 13

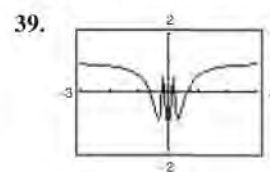
31. Limit does not exist; one-sided limits do not agree.

33. Limit does not exist; function oscillates between -2 and 2.

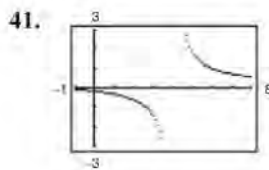
35. Limit does not exist; one-sided limits do not exist.



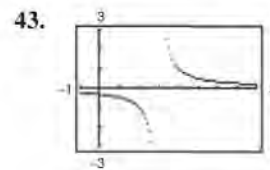
No



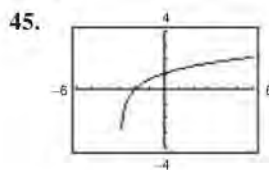
No



No



Yes



Yes

47. (a) -12 (b) 9 (c) $\frac{1}{2}$ (d) $\sqrt{3}$

49. (a) 8 (b) $\frac{3}{8}$ (c) 3 (d) $-\frac{61}{8}$

51. -15 53. 7 55. -3 57. $-\frac{9}{10}$ 59. $\frac{7}{13}$

61. 1 63. $\frac{35}{3}$ 65. e^3 67. 0 69. $\frac{\pi}{6}$

71. True 73. (a) and (b) Answers will vary.

75. (a) No. The function may approach different values from the right and left of 4.

(b) No. The function may approach 4 as x approaches 2, but the function could be undefined at $x = 2$.

77. $-\frac{1}{3}, x \neq 5$ 79. $\frac{5x+4}{5x+2}, x \neq \frac{1}{3}$

81. $\frac{x^2 - 3x + 9}{x - 2}, x \neq -3$

Section 11.2 (page 798)

Vocabulary Check (page 798)

1. dividing out technique 2. indeterminate form
-
3. one-sided limit 4. difference quotient

1. (a) 1 (b) 3 (c) 5

$$g_2(x) = -2x + 1$$

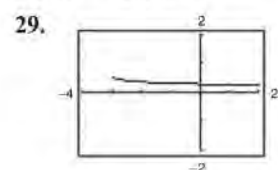
3. (a) 2 (b) 0 (c) 0

$$g_2(x) = x(x + 1)$$

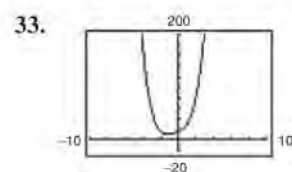
- 5.
- $\frac{1}{12}$
7. 4 9. 12 11. 0 13.
- $\frac{1}{3}$
15. 3

- 17.
- $\frac{\sqrt{5}}{10}$
- 19.
- $\frac{1}{4}$
21. -1 23. Limit does not exist.

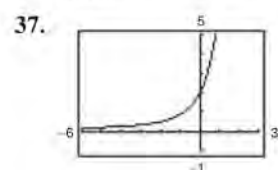
25. 0 27. 0



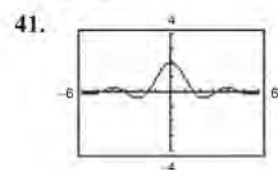
0.29



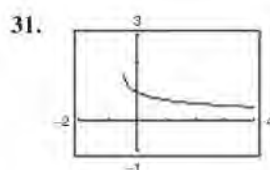
80.00



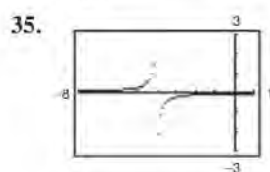
2.000



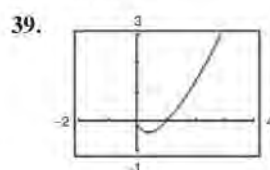
2.000



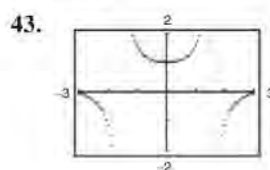
1.00



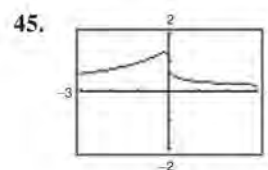
-0.06



0.000



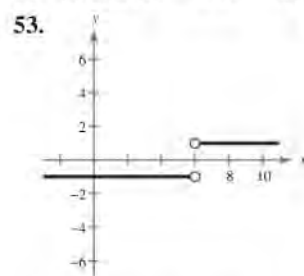
1.000



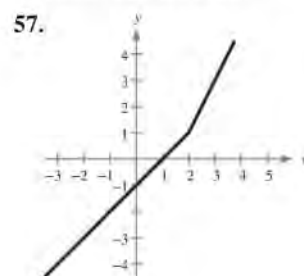
0.333

49. (a) and (b) 0.50 (c)
- $\frac{1}{2}$

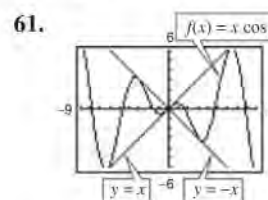
51. (a) and (b) -0.13 (c)
- $-\frac{1}{8}$



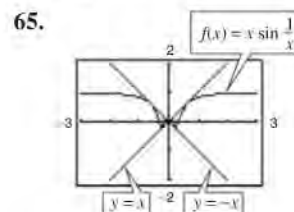
Limit does not exist.



$$\lim_{x \rightarrow 2} f(x) = 1$$

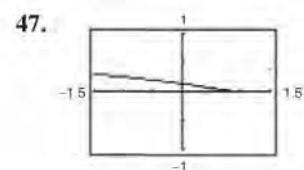


$$\lim_{x \rightarrow 0} f(x) = 0$$



$$\lim_{x \rightarrow 0} f(x) = 0$$

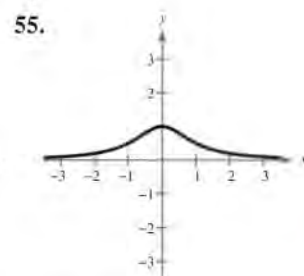
67. (a) Direct substitution; 0 (b) 1



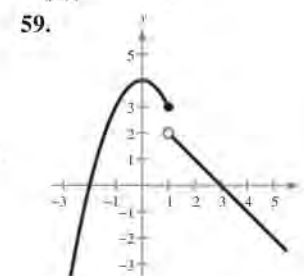
0.135

49. (a) and (b) 0.50 (c)
- $\frac{1}{2}$

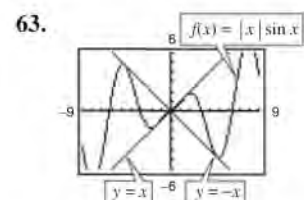
51. (a) and (b) -0.13 (c)
- $-\frac{1}{8}$



$$\lim_{x \rightarrow 1} f(x) = \frac{1}{2}$$



Limit does not exist.

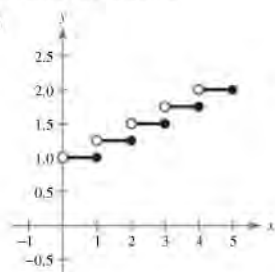


$$\lim_{x \rightarrow 0} f(x) = 0$$

69. 3 71. $\frac{1}{2\sqrt{x}}$ 73. $2x - 3$ 75. $-\frac{1}{(x+2)^2}$

77. -32 feet per second

79. (a)



(b)

t	3	3.3	3.4	3.5	3.6	3.7	4
C	1.50	1.75	1.75	1.75	1.75	1.75	1.75

$$\lim_{t \rightarrow 3.5} C(t) = 1.75$$

(c)

t	2	2.5	2.9	3	3.1	3.5	4
C	1.25	1.50	1.50	1.50	1.75	1.75	1.75

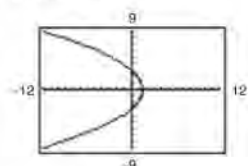
Limit does not exist; the one-sided limits do not agree.

81. Answers will vary. 83. True.

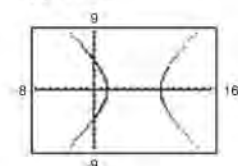
85. (a) and (b) Answers will vary.

87. $x - 2y - 26 = 0$

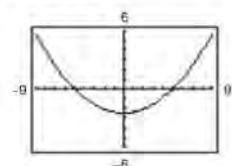
89. Parabola



91. Hyperbola



93. Parabola



95. Orthogonal

97. Parallel

Section 11.3 (page 808)

Vocabulary Check (page 808)

- Calculus
- tangent line
- secant line
- difference quotient
- derivative

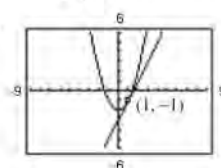
1. 0 3. $\frac{1}{2}$ 5. 2 7. -2 9. -1 11. $\frac{1}{6}$

13. $m = -2x$; (a) 0 (b) 2

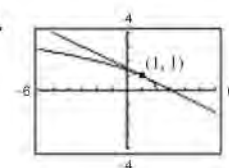
15. $m = -\frac{1}{(x+4)^2}$; (a) $-\frac{1}{16}$ (b) $-\frac{1}{4}$

17. $m = \frac{1}{2\sqrt{x-1}}$; (a) $\frac{1}{4}$ (b) $\frac{1}{6}$

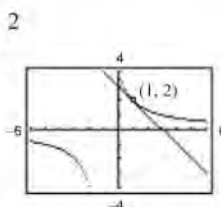
19.



21.



23.



25. 0

-1

27. $-\frac{1}{3}$ 29. $-6x$ 31. $-\frac{2}{x^3}$ 33. $\frac{1}{2\sqrt{x-4}}$

35. $-\frac{1}{(x+2)^2}$ 37. $-\frac{1}{2(x-9)^{3/2}}$

39. (a) 4

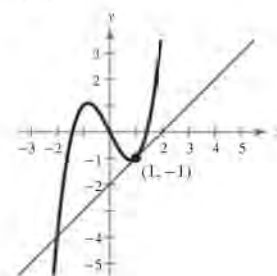
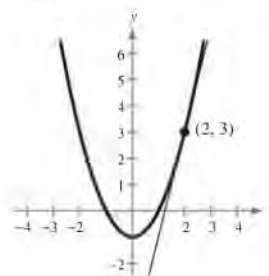
(b) $y = 4x - 5$

(c)

41. (a) 1

(b) $y = x - 2$

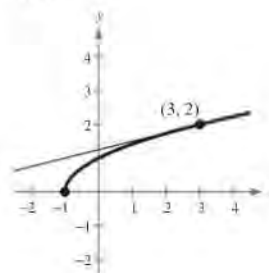
(c)



43. (a) $\frac{1}{4}$

(b) $y = \frac{1}{4}x + \frac{5}{4}$

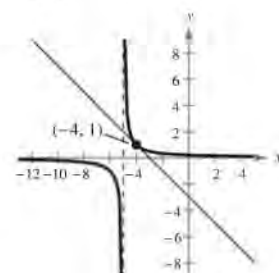
(c)



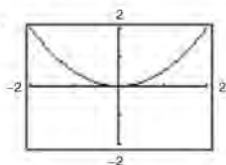
45. (a) -1

(b) $y = -x - 3$

(c)



47.

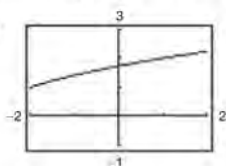


x	-2	-1.5	-1	-0.5	0
$f(x)$	2	1.125	0.5	0.125	0
$f'(x)$	-2	-1.5	-1	-0.5	0

x	0.5	1	1.5	2
$f(x)$	0.125	0.5	1.125	2
$f'(x)$	0.5	1	1.5	2

They appear to be the same.

49.



x	-2	-1.5	-1	-0.5	0
$f(x)$	1	1.225	1.414	1.581	1.732
$f'(x)$	0.5	0.408	0.354	0.316	0.289

x	0.5	1	1.5	2
$f(x)$	1.871	2	2.121	2.236
$f'(x)$	0.267	0.25	0.236	0.224

They appear to be the same.

51. $f'(x) = 2x - 4$; $(2, -1)$

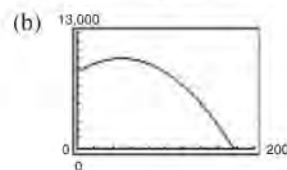
53. $f'(x) = 9x^2 - 9$; $(-1, 6)$, $(1, -6)$

55. $(-1, -1)$, $(0, 0)$, $(1, -1)$

57. $\left(\frac{\pi}{6}, \sqrt{3} + \frac{\pi}{6}\right)$, $\left(\frac{5\pi}{6}, \frac{5\pi}{6} - \sqrt{3}\right)$

59. $(0, 0)$, $(-2, 4e^{-2})$ 61. $(e^{-1}, -e^{-1})$

63. (a) $P(t) = -0.63t^2 + 63.3t + 8448$



38; The population is increasing at approximately 38,000 people per year in 2020.

(c) $f'(x) = -1.26x + 63.3$; $f'(20) = 38.1$

(d) Answers will vary.

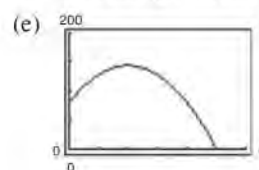
65. (a) $V'(r) = 4\pi r^2$ (b) ≈ 201.06

(c) Cubic inches per inch; The derivative is a formula for rate of change.

67. (a) $s'(t) = -32t + 64$ (b) 16 feet per second

 (c) $t = 2$ seconds; Answers will vary.

(d) -96 feet per second

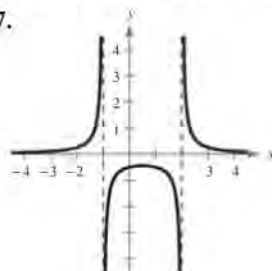


69. True. The graph of the derivative is a line, which is a one-to-one function.

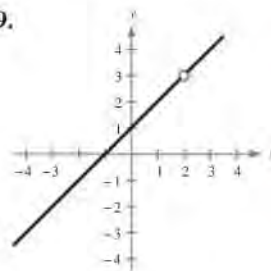
71. b 72. a 73. d 74. c

75. Answers will vary. Sample answer: A sketch of any linear function with positive slope

77.



79.



81. $\langle -2, 3, -1 \rangle$ 83. $\langle 0, 0, -36 \rangle$

Section 11.4 (page 817)

Vocabulary Check (page 817)

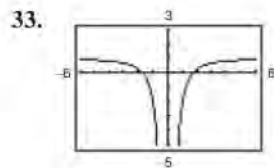
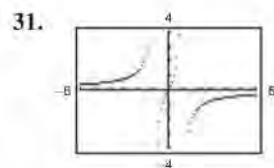
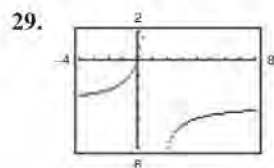
1. limit, infinity 2. converge 3. diverge

A272 Answers to Odd-Numbered Exercises and Tests

1. c 2. a 3. d 4. b 5. f 6. g 7. h
8. e 9. 0 11. -1 13. $\frac{5}{6}$ 15. -4

17. Limit does not exist. 19. $\frac{1}{3}$ 21. 2 23. -1

25. -4 27. -5

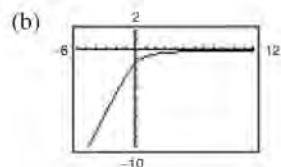


35. (a)

x	10^0	10^1	10^2	10^3
$f(x)$	-0.7321	-0.0995	-0.0100	-0.0010

x	10^4	10^5	10^6
$f(x)$	-1.0×10^{-4}	-1.0×10^{-5}	-1.0×10^{-6}

0

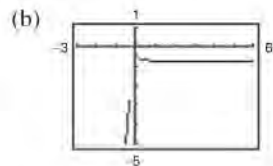


0

37. (a)

x	10^0	10^1	10^2	10^3
$f(x)$	-0.7082	-0.7454	-0.7495	-0.74995

x	10^4	10^5	10^6
$f(x)$	-0.749995	-0.7499995	-0.7500



-0.75

39. $1, \frac{3}{5}, \frac{2}{5}, \frac{5}{17}, \frac{3}{13}$

Limit: 0

41. $\frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{4}{9}, \frac{5}{11}$

Limit: $\frac{1}{2}$

43. $\frac{1}{5}, \frac{1}{2}, \frac{9}{11}, \frac{8}{7}, \frac{25}{17}$

Limit does not exist.

45. 2, 3, 4, 5, 6

Limit does not exist.

47. $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}$

Limit: 0

49.

n	10^0	10^1	10^2	10^3
a_n	2	1.55	1.505	1.5005

n	10^4	10^5	10^6
a_n	1.5001	1.5000	1.5000

1.5

$$\lim_{n \rightarrow \infty} a_n = \frac{3}{2}$$

51.

n	10^0	10^1	10^2	10^3
a_n	16	6.16	5.4136	5.3413

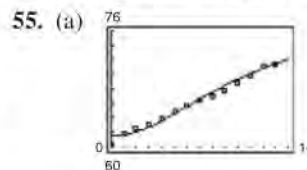
n	10^4	10^5	10^6
a_n	5.3341	5.3334	5.3333

5.33

$$\lim_{n \rightarrow \infty} a_n = \frac{16}{3}$$

53. (a) $\bar{C} = \frac{13.50x + 45,750}{x}$ (b) \$471; \$59.25

(c) \$13.50; As the number of PDAs produced gets very large, the average cost approaches \$13.50.



Answers will vary.

(b) 2004: 72,000,000; 2008: 73,800,000

(c) 78 million; As time passes, school enrollment in the United States approaches 78 million.

(d) Answers will vary.

57. False. $y = \frac{x^2}{x+1}$ does not have a horizontal asymptote.

59. True

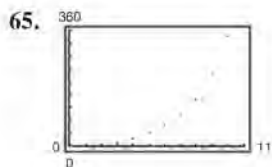
61. Answers will vary. Sample answer: Let $f(x) = \frac{1}{x^2}$,

$g(x) = \frac{1}{x^2}$, and $c = 0$. Now $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$ and

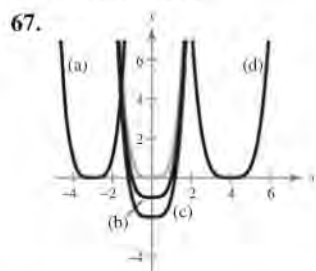
$$\lim_{x \rightarrow 0} [f(x) - g(x)] = 0.$$



Converges to 0

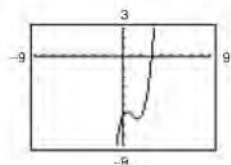
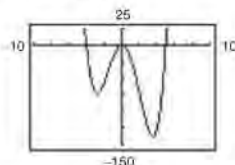


Diverges



69. $x^2 + 2x + 1$ 71. $x^3 + 5x^2 - 3 - \frac{2}{3x + 2}$

73. $-4, 5, 0, 0$ 75. 3



77. 60 79. 150

Section 11.5 (page 826)

Vocabulary Check (page 826)

1. $\frac{n(n+1)}{2}$ 2. $\frac{n^2(n+1)^2}{4}$ 3. area

1. 420 3. 44,140 5. 44,140 7. 5850

9. (a) $S(n) = \frac{n^2 + 2n + 1}{4n^2}$

n	10 ⁰	10 ¹	10 ²	10 ³	10 ⁴
S(n)	1	0.3025	0.2550	0.2505	0.2501

(c) $\lim_{n \rightarrow \infty} S(n) = \frac{1}{4}$

11. (a) $S(n) = \frac{2n^2 + 3n + 7}{2n^2}$

n	10 ⁰	10 ¹	10 ²	10 ³	10 ⁴
S(n)	6	1.185	1.0154	1.0015	1.0002

(c) $\lim_{n \rightarrow \infty} S(n) = 1$

13. (a) $S(n) = \frac{14n^2 + 3n + 1}{6n^3}$

n	10 ⁰	10 ¹	10 ²	10 ³	10 ⁴
S(n)	3	0.2385	0.0234	0.0023	0.0002

(c) $\lim_{n \rightarrow \infty} S(n) = 0$

15. (a) $S(n) = \frac{4n^2 - 3n - 1}{6n^2}$

n	10 ⁰	10 ¹	10 ²	10 ³	10 ⁴
S(n)	0	0.615	0.6617	0.6662	0.6666

(c) $\lim_{n \rightarrow \infty} S(n) = \frac{2}{3}$

17. 14.25 19. 1.27

n	4	8	20	50
Approximate area	18	21	22.8	23.52

n	4	8	20	50
Approximate area	3.5156	2.8477	2.4806	2.3409

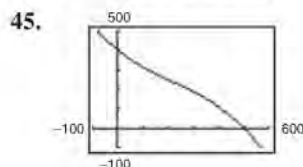
n	4	8	20	50	100	∞
Area	40	38	36.8	36.32	36.16	36

n	4	8	20	50	100	∞
Area	36	38	39.2	39.68	39.84	40

n	4	8	20	50	100	∞
Area	14.25	14.81	15.13	15.25	15.29	$\frac{46}{3}$

n	4	8	20	50	100	∞
Area	19	18.5	18.2	18.08	18.04	18

33. 3 35. 2 37. $\frac{10}{3}$ 39. $\frac{17}{4}$ 41. $\frac{3}{4}$ 43. $\frac{51}{4}$



$\approx 105,208.33$ square feet

47. True 49. Answers will vary. 51. $n\pi$

53. $\langle 24, -30 \rangle$ 55. $\sqrt{5} - 2$

Review Exercises (page 829)

1.

x	2.9	2.99	2.999	3
$f(x)$	16.4	16.94	16.994	17

x	3.001	3.01	3.1
$f(x)$	17.006	17.06	17.6

17. Yes

3.

x	-0.1	-0.01	-0.001	0
$f(x)$	1.0517	1.0050	1.0005	Error

x	0.001	0.01	0.1
$f(x)$	0.9995	0.9950	0.9516

1. No

5. 2 7. 2 9. (a) 64 (b) 7 (c) 20 (d) $\frac{4}{5}$

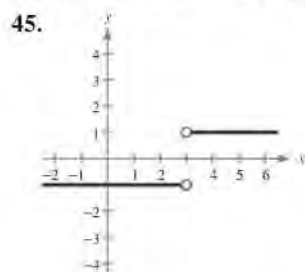
11. 5 13. 77 15. $\frac{10}{3}$ 17. -2 19. 0

21. $\frac{2}{e}$ 23. $-\frac{\pi}{6}$ 25. $-\frac{1}{4}$ 27. $\frac{1}{15}$

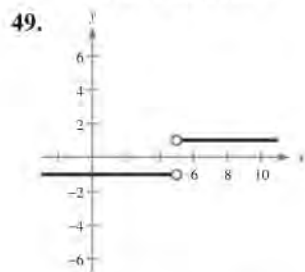
29. $-\frac{1}{3}$ 31. -1 33. $\frac{1}{4}$ 35. $\frac{1}{4}$

37. (a) and (b) 0.17 39. Limit does not exist.

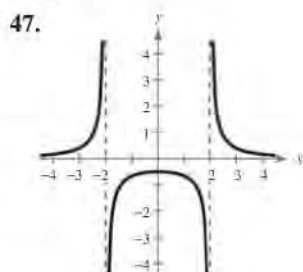
41. (a) and (b) 2 43. (a) and (b) 0.577



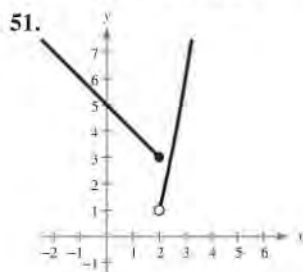
Limit does not exist.



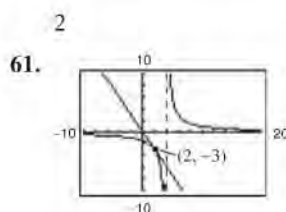
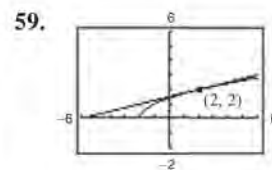
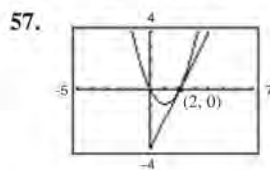
Limit does not exist.



Limit does not exist.



Limit does not exist.

53. $3 - 2x$ 55. 2

 $-\frac{3}{2}$

63. $m = 2x - 4$; (a) -4 (b) 6

65. $m = -\frac{4}{(x-6)^2}$; (a) -4 (b) -1

67. $f'(x) = 0$ 69. $h'(x) = -\frac{1}{2}$ 71. $g'(x) = 4x$

73. $f'(t) = \frac{1}{2\sqrt{t+5}}$ 75. $g'(s) = -\frac{4}{(s+5)^2}$

77. $g'(x) = -\frac{1}{2(x+4)^{3/2}}$ 79. 2 81. 0

83. Limit does not exist.

85. 3

87. $-\frac{1}{9}, \frac{1}{14}, \frac{3}{19}, \frac{5}{24}, \frac{7}{29}$ 89. $-1, \frac{1}{8}, -\frac{1}{27}, \frac{1}{64}, -\frac{1}{125}$

Limit: $\frac{2}{5}$

Limit: 0

91. $-\frac{1}{2}, -\frac{9}{8}, -\frac{7}{6}, -\frac{37}{32}, -\frac{57}{50}$

Limit: -1

93. (a) $S(n) = \frac{5n^2 + 9n + 4}{6n^2}$

(b)

n	10^0	10^1	10^2	10^3	10^4
$S(n)$	3	0.99	0.8484	0.8348	0.8335

(c) $\frac{5}{6}$

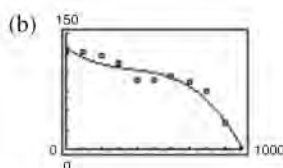
95. 675

97.

n	4	8	20	50
Approximate area	7.5	6.375	5.74	5.4944

99. 50 101. 15 103. 6 105. 34

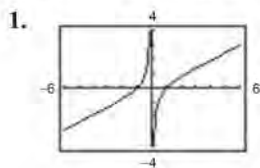
107. (a) $y = (-3.376 \times 10^{-7})x^3 + (3.753 \times 10^{-4})x^2 - 0.168x + 132$



(c) 88,700 square feet

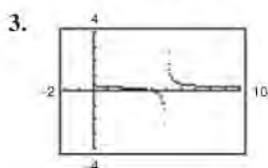
109. False. The limit of the rational function as x approaches ∞ does not exist.

Chapter Test (page 832)

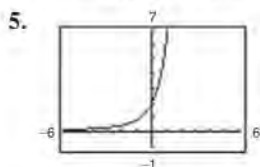


-0.75

$$\lim_{x \rightarrow -2} f(x) = -\frac{3}{4}$$



Limit does not exist.



2.0000

6. (a) $m = 6x - 5$; 7 (b) $m = 6x^2 + 6$; 12

7. $f'(x) = -\frac{2}{5}$ 8. $f'(x) = 4x + 4$

9. $f'(x) = -\frac{1}{(x+3)^2}$ 10. 0 11. -3

12. Limit does not exist.

13. $0, \frac{3}{4}, \frac{14}{19}, \frac{12}{17}, \frac{36}{53}$

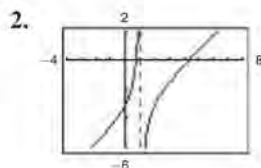
Limit: $\frac{1}{2}$

14. $0, 1, 0, \frac{1}{2}, 0$

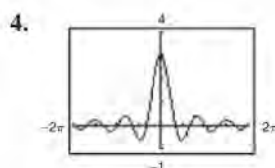
Limit: 0

15. 12.5 16. 8 17. $\frac{16}{3}$

18. (a) $y = 8.79x^2 - 6.2x - 0.4$ (b) 81.7 feet per second



Limit does not exist.



3.0000

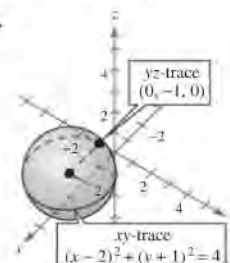
Cumulative Test for Chapters 10 and 11 (page 833)

1. $(-6, 1, 3)$ 2. $(0, -4, 0)$ 3. $\sqrt{149}$

4. 3, 4, 5 5. $(-1, 2, \frac{1}{2})$

6. $(x-2)^2 + (y-2)^2 + (z-4)^2 = 24$

7.



8. $\mathbf{u} \cdot \mathbf{v} = -38$

$\mathbf{u} \times \mathbf{v} = \langle -18, -6, -14 \rangle$

9. Neither 10. Orthogonal 11. Parallel 12. 12

13. (a) $x = -2 + 7t, y = 3 + 5t, z = 25t$

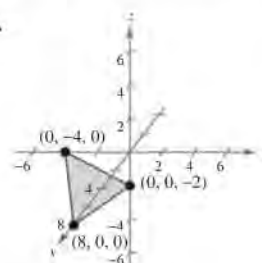
(b) $\frac{x+2}{7} = \frac{y-3}{5} = \frac{z}{25}$

14. $x = -1 + 2t, y = 2 - 4t, z = t$

15. $75x + 50y - 31z = 0$

16.

17. $\frac{\sqrt{30}}{2}$



18. 84.26° 19. 4 20. $-\frac{1}{3}$ 21. $\frac{1}{14}$ 22. $\frac{1}{4}$

23. -1 24. Limit does not exist. 25. $-\frac{1}{4}$ 26. $\frac{1}{2}$

27. $\frac{1}{4}$ 28. $m = -2x; 0$ 29. $m = \frac{1}{2}(x+3)^{-1/2}; \frac{1}{2}$

30. $m = -(x+3)^{-2}; -\frac{1}{16}$ 31. $m = 2x - 1; 1$

32. Limit does not exist. 33. -7 34. 3

35. 0 36. 0 37. Limit does not exist.

38. -42,875 39. 8190 40. 672,880

41. 10.5 42. 8.13 43. 2.69 44. 1.57 45. $\frac{5}{2}$

46. 64 47. 28 48. $\frac{76}{3}$ 49. $\frac{16}{3}$ 50. $\frac{3}{4}$

Appendices

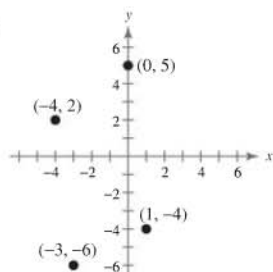
Appendix B.1 (page A32)

Vocabulary Check (page A32)

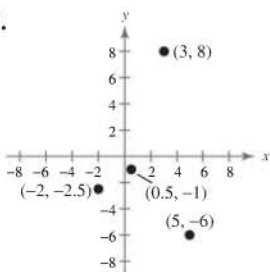
1. (a) iii (b) vi (c) i (d) iv (e) v (f) ii
 2. Cartesian 3. Distance Formula
 4. Midpoint Formula
 5. $(x - h)^2 + (y - k)^2 = r^2$, center, radius

1. A: (2, 6); B: (-6, -2); C: (4, -4); D: (-3, 2)

3.



5.

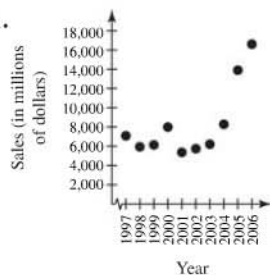


7. (-5, 4) 9. (-6, -6) 11. Quadrant IV

13. Quadrant II 15. Quadrant III or IV

17. Quadrant III 19. Quadrant I or III

21.



23. 8 25. 5 27. 13 29.
- $\frac{\sqrt{277}}{6}$
- 31.
- $\sqrt{71.78}$

33. (a) 4, 3, 5 (b)
- $4^2 + 3^2 = 5^2$

35. (a) 10, 3,
- $\sqrt{109}$
- (b)
- $10^2 + 3^2 = (\sqrt{109})^2$

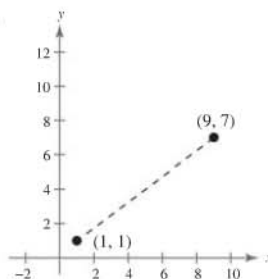
- 37.
- $(\sqrt{5})^2 + (\sqrt{45})^2 = (\sqrt{50})^2$

39. Two equal sides of length
- $\sqrt{29}$

41. Opposite sides have equal lengths of
- $2\sqrt{5}$
- and
- $\sqrt{85}$
- .

43. The diagonals are of equal length ($\sqrt{58}$). The slope of the line between (-5, 6) and (0, 8) is $\frac{2}{5}$. The slope of the line between (-5, 6) and (-3, 1) is $-\frac{5}{2}$. The slopes are negative reciprocals, making them perpendicular lines, which form a right angle.

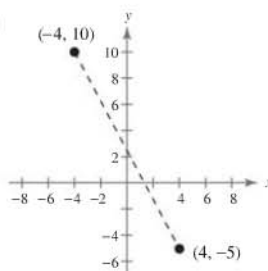
45. (a)



(b) 10

(c) (5, 4)

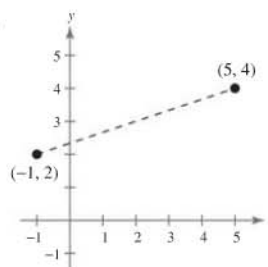
47. (a)



(b) 17

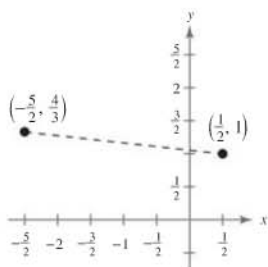
(c) $(0, \frac{5}{2})$

49. (a)

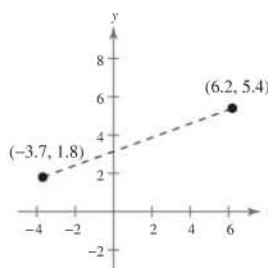
(b) $2\sqrt{10}$

(c) (2, 3)

51. (a)

(b) $\frac{\sqrt{82}}{3}$ (c) $(-1, \frac{7}{6})$

53. (a)

(b) $\sqrt{110.97}$

(c) (1.25, 3.6)

55. \$3,093.5 million

57. $(2x_m - x_1, 2y_m - y_1)$; (a) (7, 0) (b) (9, -3)

59. $x^2 + y^2 = 9$ 61. $(x - 2)^2 + (y + 1)^2 = 16$

63. $(x + 1)^2 + (y - 2)^2 = 5$

65. $(x - 3)^2 + (y - 4)^2 = 25$

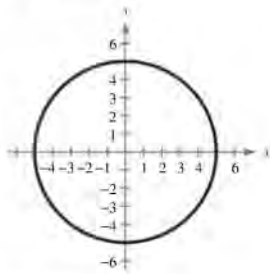
67. $(x + 2)^2 + (y - 1)^2 = 1$

69. $(x - 3)^2 + (y + 6)^2 = 16$

71. $(x - 2)^2 + (y + 1)^2 = 16$

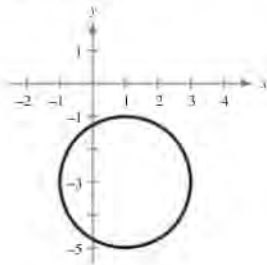
73. Center: $(0, 0)$

Radius = 5



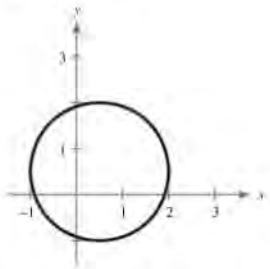
75. Center: $(1, -3)$

Radius = 2



77. Center: $(\frac{1}{2}, \frac{1}{2})$

Radius = $\frac{3}{2}$



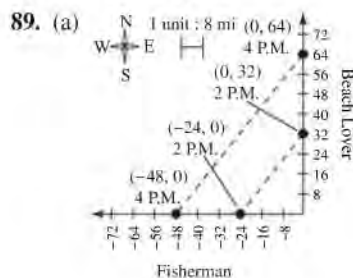
79. $(0, 1), (4, 2), (1, 4)$ 81. $(-1, 5), (2, 8), (4, 5), (1, 2)$

83. 65

85. (a) Answers will vary. Sample answer: The number of artists elected each year seems to be nearly steady except for the first few years. Estimate: From 5 to 7 new members in 2007

(b) Answers will vary. Sample answer: The Rock and Roll Hall of Fame was opened in 1986.

87. $5\sqrt{74} \approx 43$ yards



(b) 2 P.M.: 40 miles; 4 P.M.: 80 miles; The yachts are twice as far from each other at 4 P.M. as they were at 2 P.M.

91. The distance between $(2, 6)$ and $(2 + 2\sqrt{3}, 0)$ is $4\sqrt{3}$. The distance between $(2 + 2\sqrt{3}, 0)$ and $(2 - 2\sqrt{3}, 0)$ is $4\sqrt{3}$. The distance between $(2, 6)$ and $(2 - 2\sqrt{3}, 0)$ is $4\sqrt{3}$. Because the distance between each set of points is $4\sqrt{3}$, the sides connecting those points are all the same length, making the coordinates the vertices of an equilateral triangle.

93. False. You would have to use the Midpoint Formula 15 times.

95. False. It could be a rhombus.

97. No. The scales depend on the magnitudes of the quantities measured.

Appendix B.2 (page A43)

Vocabulary Check (page A43)

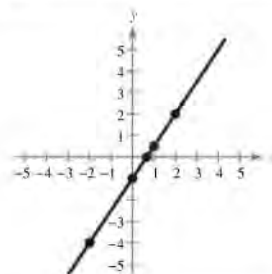
1. solution point 2. graph 3. intercepts

1. (a) Yes (b) Yes 3. (a) No (b) Yes

5. (a) No (b) Yes

7.

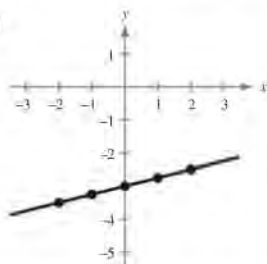
x	-2	0	$\frac{2}{3}$	1	2
y	-4	-1	0	$\frac{1}{2}$	2
Solution point	$(-2, -4)$	$(0, -1)$	$(\frac{2}{3}, 0)$	$(1, \frac{1}{2})$	$(2, 2)$



9. (a)

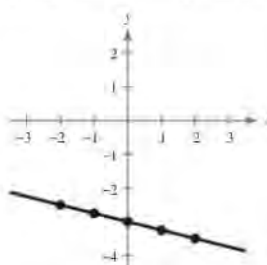
x	-2	-1	0	1	2
y	$-\frac{7}{2}$	$-\frac{13}{4}$	-3	$-\frac{11}{4}$	$-\frac{5}{2}$

(b)



(c)

x	-2	-1	0	1	2
y	$-\frac{5}{2}$	$-\frac{11}{4}$	-3	$-\frac{13}{4}$	$-\frac{7}{2}$



The lines have opposite slopes but the same y-intercept.

11. e

12. f

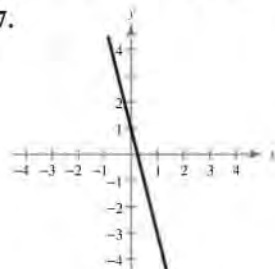
13. b

14. d

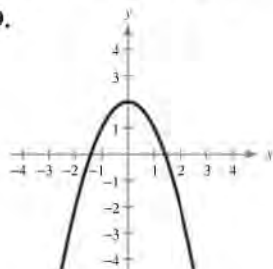
15. c

16. a

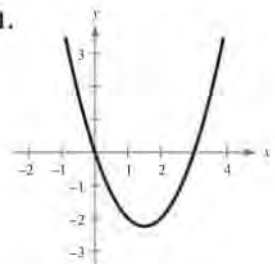
17.



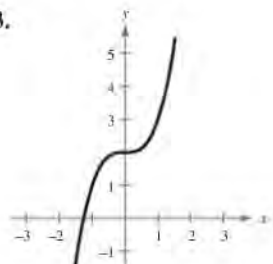
19.



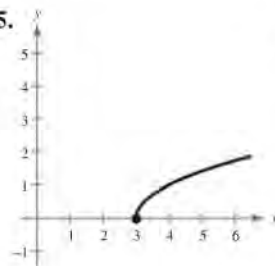
21.



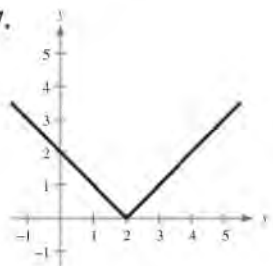
23.



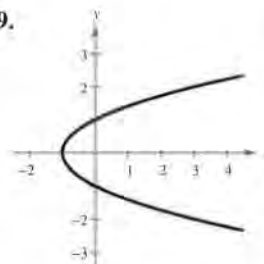
25.



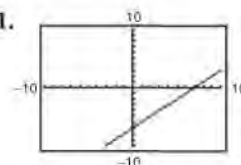
27.



29.

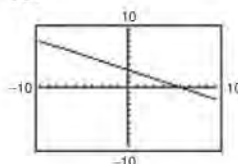


31.



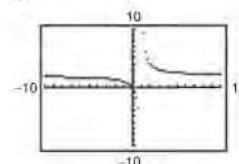
Intercepts: (7, 0), (0, -7)

33.



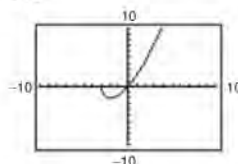
Intercepts: (6, 0), (0, 3)

35.



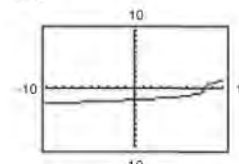
Intercept: (0, 0)

37.



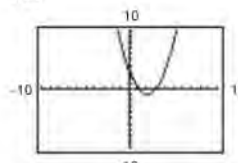
Intercepts: (-3, 0), (0, 0)

39.



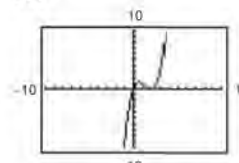
Intercepts: (0, -2), (8, 0)

41.



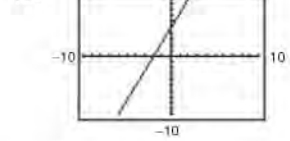
Intercepts: (0, 3), (1, 0), (3, 0)

43.

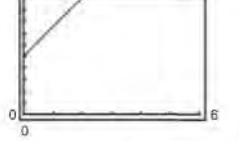


Intercepts: (0, 0), (2, 0)

45.

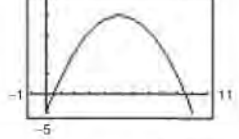
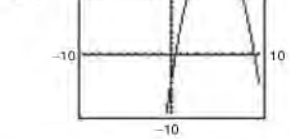


47.



The standard setting gives a more complete graph.

49.



The specified setting gives a more complete graph.

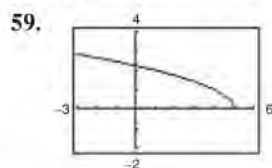
49. $X_{\min} = -10$
 $X_{\max} = 10$
 $X_{\text{scl}} = 2$
 $Y_{\min} = -50$
 $Y_{\max} = 100$
 $Y_{\text{scl}} = 25$

51. $X_{\min} = -5$
 $X_{\max} = 1$
 $X_{\text{scl}} = 1$
 $Y_{\min} = -3$
 $Y_{\max} = 1$
 $Y_{\text{scl}} = 1$

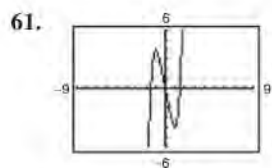
53. $X_{\min} = -30$
 $X_{\max} = 30$
 $X_{\text{scl}} = 5$
 $Y_{\min} = -10$
 $Y_{\max} = 50$
 $Y_{\text{scl}} = 5$

55. The graphs are identical.
 Distributive Property

57. The graphs are identical. Associative Property
 of Multiplication

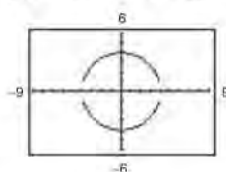


- (a) (2, 1.73)
 (b) (-4, 3)

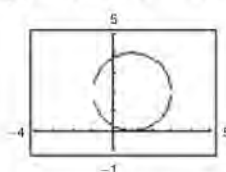


- (a) (-0.5, 2.47)
 (b) (1, -4), (-1.65, -4)

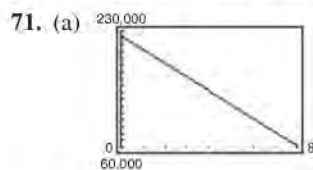
63. $y_1 = \sqrt{16 - x^2}$, $y_2 = -\sqrt{16 - x^2}$



65. $y_1 = \sqrt{4 - (x - 1)^2} + 2$, $y_2 = -\sqrt{4 - (x - 1)^2} + 2$



67. a 68. c 69. b, c, d 70. c, d



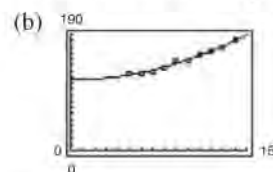
(b) \$109,000 (c) \$178,000

73. (a)

Year	1995	1996	1997	1998	1999
Median sales price (in thousands of dollars)	123.4	127.1	131.5	136.5	142.3

Year	2000	2001	2002	2003	2004
Median sales price (in thousands of dollars)	148.6	155.5	163.0	171.1	179.6

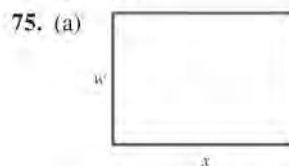
The model fits the data fairly well.



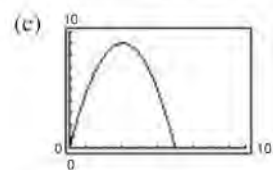
Answers will vary. Sample answer: Because the graphs overlap and have little divergence from one another, the model fits the data well.

(c) 2008: \$218,155; 2010: \$239,700; Answers will vary.

(d) 2000



(b) Answers will vary.



(d) $A \approx 5.4$ (e) $x = 3$, $w = 3$

77. False. $y = x^2 - 1$ has two x -intercepts.

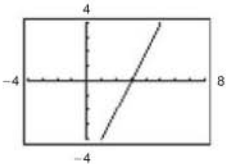
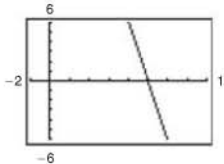
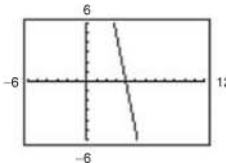
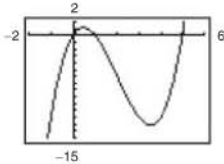
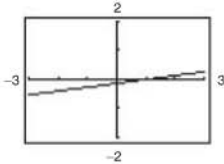
79. Answers will vary. Sample answer: use the *ZoomFit* and *Zoom out* features as needed.

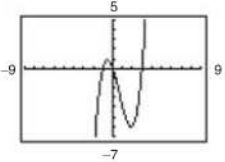
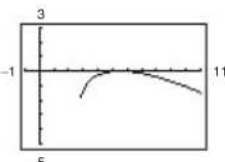
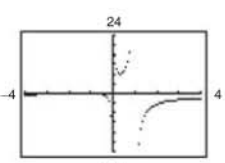
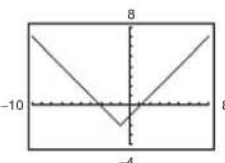
81. Answers will vary. Sample answer: $y = 250x + 1000$ could represent the amount of money in someone's checking account after time x if they deposited an initial \$1000 and added \$250 every x time increment (x could potentially stand for a month, for example).

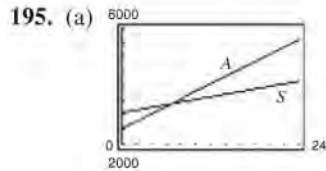
Appendix B.3 (page A59)

Vocabulary Check (page A59)

1. equation 2. solve 3. extraneous
4. x-intercept, y-intercept 5. point of intersection

1. (a) Yes (b) No (c) No (d) No
3. (a) Yes (b) No (c) No (d) No
5. (a) No (b) No (c) No (d) Yes
7. Identity 9. Identity 11. Conditional equation
13. $x = -\frac{96}{23}$; Answers will vary.
15. $x = \frac{20}{81}$; Answers will vary. 17. $x = 12$ 19. $y = 1$
21. $y = -9$ 23. $x = -10$ 25. $z = -\frac{6}{5}$
27. $z = \frac{17}{48}$ 29. $u = 10$ 31. $x = 4$ 33. $x = 5$
35. $x = \frac{11}{6}$ 37. $x = \frac{5}{3}$ 39. No solution
41. (5, 0), (0, -5) 43. (-2, 0), (1, 0), (0, -2)
45. (-2, 0), (0, 0) 47. No intercepts
49. (-2, 0), (6, 0), (0, -2) 51. (1, 0), (0, $\frac{1}{2}$)
53. 
(3, 0)
55. 
(10, 0)
57. $5(4 - 4) = 0$

59. $(0)^3 - 6(0)^2 + 5(0) = 0$
 $(5)^3 - 6(5)^2 + 5(5) = 0$
 $(1)^3 - 6(1)^2 + 5(1) = 0$

61. $\frac{1+2}{3} - \frac{1-1}{5} - 1 = 0$

63. $\frac{12}{23}$ 65. $\frac{89}{13}$ 67. 6 69. -194 71. 3, 12
73. 1, -1.2 75. $-\frac{3}{10}$ 77. 0.5, -3, 3 79. ± 1

81. -1.333 83. -1, 7 85. 11 87. (1, 1)
89. (-1, 3), (2, 6) 91. (-1, 3) 93. (4, 1)
95. (1.449, 1.899), (-3.449, -7.899)
97. (0, 0), (-2, 8), (2, 8) 99. $0, -\frac{1}{2}$ 101. 4, -2
103. 3, $-\frac{1}{2}$ 105. 2, -6 107. $-a \pm b$ 109. ± 7
111. 16, 8 113. No solution
115. $\frac{1}{2} \pm \sqrt{3}$; 2.23, -1.23 117. 2 119. -8, 4
121. $-3 \pm \sqrt{7}$ 123. $1 \pm \frac{\sqrt{6}}{3}$ 125. No solution
127. $\frac{-5 \pm \sqrt{89}}{4}$ 129. $1 \pm \sqrt{3}$ 131. $-4 \pm 2\sqrt{5}$
133. No solution 135. $\frac{2}{7}$ 137. $-\frac{8}{3} \pm \frac{\sqrt{13}}{3}$
139. $1 \pm \sqrt{2}$ 141. 6, -12 143. 7 145. $\frac{1}{2} \pm \sqrt{3}$
147. $-\frac{1}{2}$ 149. $\pm 2, 0$ 151. -3, 0 153. $0, \pm \frac{3\sqrt{2}}{2}$
155. $\pm 1, \pm \sqrt{3}$ 157. 3, 1, -1 159. $\pm \frac{1}{2}, \pm 4$
161. $-\frac{1}{5}, -\frac{1}{3}$ 163. 2, $-\frac{3}{5}$ 165. $\frac{1}{4}$ 167. 26
169. 0 171. -256.5 173. 9 175. -59, 69
177. 1 179. $\frac{-3 \pm \sqrt{21}}{6}$ 181. $2, -\frac{3}{2}$
183. 3, -2 185. $\sqrt{3}, -3$
187. (a) 
(b) and (c)
 $x = 0, 3, -1$
(d) They are the same.
189. (a) 
(b) and (c)
 $x = 5, 6$
(d) They are the same.
191. (a) 
(b) and (c)
 $x = -1$
(d) They are the same.
193. (a) 
(b) and (c)
 $x = 1, -3$
(d) They are the same.



(6.6, 3386.5); The x -coordinate represents the year and the y -coordinate represents the population. In 1986, the populations were the same at about 3,386,500 people.

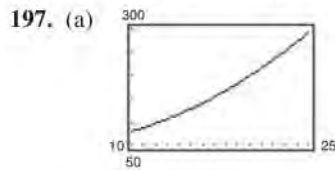
(b) (6.6, 3386.5); The point of intersection represents the point in time at which the populations of the two cities were the same.

(c) South Carolina increases by 45,200 people per year while Arizona increases by 128,200 people per year. Arizona has a greater population growth rate than South Carolina.

(d) South Carolina: 4,443,000 people

Arizona: 6,397,000 people

Answers will vary.



(b) 16.8°C (c) 2.5

199. False. The lines could be identical.

201. $c = \frac{5}{8}$ 203. (a) 0, $-\frac{b}{a}$ (b) 0, 1

Appendix B.4 (page A72)

Vocabulary Check (page A72)

1. negative 2. double 3. $-a \leq x \leq a$
4. $x \leq -a$, $x \geq a$ 5. zeros, undefined values

1. f 2. a 3. d 4. b 5. e 6. c

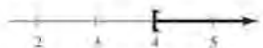
7. (a) Yes (b) No (c) Yes (d) No

9. (a) No (b) Yes (c) Yes (d) No

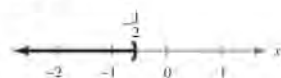
11. $x > -4$



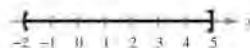
15. $x \geq 4$



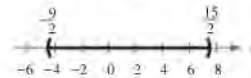
13. $x < -\frac{1}{2}$



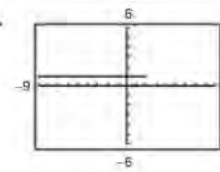
17. $-2 < x \leq 5$



19. $-\frac{9}{2} < x < \frac{15}{2}$

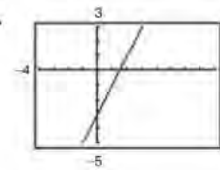


21.



$x \leq 2$

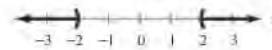
25.



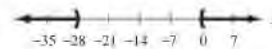
(a) $x \geq 2$

(b) $x \leq \frac{3}{2}$

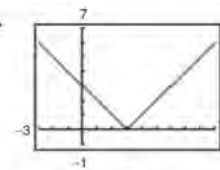
29. $x < -2$, $x > 2$



33. $x < -28$, $x > 0$



37.



(a) $1 \leq x \leq 5$

(b) $x \leq -1$, $x \geq 7$

41. $|x| > 3$ 43. $|x - 7| \leq 10$ 45. $|x - 3| \geq 5$

47. Positive on: $(-\infty, -1) \cup (5, \infty)$

Negative on: $(-1, 5)$

49. Positive on: $\left(-\infty, \frac{2 - \sqrt{10}}{2}\right) \cup \left(\frac{2 + \sqrt{10}}{2}, \infty\right)$

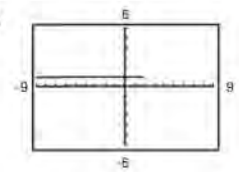
Negative on: $\left(\frac{2 - \sqrt{10}}{2}, \frac{2 + \sqrt{10}}{2}\right)$

51. Positive on: $(-\infty, \infty)$

53. $(-7, 3)$

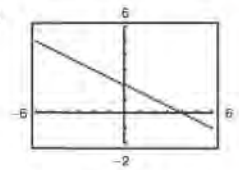


23.



$x < 2$

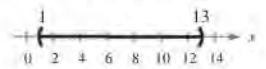
27.



(a) $-2 \leq x \leq 4$

(b) $x \leq 4$

31. $1 < x < 13$



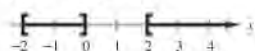
35. $\frac{1}{4} < x < \frac{3}{4}$



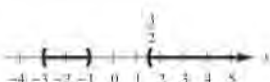
39. $|x| \leq 3$



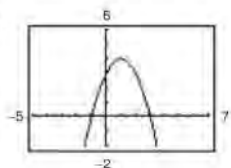
57. $[-2, 0], [2, \infty)$



61. $(-3, -1), (\frac{3}{2}, \infty)$



65.



(a) $x \leq -1, x \geq 3$

(b) $0 \leq x \leq 2$

67. $(-\infty, -1), (0, 1)$



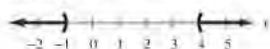
59. No solution

63. (a) $x = 1$

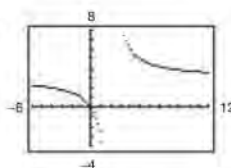
(b) $x \geq 1$

(c) $x > 1$

69. $(-\infty, -1), (4, \infty)$



71.



(a) $0 \leq x < 2$

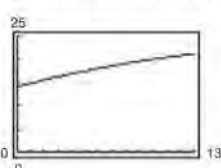
(b) $2 < x \leq 4$

73. $[5, \infty)$ 75. $(-\infty, \infty)$ 77. $(-\infty, -2], [2, \infty)$

79. (a) 1994 (b) (1990, 1994); (1994, 2004)

81. (a) 10 seconds (b) (4, 6)

83. (a)



(b) (1991, 2000)

(c) $1.28 < t < 10.09$

(d) No; Answers will vary.

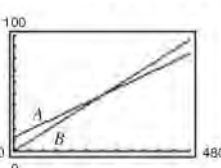
85. $t \geq 2.11$; Answers will vary.

87. 20.70; Answers will vary.

89. $333\frac{1}{3}$ vibrations per second 91. $1.2 < t < 2.4$

93. (a) $A = 12 + 0.15x, B = 0.20x$

(b)



(c) Option B is the better option for monthly usage of up to 240 minutes. For more than 240 minutes, option A is the better option.

(d) Sample answer: I would choose option A because I normally use my cell phone more than 240 minutes per month.

95. False, $10 \geq -x$ 97. a, b 99. iv, ii, iii, i

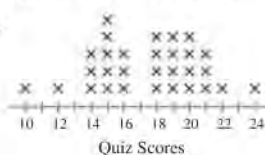
Appendix B.5 (page A81)

Vocabulary Check (page A81)

- Statistics
- Line plots
- histogram
- frequency distribution
- bar graph
- Line graphs

1. (a) \$2.569 (b) 0.19

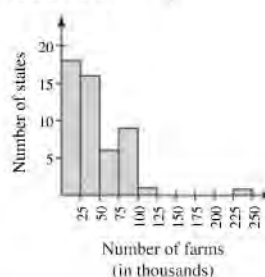
3.



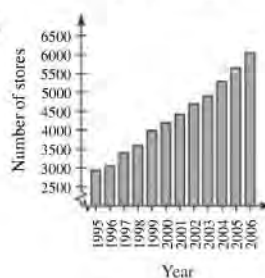
15

5. Sample answer:

Interval	Tally
[0, 25)	
[25, 50)	
[50, 75)	
[75, 100)	
[100, 125)	
[125, 150)	
[150, 175)	
[175, 200)	
[200, 225)	
[225, 250)	



7.



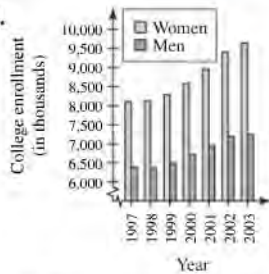
Answers will vary. Sample answer: As time progresses from 1995 to 2006, the number of Wal-Mart stores increases at a fairly constant rate.

9.

Year	1999	2000	2001
Difference in tuition charges (in dollars)	10,998	11,575	12,438

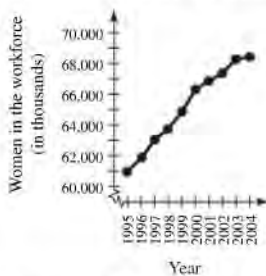
Year	2002	2003	2004
Difference in tuition charges (in dollars)	13,042	13,480	14,129

11.



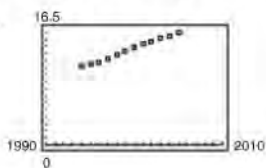
13. 118% 15. \$2.59; January

17.

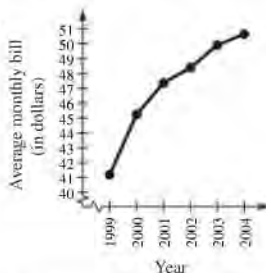


Answers will vary. Sample answer: As time progresses from 1995 to 2004, the number of women in the workforce increases at a fairly constant rate.

19.

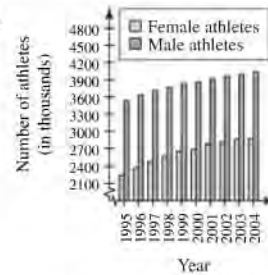


21.



Answers will vary. Sample answer: A line graph is best because the data are amounts that increased or decreased from year to year.

23.



Answers will vary. Sample answer: A double bar graph is best because there are two different sets of data within the same time interval that do not deal primarily with increasing or decreasing behavior.

25. Line plots are useful for ordering small sets.

Histograms or bar graphs can be used to organize larger sets.

Line graphs are used to show trends over periods of time.

Appendix C.1 (page A91)

Vocabulary Check (page A91)

- measure, central tendency
- modes, bimodal
- variance, standard deviation
- Quartiles

- Mean: ≈ 8.86 ; median: 8; mode: 7
- Mean: ≈ 10.29 ; median: 8; mode: 7
- Mean: 9; median: 8; mode: 7
- (a) The mean is sensitive to extreme values.
(b) Mean: ≈ 14.86 ; median: 14; mode: 13
Each is increased by 6.

- (c) Each will increase by k .
- Mean: 320; median: 320; mode: 320
- (a) Jay: 199.67, Hank: 199.67, Buck: 229.33
(b) 209.56 (c) 202

13. Answers will vary. Sample answer: {4, 4, 10}

15. The median gives the most representative description.

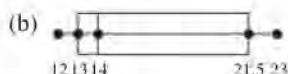
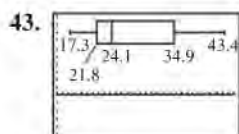
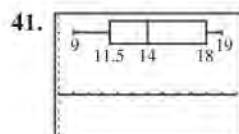
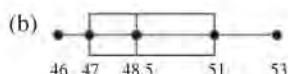
- (a) $\bar{x} = 12$; $\sigma \approx 2.83$ (b) $\bar{x} = 20$; $\sigma \approx 2.83$
(c) $\bar{x} = 12$; $\sigma \approx 1.41$ (d) $\bar{x} = 9$; $\sigma \approx 1.41$

19. $\bar{x} = 6$, $v = 10$, $\sigma \approx 3.16$ 21. $\bar{x} = 2$, $v = \frac{4}{3}$, $\sigma \approx 1.15$ 23. $\bar{x} = 4$, $v = 4$, $\sigma = 2$ 25. $\bar{x} = 47$, $v = 226$, $\sigma \approx 15.03$ 27. ≈ 3.42 29. ≈ 1.65 31. $\bar{x} = 12$ and $|x_i - 12| = 8$ for all x_i .

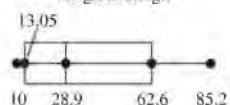
33. The mean will increase by 5, but the standard deviation will not change.

35. $[179, 291]; [151, 319]$

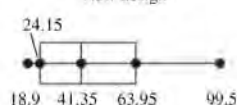
$[203, 267]; [187, 283]$

37. (a) Upper quartile: 21.5
Lower quartile: 13

39. (a) Upper quartile: 51
Lower quartile: 47


45. Original design



New design



From the plots, you can see that the lifetimes of the sample units made by the new design are greater than the lifetimes of the sample units made by the original design. (The median lifetime increased by more than 12 months.)

Appendix C.2 (page A95)

1. $y = 1.6x + 7.5$ 3. $y = 0.262x + 1.93$

Appendix D (page A100)

Vocabulary Check (page A100)

1. directly proportional
2. constant, variation
3. directly proportional
4. inverse
5. combined
6. jointly proportional

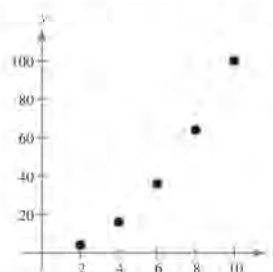
1. $y = \frac{12}{5}x$ 3. $y = 205x$

5. Model: $y = \frac{33}{13}x$; 25.38 centimeters, 50.77 centimeters

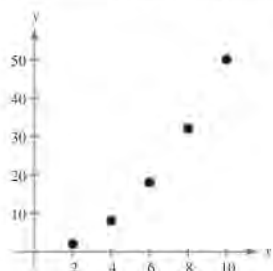
7. $y = 0.0368x$; \$7360

9. (a) 0.05 meter (b) $176\frac{2}{3}$ newtons

x	2	4	6	8	10
$y = kx^2$	4	16	36	64	100

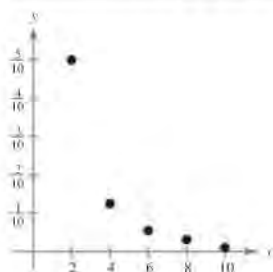


x	2	4	6	8	10
$y = kx^2$	2	8	28	32	50



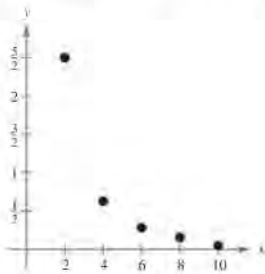
15. 0.61 mile per hour

x	2	4	6	8	10
$y = k/x^2$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{18}$	$\frac{1}{32}$	$\frac{1}{50}$



19.

$\bar{x}$	2	4	6	8	10
$y = k/x^2$	$\frac{5}{2}$	$\frac{5}{8}$	$\frac{5}{18}$	$\frac{5}{32}$	$\frac{1}{10}$



21. $y = \frac{5}{x}$ 23. $y = -\frac{7}{10}x$ 25. $A = kr^2$

27. $y = \frac{k}{x^2}$ 29. $F = \frac{kg}{r^2}$ 31. $P = \frac{k}{V}$

33. $R = k(T_e - T)$

35. The area of a triangle is jointly proportional to its base and height.

37. The volume of a sphere varies directly as the cube of its radius.

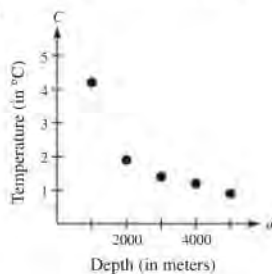
39. Average speed is directly proportional to the distance and inversely proportional to the time.

41. $A = \pi r^2$ 43. $y = \frac{28}{x}$ 45. $F = 14rs^3$

47. $z = \frac{2x^2}{3y}$ 49. 506.27 feet 51. 1470 joules

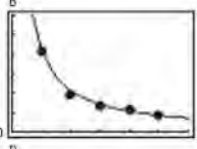
53. The velocity is increased by 4.

55. (a)



(b) Yes. $k_1 = 4200$, $k_2 = 3800$, $k_3 = 4200$,
 $k_4 = 4800$, $k_5 = 4500$

(c) $C = \frac{4300}{d}$

(d)  (e) ≈ 1433 meters

57. False. y will increase if k is positive and y will decrease if k is negative.

59. Inversely

Appendix E (page A105)

Vocabulary Check (page A105)

1. linear 2. equivalent inequalities

1. 4 3. 7 5. 4 7. 20 9. 4 11. 3

13. 5 15. -10 17. No solution 19. $-\frac{6}{5}$

21. 9 23. $x < 2$ 25. $x < 9$ 27. $x \leq -14$

29. $x > 10$ 31. $x < 4$ 33. $x < 3$ 35. $x \geq 2$

37. $x \leq -5$ 39. $x < 6$ 41. $x \geq 4$ 43. $x \geq -4$

Appendix F.1 (page A113)

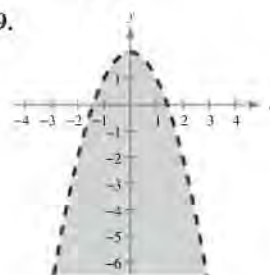
Vocabulary Check (page A113)

1. solution 2. graph 3. linear
4. point, equilibrium

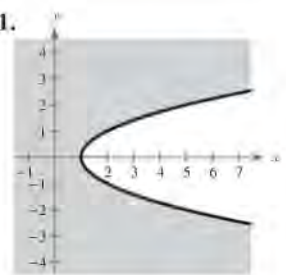
1. g 2. d 3. a 4. h 5. e 6. b

7. f 8. c

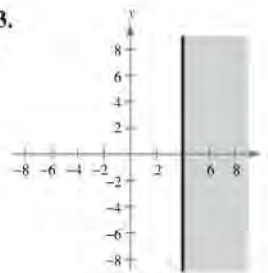
9.



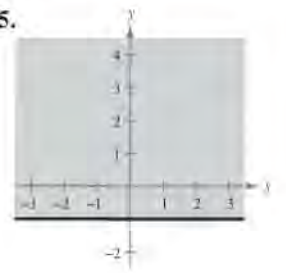
11.

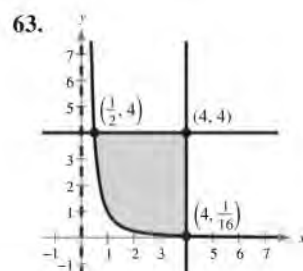
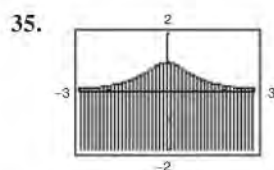
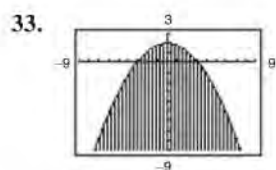
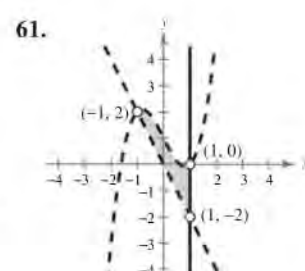
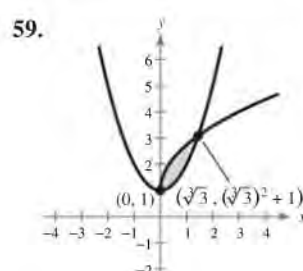
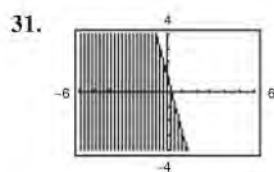
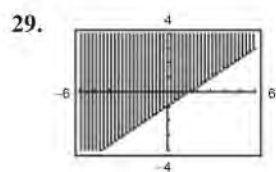
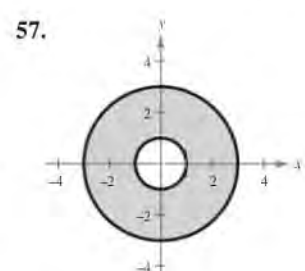
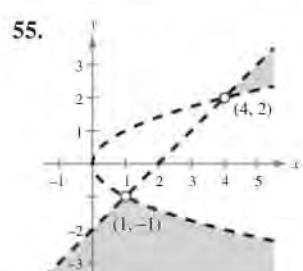
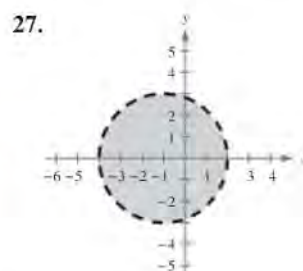
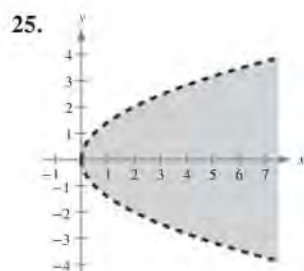
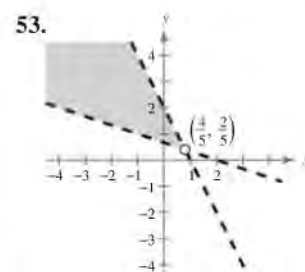
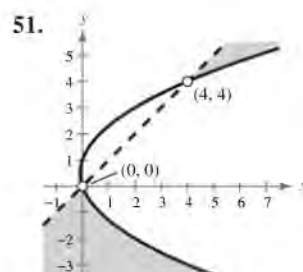
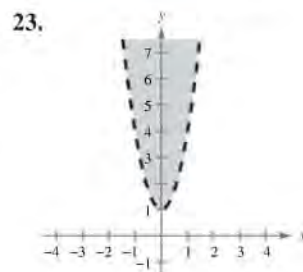
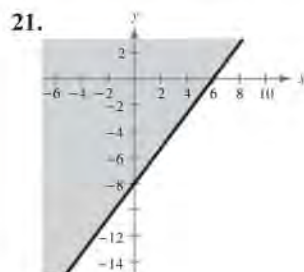
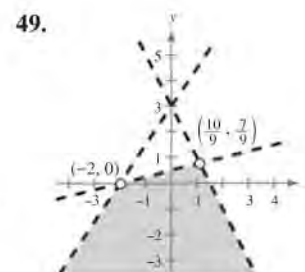
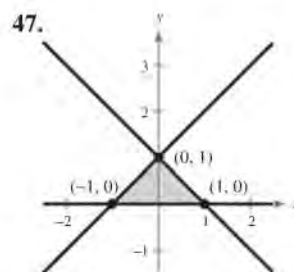
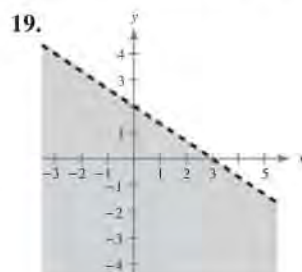
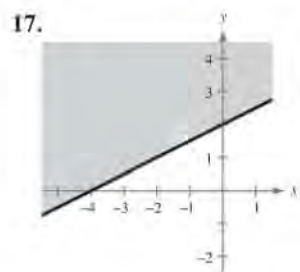


13.



15.





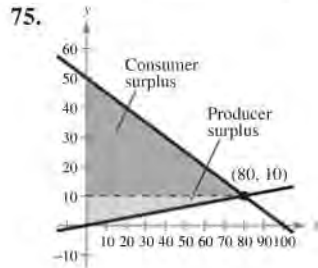
65.
$$\begin{cases} \frac{1}{4}x + \frac{1}{4}y < 1 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

41. $\frac{x}{3} + \frac{y}{2} > 1$ 43. $x^2 + y^2 \leq 9$

45. (a) Yes (b) No (c) No (d) No

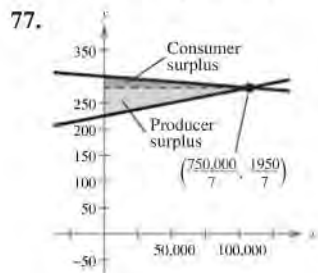
67.
$$\begin{cases} y \leq 4 - x \\ y \leq 2 - \frac{1}{4}x \\ x \geq 0 \\ y \geq 0 \end{cases}$$
 69. $x^2 + (y - 2)^2 \leq 4$

71. $\begin{cases} 2 \leq x \leq 5 \\ 1 \leq y \leq 7 \end{cases}$ 73. $\begin{cases} y \leq \frac{3}{2}x \\ y \leq -x + 5 \\ y \geq 0 \end{cases}$



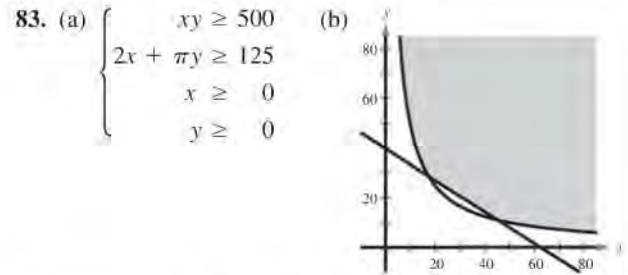
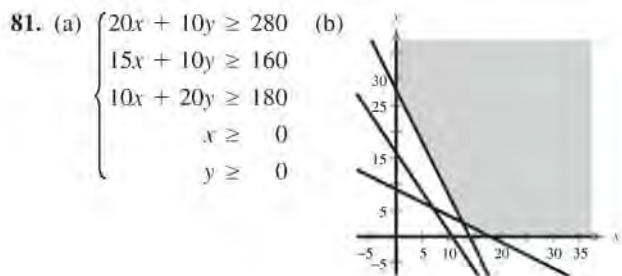
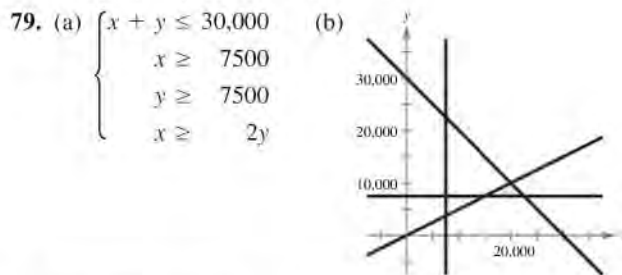
Consumer surplus: 1600

Producer surplus: 400



Consumer surplus: $\approx 1,147,959.18$

Producer surplus: $\approx 2,869,897.96$



85. True 87. Test a point on either side.

Appendix F.2 (page A122)

Vocabulary Check (page A122)

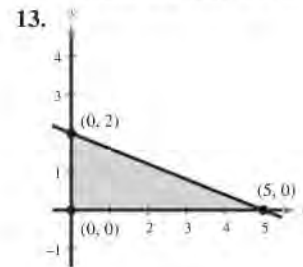
1. optimization
2. objective function
3. constraints, feasible solutions

1. Minimum at (0, 0): 0 3. Minimum at (0, 0): 0
Maximum at (0, 6): 30 Maximum at (6, 0): 60

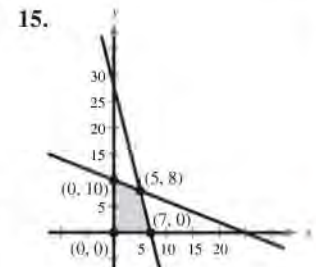
5. Minimum at (0, 0): 0 7. Minimum at (0, 0): 0
Maximum at (3, 4): 17 Maximum at (4, 0): 20

9. Minimum at (0, 0): 0
Maximum at (60, 20): 740

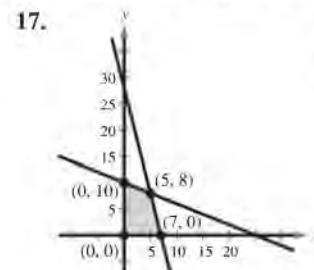
11. Minimum at (0, 0): 0
Maximum at any point on the line segment connecting (60, 20) and (30, 45): 2100



Minimum at (0, 0): 0
Maximum at (5, 0): 30

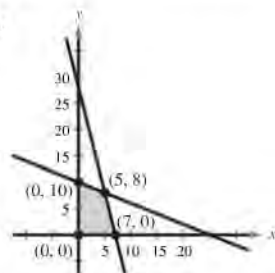


Minimum at (0, 0): 0
Maximum at (5, 8): 47



Minimum at (0, 0): 0
Maximum at (5, 8): 21

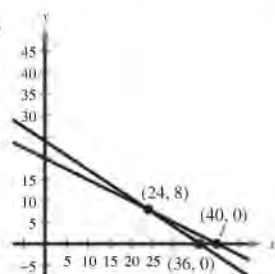
19.



Minimum at any point on the line segment connecting $(0, 0)$ and $(0, 10)$: 0

Maximum at $(7, 0)$: 14

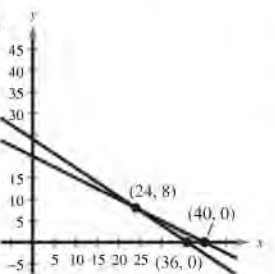
21.



Minimum at $(24, 8)$: 104

Maximum at $(40, 0)$: 160

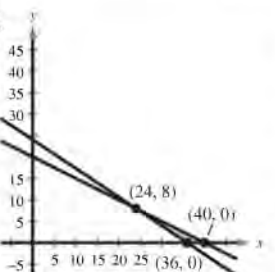
23.



Minimum at $(36, 0)$: 36

Maximum at $(24, 8)$: 56

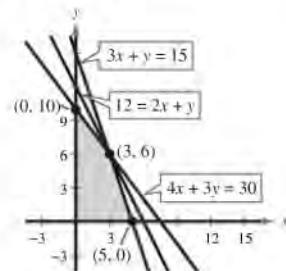
25.



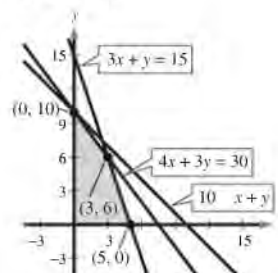
Minimum at any point on the line segment connecting $(24, 8)$ and $(36, 0)$: 72

Maximum at $(40, 0)$: 80

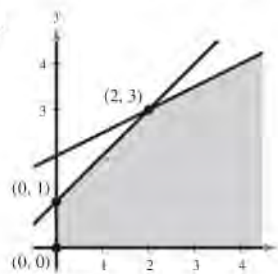
27. (a) and (b)

(c) $(3, 6)$


29. (a) and (b)

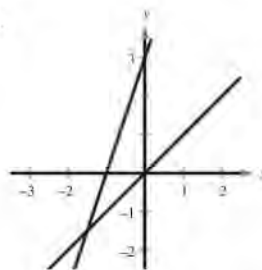
(c) $(0, 10)$


31.



The constraints do not form a closed set of points. Therefore, $z = x + y$ is unbounded.

33.



The feasible set is empty.

35. (a) Four audits, 32 tax returns

(b) Maximum revenue: \$17,600

37. (a) Three bags of brand X, six bags of brand Y

(b) Minimum cost: \$195

39. True 41. $z = x + 5y$ 43. $z = 4x + y$

45. (a) $t > 9$ (b) $\frac{3}{4} < t < 9$